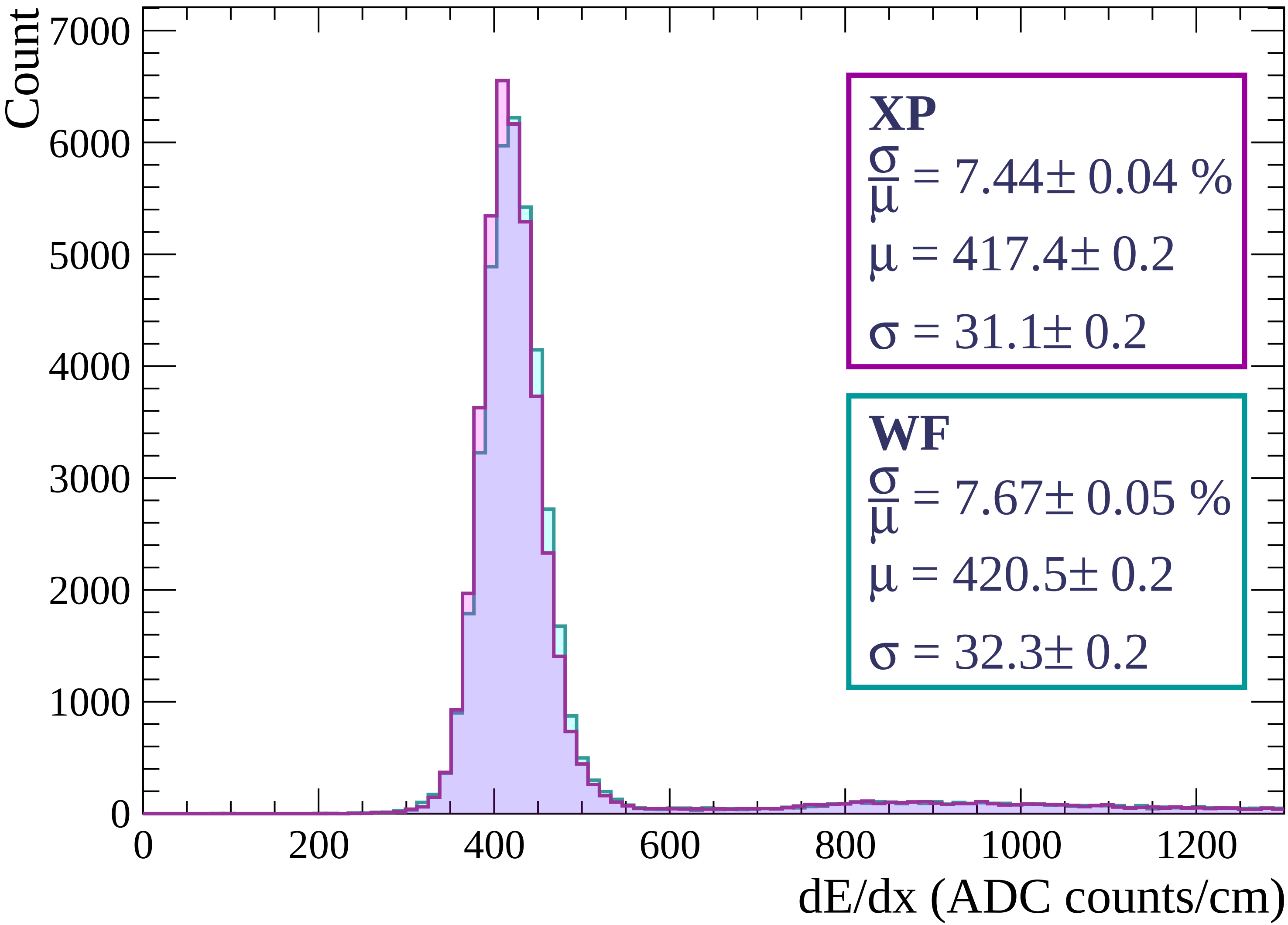
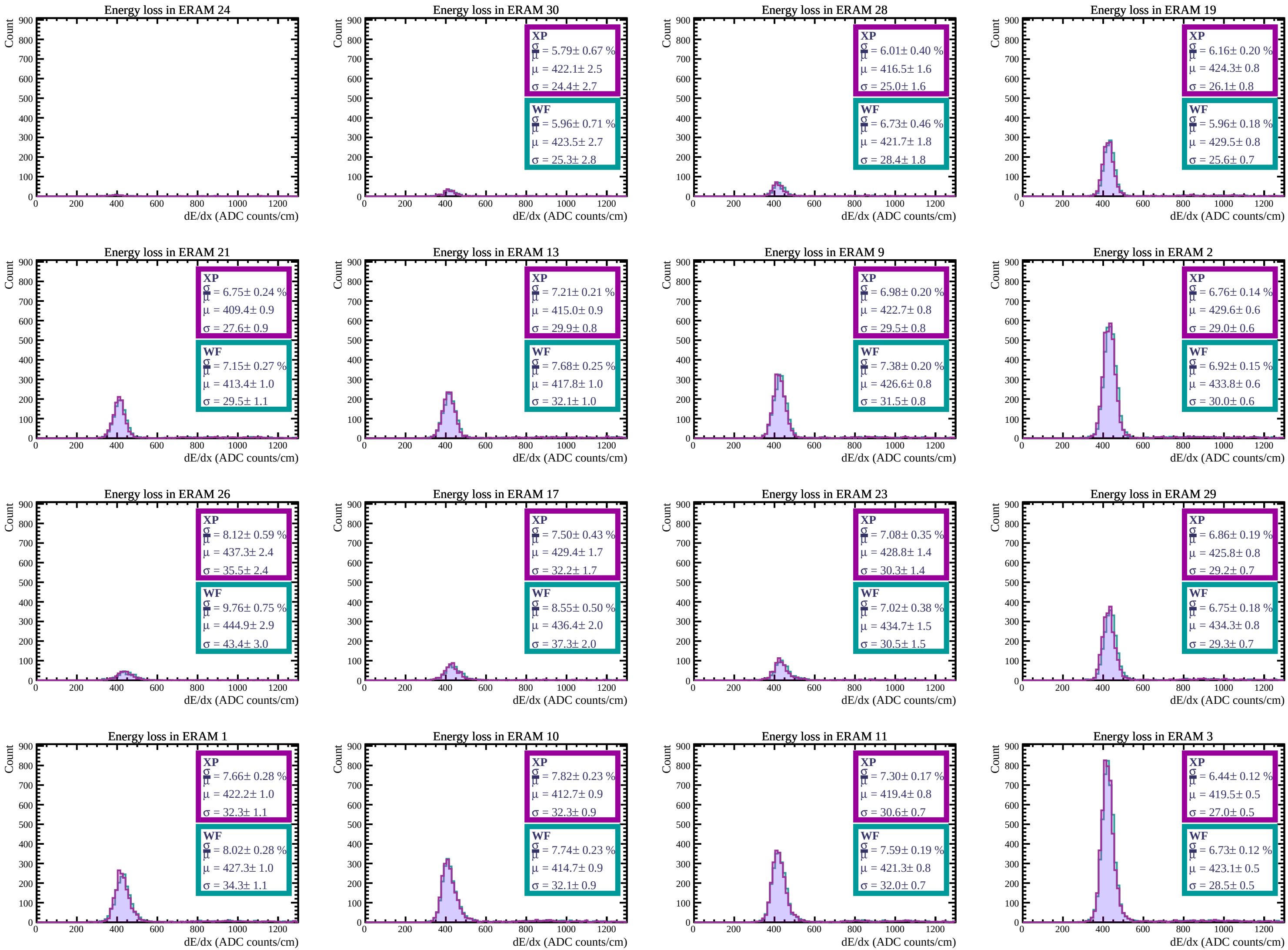
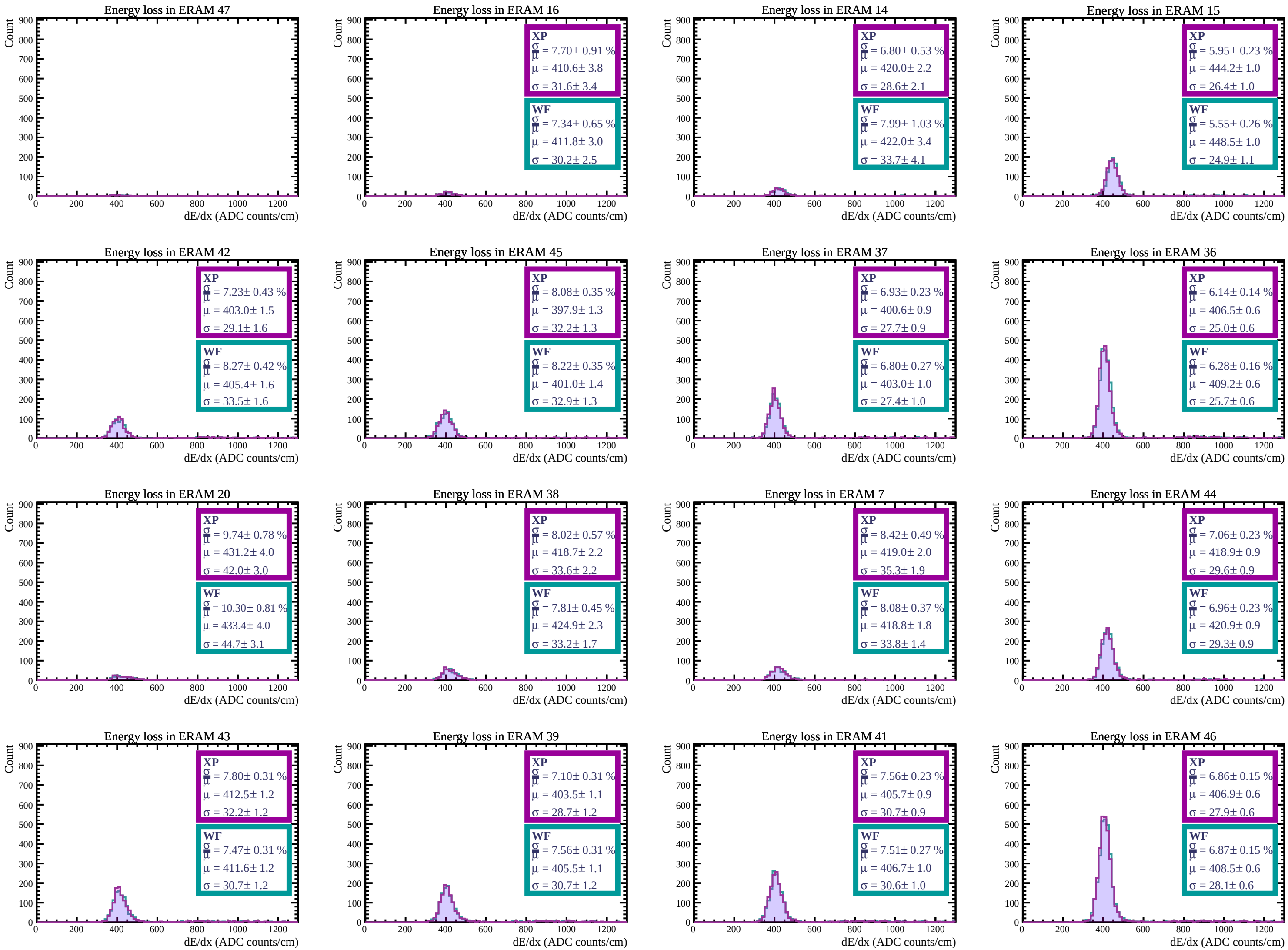
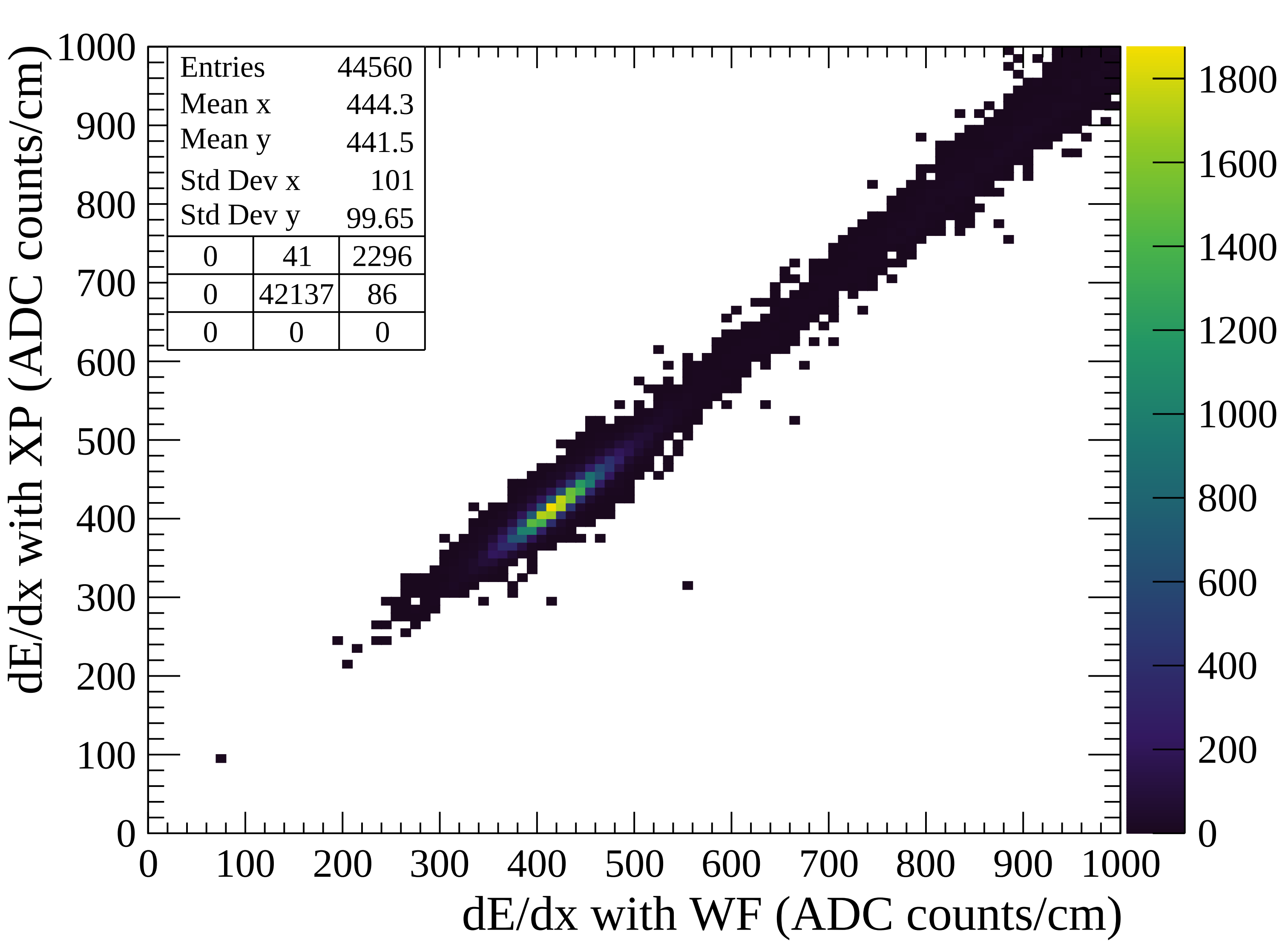


Count
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.44± 0.04 %
μ = 417.4± 0.2
σ = 31.1± 0.2
WF
$\frac{\sigma}{\mu}$ = 7.67± 0.05 %
μ = 420.5± 0.2
σ = 32.3± 0.2

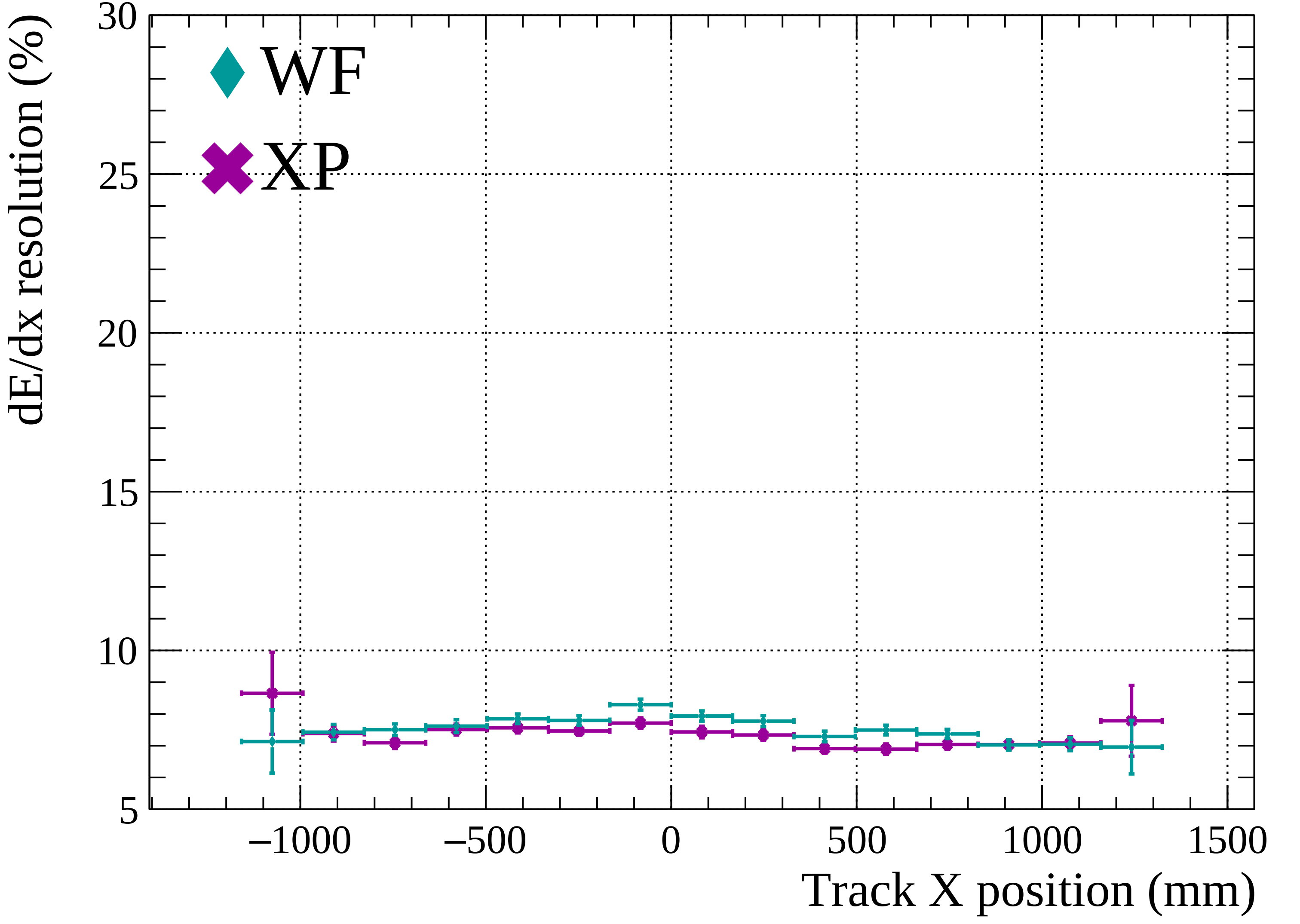

Energy loss in ERAM 24
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 30
XP
σ/μ = 5.79± 0.67 %
μ = 422.1± 2.5
σ = 24.4± 2.7
WF
σ/μ = 5.96± 0.71 %
μ = 423.5± 2.7
σ = 25.3± 2.8
Energy loss in ERAM 28
XP
σ/μ = 6.01± 0.40 %
μ = 416.5± 1.6
σ = 25.0± 1.6
WF
σ/μ = 6.73± 0.46 %
μ = 421.7± 1.8
σ = 28.4± 1.8
Energy loss in ERAM 19
XP
σ/μ = 6.16± 0.20 %
μ = 424.3± 0.8
σ = 26.1± 0.8
WF
σ/μ = 5.96± 0.18 %
μ = 429.5± 0.8
σ = 25.6± 0.7
Energy loss in ERAM 21
XP
σ/μ = 6.75± 0.24 %
μ = 409.4± 0.9
σ = 27.6± 0.9
WF
σ/μ = 7.15± 0.27 %
μ = 413.4± 1.0
σ = 29.5± 1.1
Energy loss in ERAM 13
XP
σ/μ = 7.21± 0.21 %
μ = 415.0± 0.9
σ = 29.9± 0.8
WF
σ/μ = 7.68± 0.25 %
μ = 417.8± 1.0
σ = 32.1± 1.0
Energy loss in ERAM 9
XP
σ/μ = 6.98± 0.20 %
μ = 422.7± 0.8
σ = 29.5± 0.8
WF
σ/μ = 7.38± 0.20 %
μ = 426.6± 0.8
σ = 31.5± 0.8
Energy loss in ERAM 2
XP
σ/μ = 6.76± 0.14 %
μ = 429.6± 0.6
σ = 29.0± 0.6
WF
σ/μ = 6.92± 0.15 %
μ = 433.8± 0.6
σ = 30.0± 0.6
Energy loss in ERAM 26
XP
σ/μ = 8.12± 0.59 %
μ = 437.3± 2.4
σ = 35.5± 2.4
WF
σ/μ = 9.76± 0.75 %
μ = 444.9± 2.9
σ = 43.4± 3.0
Energy loss in ERAM 17
XP
σ/μ = 7.50± 0.43 %
μ = 429.4± 1.7
σ = 32.2± 1.7
WF
σ/μ = 8.55± 0.50 %
μ = 436.4± 2.0
σ = 37.3± 2.0
Energy loss in ERAM 23
XP
σ/μ = 7.08± 0.35 %
μ = 428.8± 1.4
σ = 30.3± 1.4
WF
σ/μ = 7.02± 0.38 %
μ = 434.7± 1.5
σ = 30.5± 1.5
Energy loss in ERAM 29
XP
σ/μ = 6.86± 0.19 %
μ = 425.8± 0.8
σ = 29.2± 0.7
WF
σ/μ = 6.75± 0.18 %
μ = 434.3± 0.8
σ = 29.3± 0.7
Energy loss in ERAM 1
XP
σ/μ = 7.66± 0.28 %
μ = 422.2± 1.0
σ = 32.3± 1.1
WF
σ/μ = 8.02± 0.28 %
μ = 427.3± 1.0
σ = 34.3± 1.1
Energy loss in ERAM 10
XP
σ/μ = 7.82± 0.23 %
μ = 412.7± 0.9
σ = 32.3± 0.9
WF
σ/μ = 7.74± 0.23 %
μ = 414.7± 0.9
σ = 32.1± 0.9
Energy loss in ERAM 11
XP
σ/μ = 7.30± 0.17 %
μ = 419.4± 0.8
σ = 30.6± 0.7
WF
σ/μ = 7.59± 0.19 %
μ = 421.3± 0.8
σ = 32.0± 0.7
Energy loss in ERAM 3
XP
σ/μ = 6.44± 0.12 %
μ = 419.5± 0.5
σ = 27.0± 0.5
WF
σ/μ = 6.73± 0.12 %
μ = 423.1± 0.5
σ = 28.5± 0.5



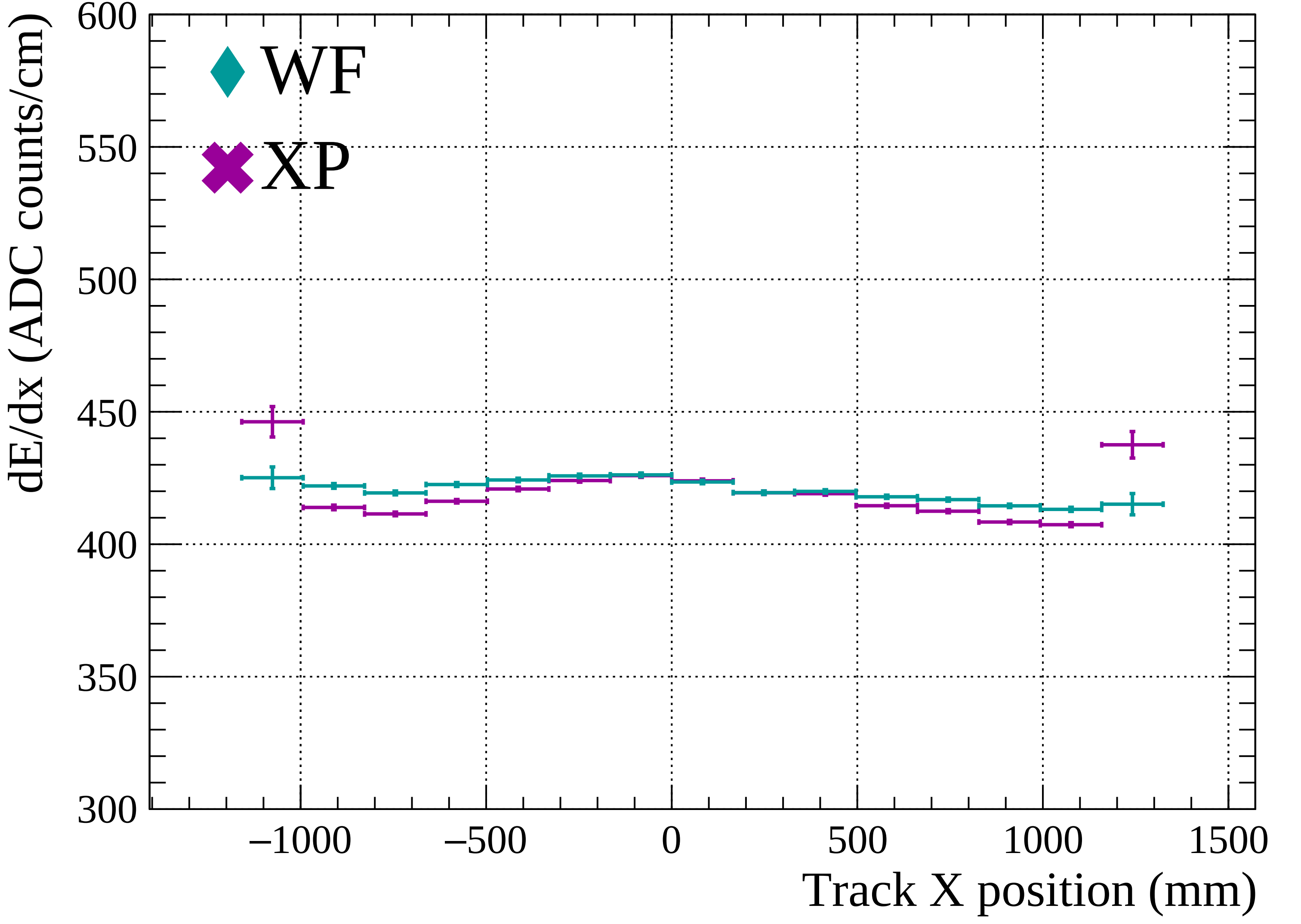
Energy loss in ERAM 47
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 16
XP
σ/μ = 7.70± 0.91 %
μ = 410.6± 3.8
σ = 31.6± 3.4
WF
σ/μ = 7.34± 0.65 %
μ = 411.8± 3.0
σ = 30.2± 2.5
Energy loss in ERAM 14
XP
σ/μ = 6.80± 0.53 %
μ = 420.0± 2.2
σ = 28.6± 2.1
WF
σ/μ = 7.99± 1.03 %
μ = 422.0± 3.4
σ = 33.7± 4.1
Energy loss in ERAM 15
XP
σ/μ = 5.95± 0.23 %
μ = 444.2± 1.0
σ = 26.4± 1.0
WF
σ/μ = 5.55± 0.26 %
μ = 448.5± 1.0
σ = 24.9± 1.1
Energy loss in ERAM 42
XP
σ/μ = 7.23± 0.43 %
μ = 403.0± 1.5
σ = 29.1± 1.6
WF
σ/μ = 8.27± 0.42 %
μ = 405.4± 1.6
σ = 33.5± 1.6
Energy loss in ERAM 45
XP
σ/μ = 8.08± 0.35 %
μ = 397.9± 1.3
σ = 32.2± 1.3
WF
σ/μ = 8.22± 0.35 %
μ = 401.0± 1.4
σ = 32.9± 1.3
Energy loss in ERAM 37
XP
σ/μ = 6.93± 0.23 %
μ = 400.6± 0.9
σ = 27.7± 0.9
WF
σ/μ = 6.80± 0.27 %
μ = 403.0± 1.0
σ = 27.4± 1.0
Energy loss in ERAM 36
XP
σ/μ = 6.14± 0.14 %
μ = 406.5± 0.6
σ = 25.0± 0.6
WF
σ/μ = 6.28± 0.16 %
μ = 409.2± 0.6
σ = 25.7± 0.6
Energy loss in ERAM 20
XP
σ/μ = 9.74± 0.78 %
μ = 431.2± 4.0
σ = 42.0± 3.0
WF
σ/μ = 10.30± 0.81 %
μ = 433.4± 4.0
σ = 44.7± 3.1
Energy loss in ERAM 38
XP
σ/μ = 8.02± 0.57 %
μ = 418.7± 2.2
σ = 33.6± 2.2
WF
σ/μ = 7.81± 0.45 %
μ = 424.9± 2.3
σ = 33.2± 1.7
Energy loss in ERAM 7
XP
σ/μ = 8.42± 0.49 %
μ = 419.0± 2.0
σ = 35.3± 1.9
WF
σ/μ = 8.08± 0.37 %
μ = 418.8± 1.8
σ = 33.8± 1.4
Energy loss in ERAM 44
XP
σ/μ = 7.06± 0.23 %
μ = 418.9± 0.9
σ = 29.6± 0.9
WF
σ/μ = 6.96± 0.23 %
μ = 420.9± 0.9
σ = 29.3± 0.9
Energy loss in ERAM 43
XP
σ/μ = 7.80± 0.31 %
μ = 412.5± 1.2
σ = 32.2± 1.2
WF
σ/μ = 7.47± 0.31 %
μ = 411.6± 1.2
σ = 30.7± 1.2
Energy loss in ERAM 39
XP
σ/μ = 7.10± 0.31 %
μ = 403.5± 1.1
σ = 28.7± 1.2
WF
σ/μ = 7.56± 0.31 %
μ = 405.5± 1.1
σ = 30.7± 1.2
Energy loss in ERAM 41
XP
σ/μ = 7.56± 0.23 %
μ = 405.7± 0.9
σ = 30.7± 0.9
WF
σ/μ = 7.51± 0.27 %
μ = 406.7± 1.0
σ = 30.6± 1.0
Energy loss in ERAM 46
XP
σ/μ = 6.86± 0.15 %
μ = 406.9± 0.6
σ = 27.9± 0.6
WF
σ/μ = 6.87± 0.15 %
μ = 408.5± 0.6
σ = 28.1± 0.6



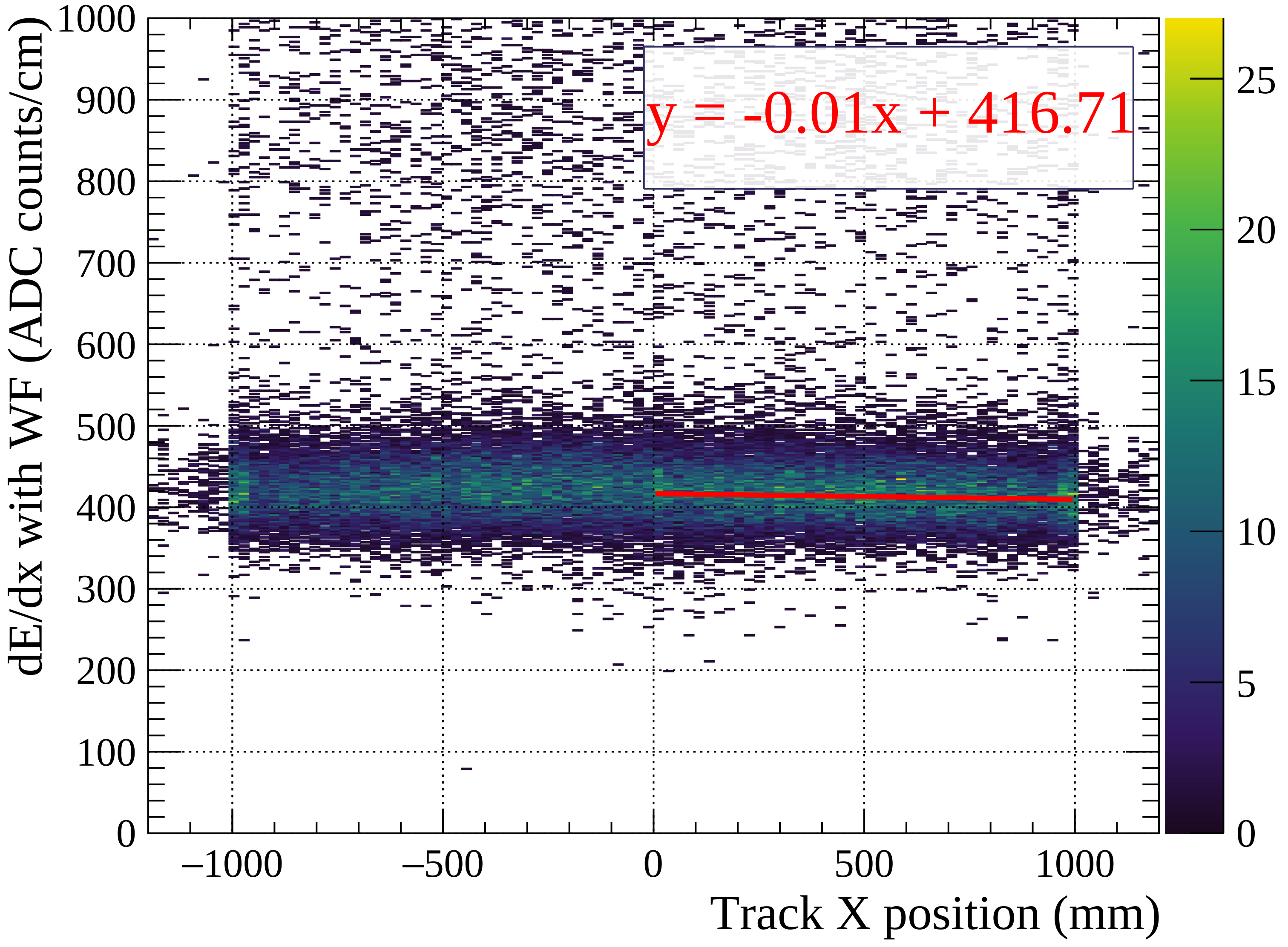
Entries 44560
Mean x 444.3
Mean y 441.5
Std Dev x 101
Std Dev y 99.65
0 41 2296
0 42137 86
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1800
1600
1400
1200
1000
800
600
400
200
0



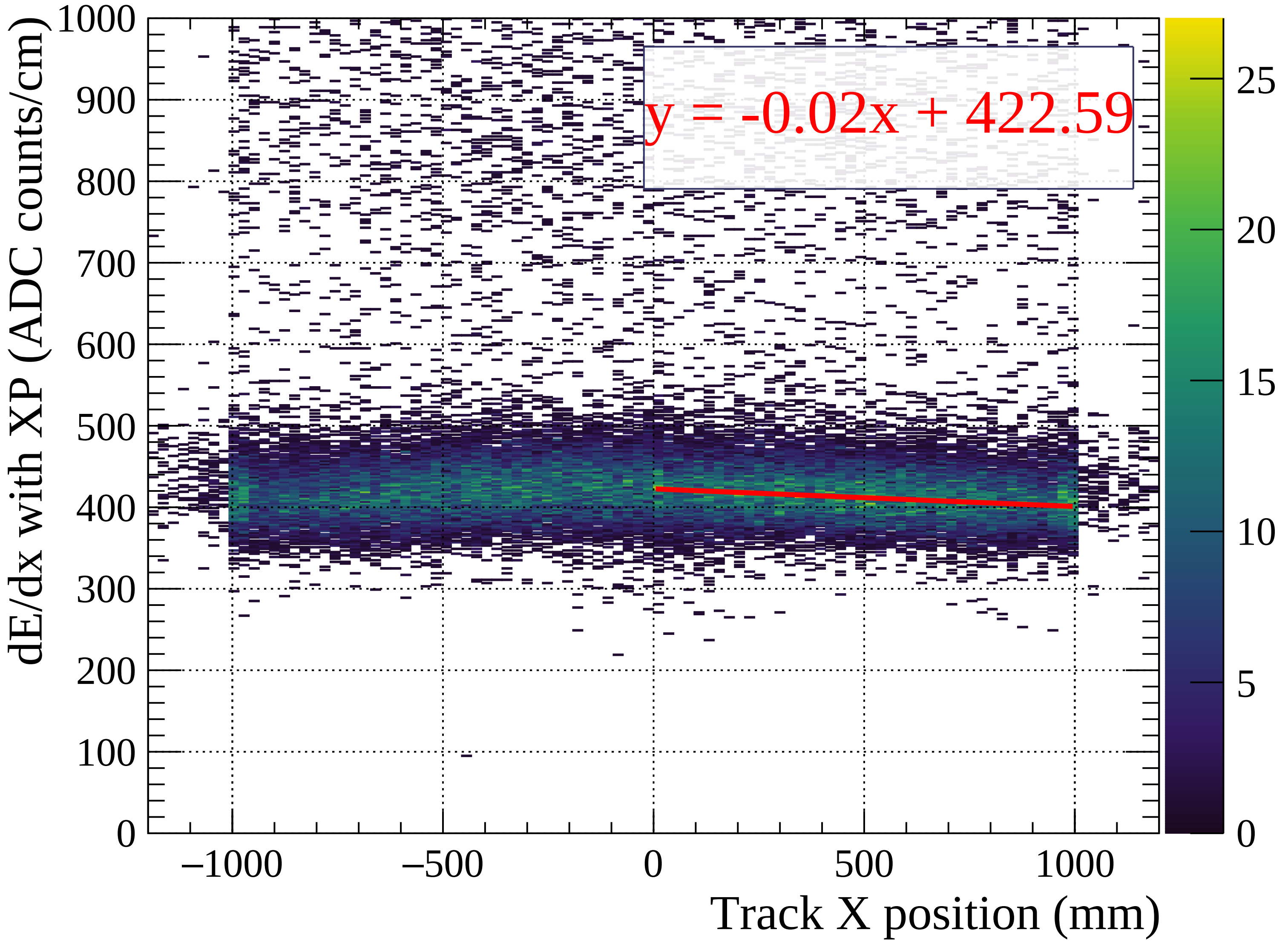
WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000
1500



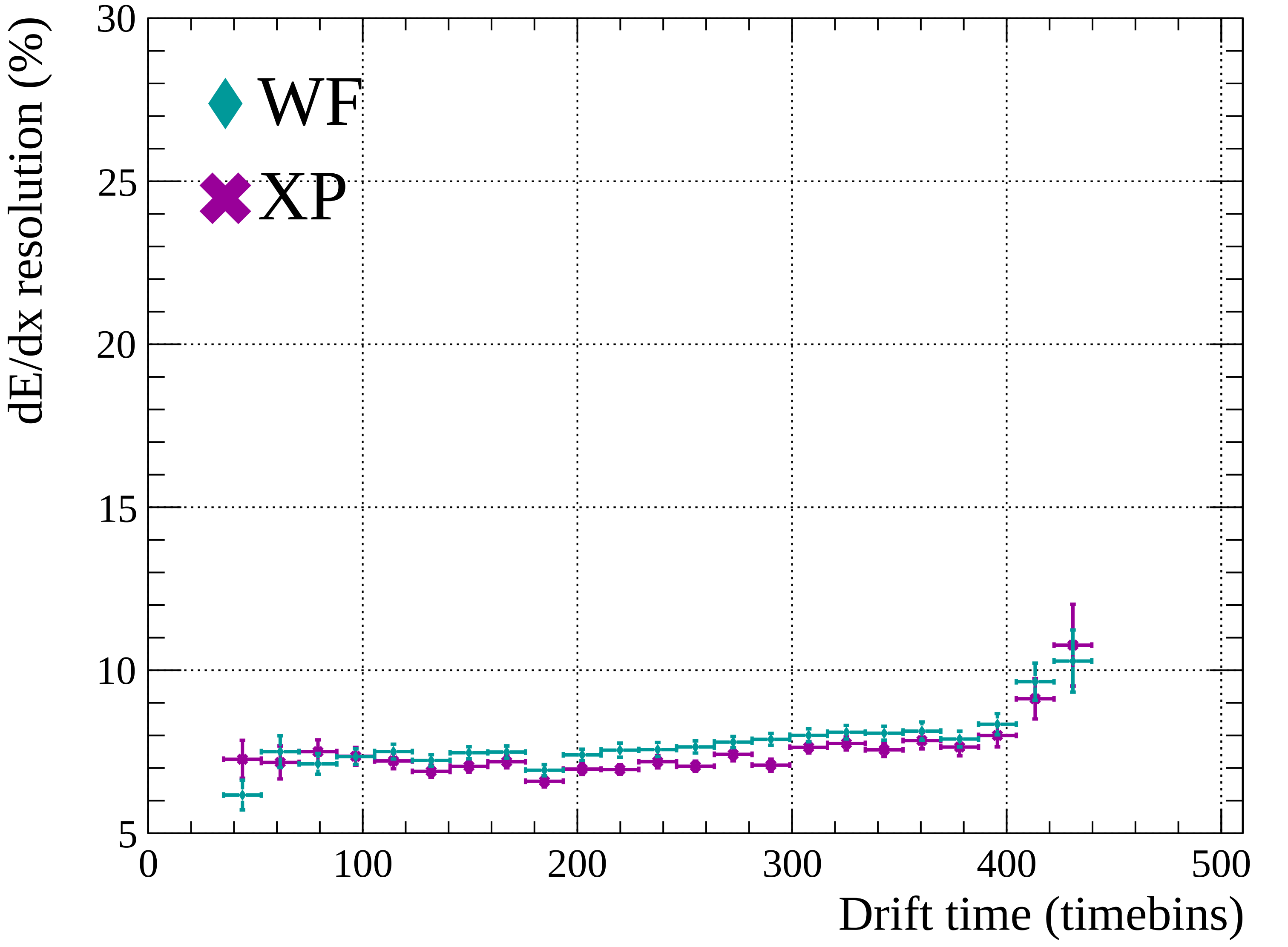
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
1500
Track X position (mm)



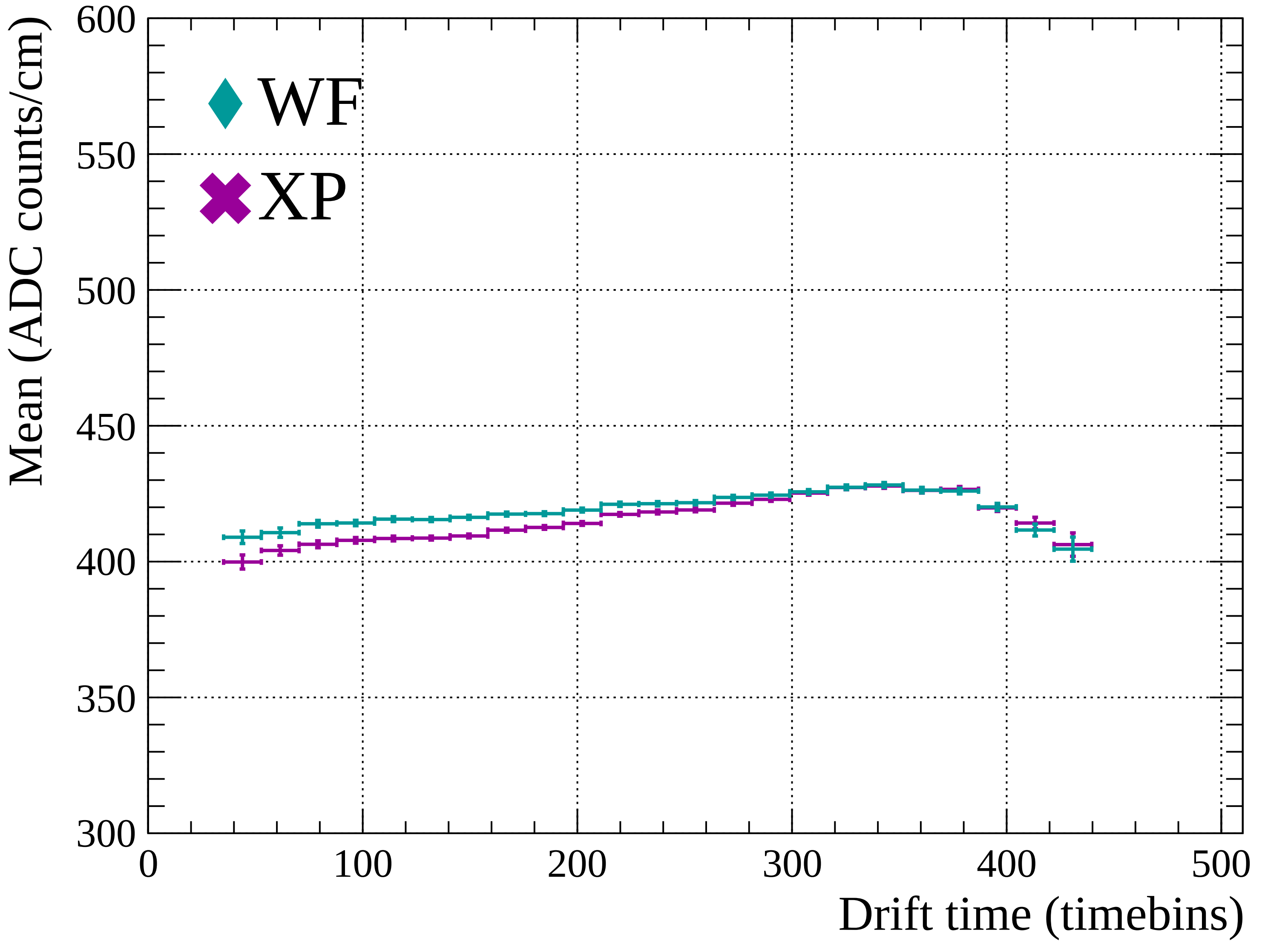
y = -0.01x + 416.71
dE/dx with WF (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
25
20
15
10
5
0

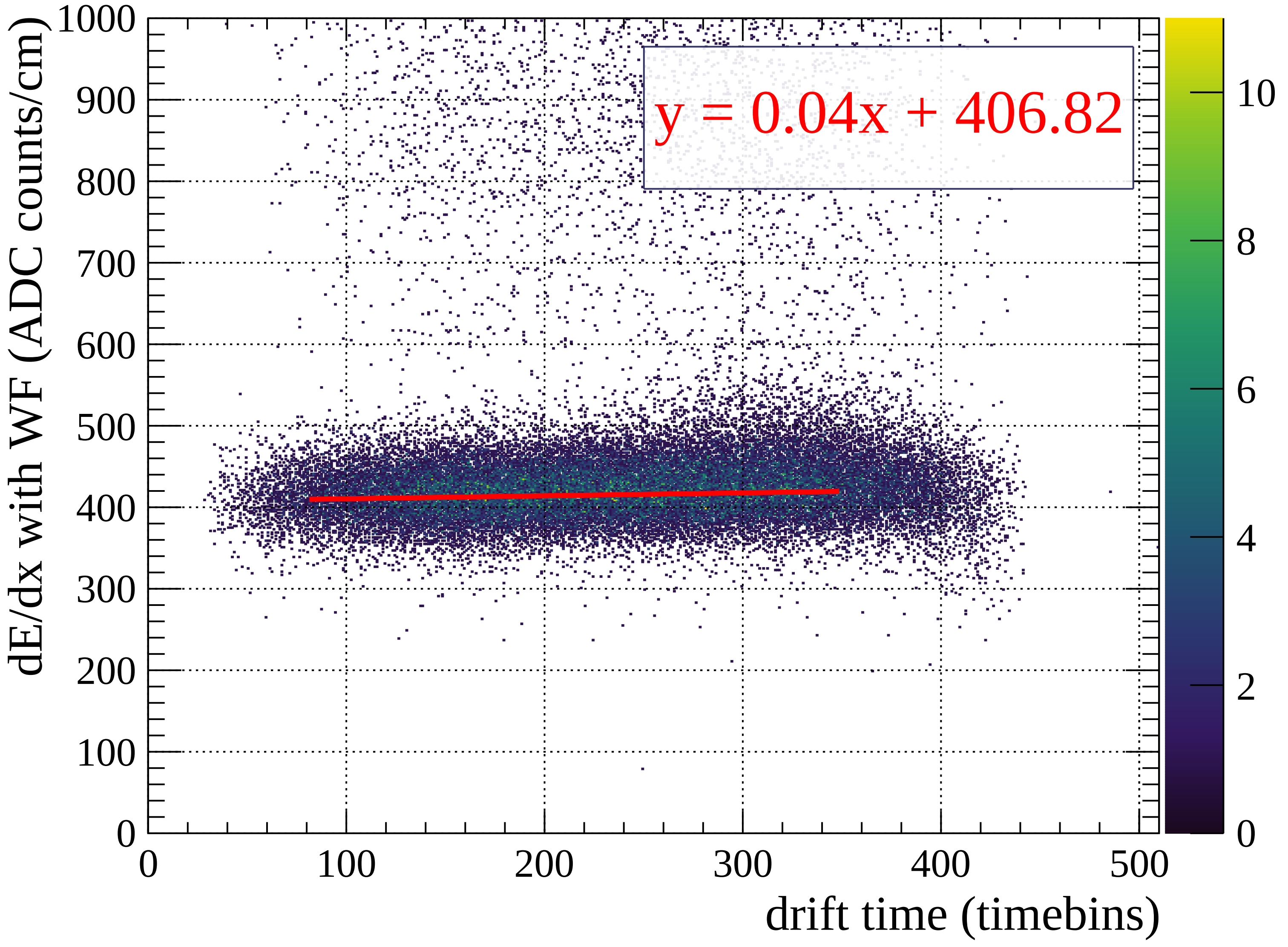


y = -0.02x + 422.59
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
5
10
15
20
25

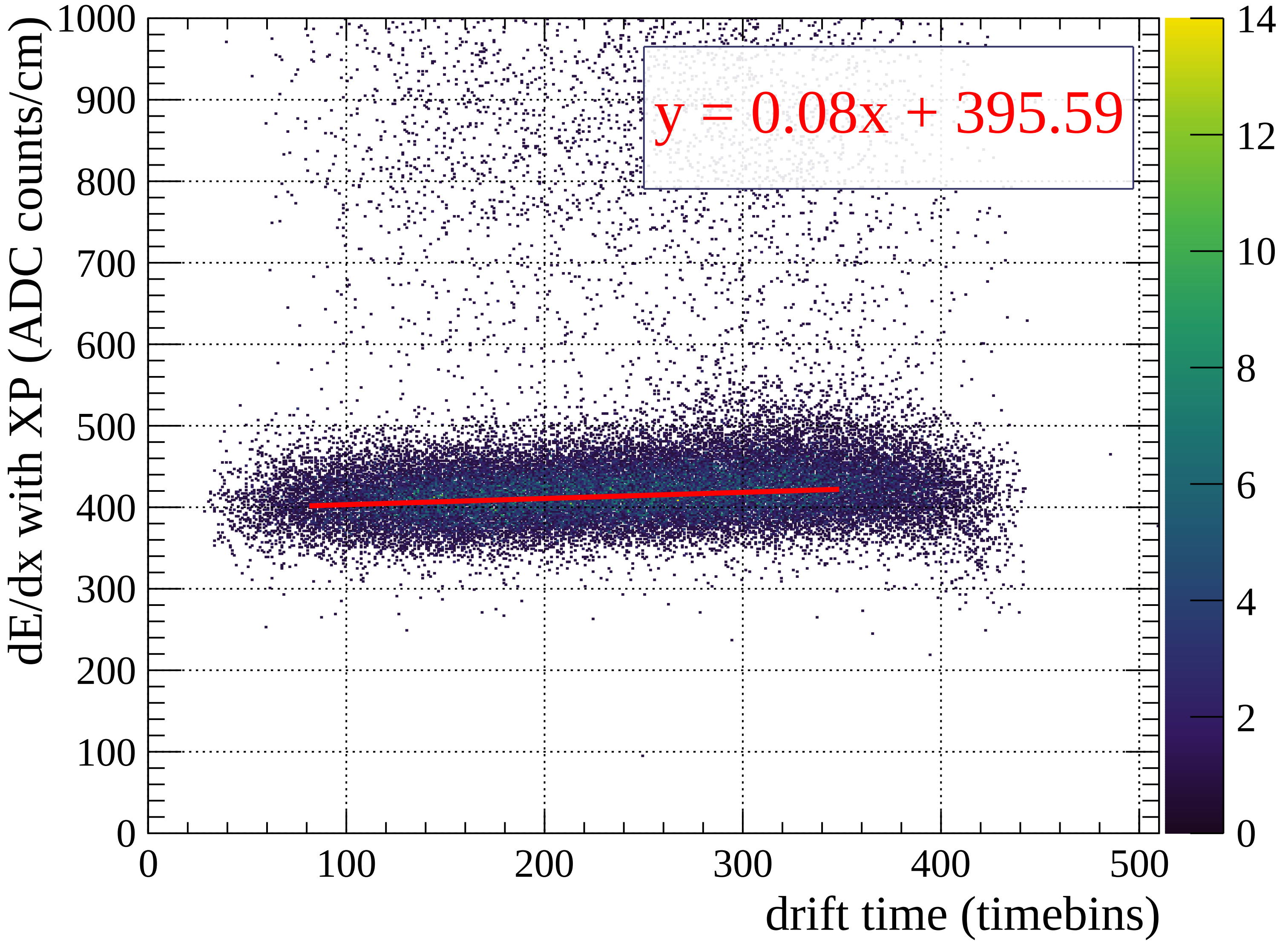


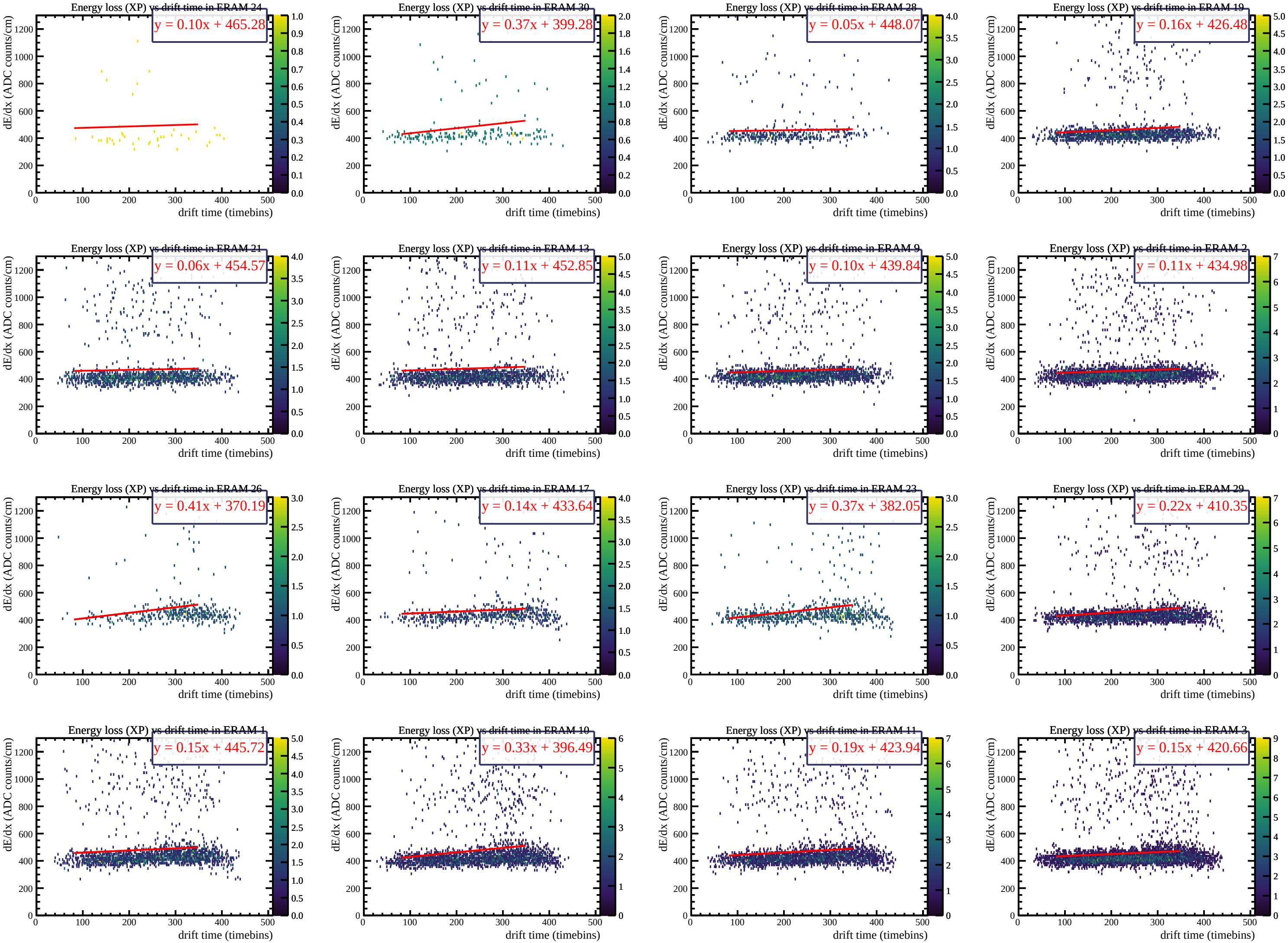
dE/dx resolution (%)
30
25
20
15
10
5
0
100
200
300
400
500
Drift time (timebins)
WF
XP

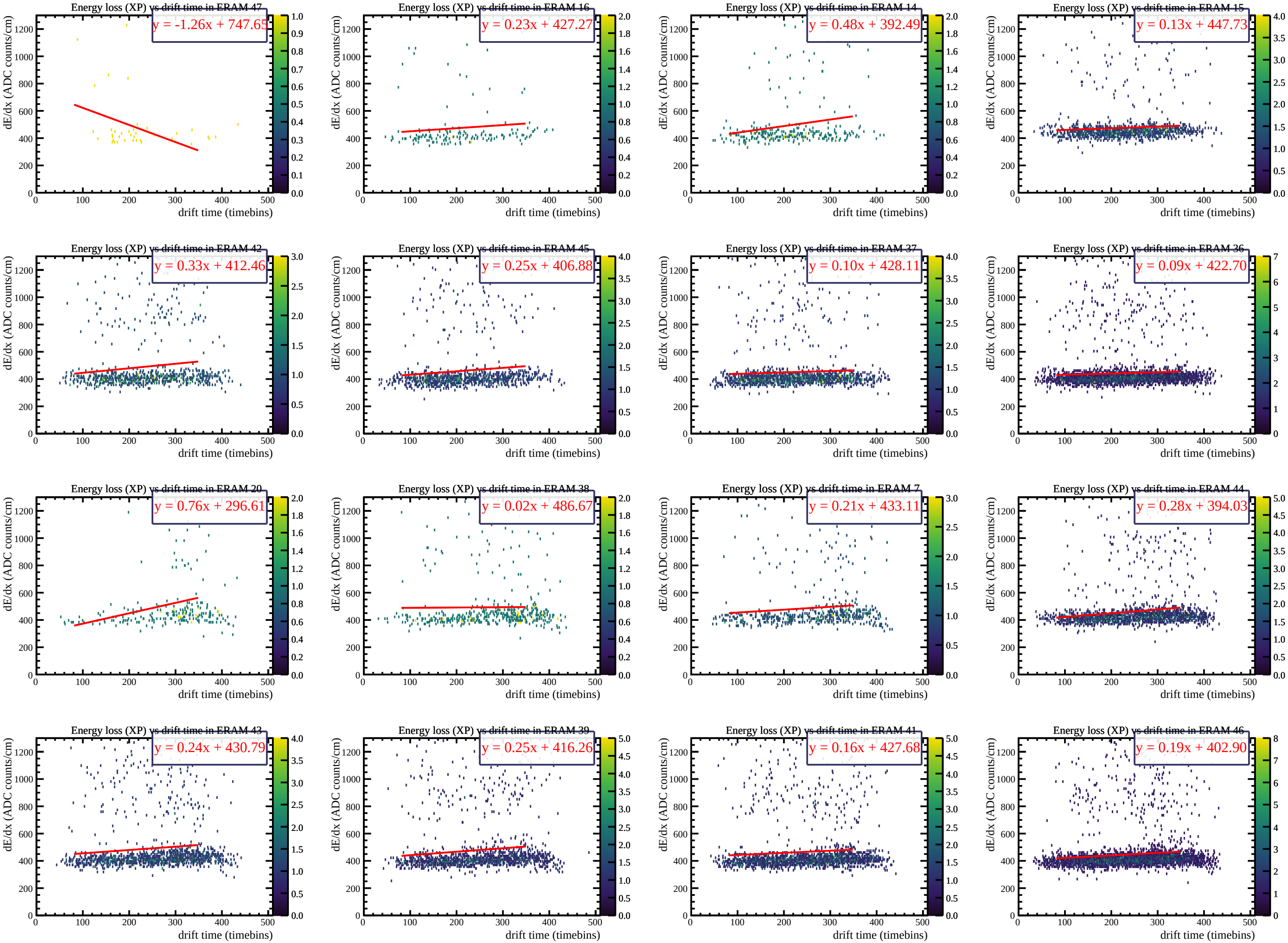

WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
300
350
400
450
500
550
600
0
100
200
300
400
500



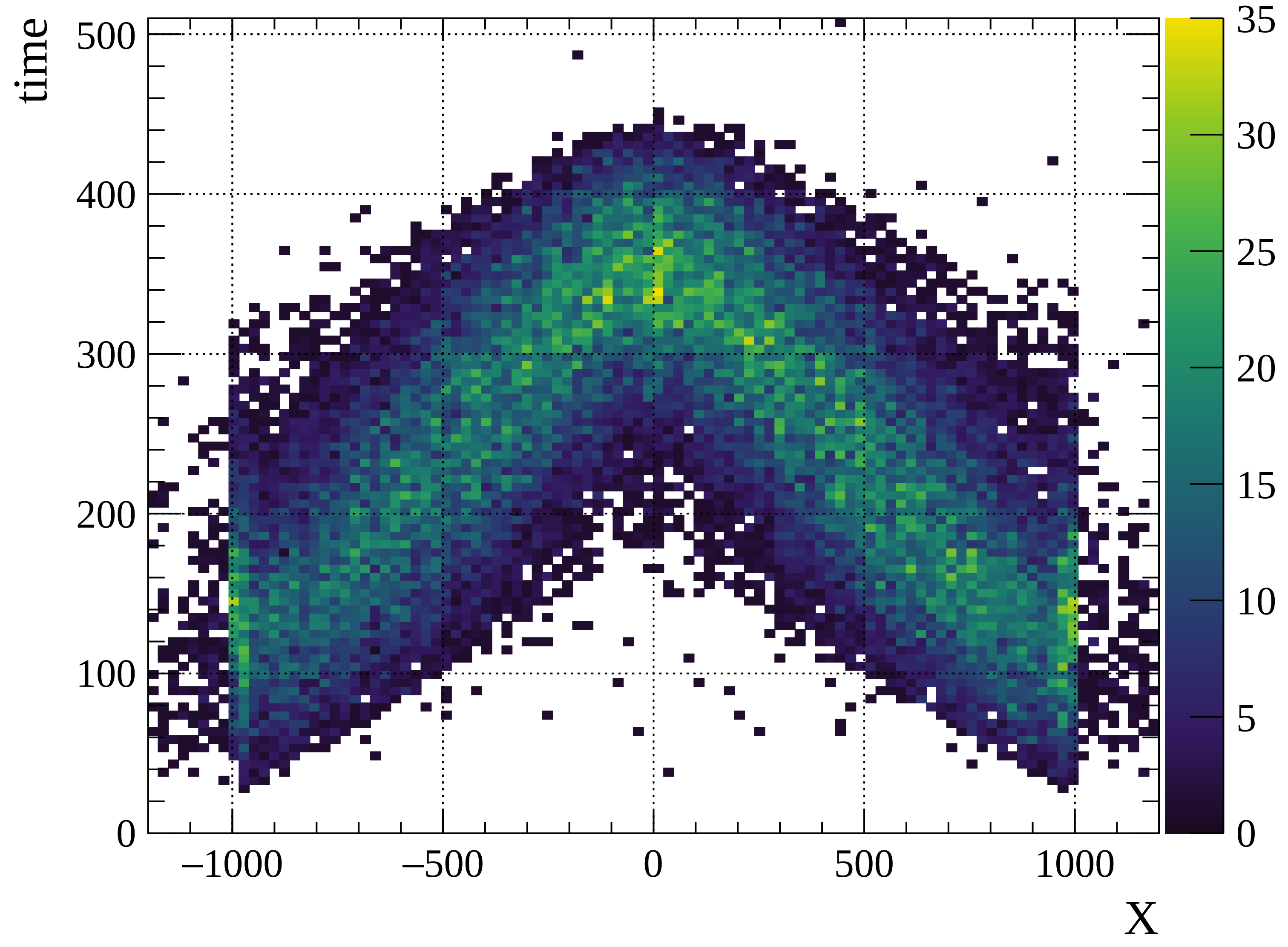
y = 0.04x + 406.82
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10


y = 0.08x + 395.59
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

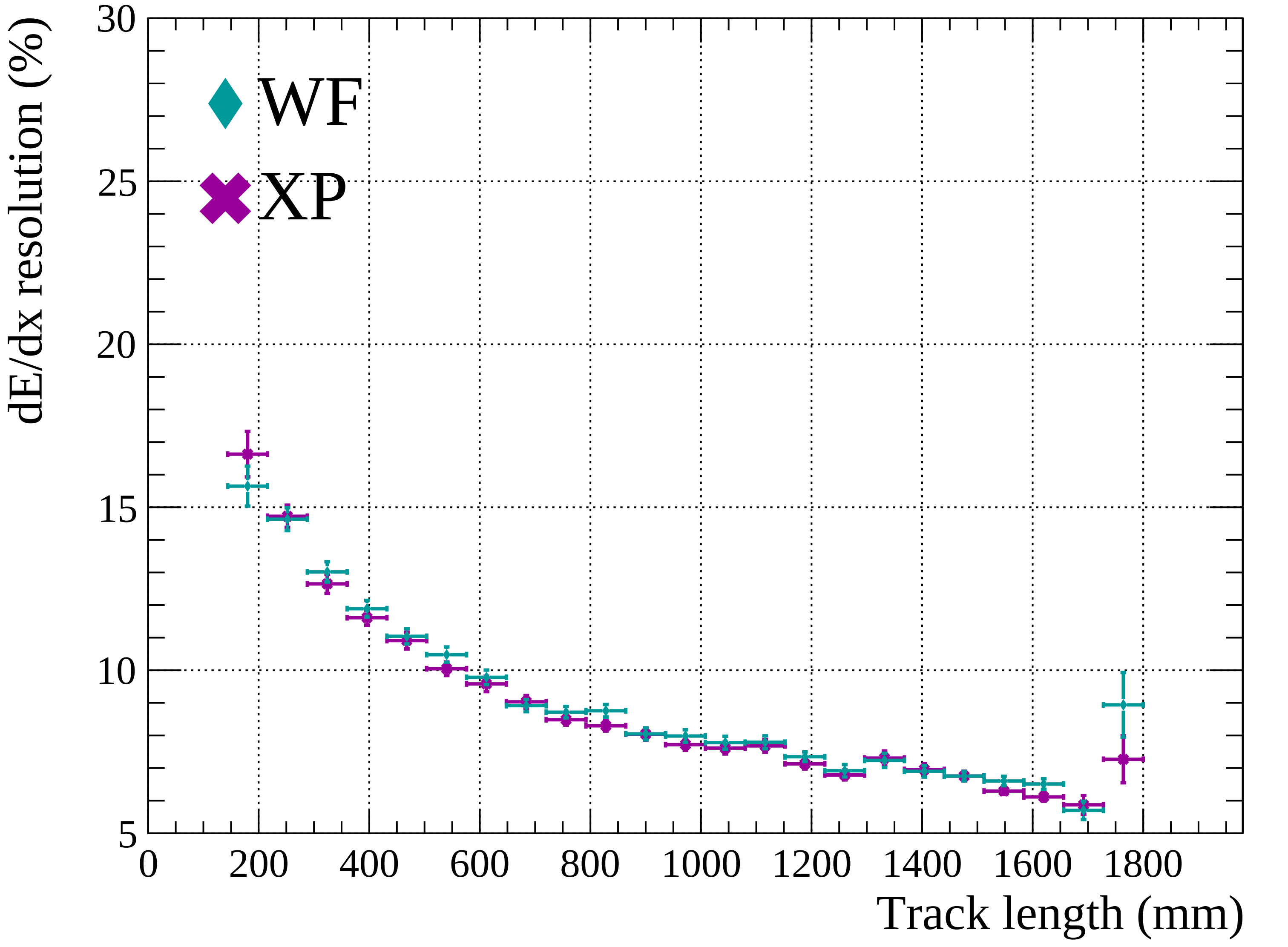

Energy loss (XP) vs drift time in ERAM 24
y = 0.10x + 465.28
Energy loss (XP) vs drift time in ERAM 30
y = 0.37x + 399.28
Energy loss (XP) vs drift time in ERAM 28
y = 0.05x + 448.07
Energy loss (XP) vs drift time in ERAM 19
y = 0.16x + 426.48
Energy loss (XP) vs drift time in ERAM 21
y = 0.06x + 454.57
Energy loss (XP) vs drift time in ERAM 13
y = 0.11x + 452.85
Energy loss (XP) vs drift time in ERAM 9
y = 0.10x + 439.84
Energy loss (XP) vs drift time in ERAM 2
y = 0.11x + 434.98
Energy loss (XP) vs drift time in ERAM 26
y = 0.41x + 370.19
Energy loss (XP) vs drift time in ERAM 17
y = 0.14x + 433.64
Energy loss (XP) vs drift time in ERAM 23
y = 0.37x + 382.05
Energy loss (XP) vs drift time in ERAM 29
y = 0.22x + 410.35
Energy loss (XP) vs drift time in ERAM 1
y = 0.15x + 445.72
Energy loss (XP) vs drift time in ERAM 10
y = 0.33x + 396.49
Energy loss (XP) vs drift time in ERAM 11
y = 0.19x + 423.94
Energy loss (XP) vs drift time in ERAM 3
y = 0.15x + 420.66
dE/dx (ADC counts/cm)
drift time (timebins)



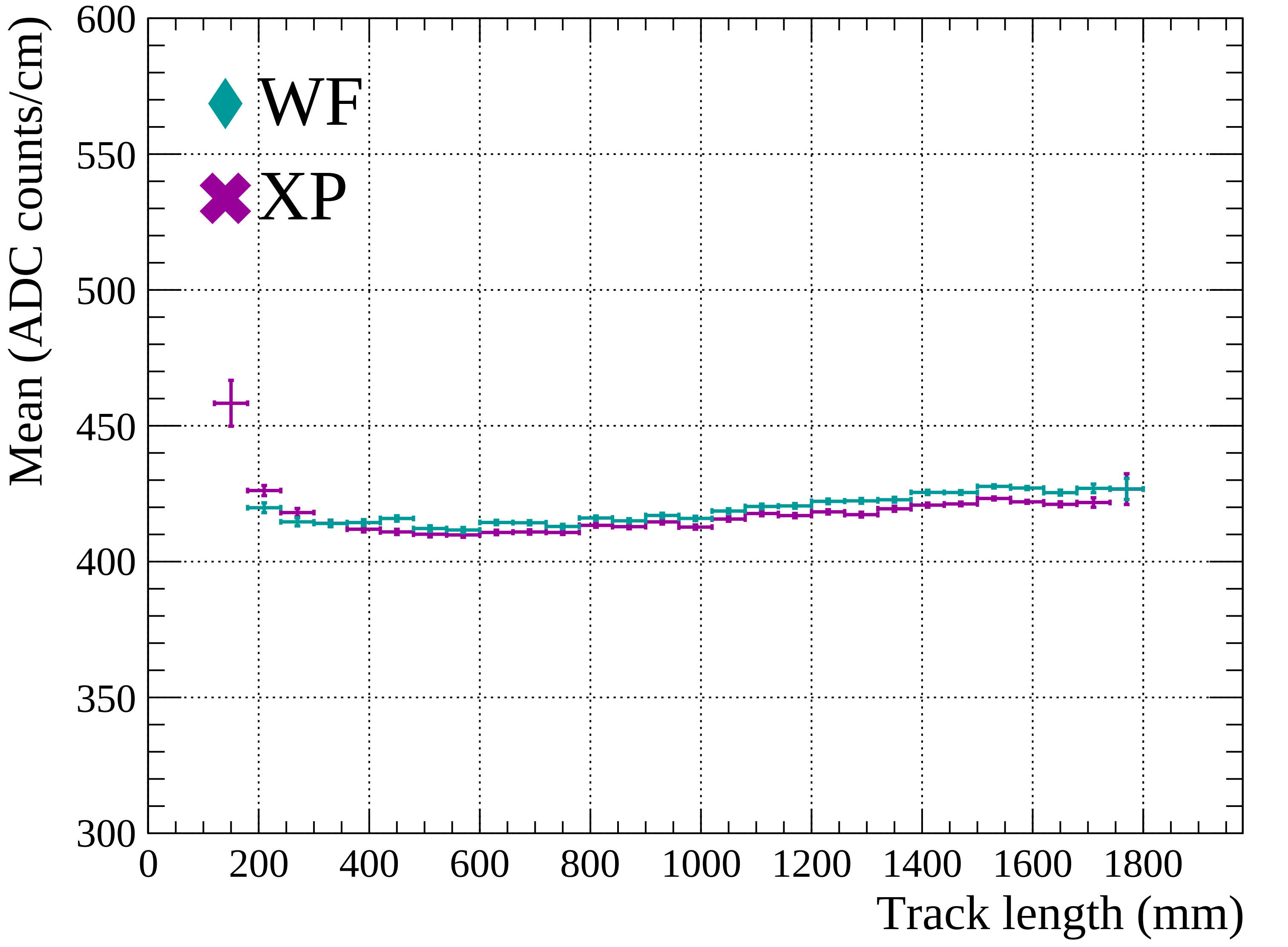
Energy loss (XP) vs drift time in ERAM 47
y = -1.26x + 747.65
Energy loss (XP) vs drift time in ERAM 16
y = 0.23x + 427.27
Energy loss (XP) vs drift time in ERAM 14
y = 0.48x + 392.49
Energy loss (XP) vs drift time in ERAM 15
y = 0.13x + 447.73
Energy loss (XP) vs drift time in ERAM 42
y = 0.33x + 412.46
Energy loss (XP) vs drift time in ERAM 45
y = 0.25x + 406.88
Energy loss (XP) vs drift time in ERAM 37
y = 0.10x + 428.11
Energy loss (XP) vs drift time in ERAM 36
y = 0.09x + 422.70
Energy loss (XP) vs drift time in ERAM 20
y = 0.76x + 296.61
Energy loss (XP) vs drift time in ERAM 38
y = 0.02x + 486.67
Energy loss (XP) vs drift time in ERAM 7
y = 0.21x + 433.11
Energy loss (XP) vs drift time in ERAM 44
y = 0.28x + 394.03
Energy loss (XP) vs drift time in ERAM 43
y = 0.24x + 430.79
Energy loss (XP) vs drift time in ERAM 39
y = 0.25x + 416.26
Energy loss (XP) vs drift time in ERAM 41
y = 0.16x + 427.68
Energy loss (XP) vs drift time in ERAM 46
y = 0.19x + 402.90
dE/dx (ADC counts/cm)
drift time (timebins)



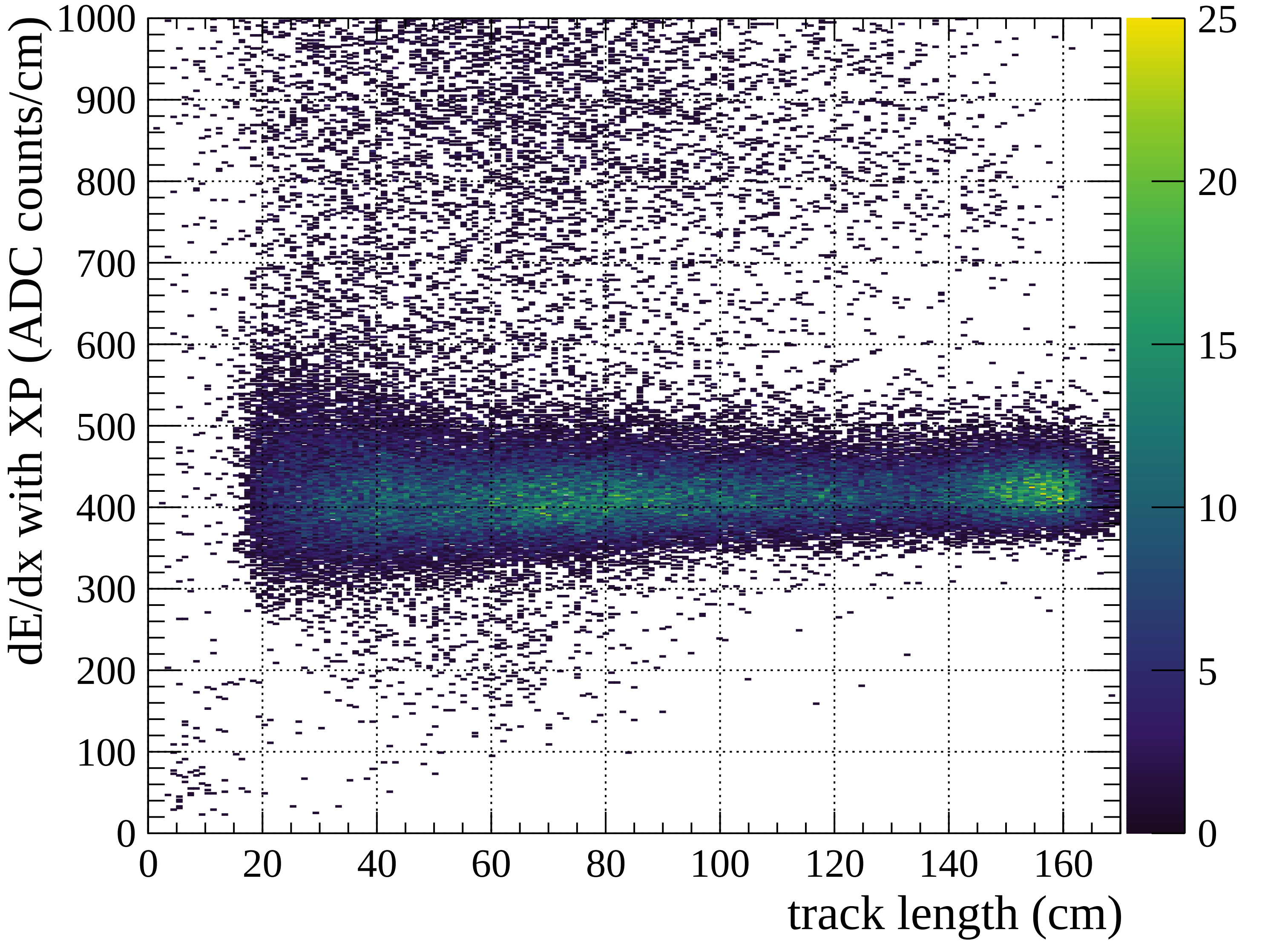
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
35
30
25
20
15
10
5
0

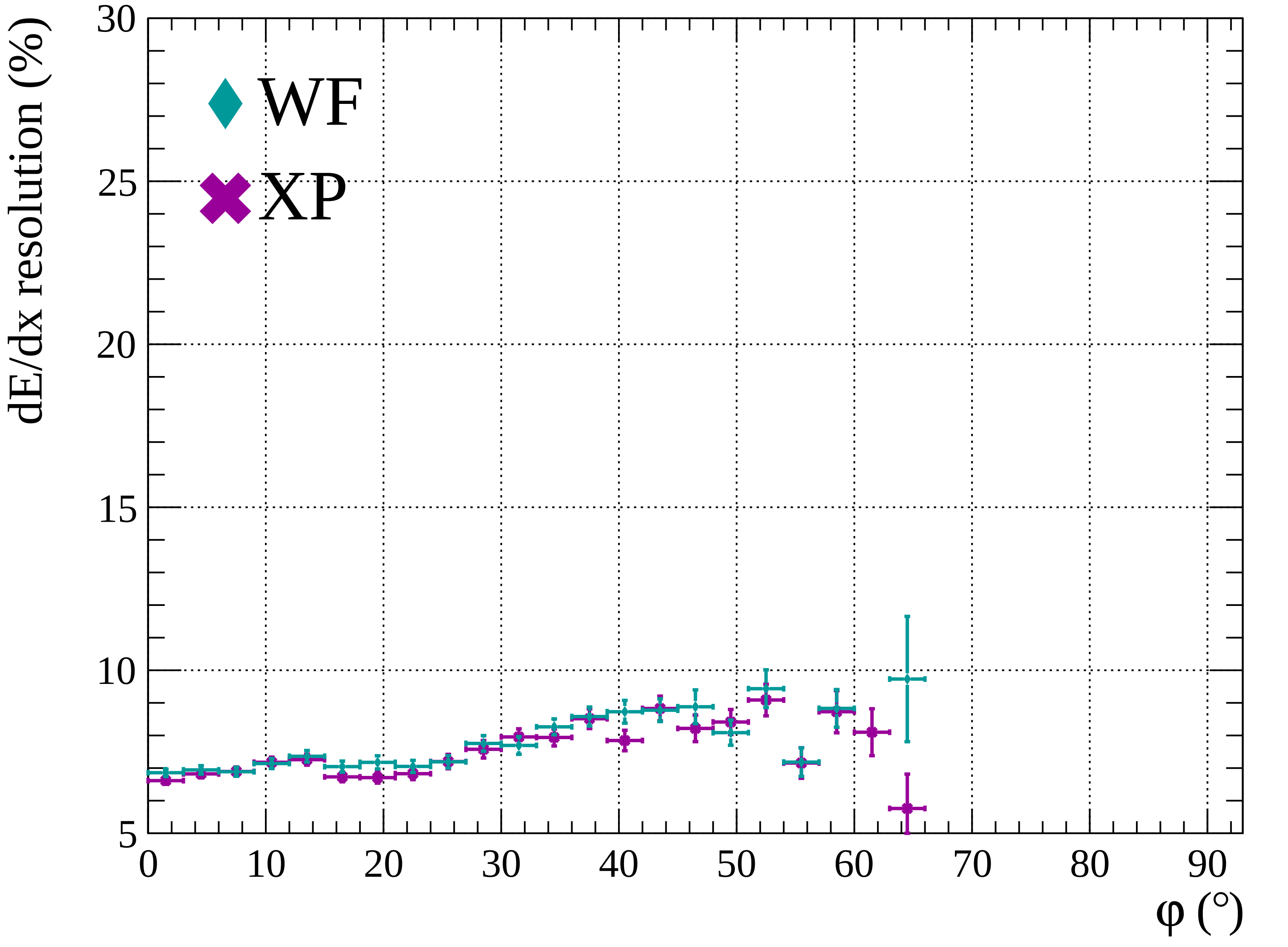


dE/dx resolution (%)
WF
XP
Track length (mm)
0
200
400
600
800
1000
1200
1400
1600
1800
5
10
15
20
25
30

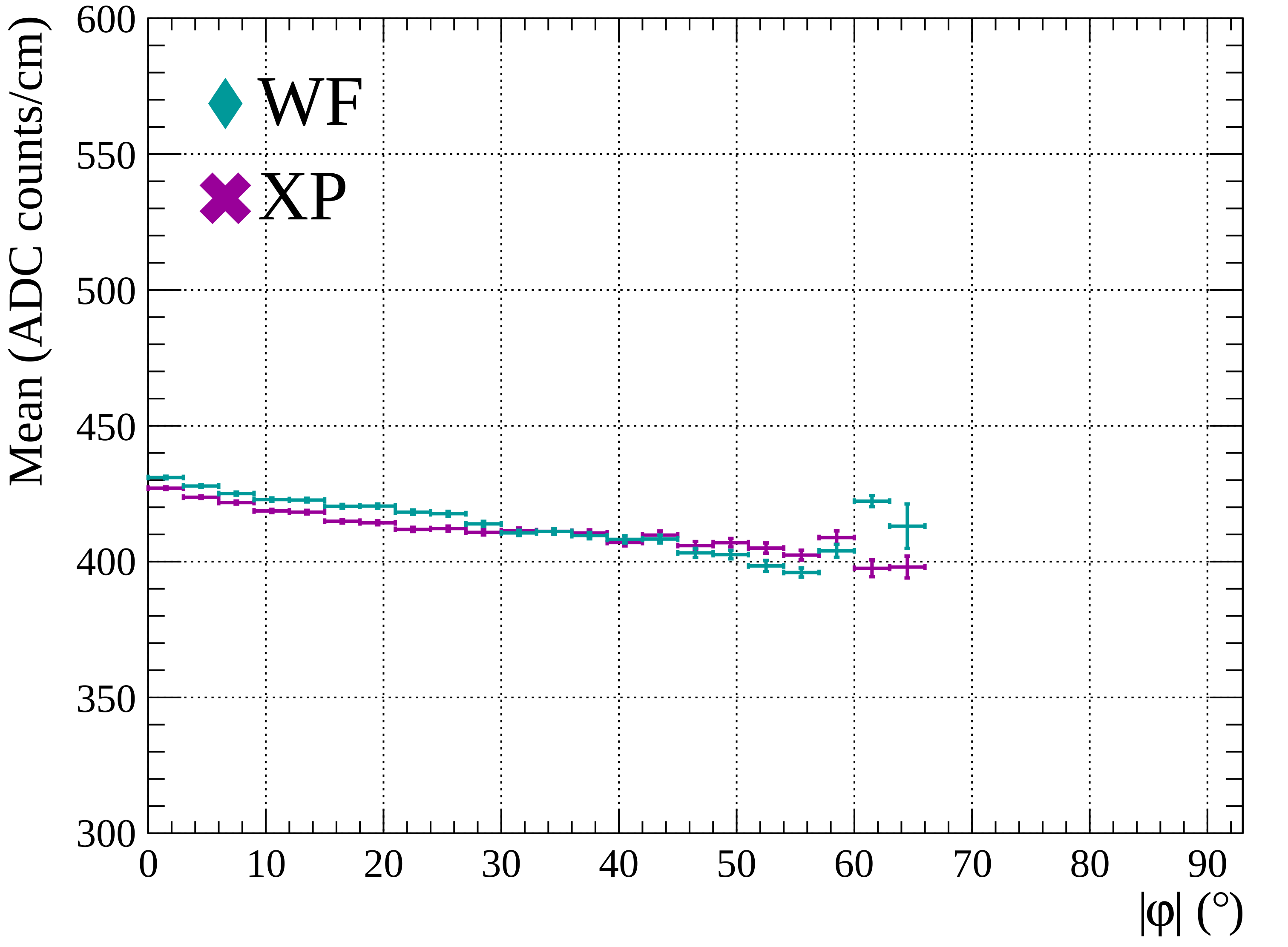


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)

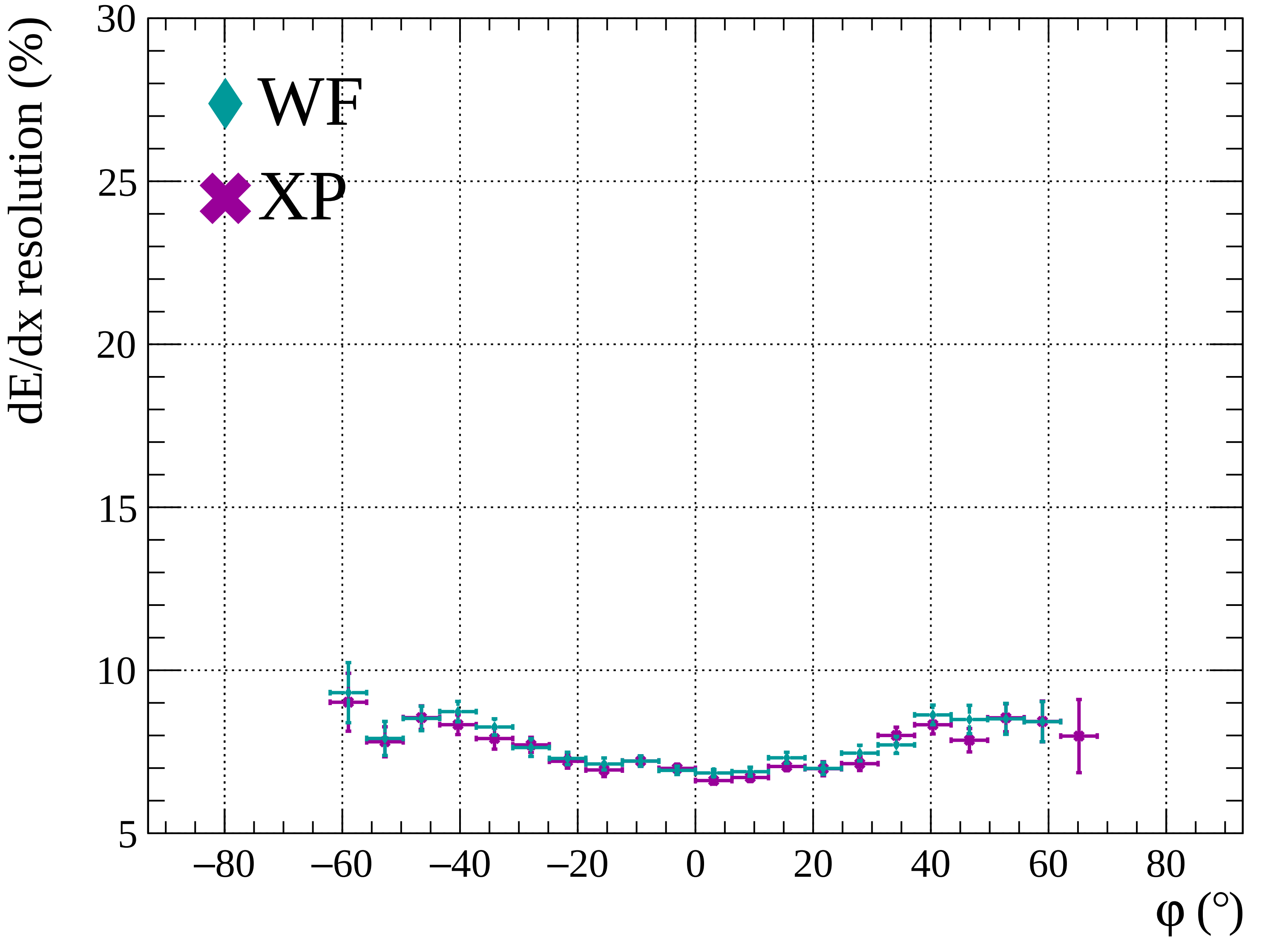

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
5
10
15
20
25



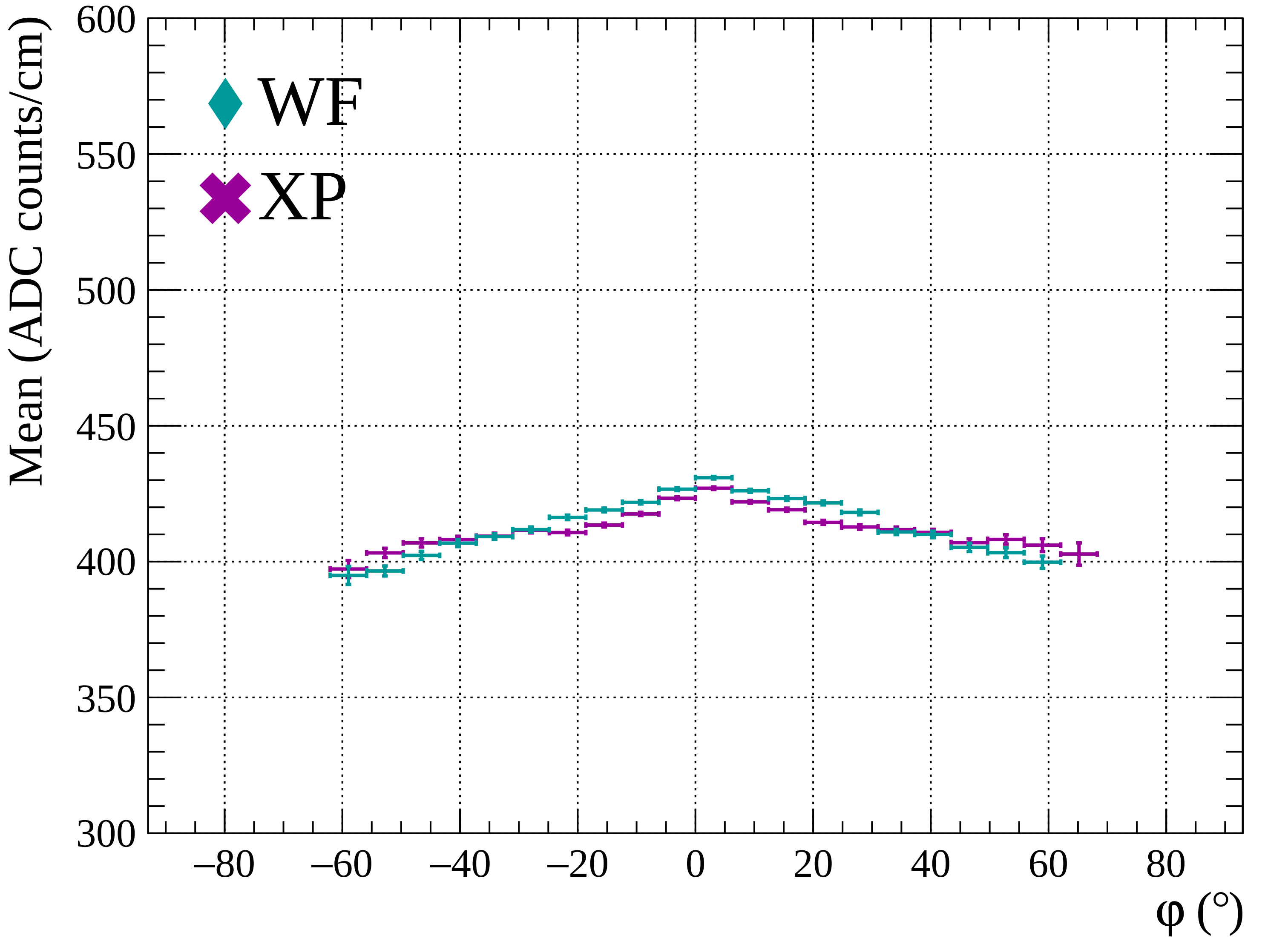
WF
XP
dE/dx resolution (%)
φ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



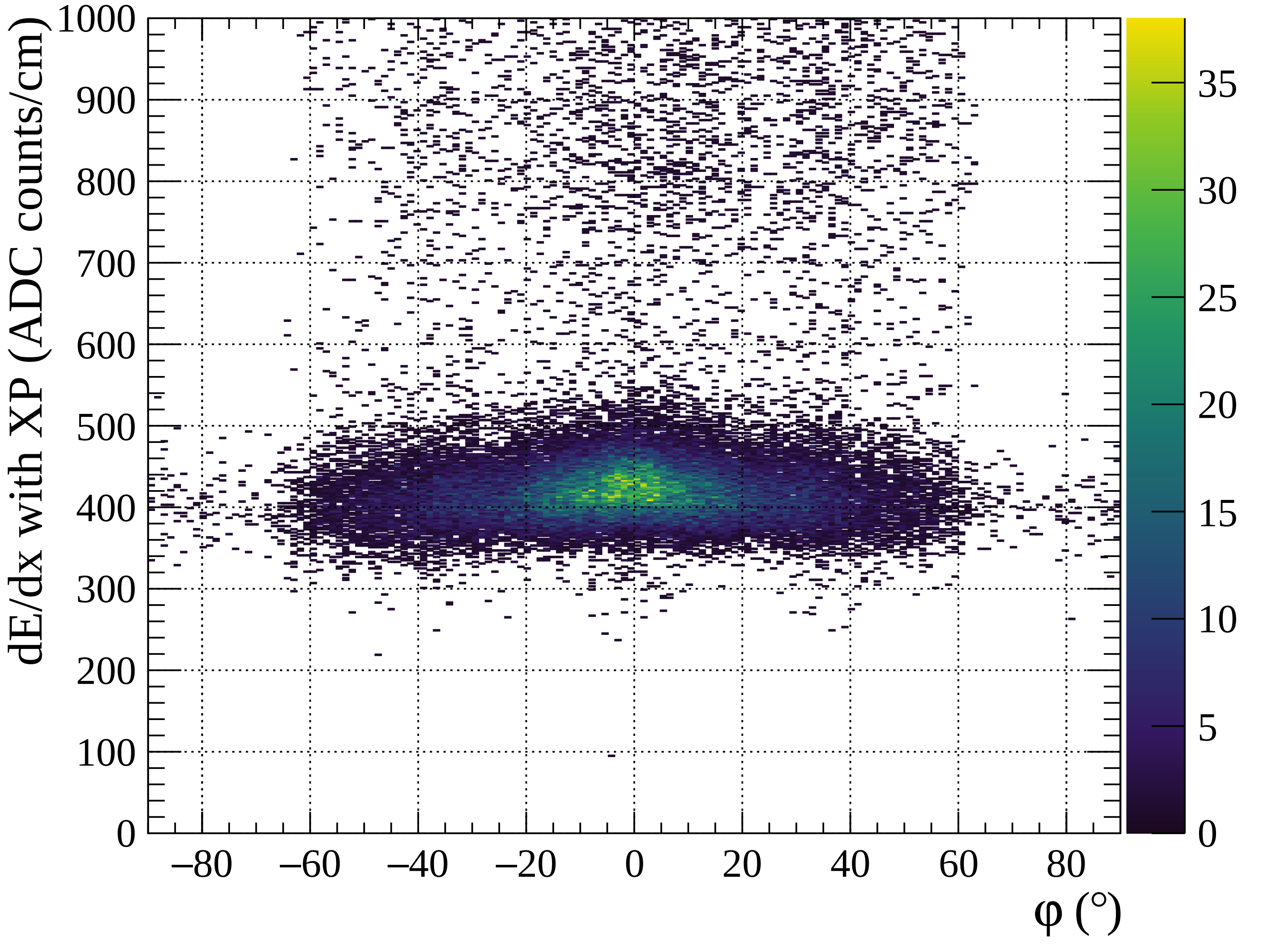
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
|φ| (°)
WF
XP

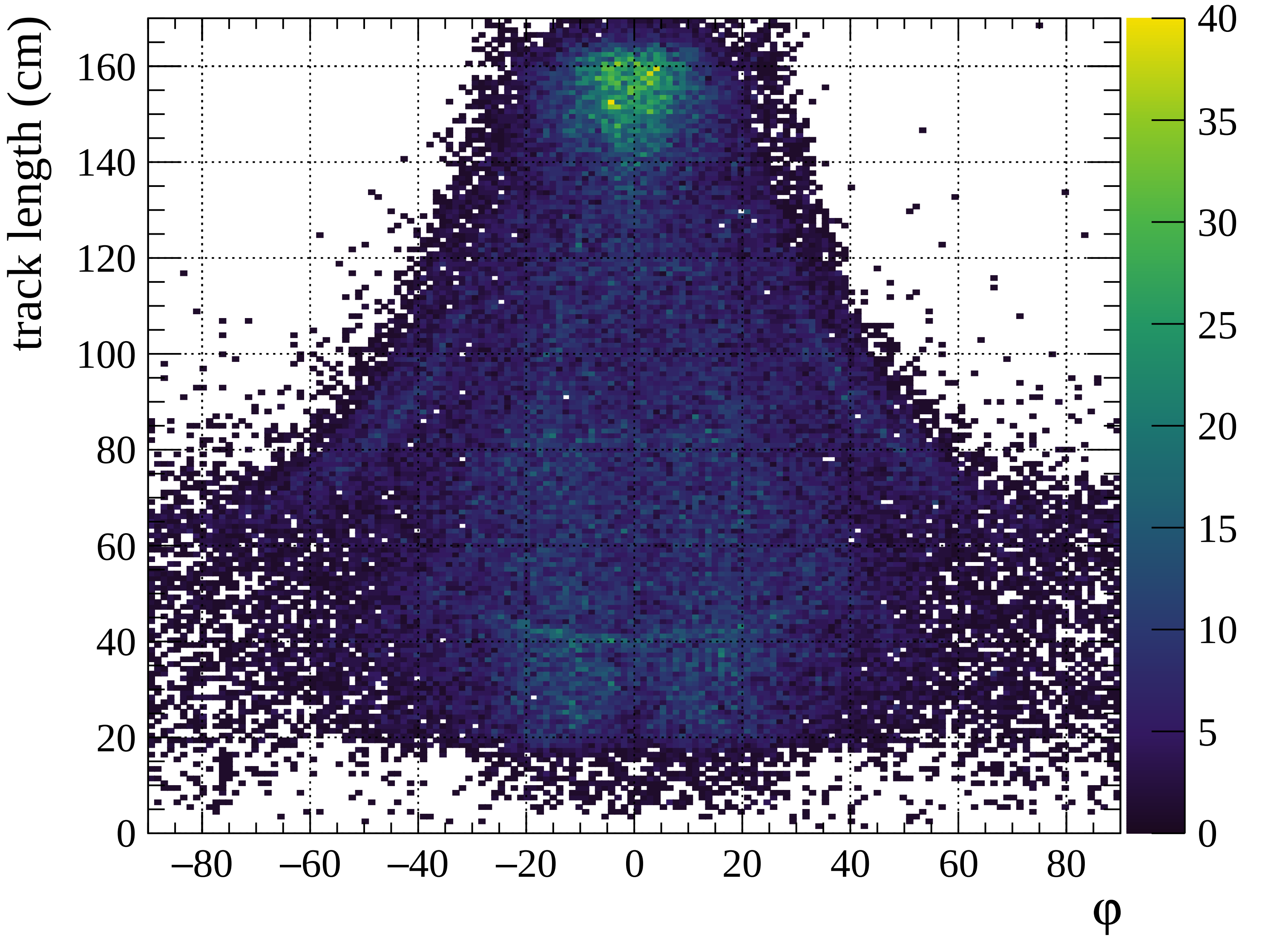


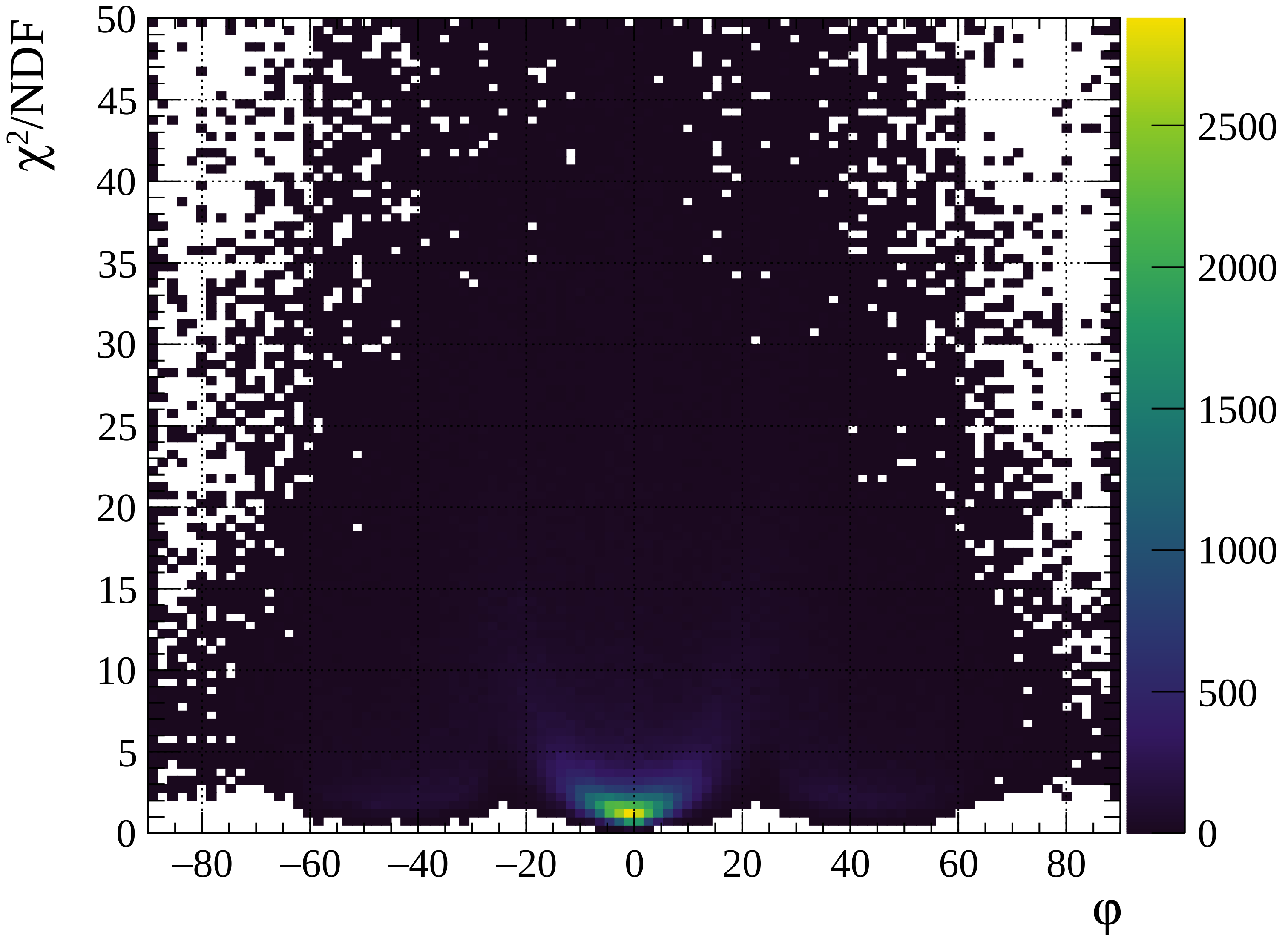
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)



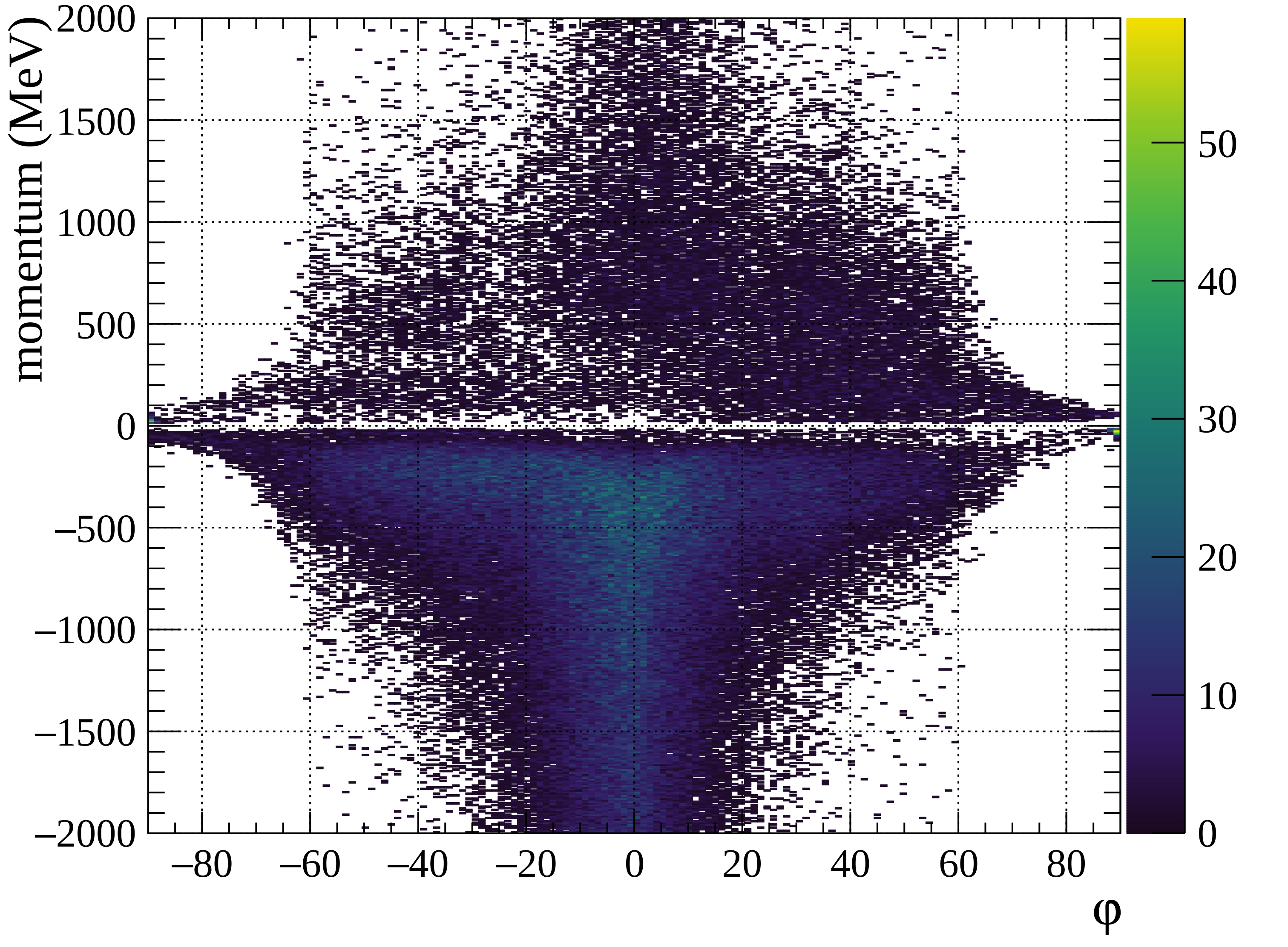
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)

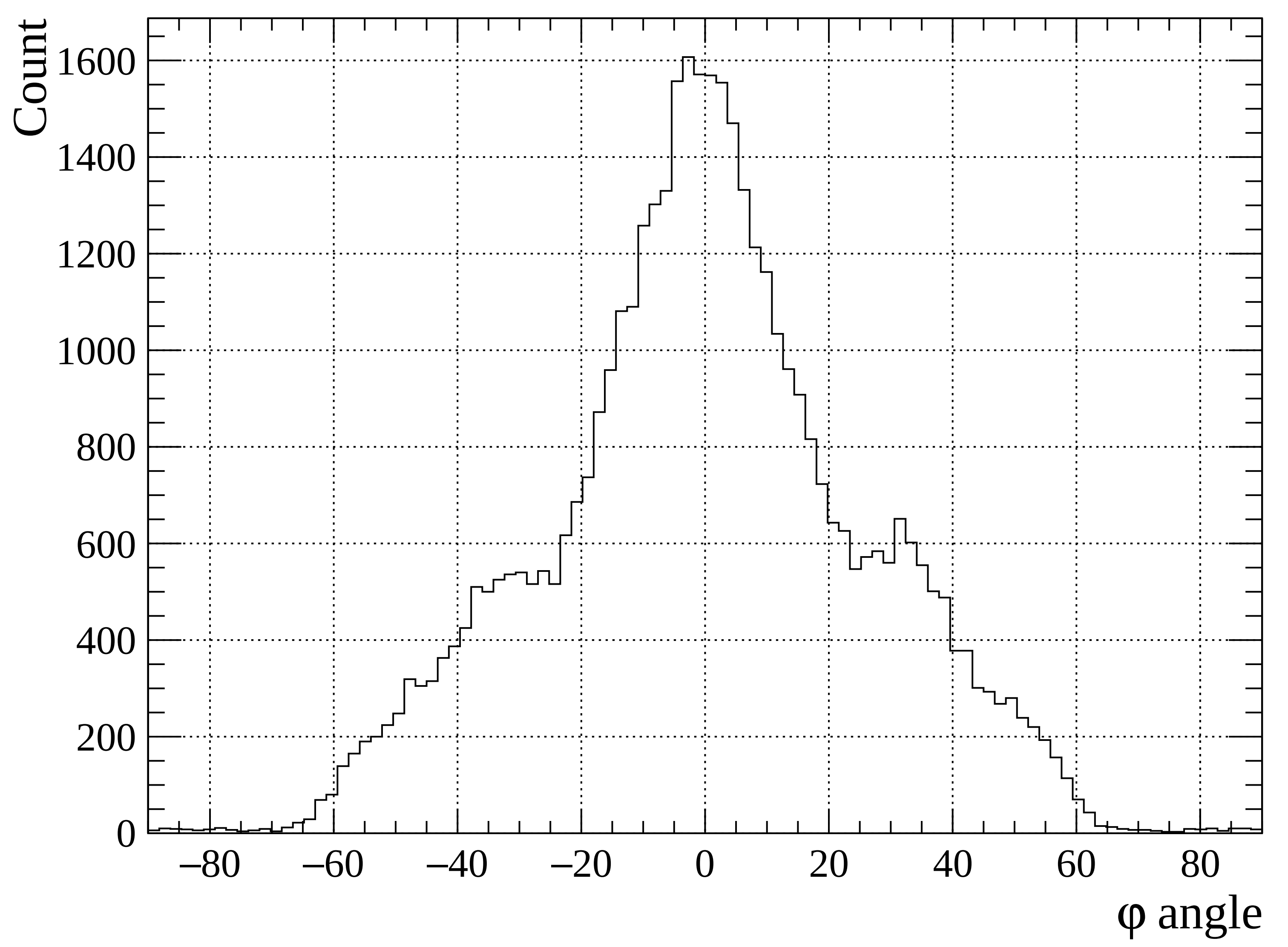

dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25
30
35


track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
5
10
15
20
25
30
35
40

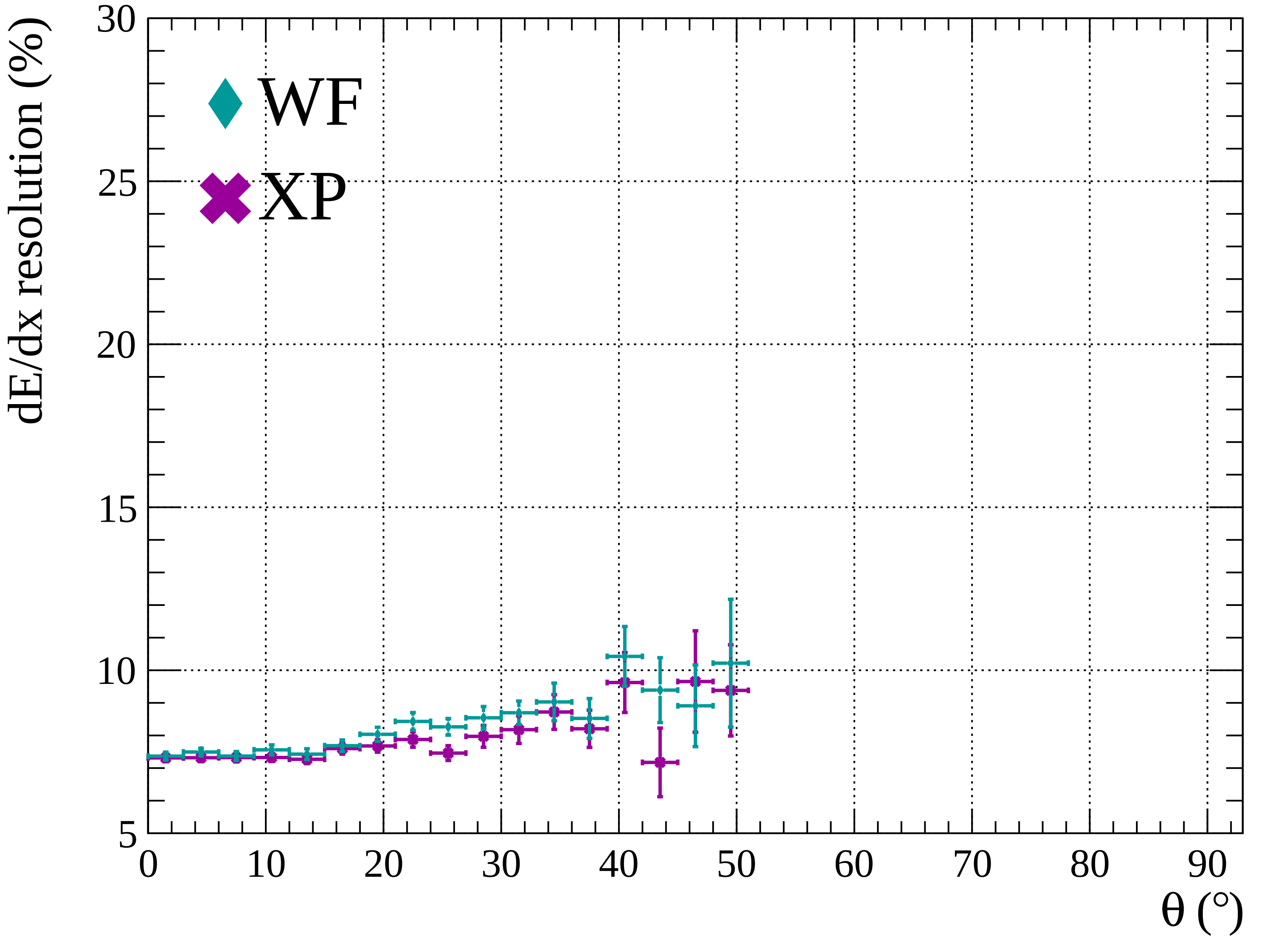


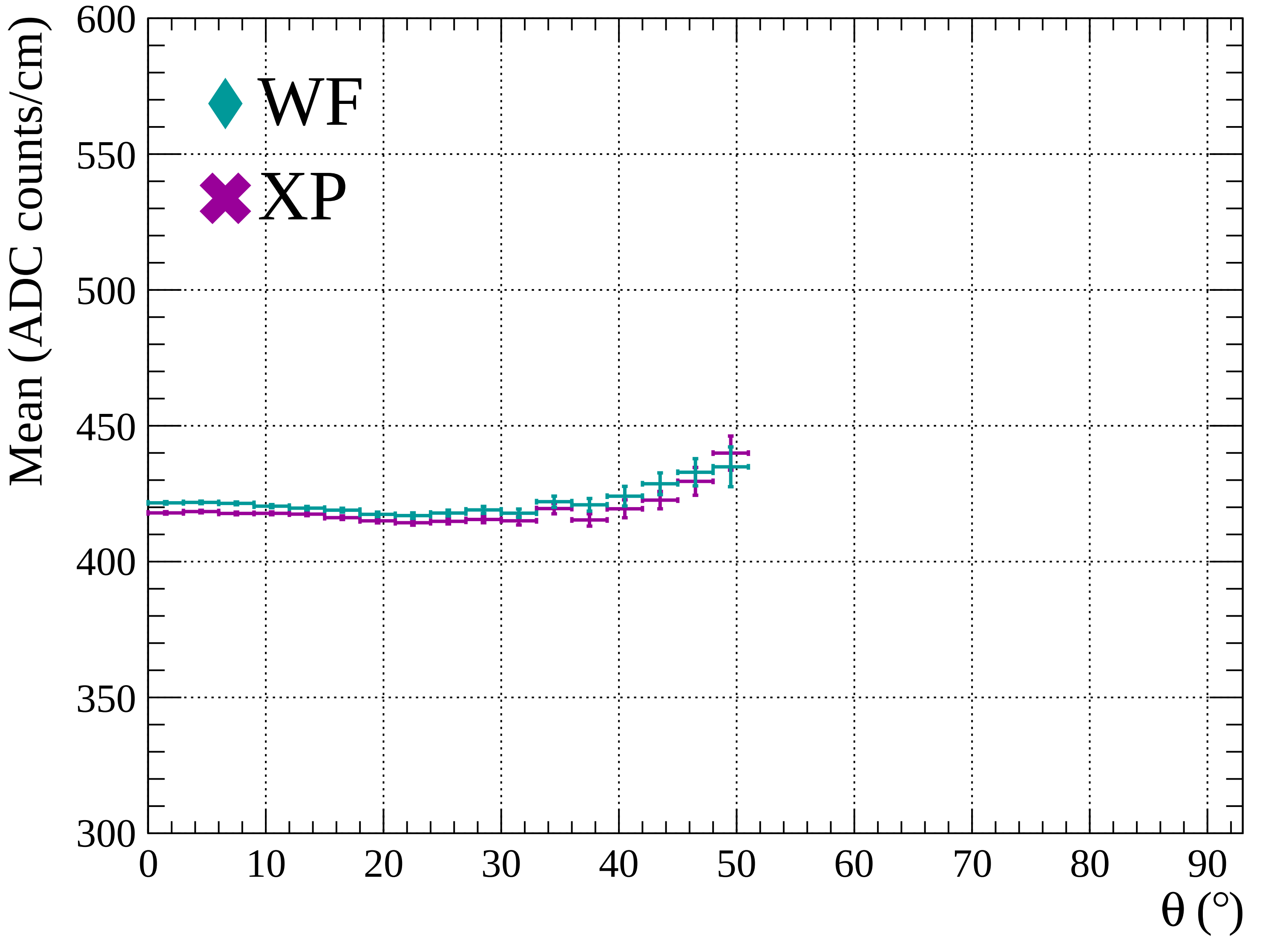
χ^2/NDF
φ
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25
30
35
40
45
50
0
500
1000
1500
2000
2500


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
50
40
30
20
10
0

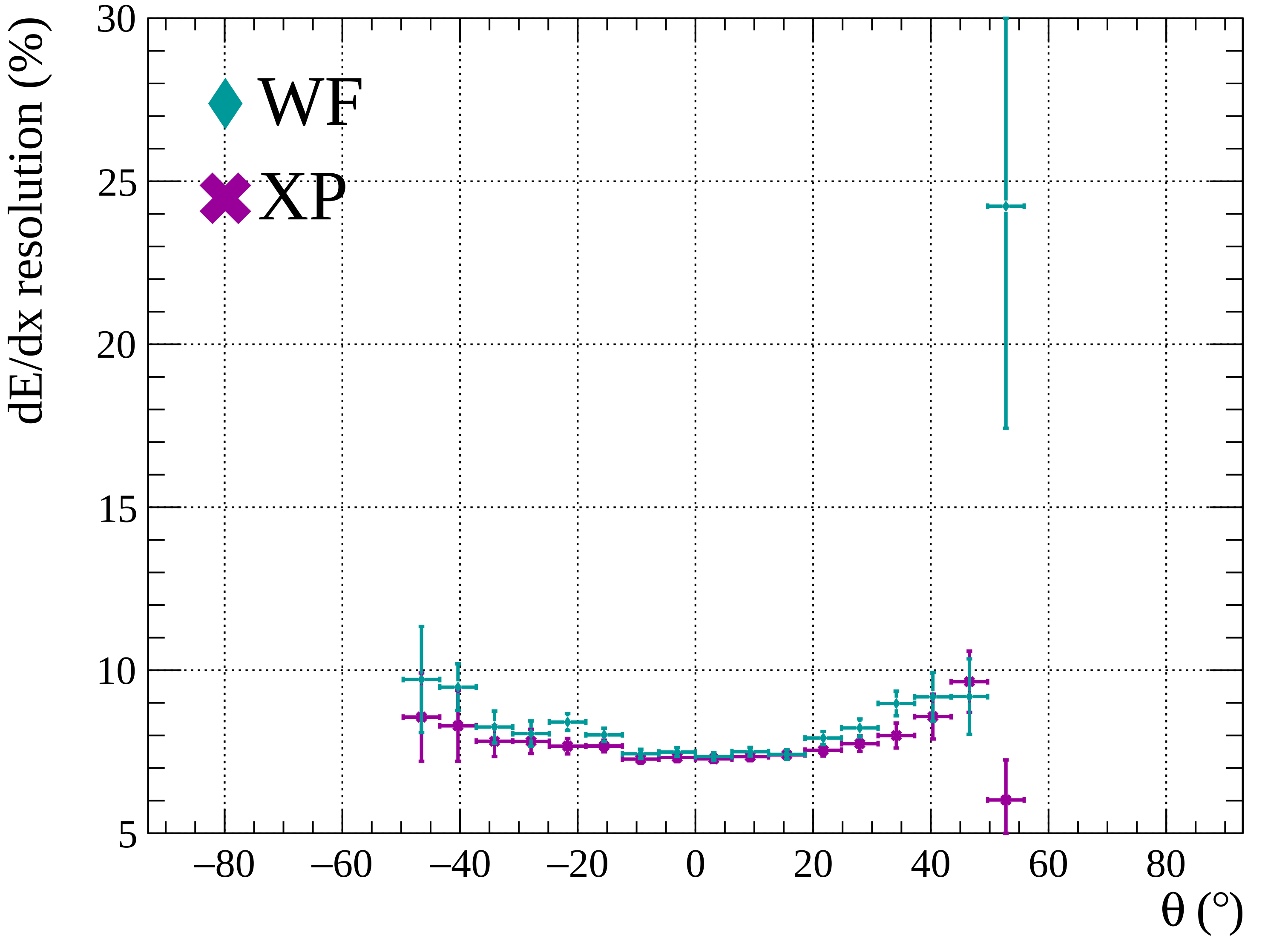


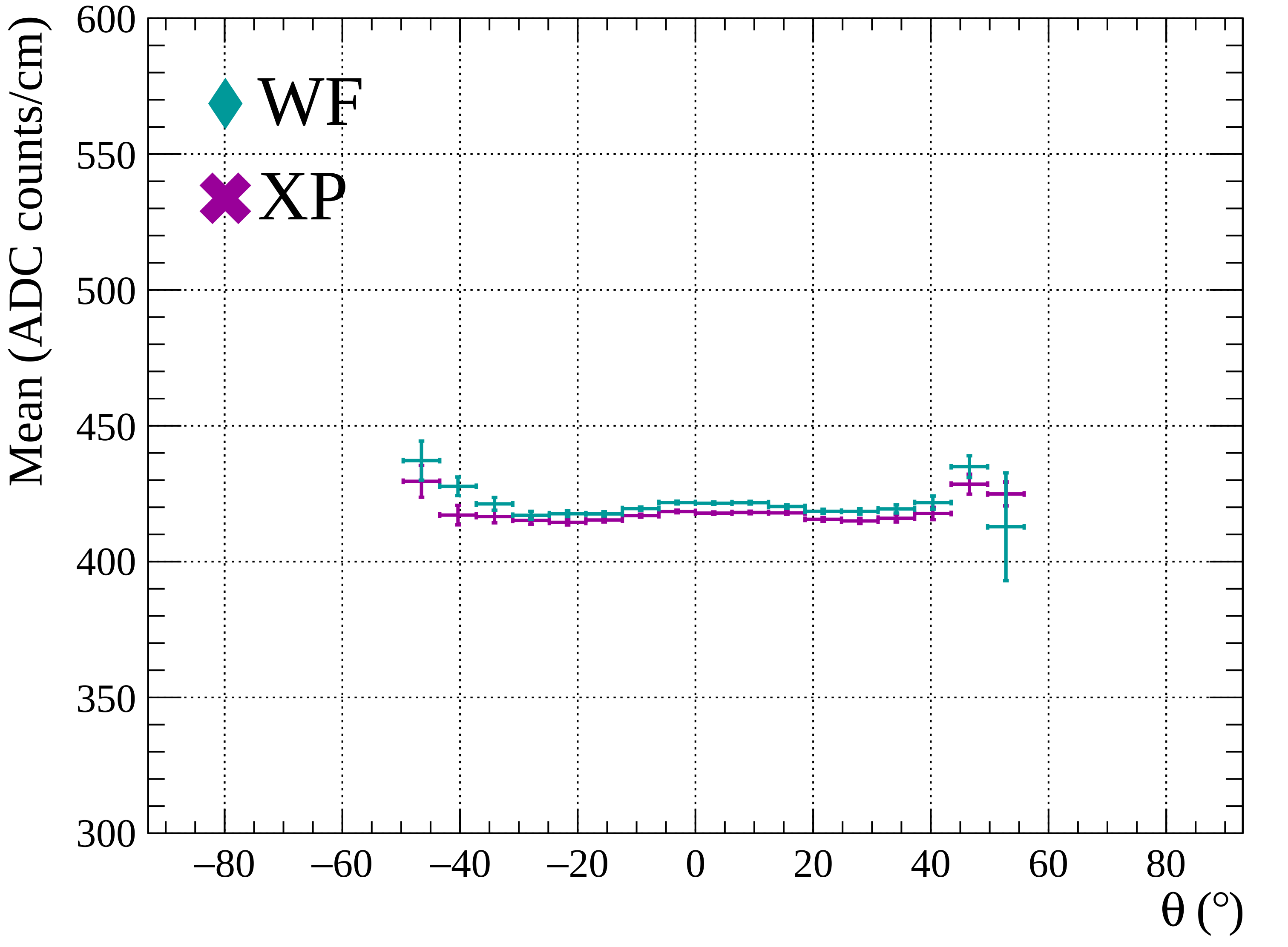
Count
1600
1400
1200
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

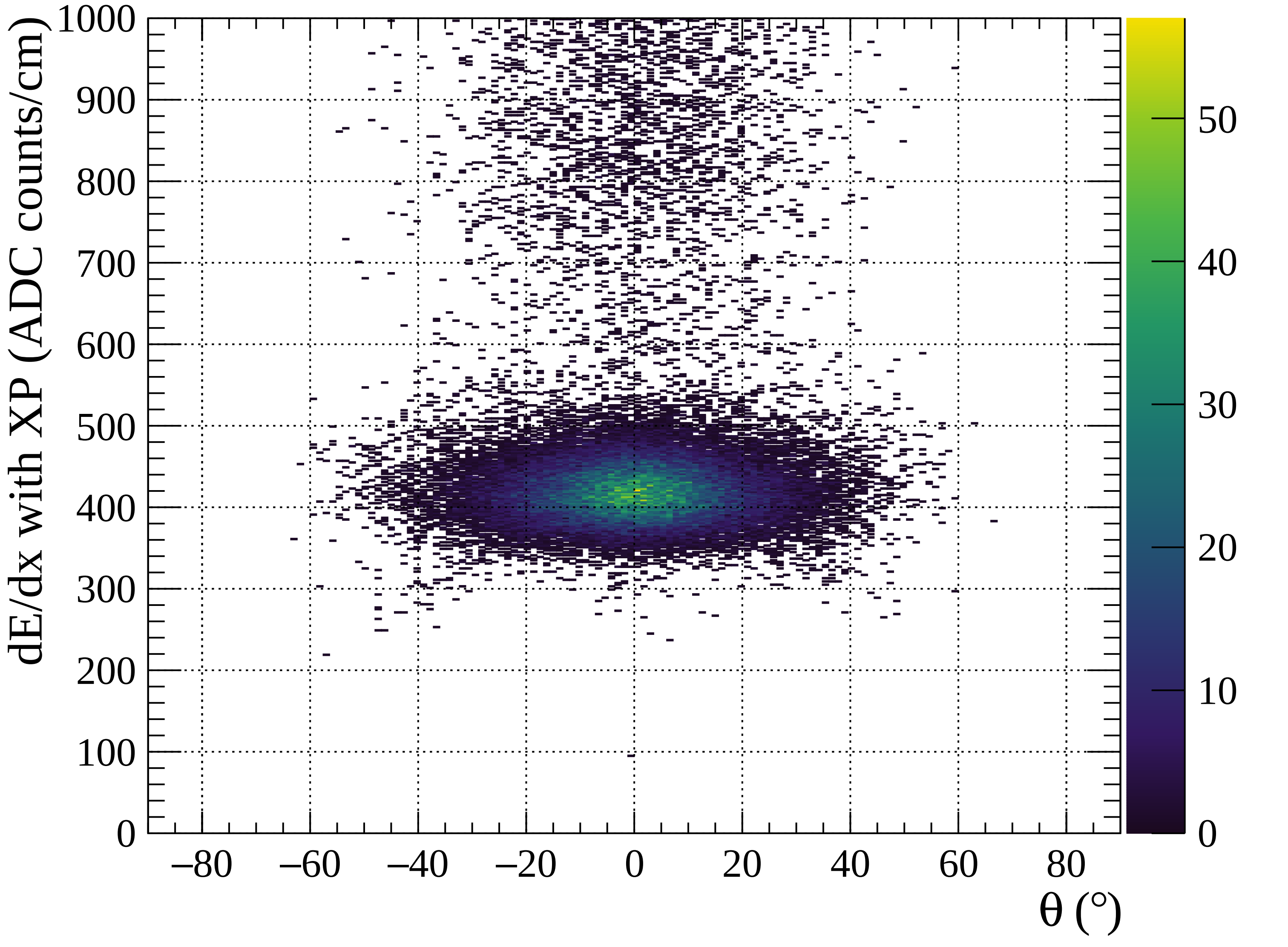

WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



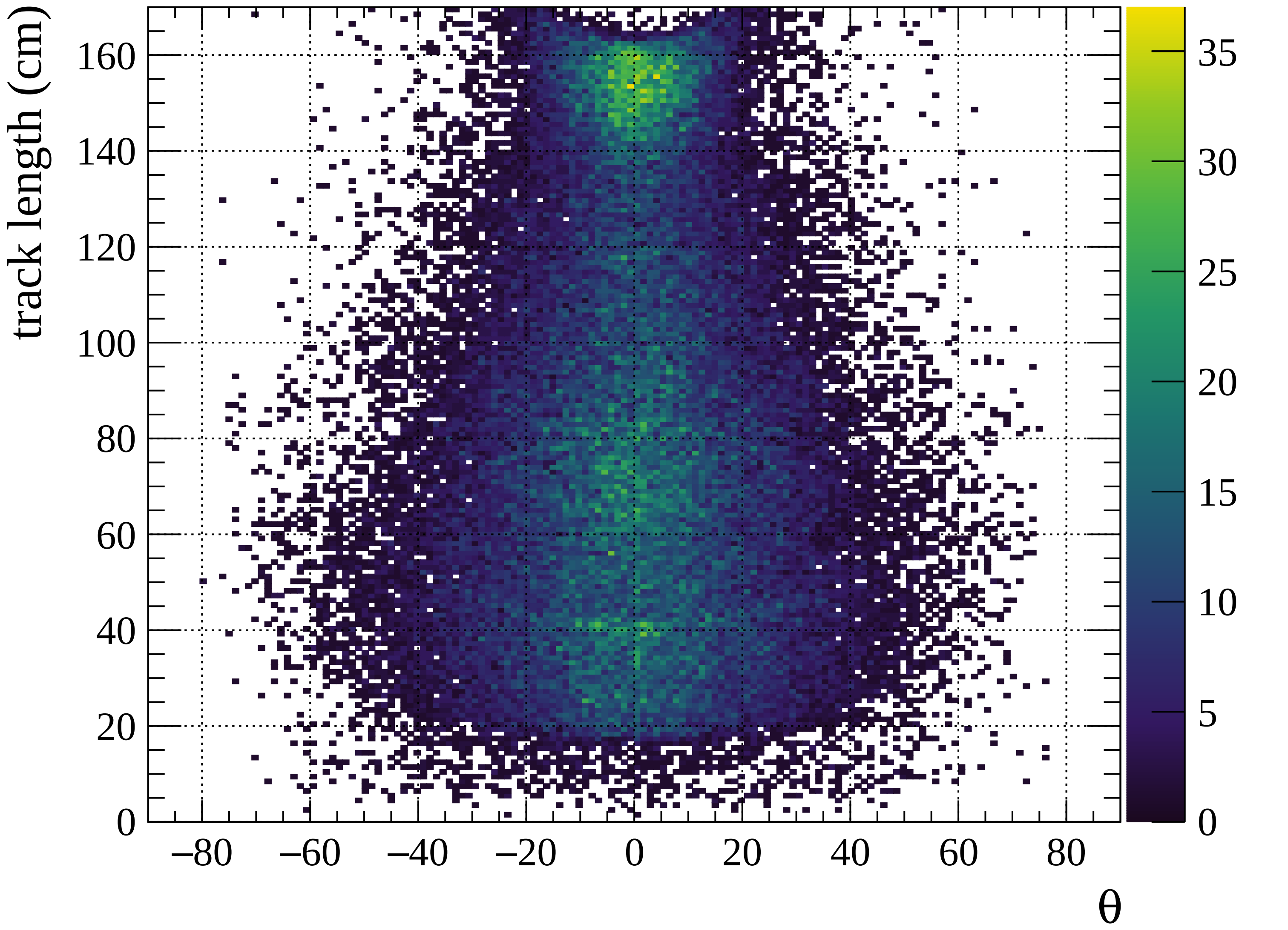
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)

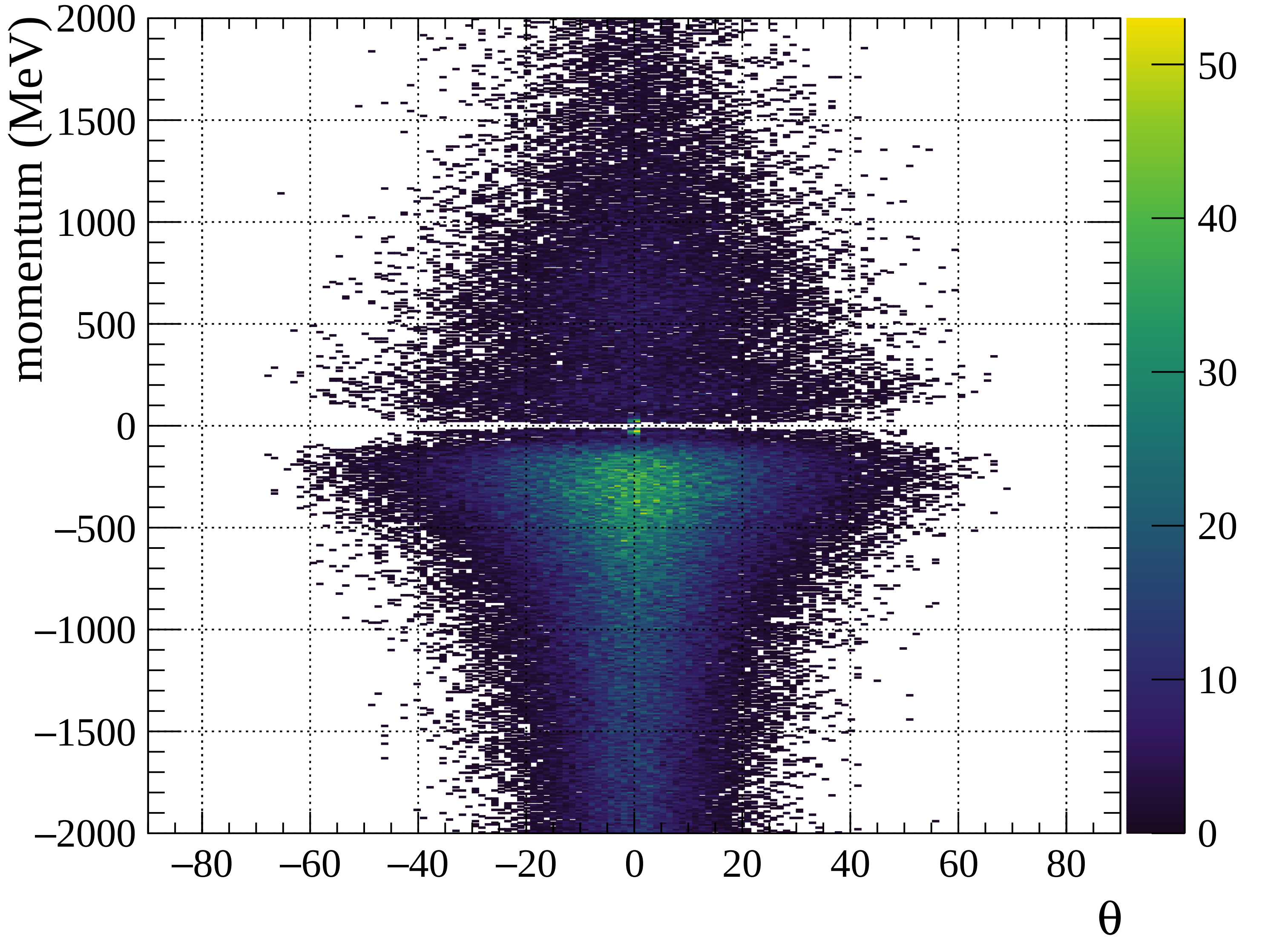

WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)

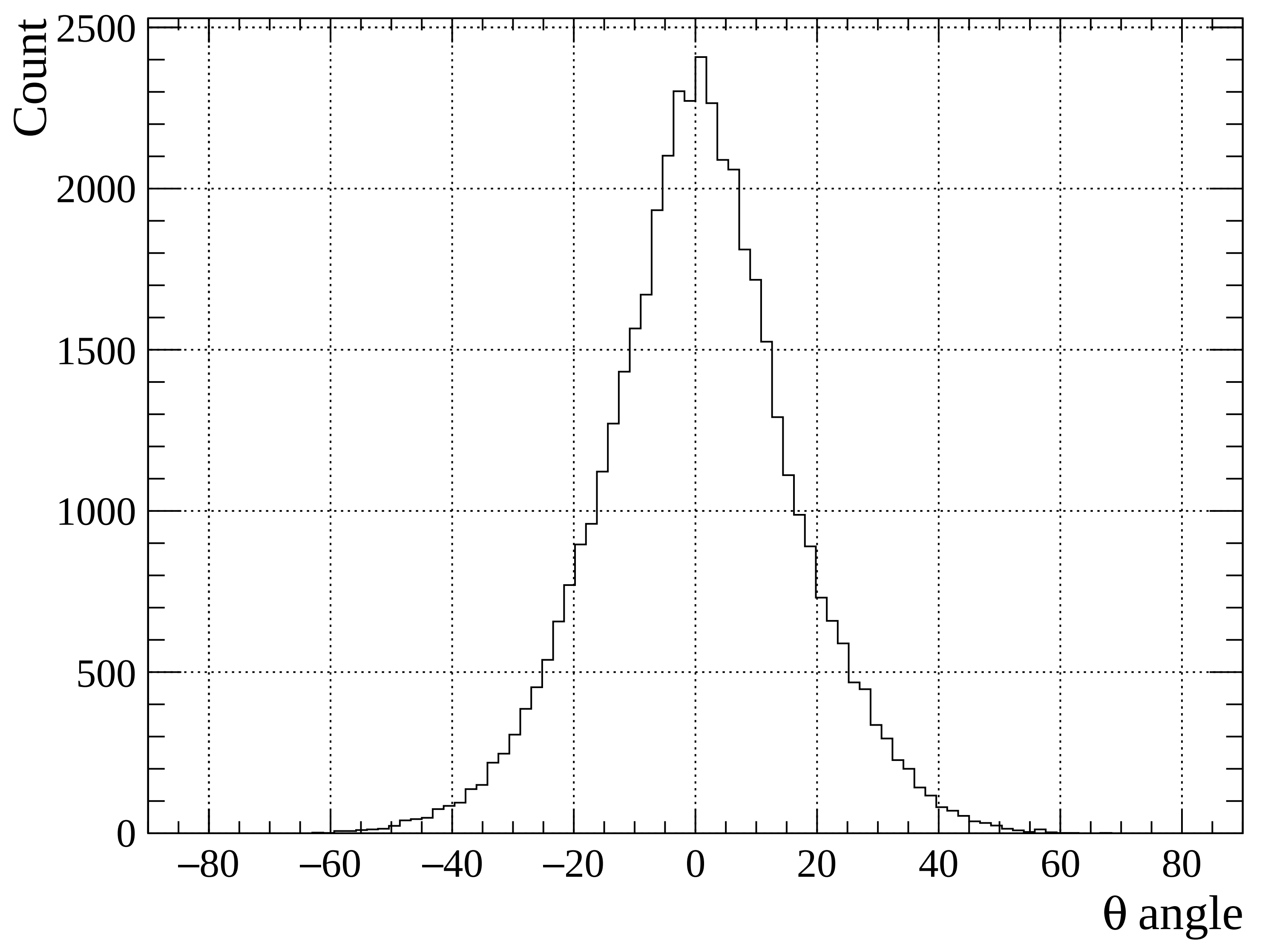


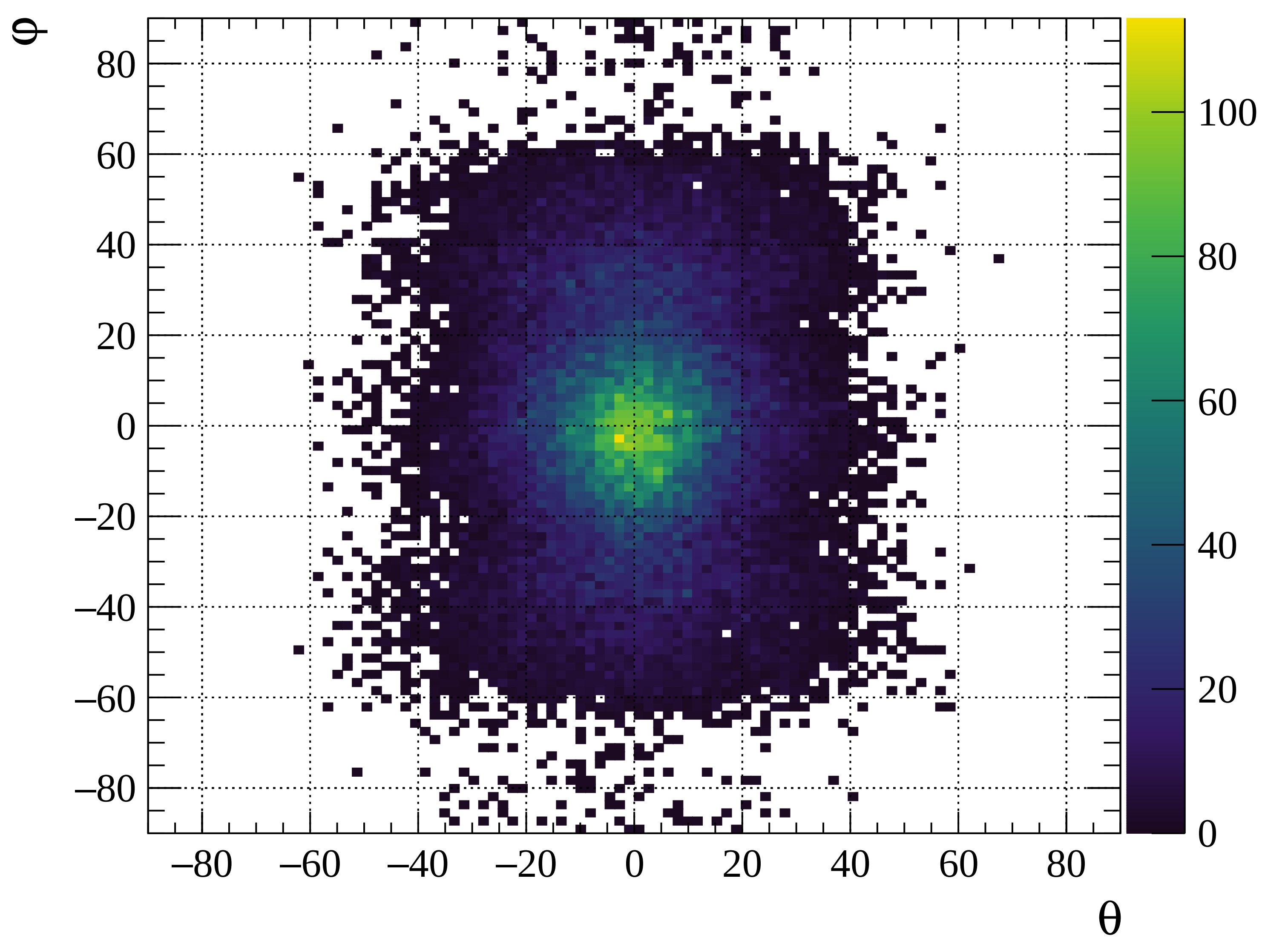
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
50
40
30
20
10
0

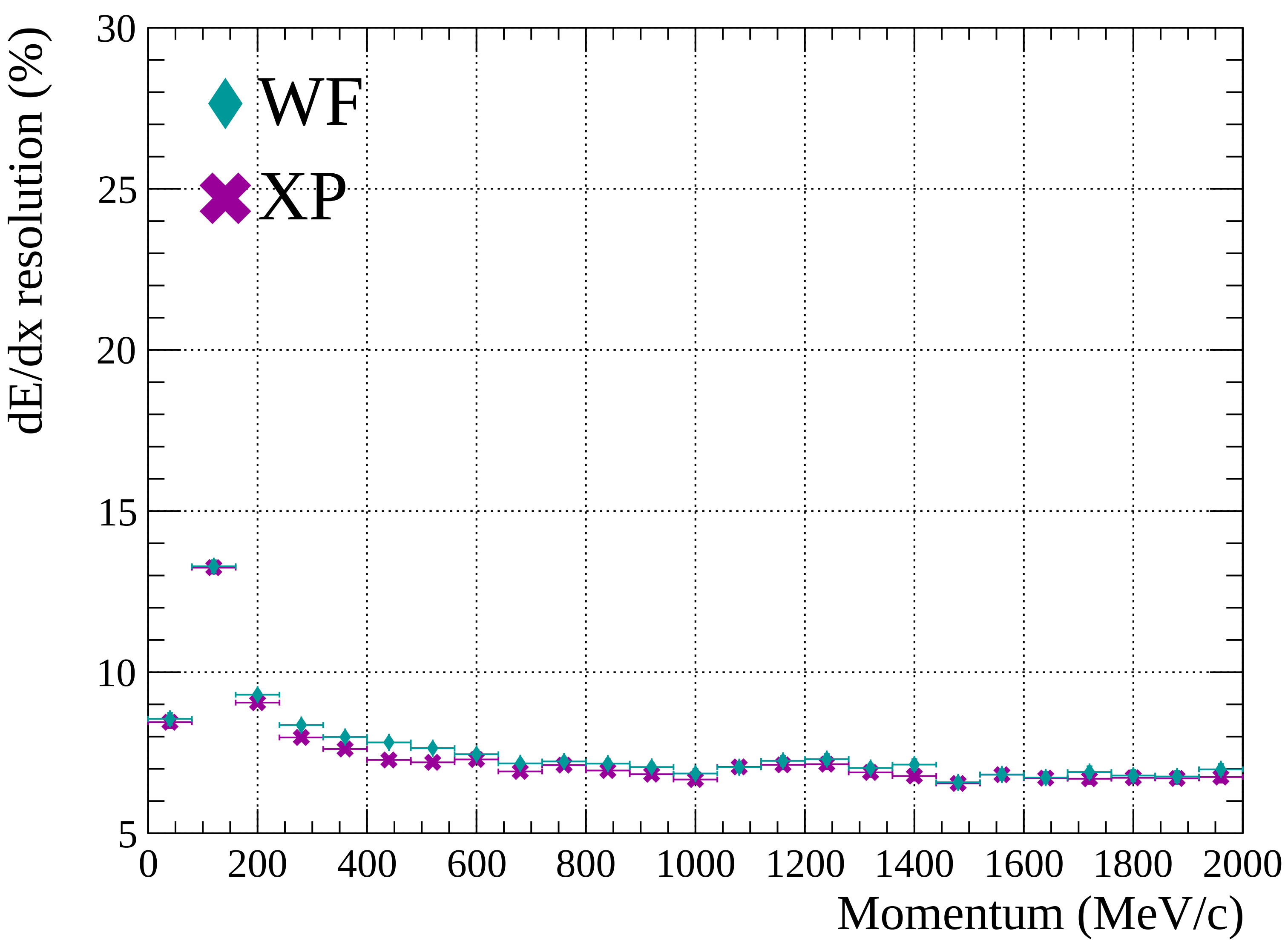

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
35
30
25
20
15
10
5
0

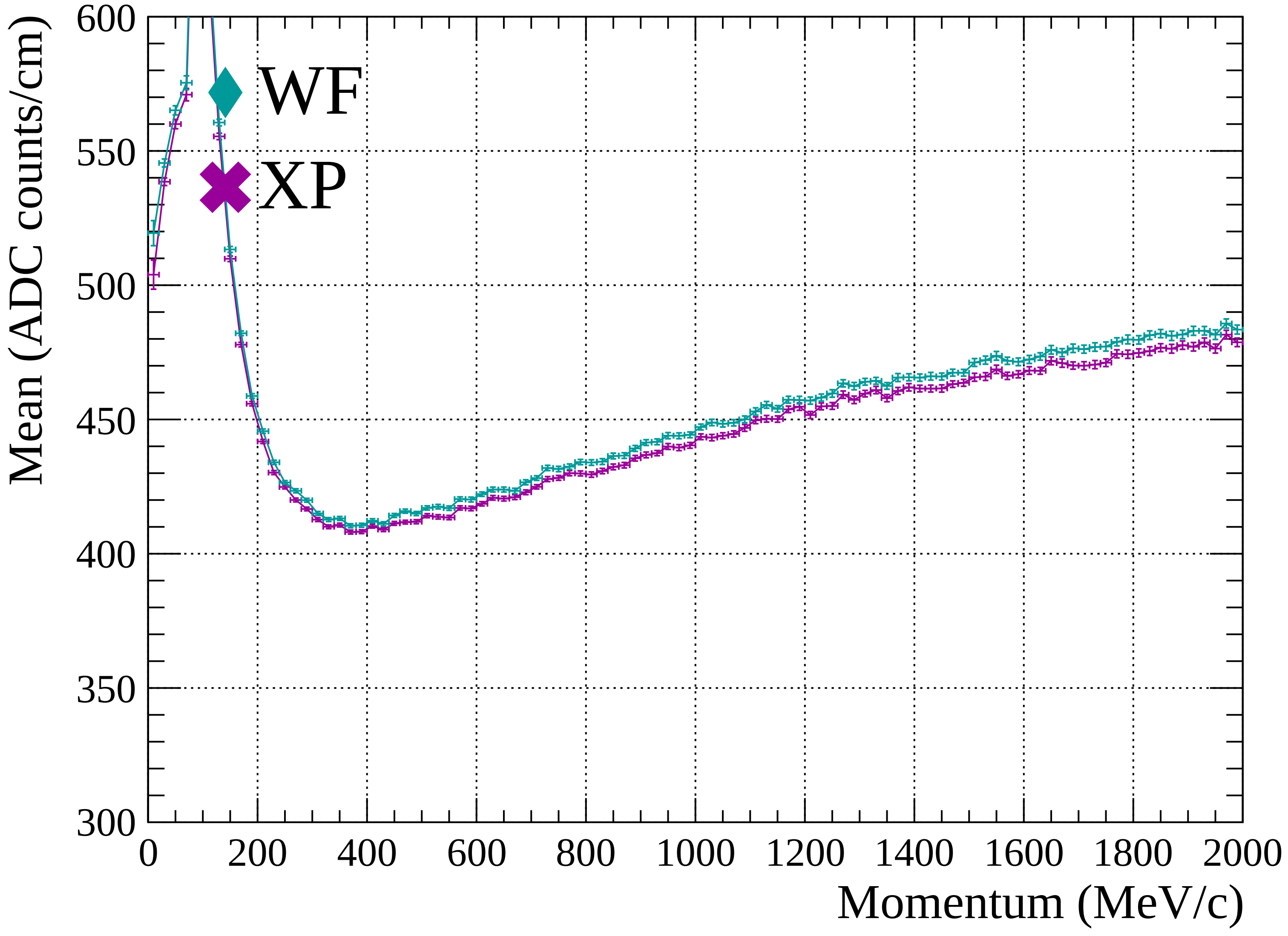


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
50
40
30
20
10
0

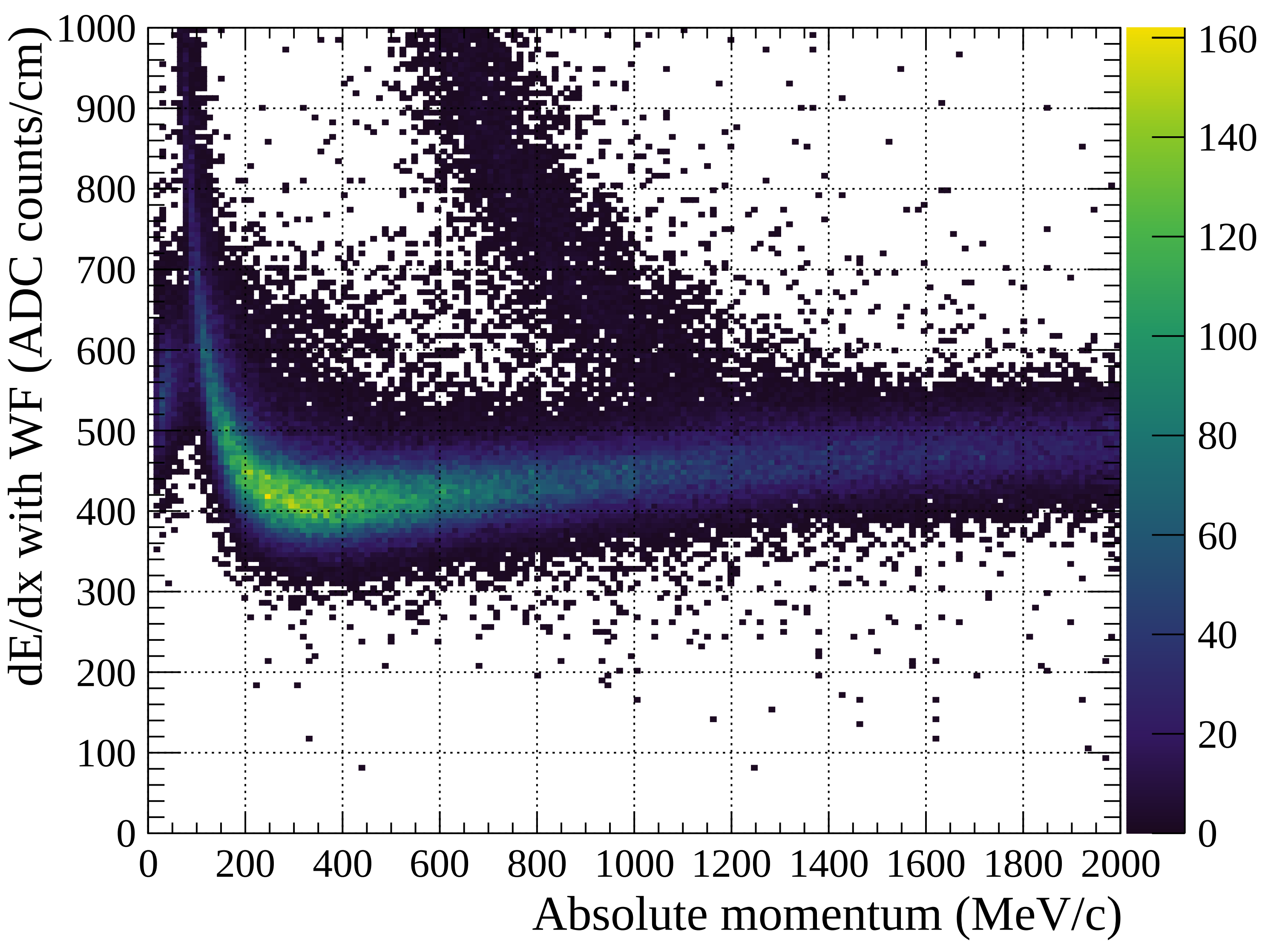

Count
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

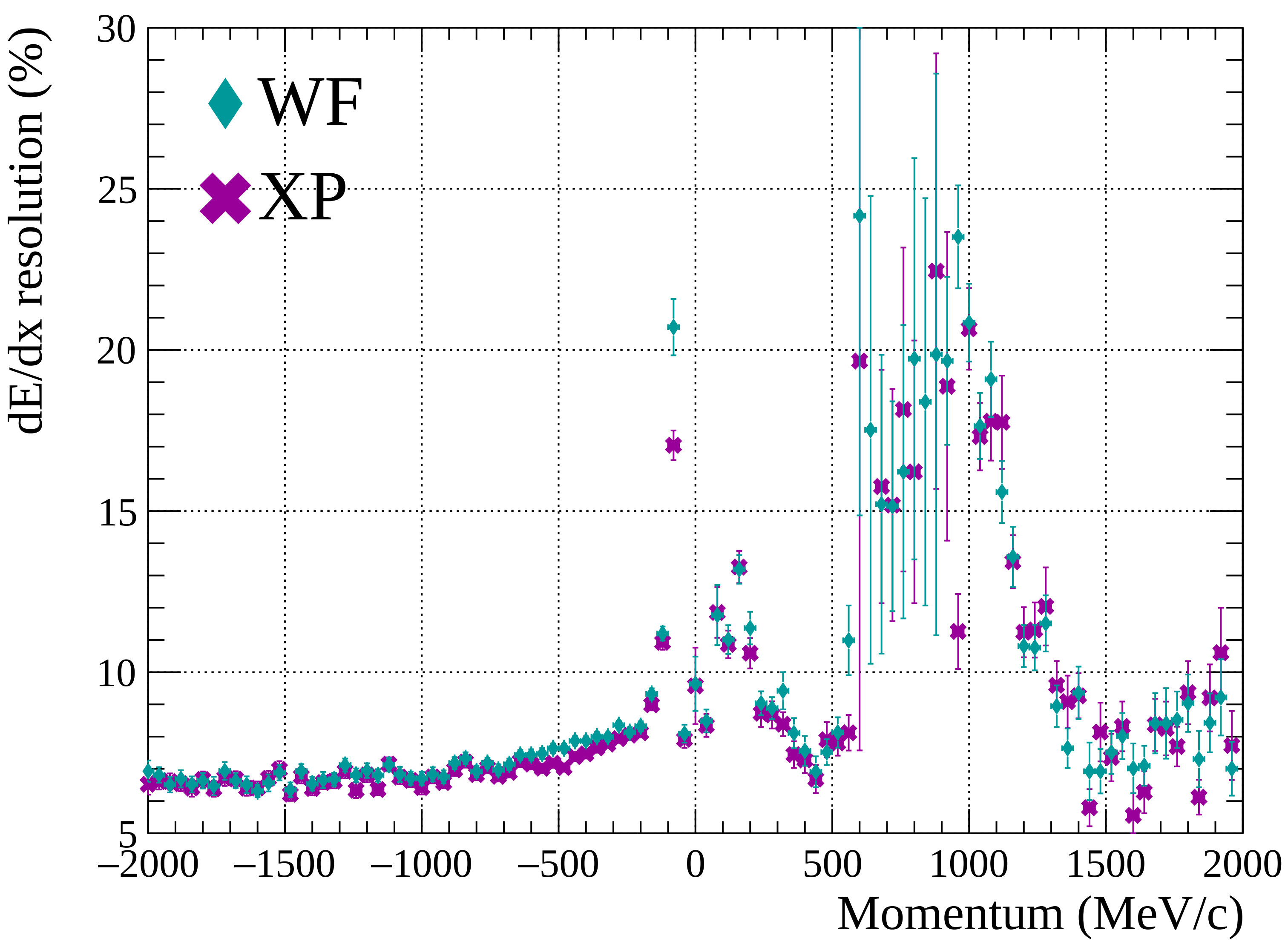

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
100
80
60
40
20
0


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

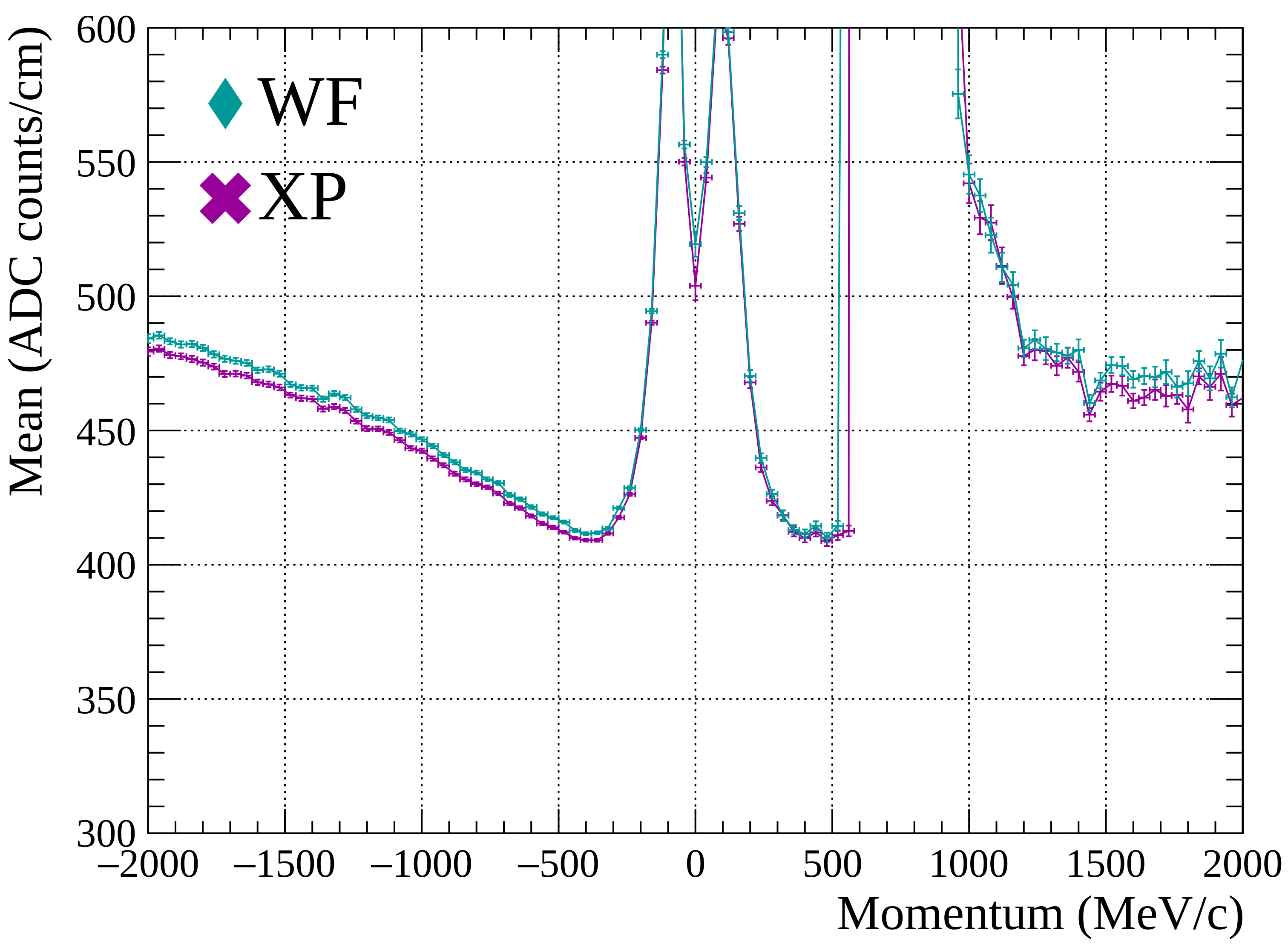


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
300
350
400
450
500
550
600
0
200
400
600
800
1000
1200
1400
1600
1800
2000

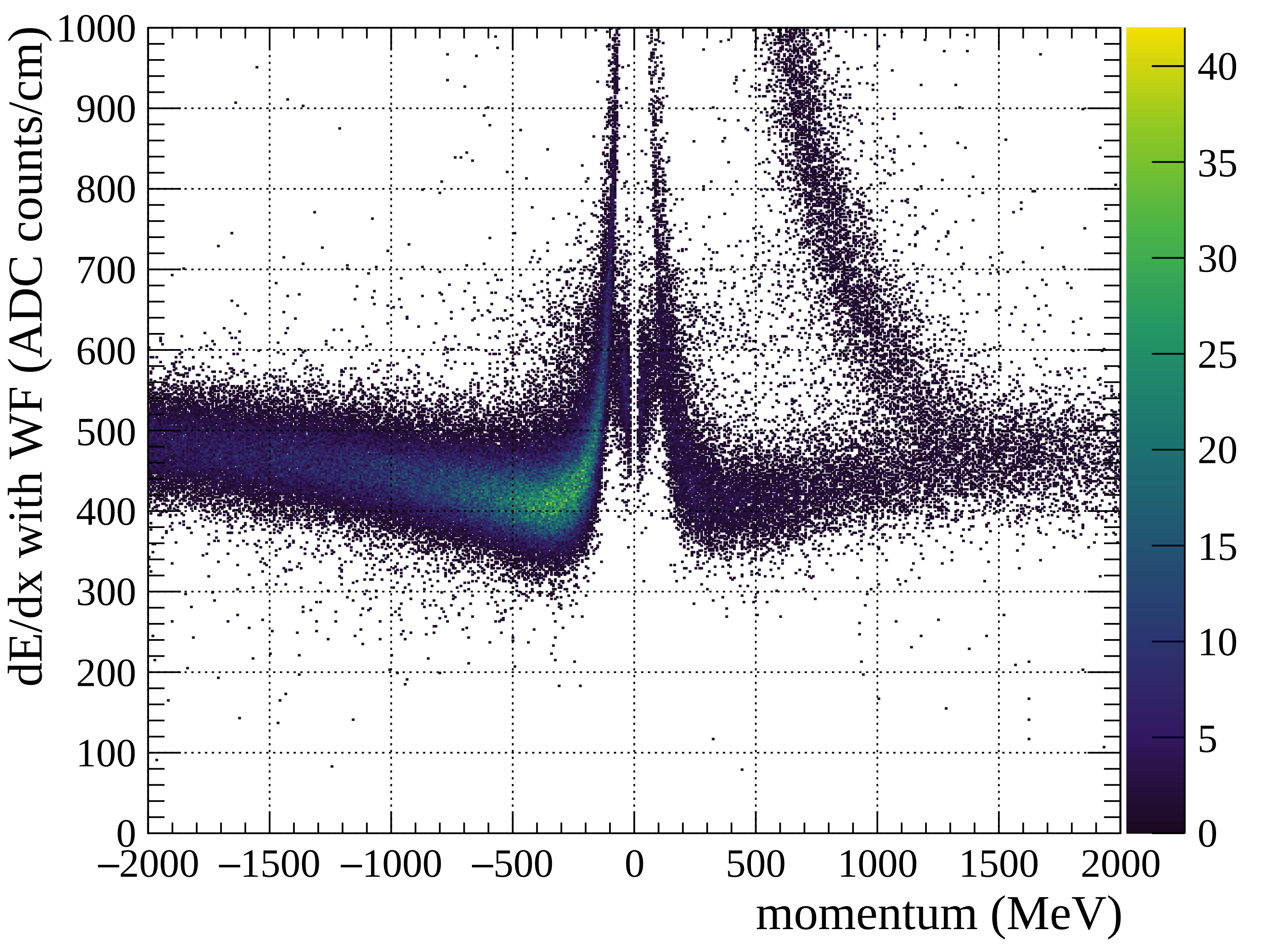

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)
160
140
120
100
80
60
40
20
0



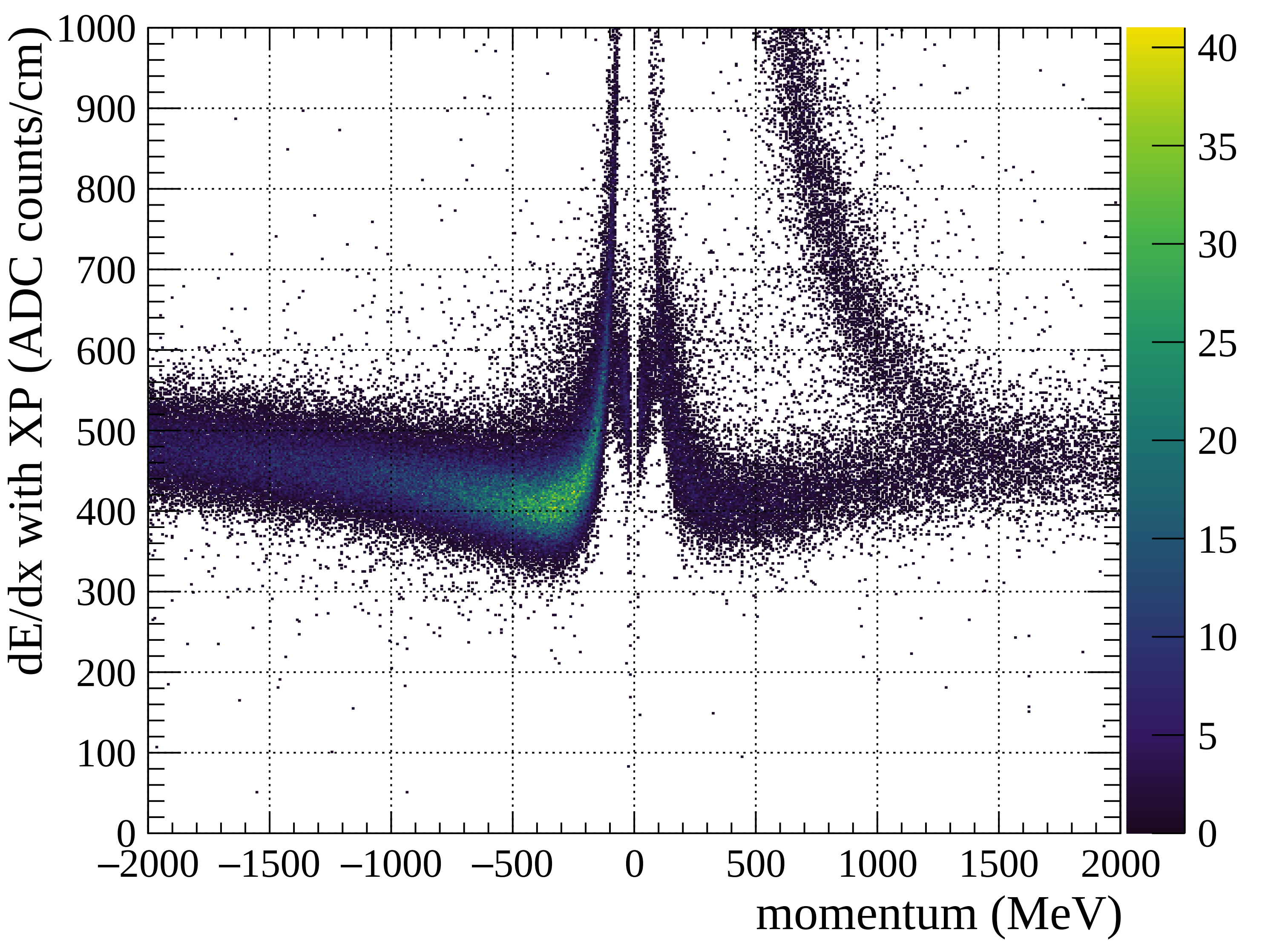
dE/dx resolution (%)
WF
XP
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

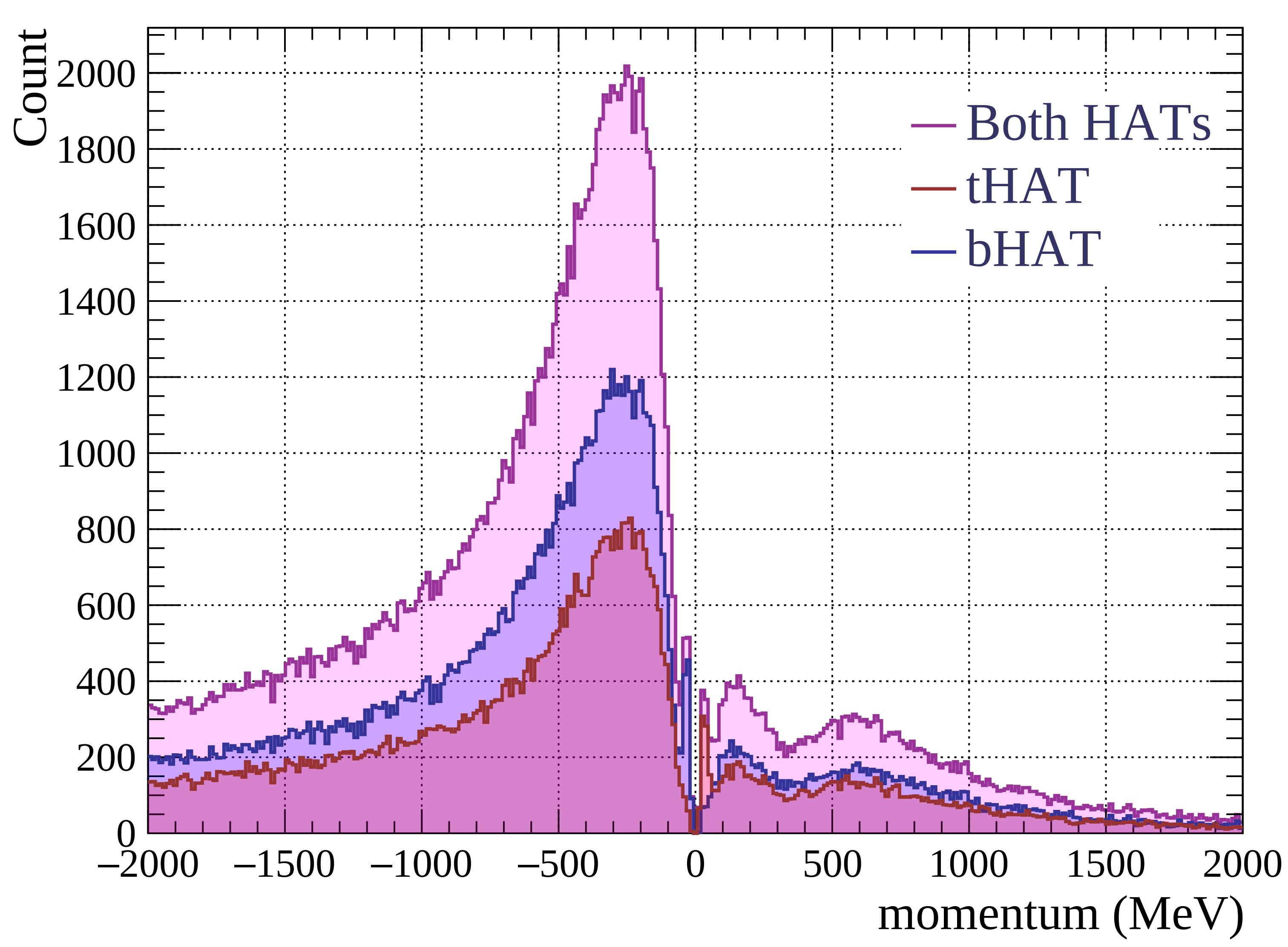


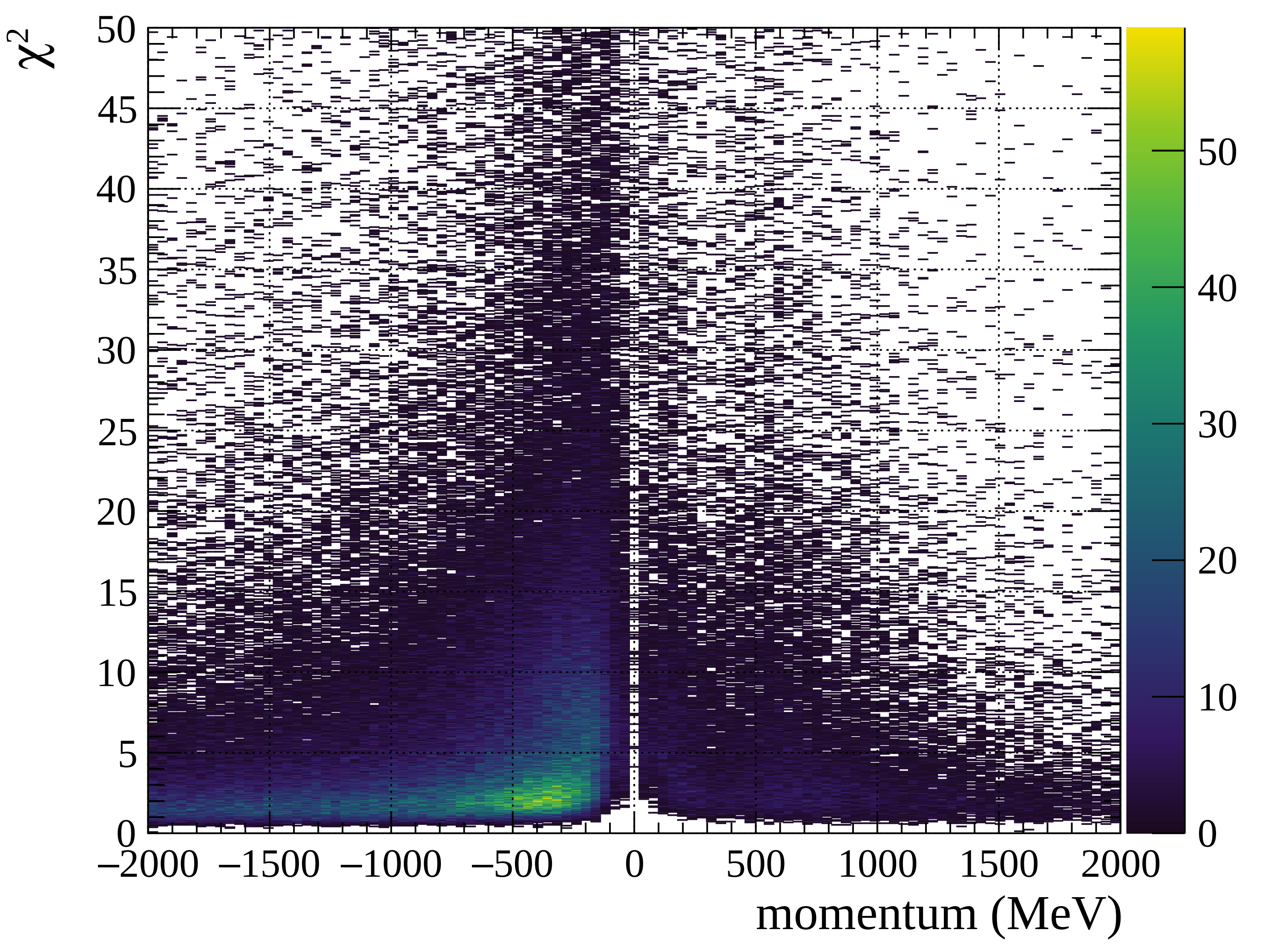
Mean (ADC counts/cm)
Momentum (MeV/c)
WF
XP
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000



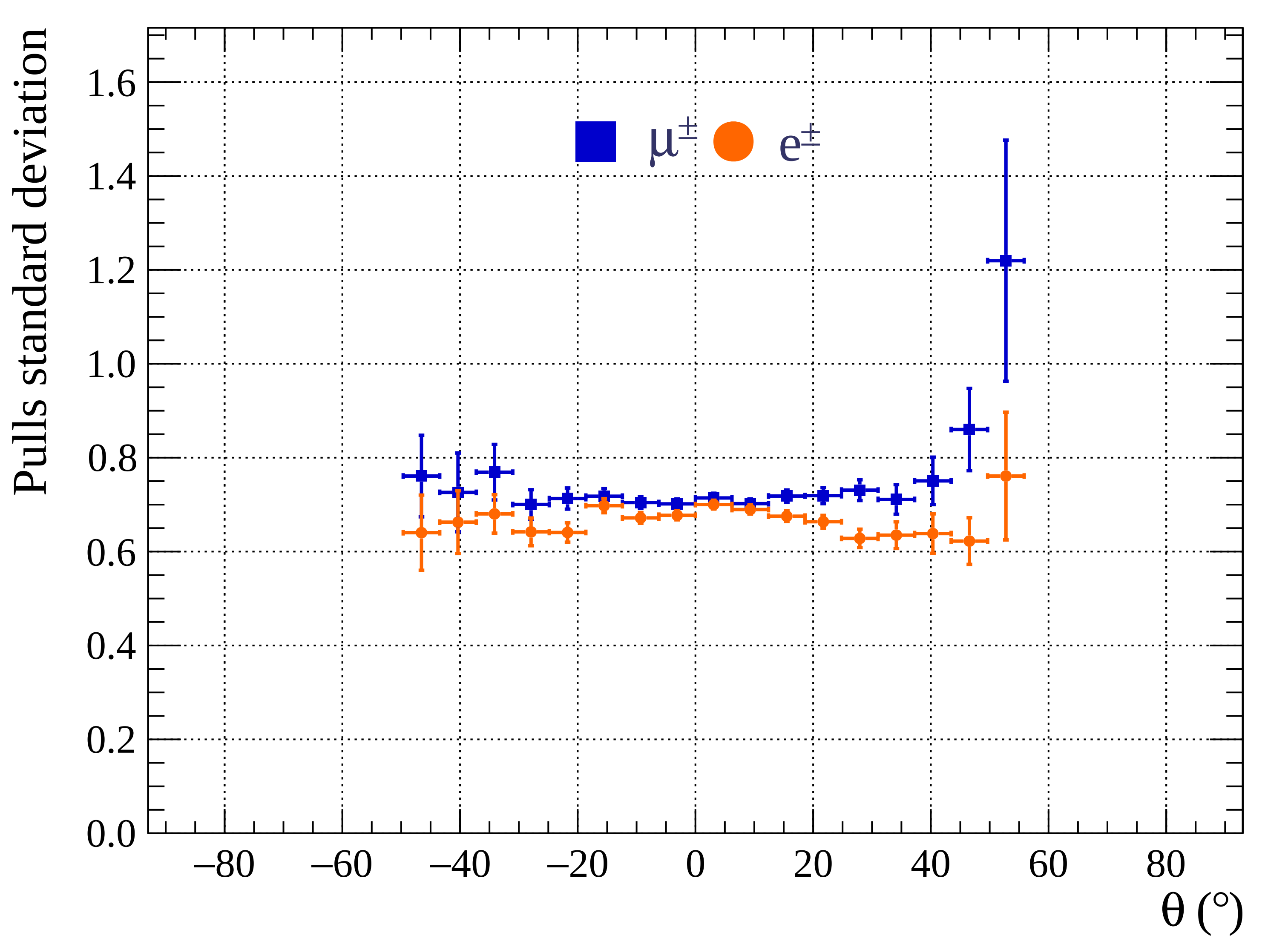
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
40
35
30
25
20
15
10
5
0


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
40
35
30
25
20
15
10
5
0

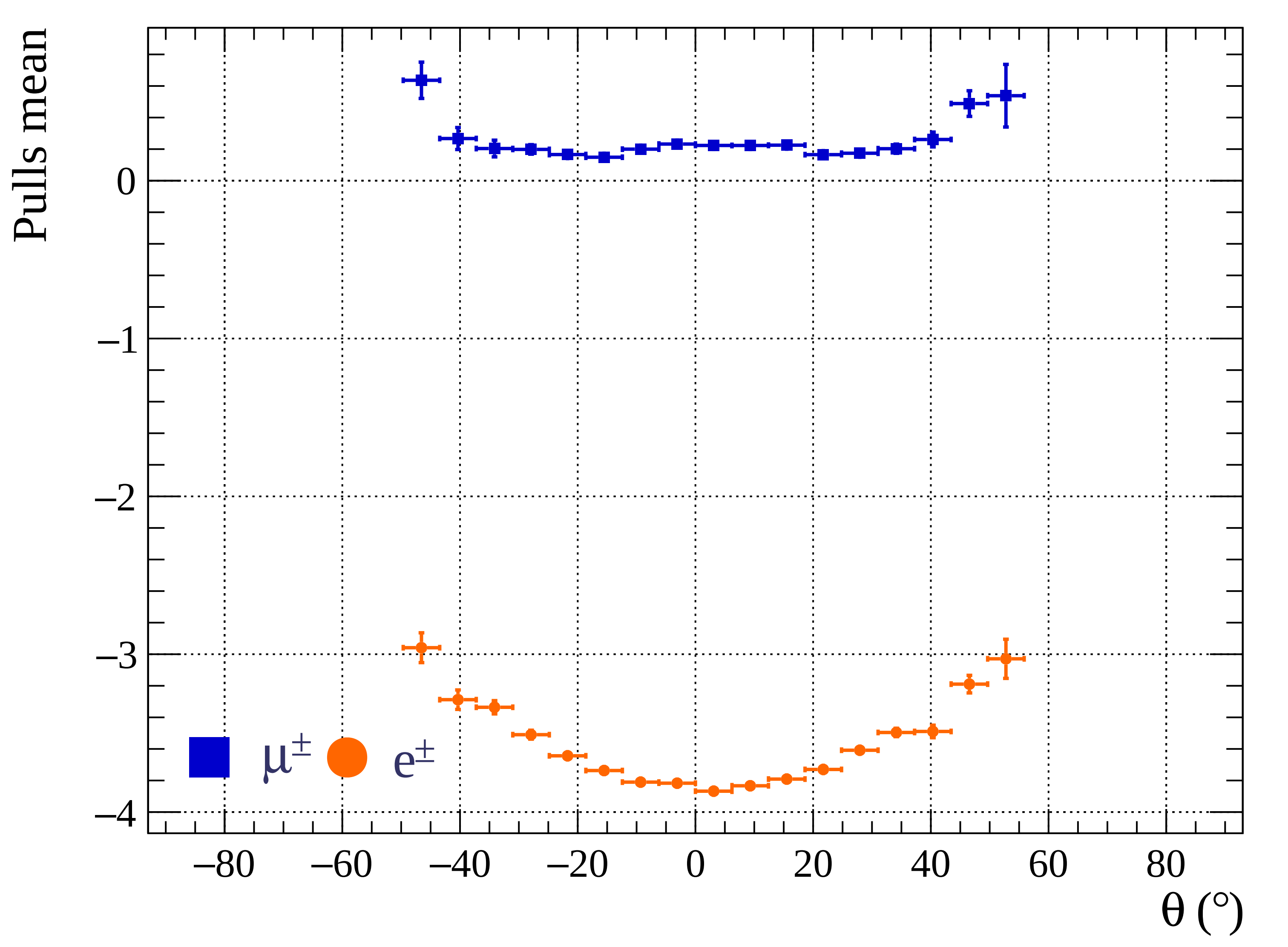

Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



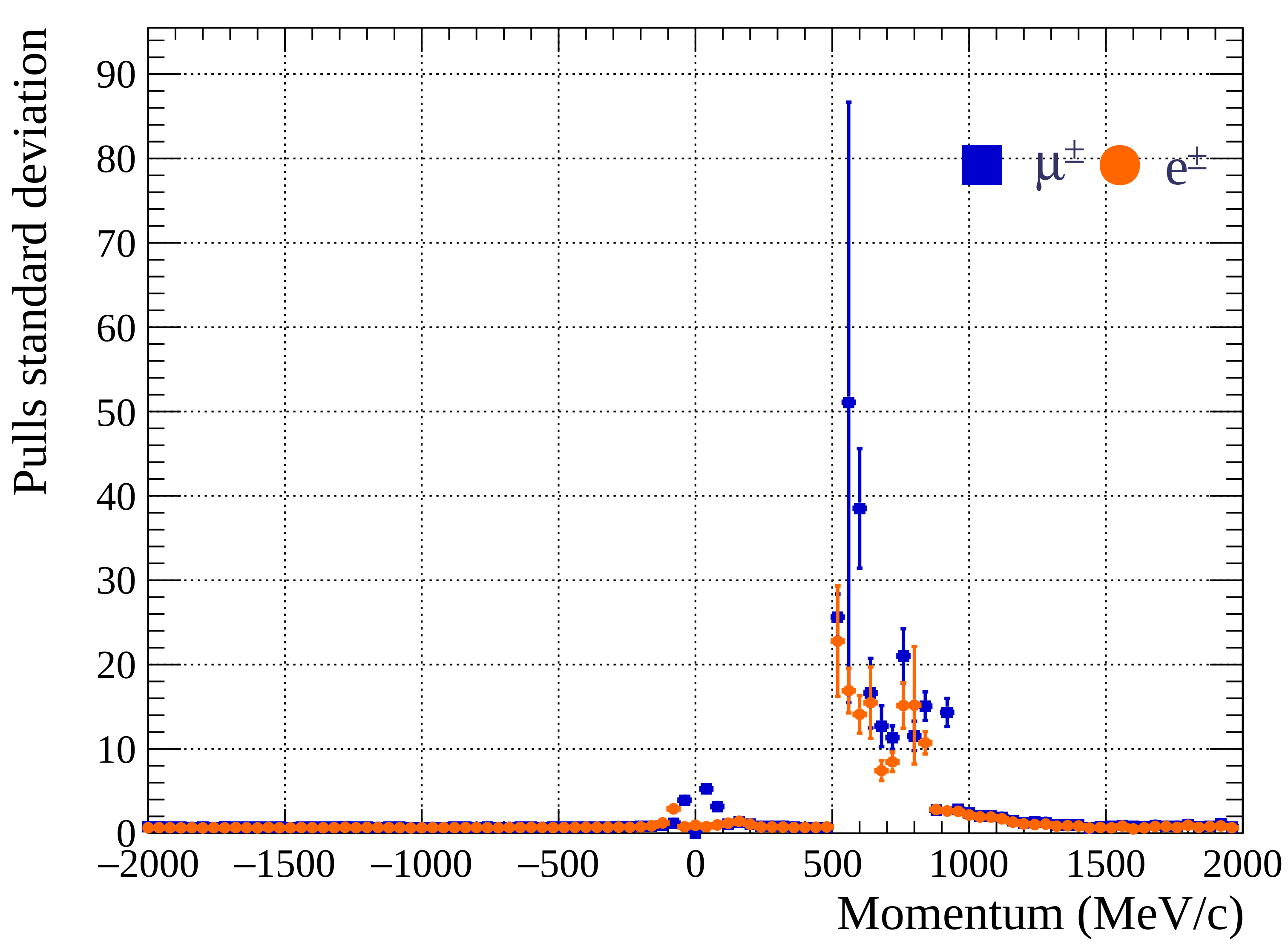
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
50
40
30
20
10
0



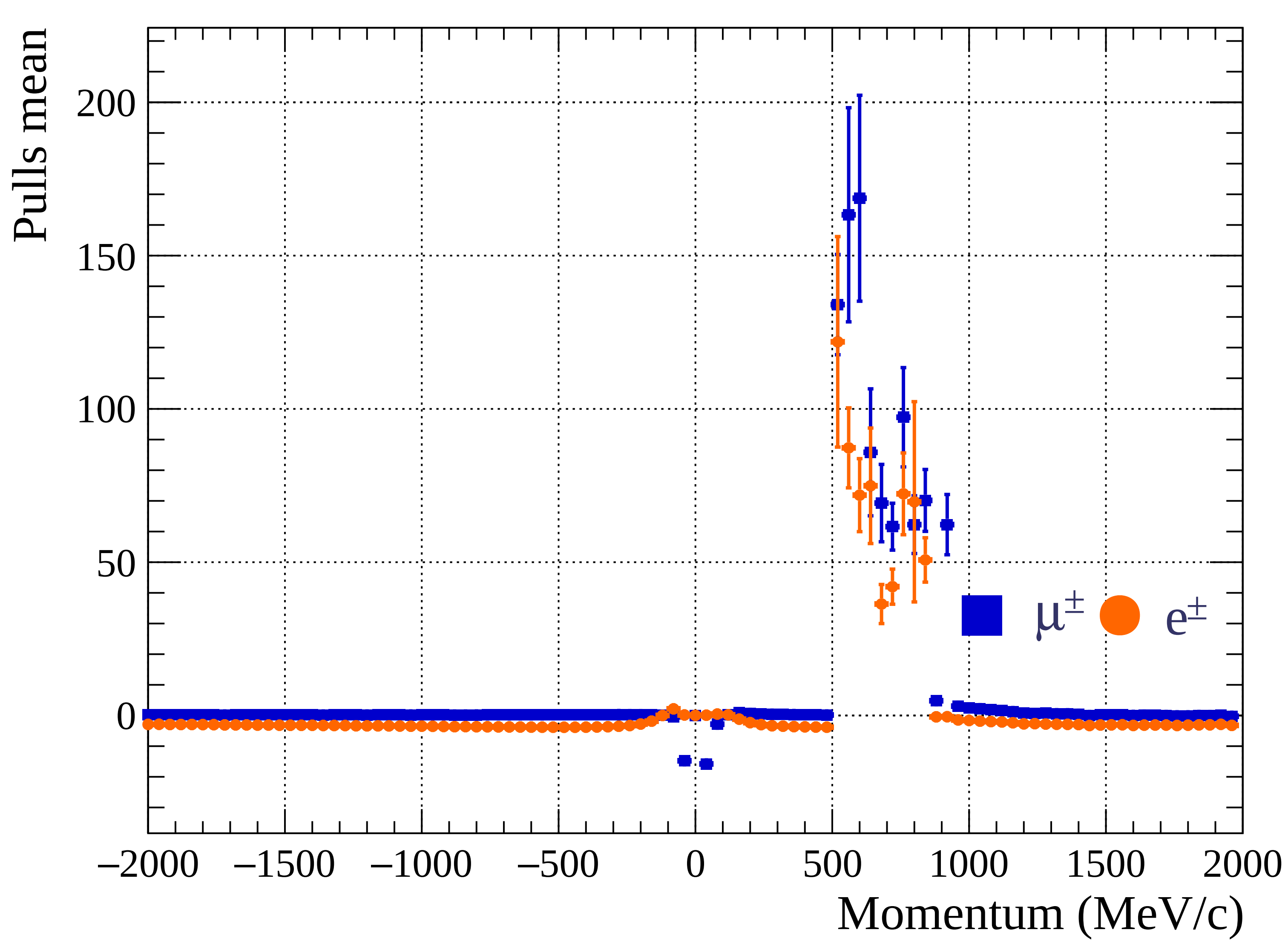
Pulls standard deviation
μ±
e±
θ (°)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
−80
−60
−40
−20
0
20
40
60
80

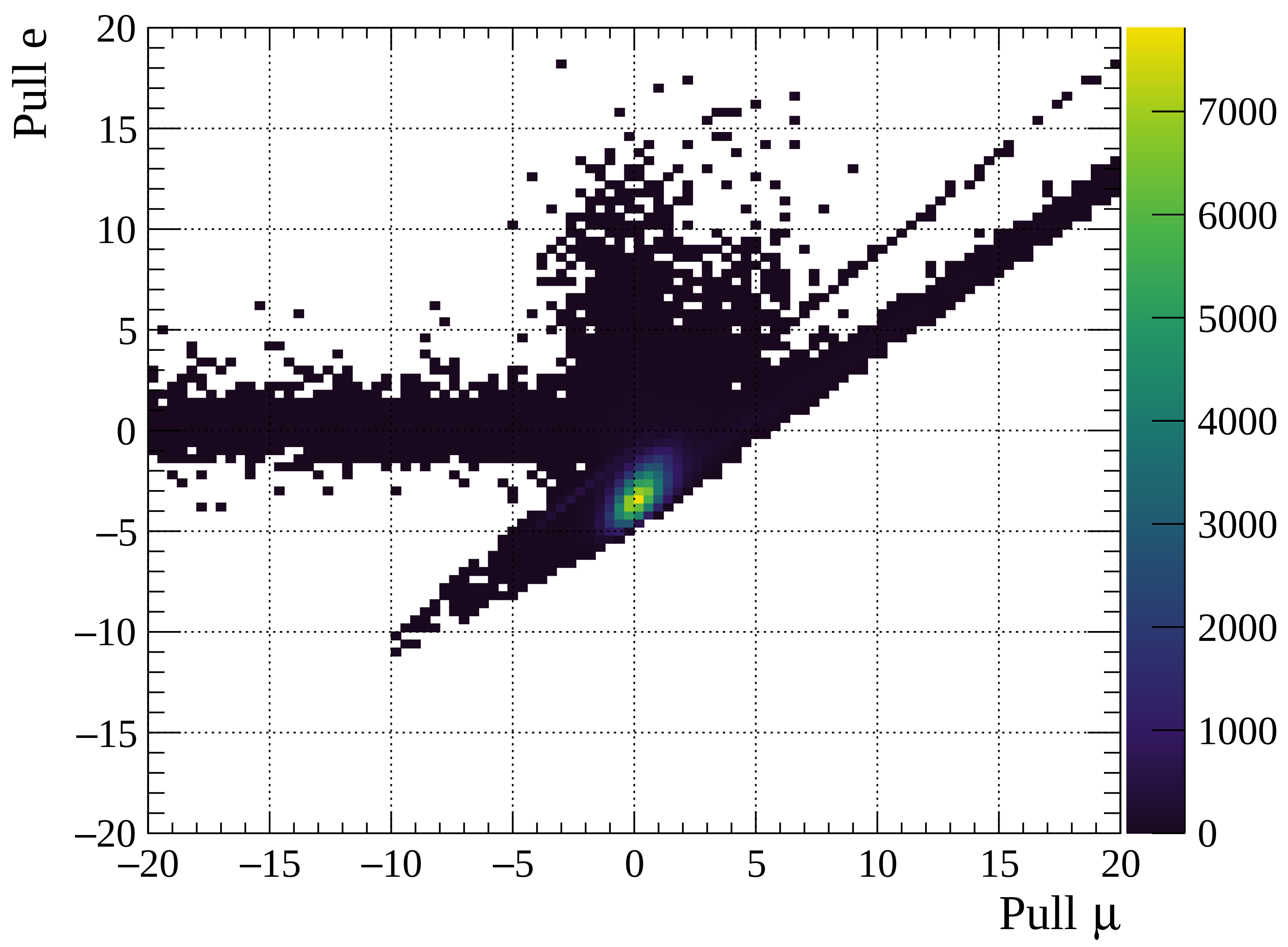


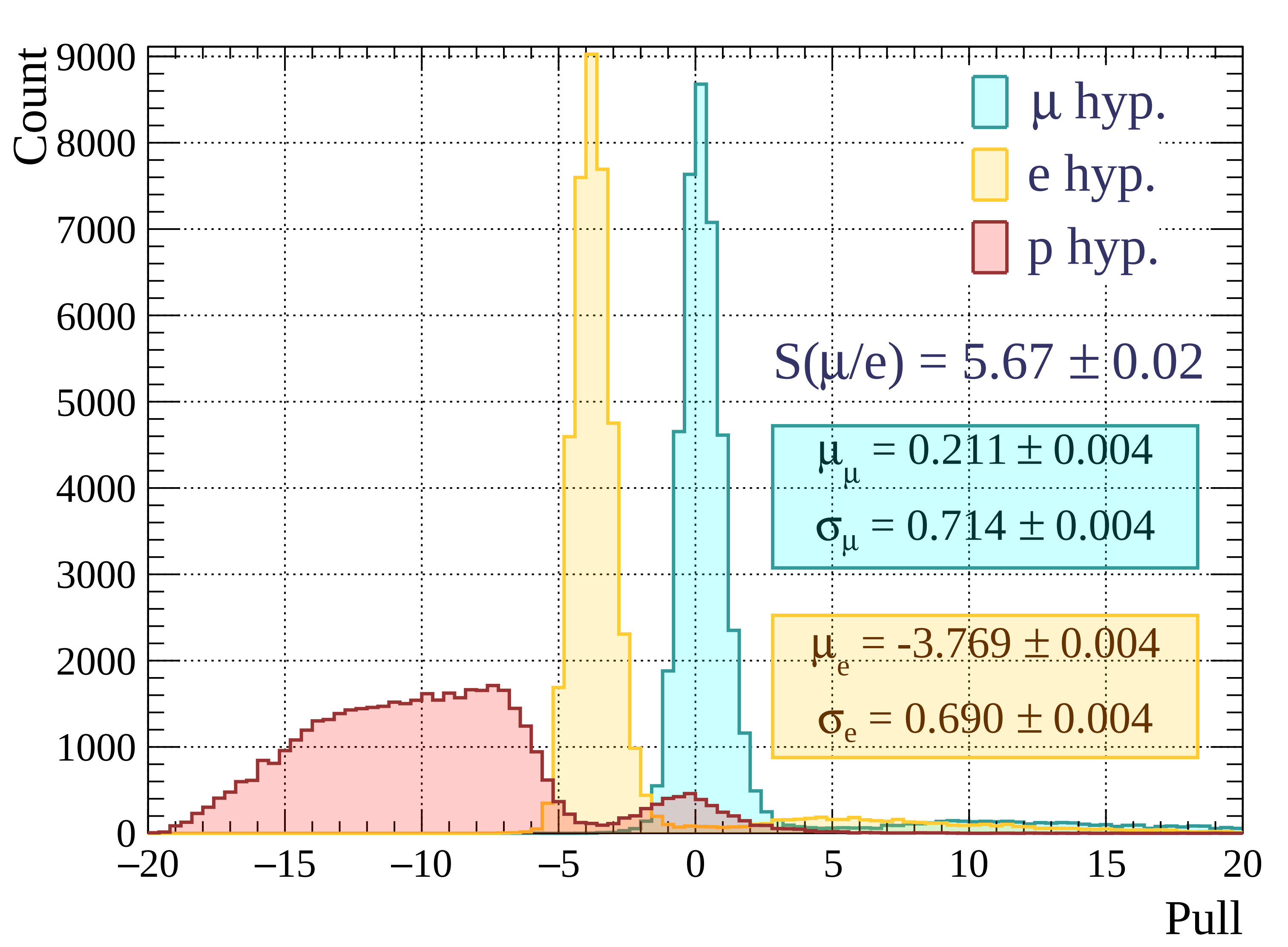
Pulls mean
0
−1
−2
−3
−4
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

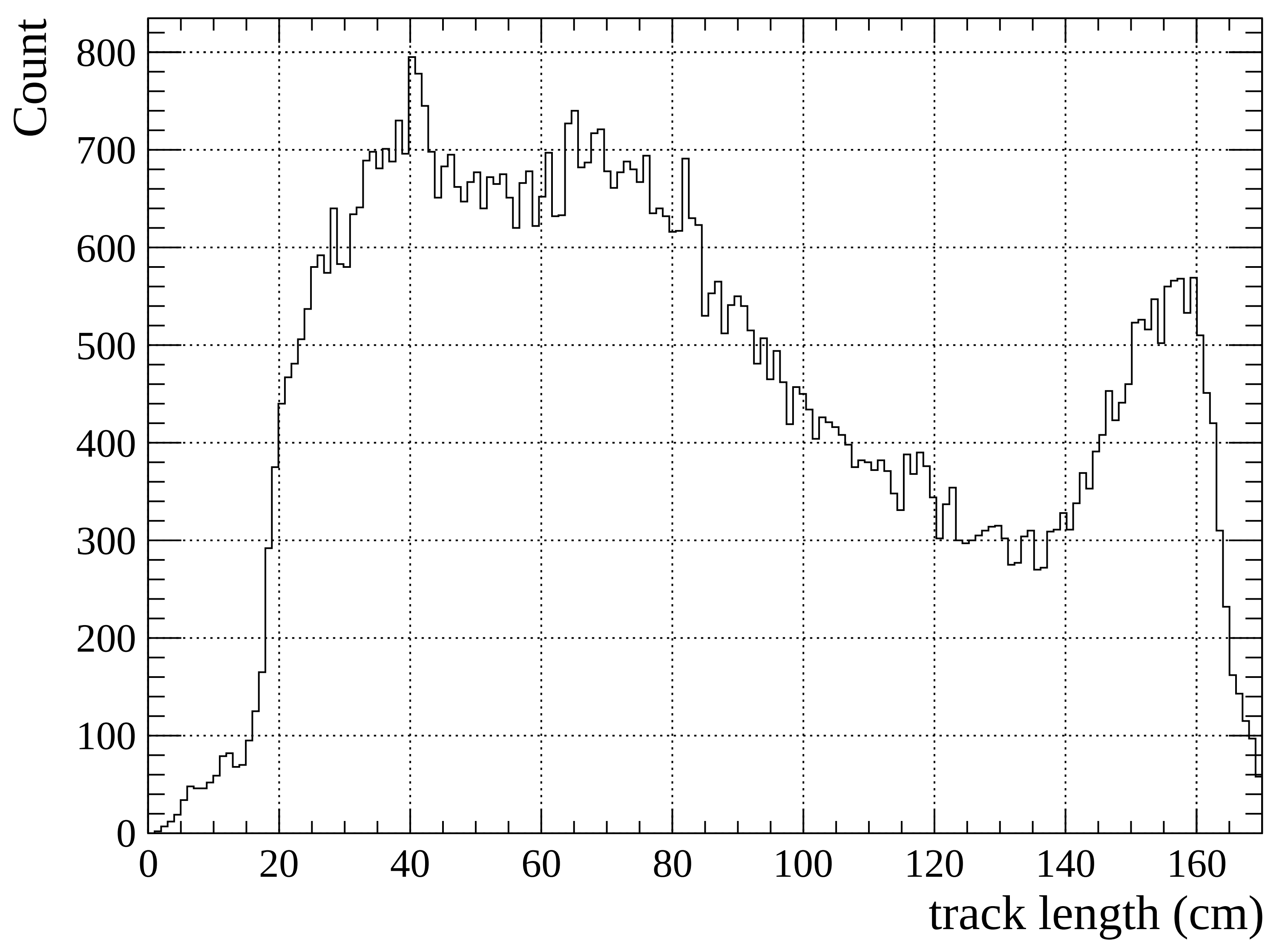


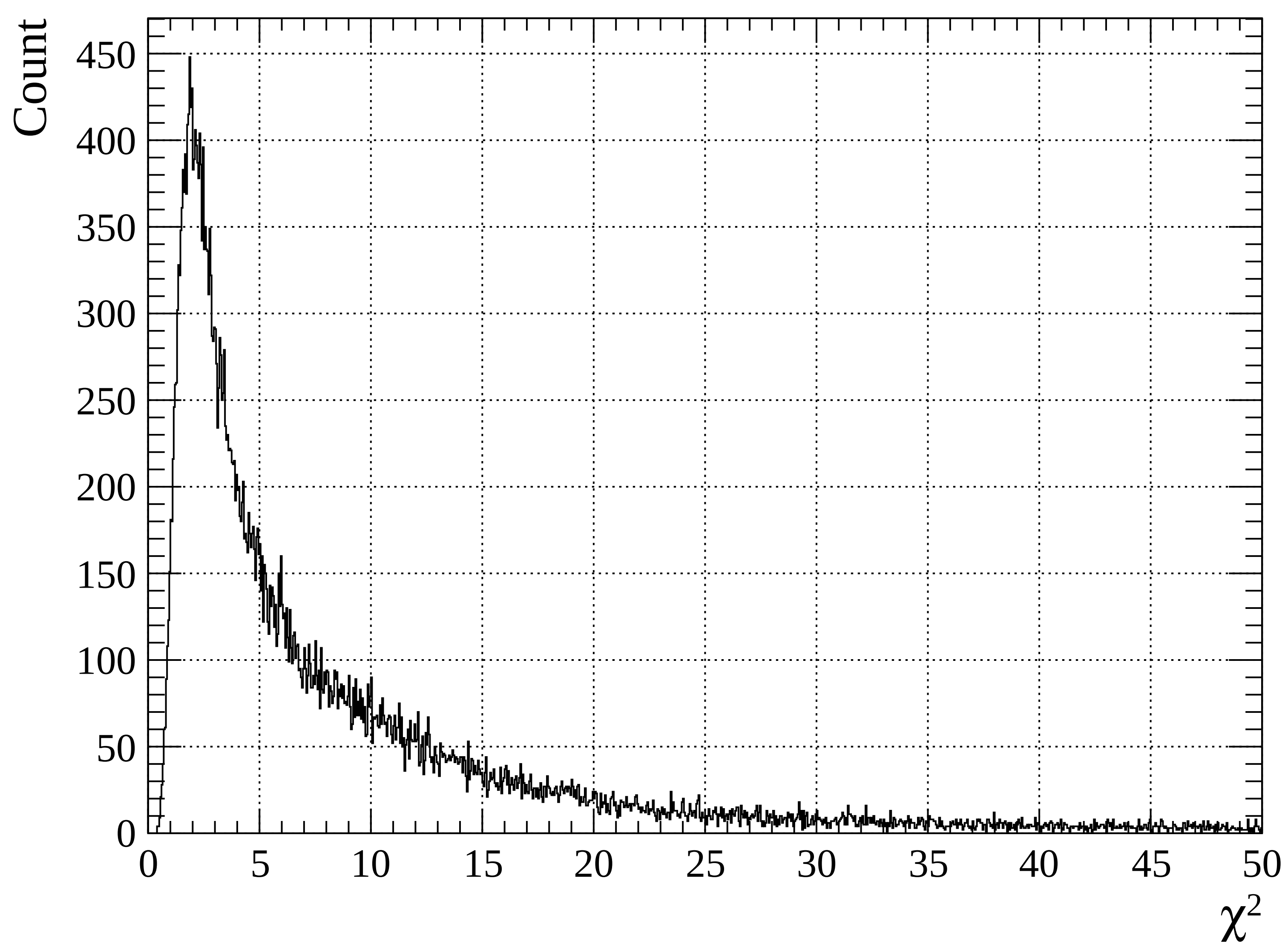
Pulls standard deviation
90
80
70
60
50
40
30
20
10
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

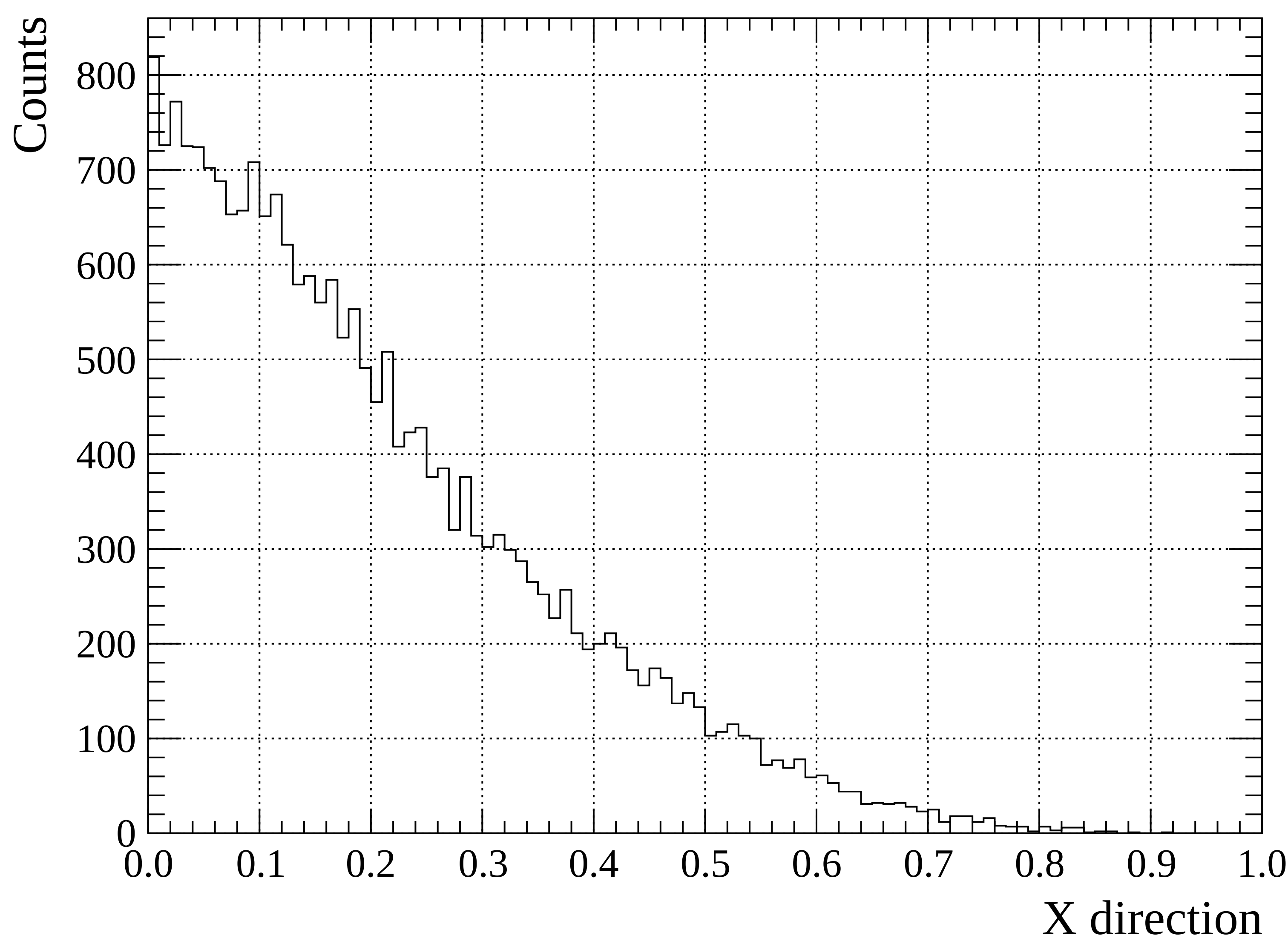

Pulls mean
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

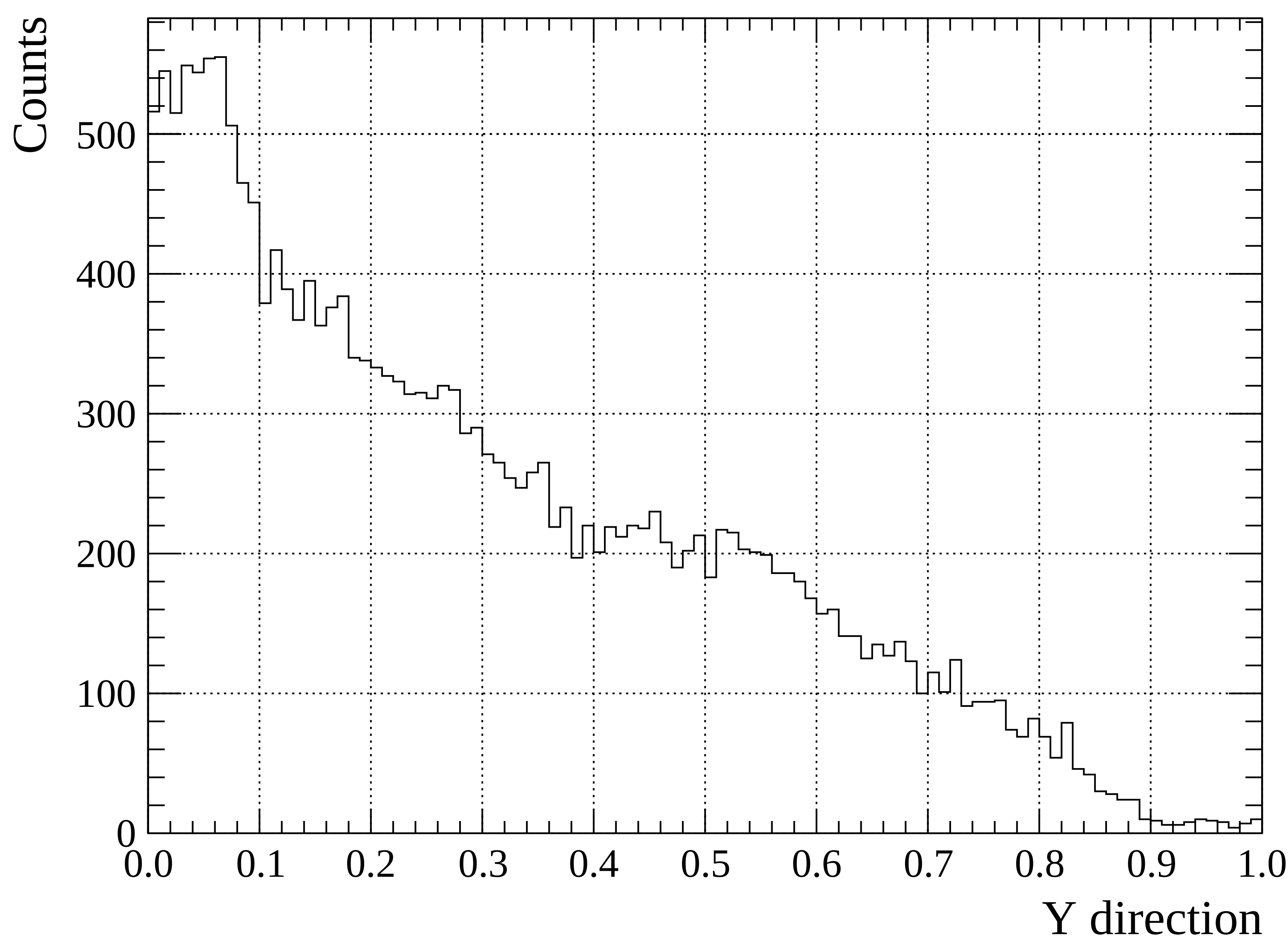

Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
7000
6000
5000
4000
3000
2000
1000
0

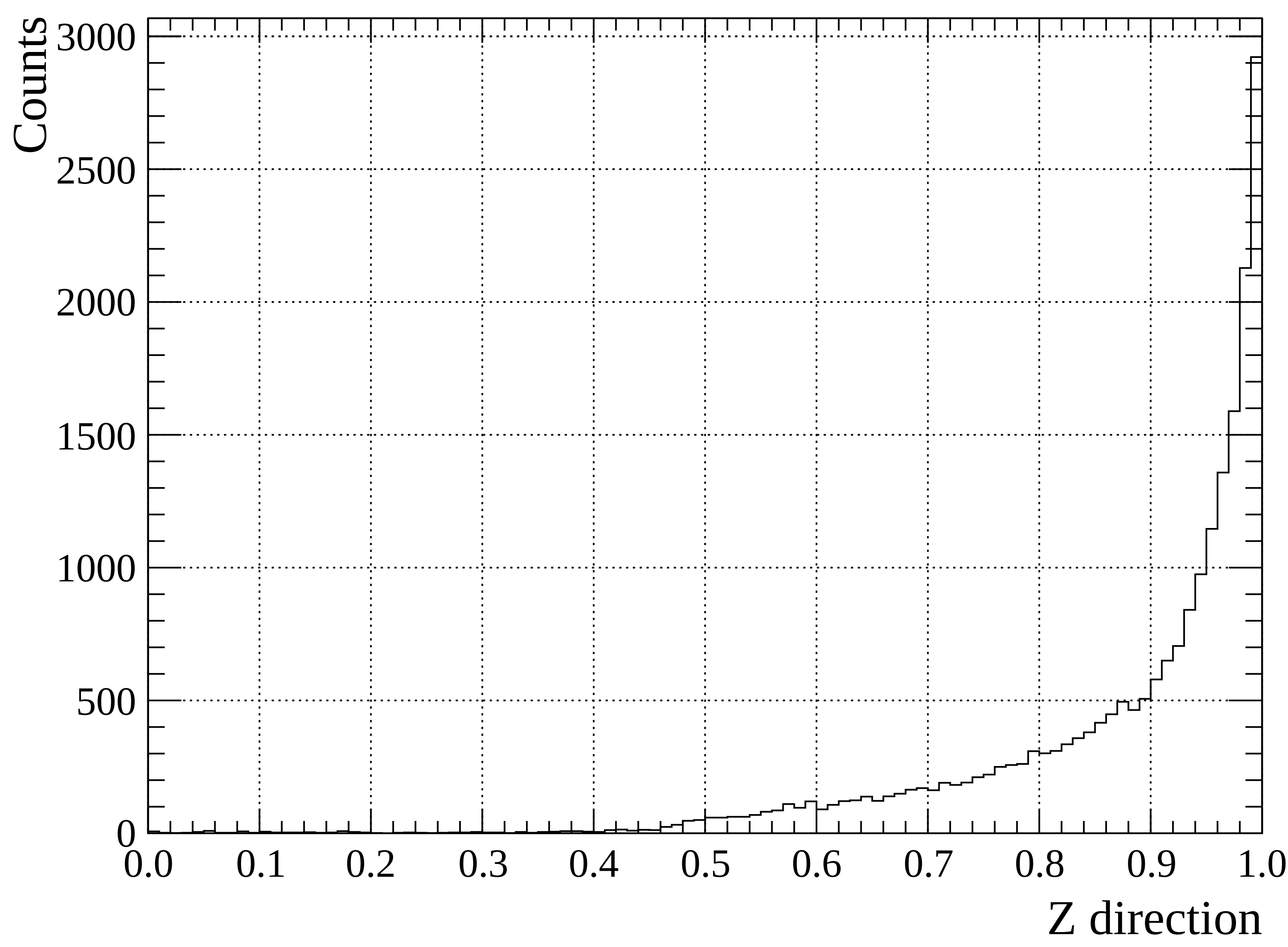

Count
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.67 ± 0.02
μ_μ = 0.211 ± 0.004
σ_μ = 0.714 ± 0.004
μ_e = -3.769 ± 0.004
σ_e = 0.690 ± 0.004
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
−20
−15
−10
−5
0
5
10
15
20

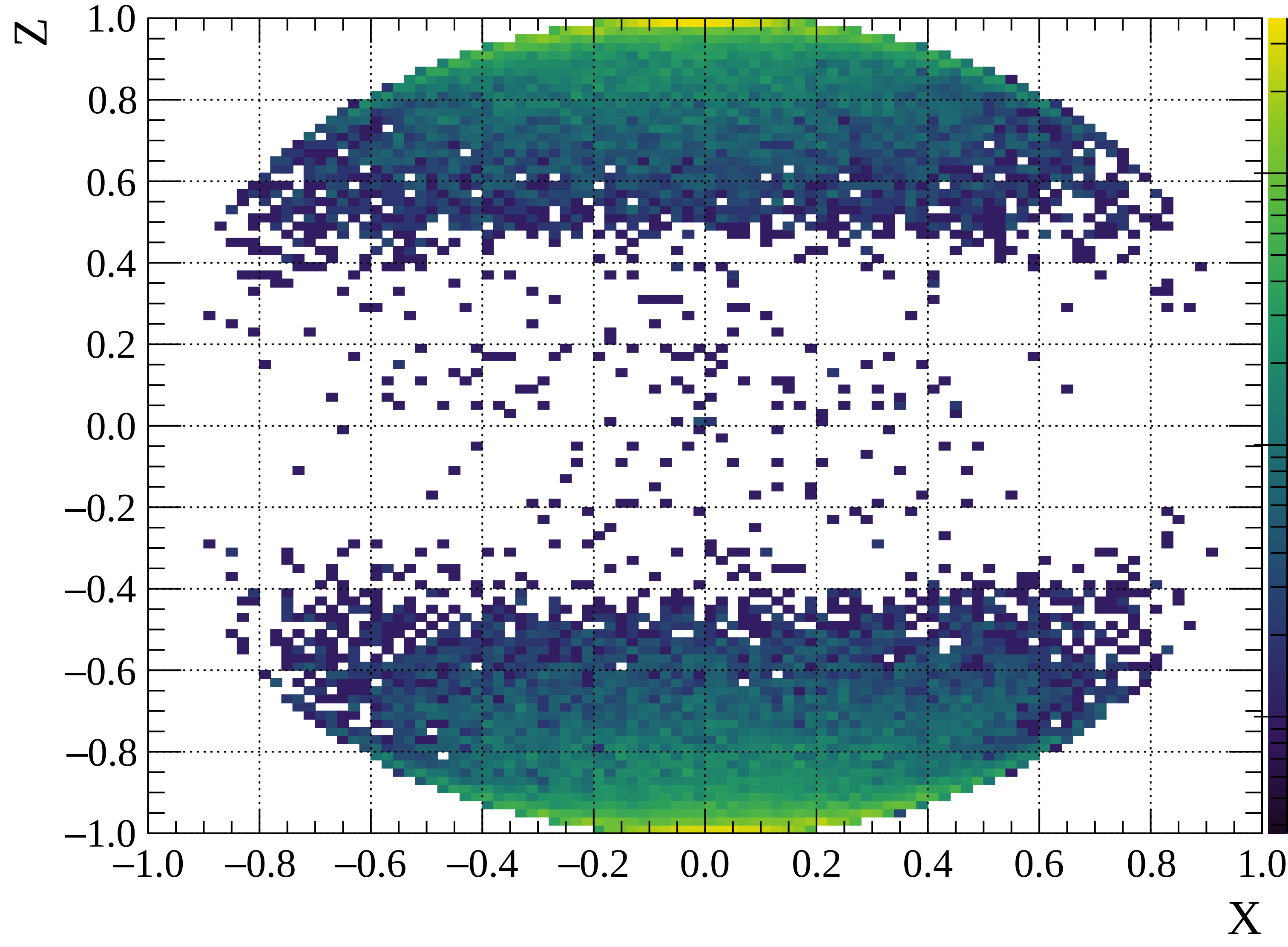

Count
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)

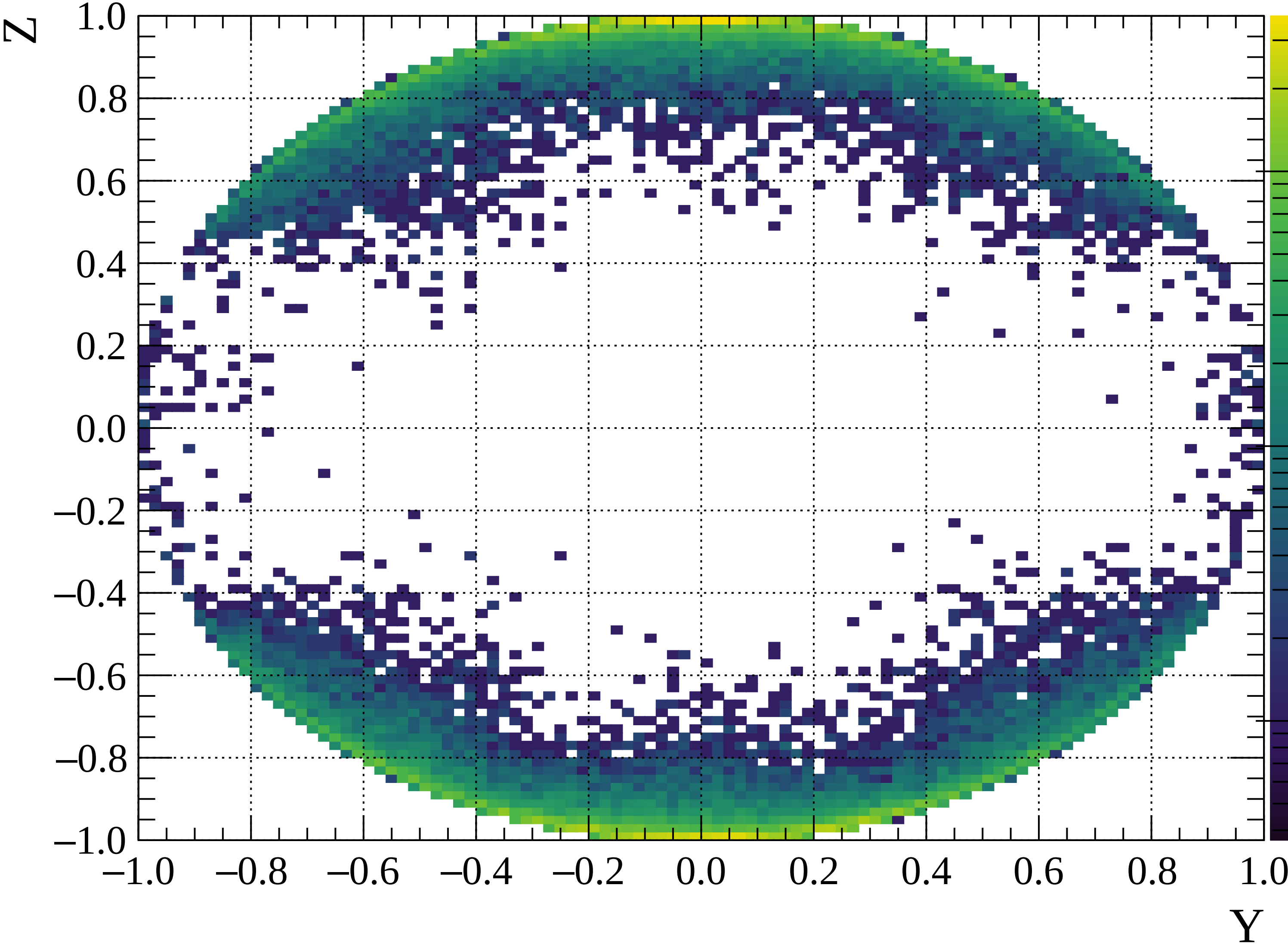

Count
450
400
350
300
250
200
150
100
50
0
0
5
10
15
20
25
30
35
40
45
50
χ^2


Counts
800
700
600
500
400
300
200
100
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

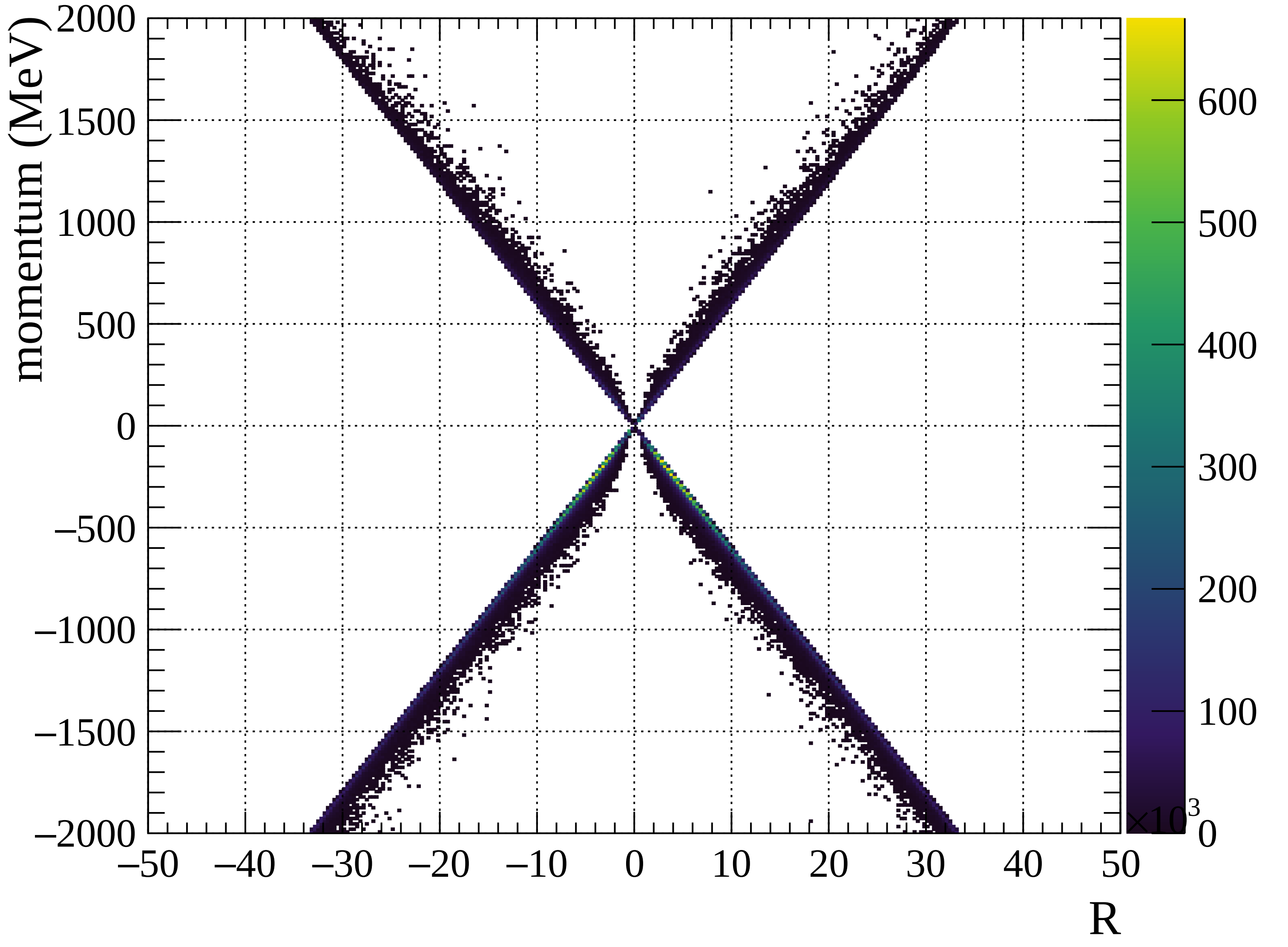

Counts
500
400
300
200
100
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction

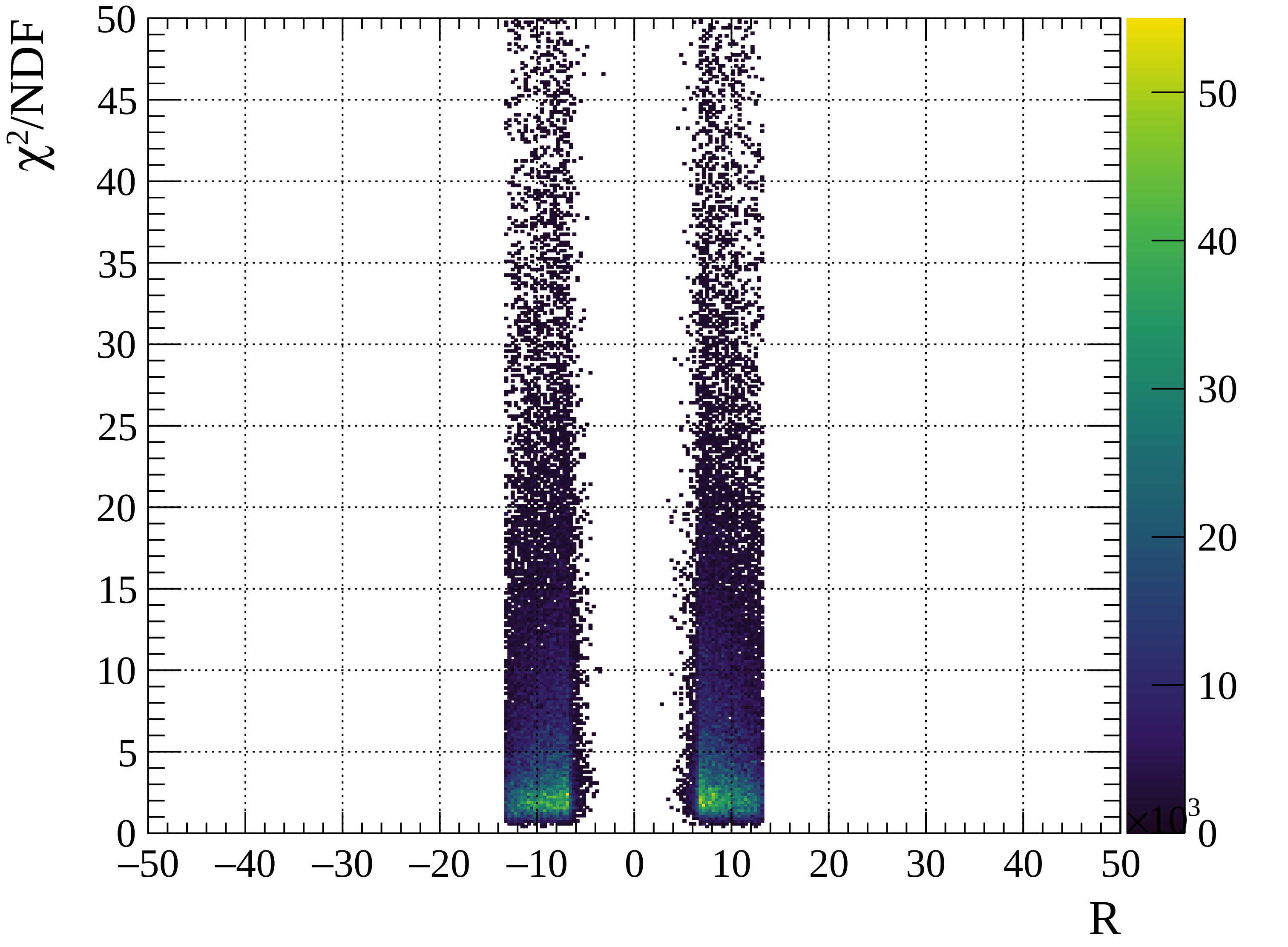

Counts
3000
2500
2000
1500
1000
500
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

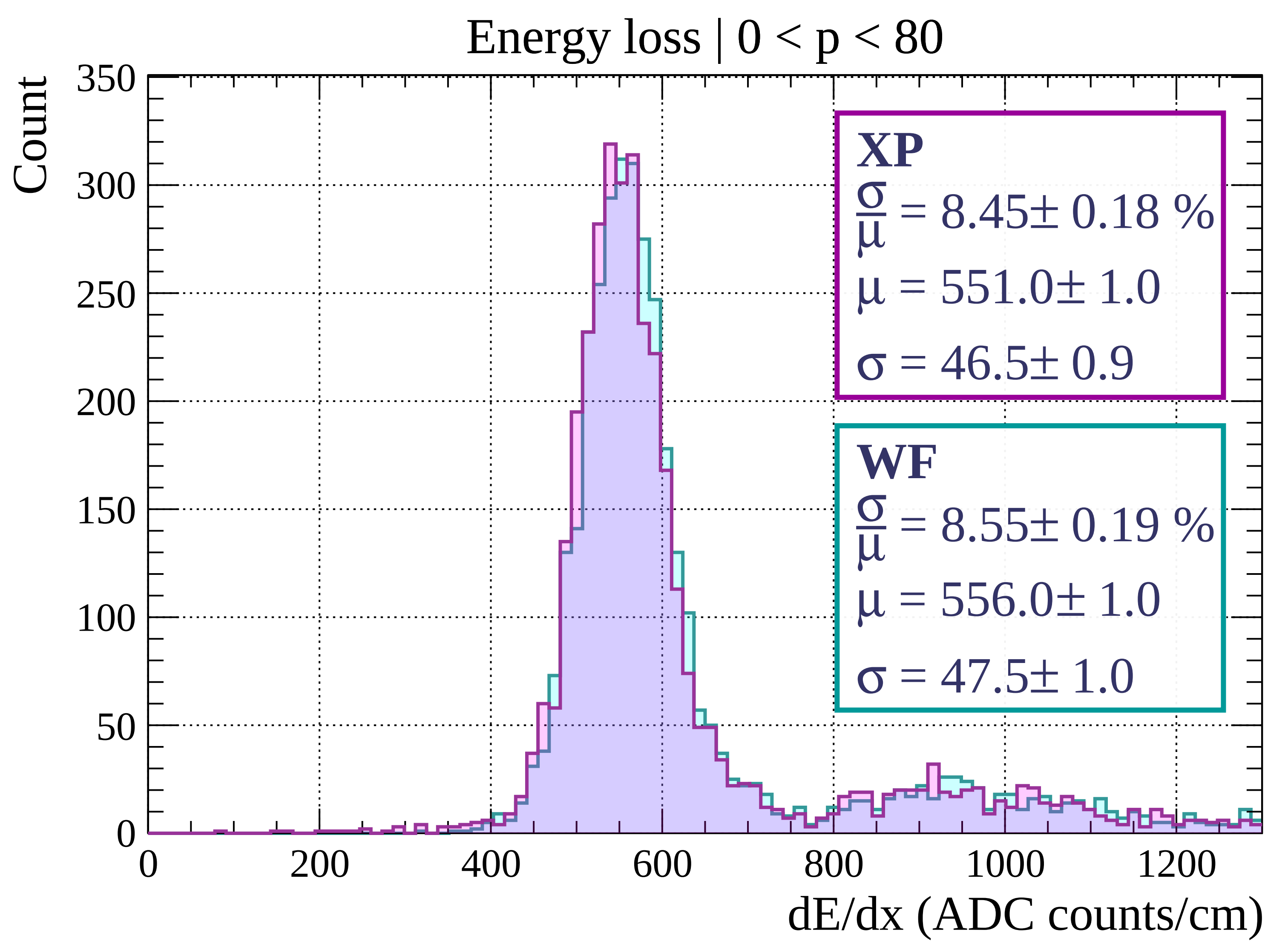


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y

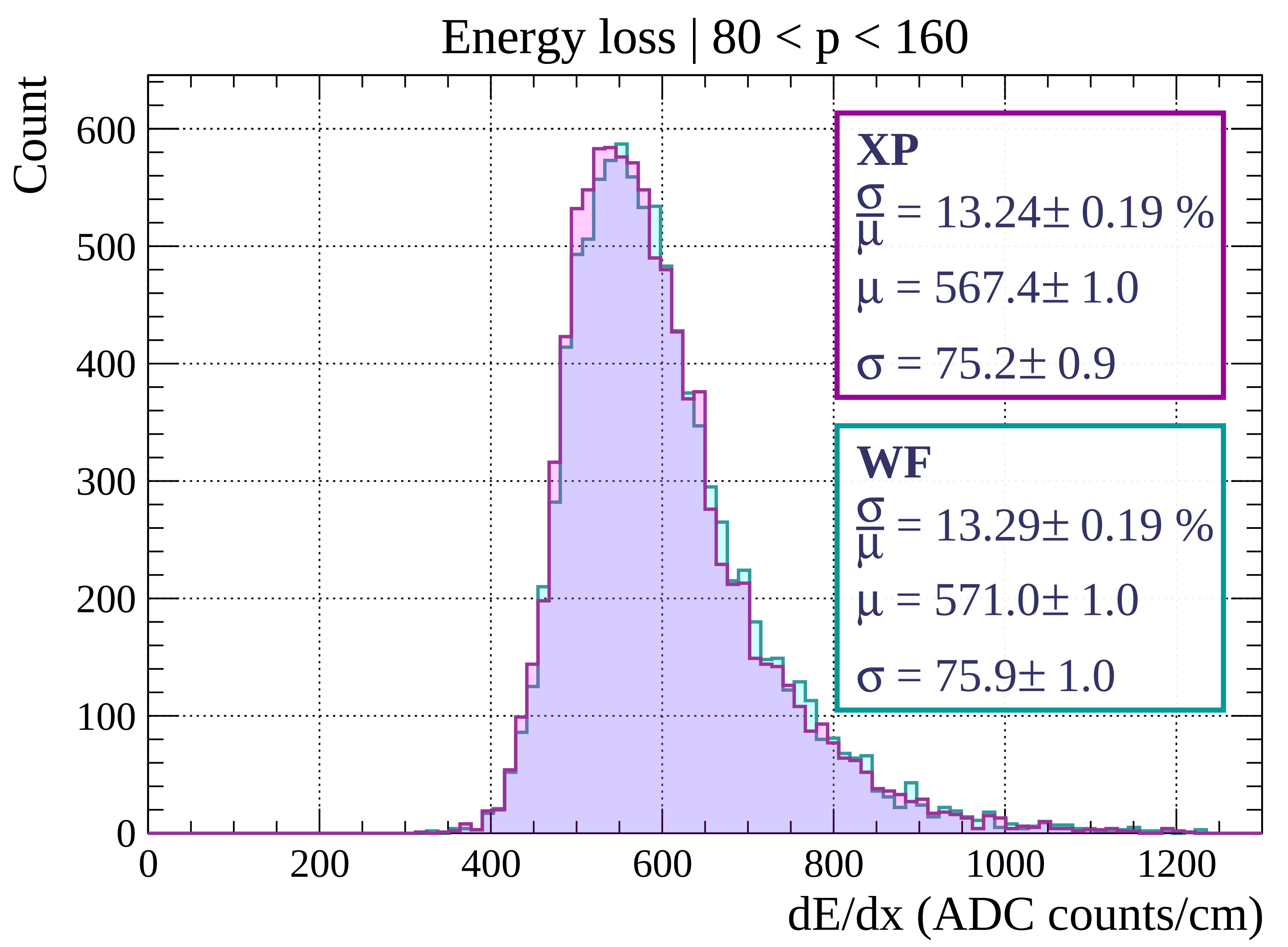

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
600
500
400
300
200
100
0
×10^3



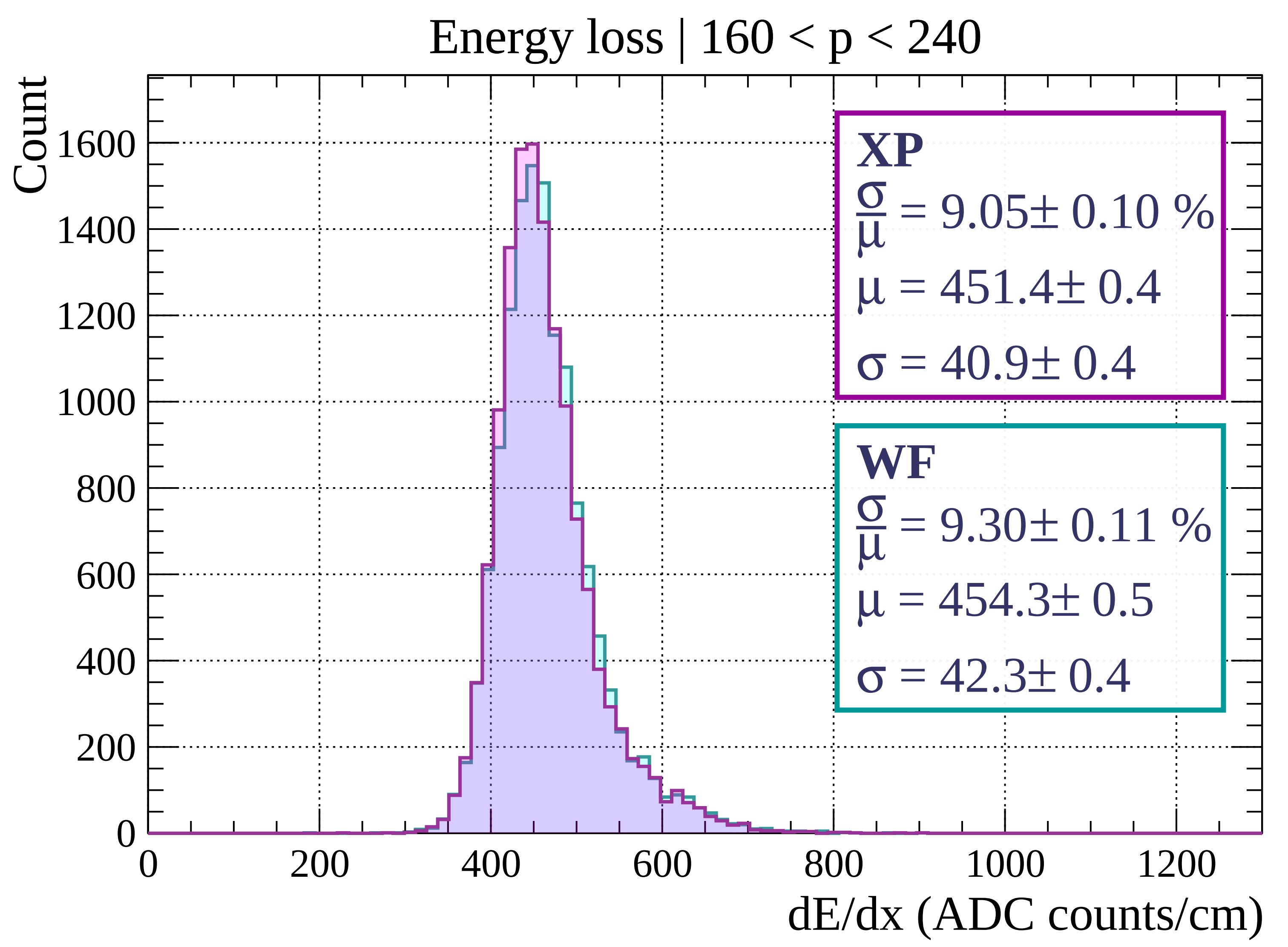
χ^2/NDF
R
$\times 10^3$
0
5
10
15
20
25
30
35
40
45
50
−50
−40
−30
−20
−10
0
10
20
30
40
50

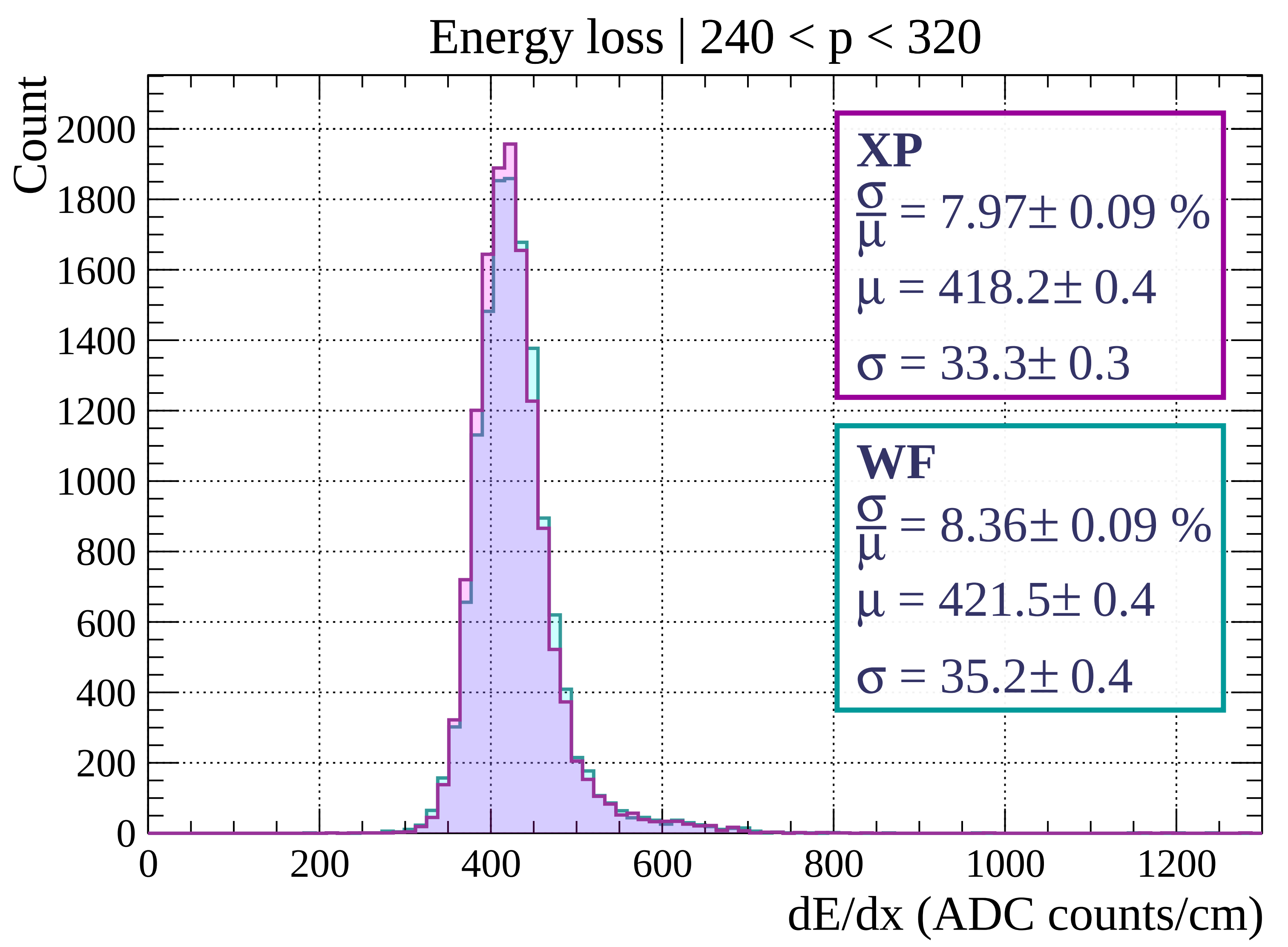


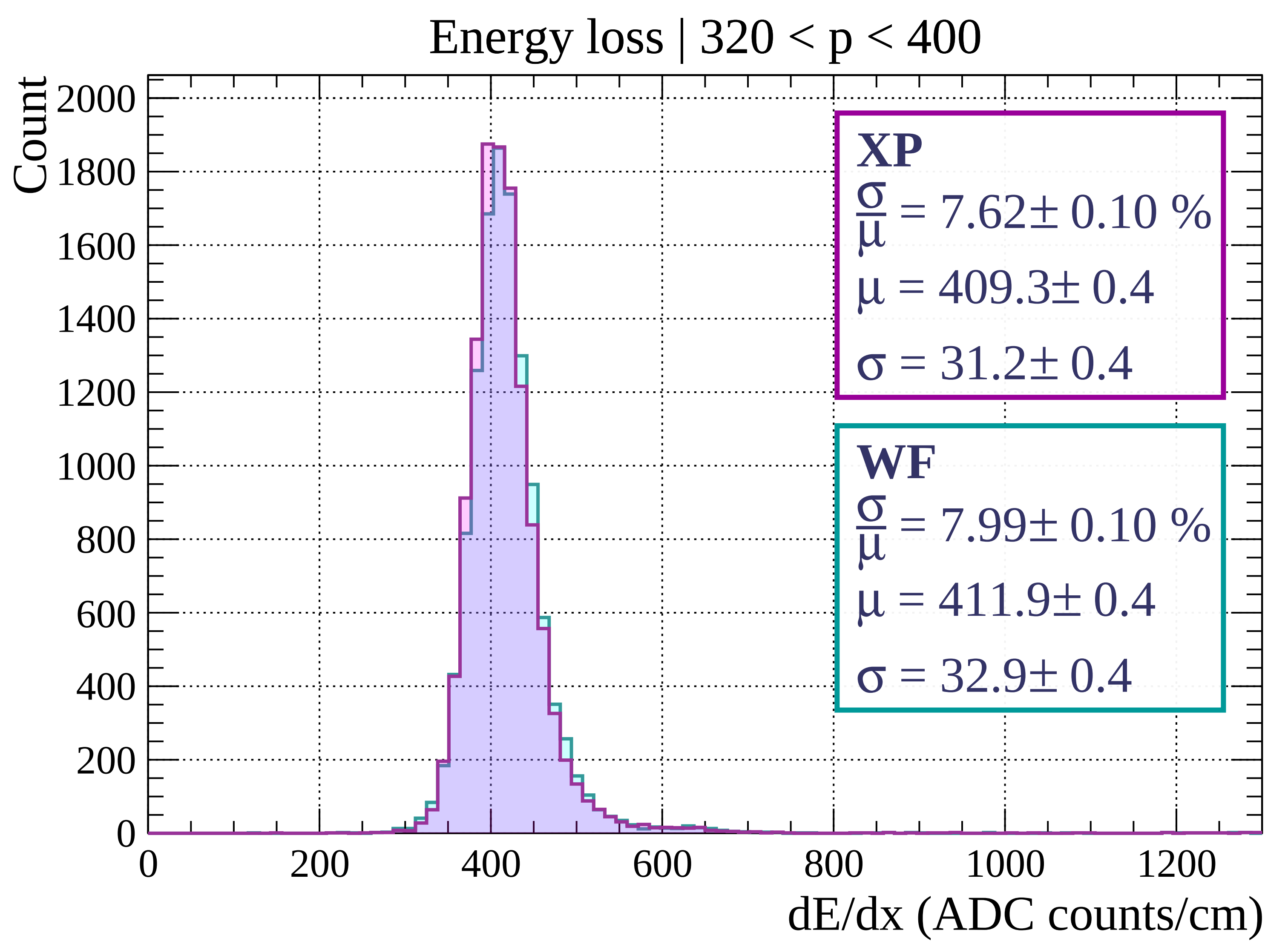
Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.45± 0.18 %
μ = 551.0± 1.0
σ = 46.5± 0.9
WF
σ/μ = 8.55± 0.19 %
μ = 556.0± 1.0
σ = 47.5± 1.0



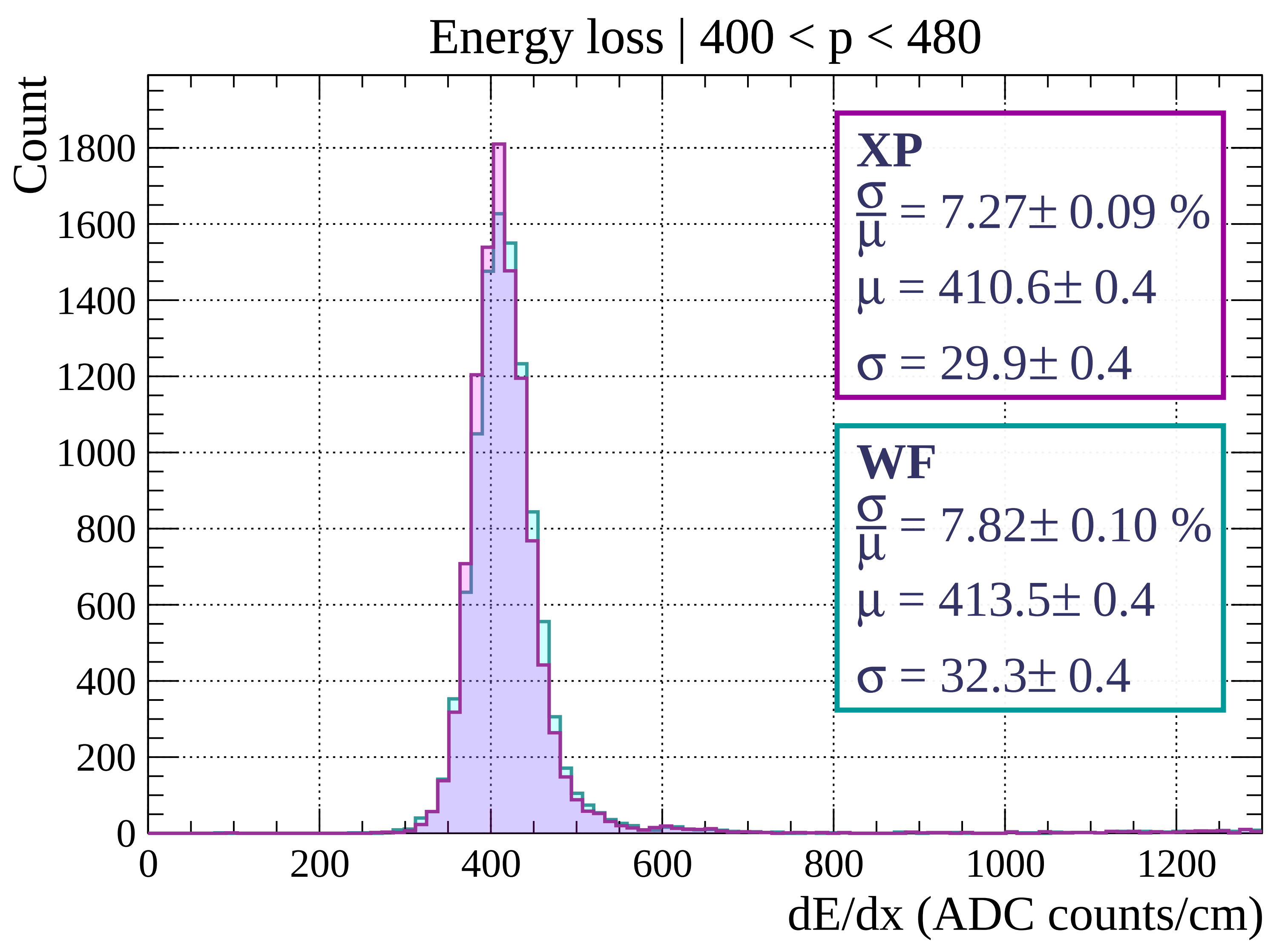
Energy loss | 80 < p < 160
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.24± 0.19 %
μ = 567.4± 1.0
σ = 75.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 13.29± 0.19 %
μ = 571.0± 1.0
σ = 75.9± 1.0


Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.05± 0.10 %
μ = 451.4± 0.4
σ = 40.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.30± 0.11 %
μ = 454.3± 0.5
σ = 42.3± 0.4

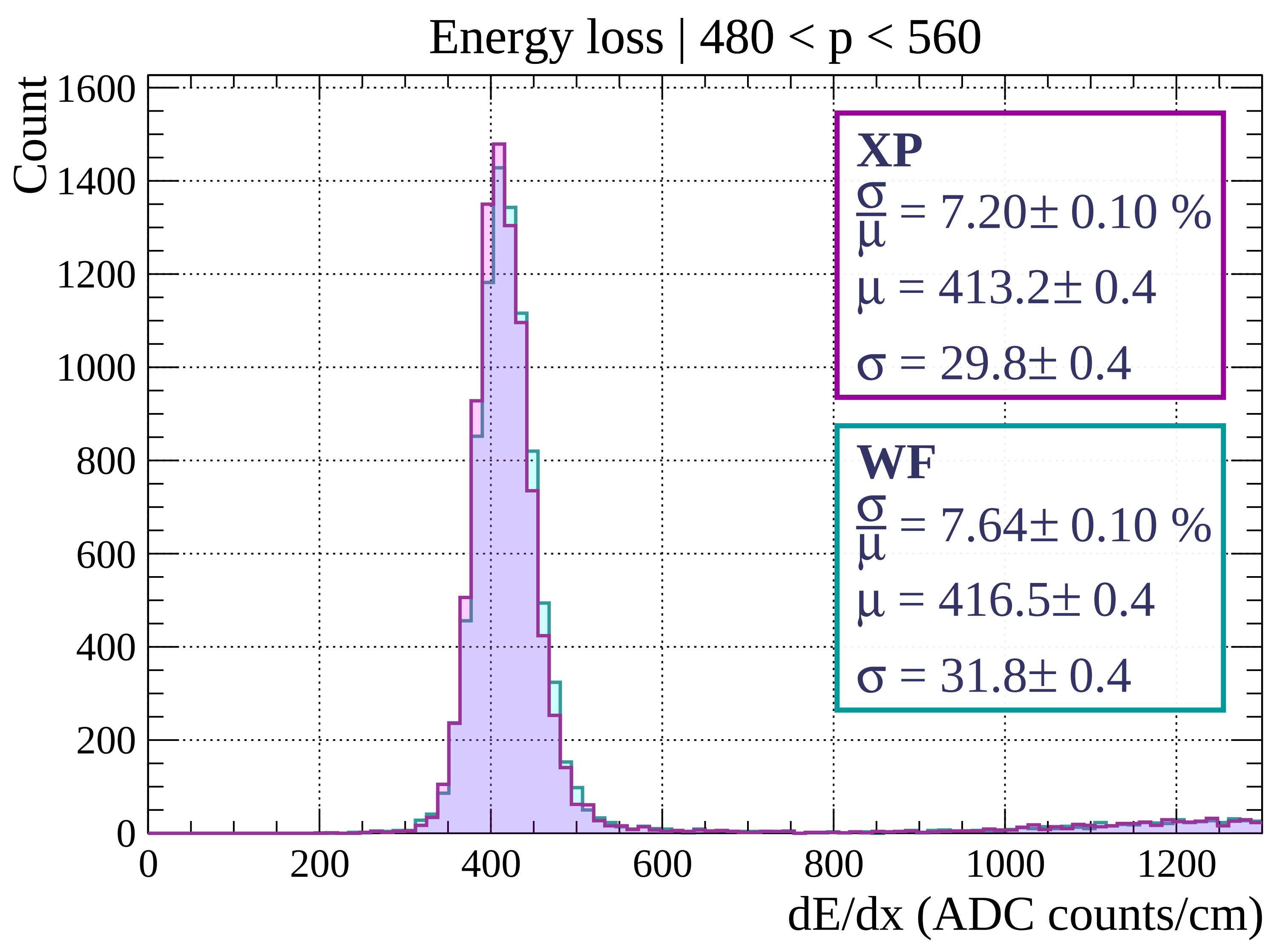

Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
σ/μ = 7.97± 0.09 %
μ = 418.2± 0.4
σ = 33.3± 0.3
WF
σ/μ = 8.36± 0.09 %
μ = 421.5± 0.4
σ = 35.2± 0.4



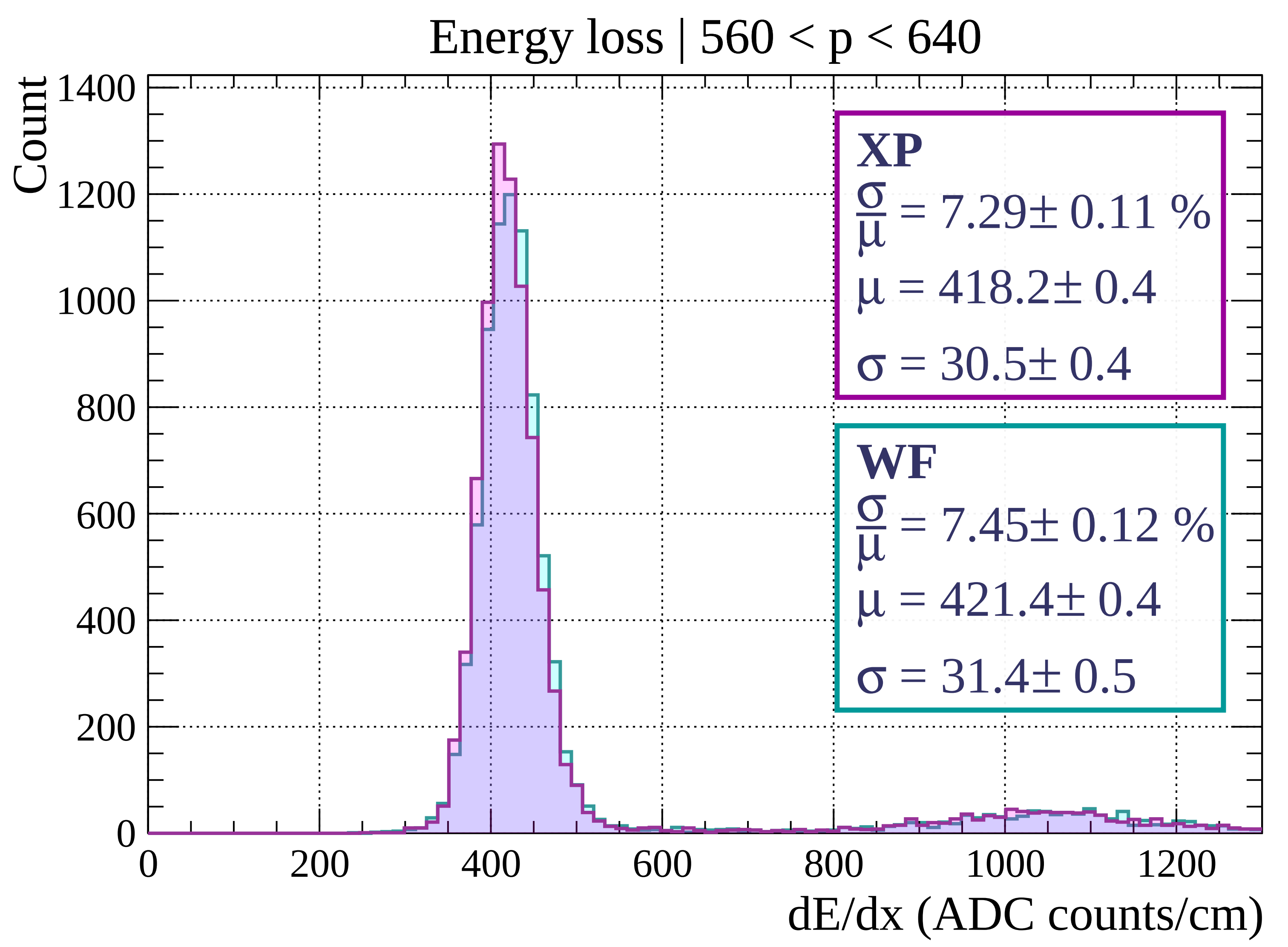
Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP
σ/μ = 7.62± 0.10 %
μ = 409.3± 0.4
σ = 31.2± 0.4
WF
σ/μ = 7.99± 0.10 %
μ = 411.9± 0.4
σ = 32.9± 0.4



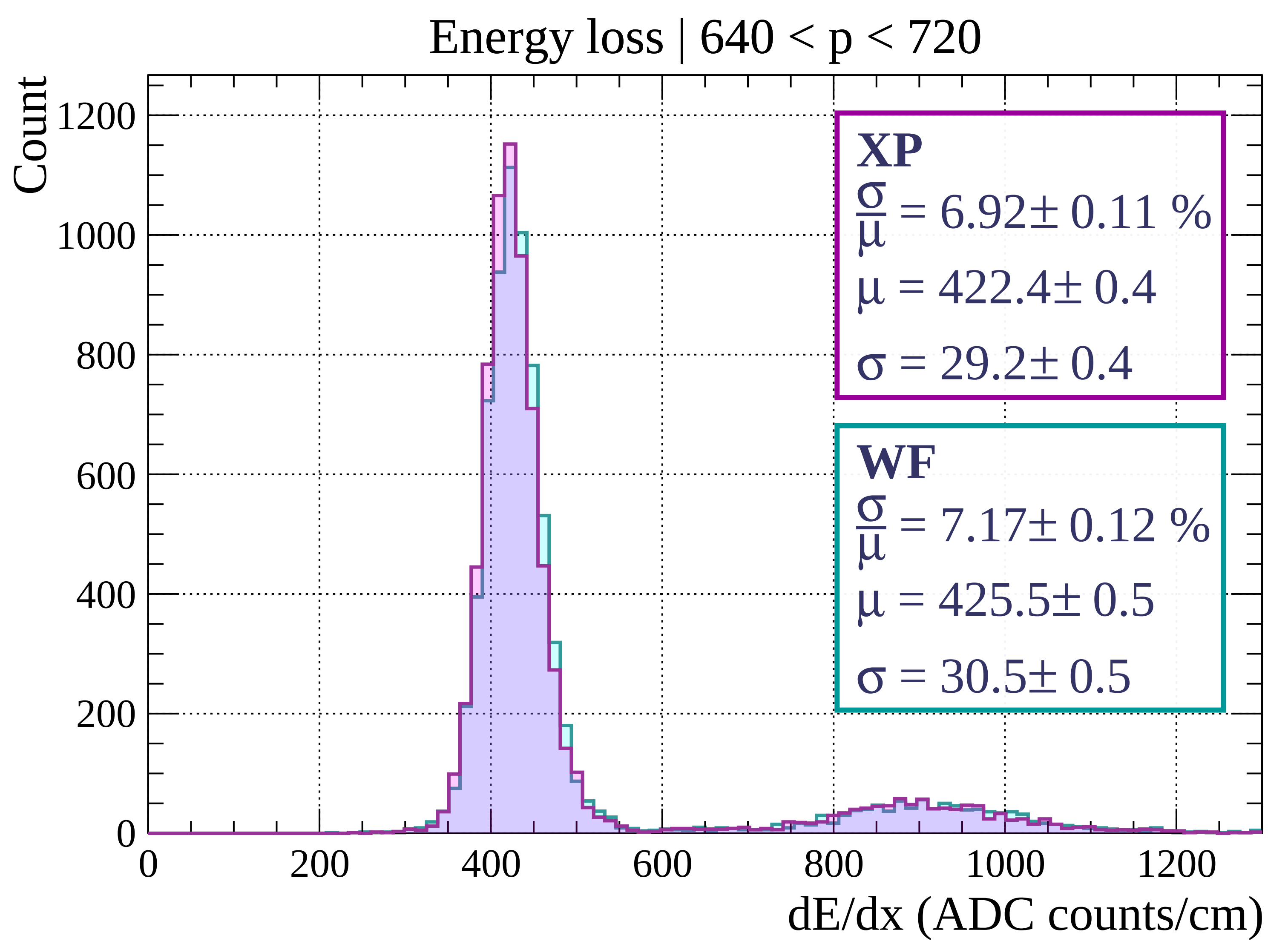
Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
XP
σ/μ = 7.27± 0.09 %
μ = 410.6± 0.4
σ = 29.9± 0.4
WF
σ/μ = 7.82± 0.10 %
μ = 413.5± 0.4
σ = 32.3± 0.4



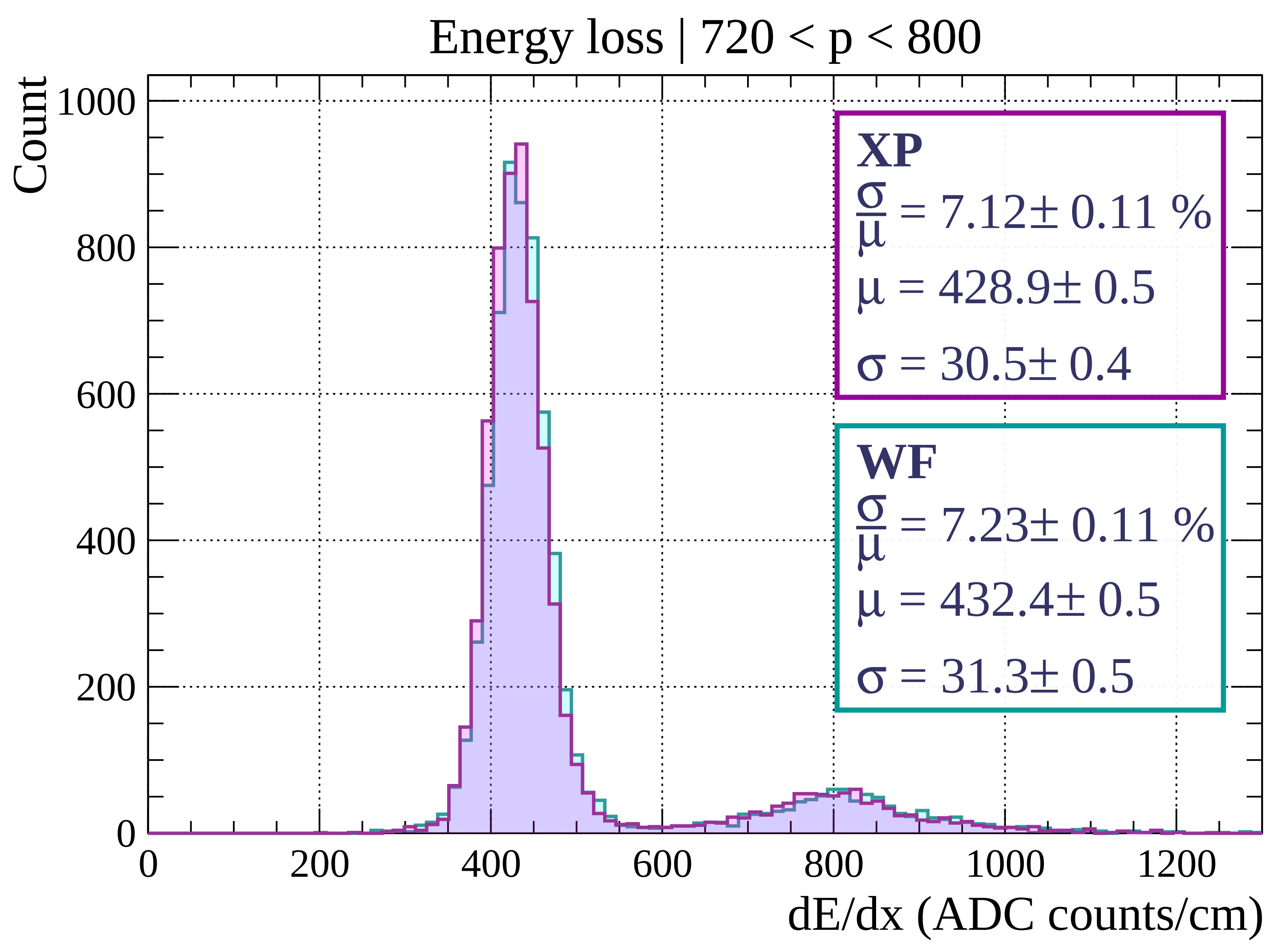
Energy loss | 480 < p < 560
Count
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.20± 0.10 %
μ = 413.2± 0.4
σ = 29.8± 0.4
WF
σ/μ = 7.64± 0.10 %
μ = 416.5± 0.4
σ = 31.8± 0.4



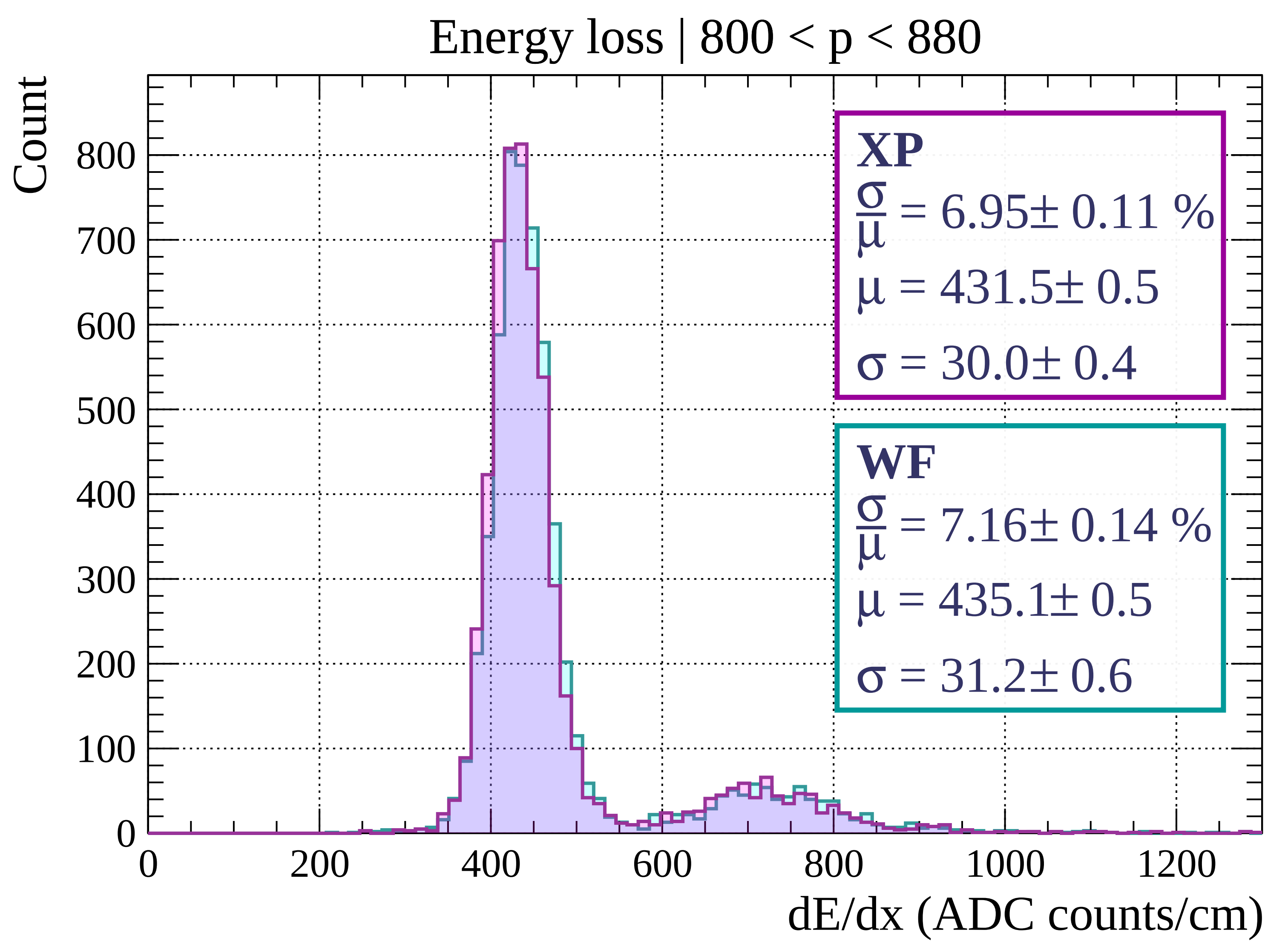
Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
$\frac{\sigma}{\mu}$ = 7.29± 0.11 %
μ = 418.2± 0.4
σ = 30.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 7.45± 0.12 %
μ = 421.4± 0.4
σ = 31.4± 0.5



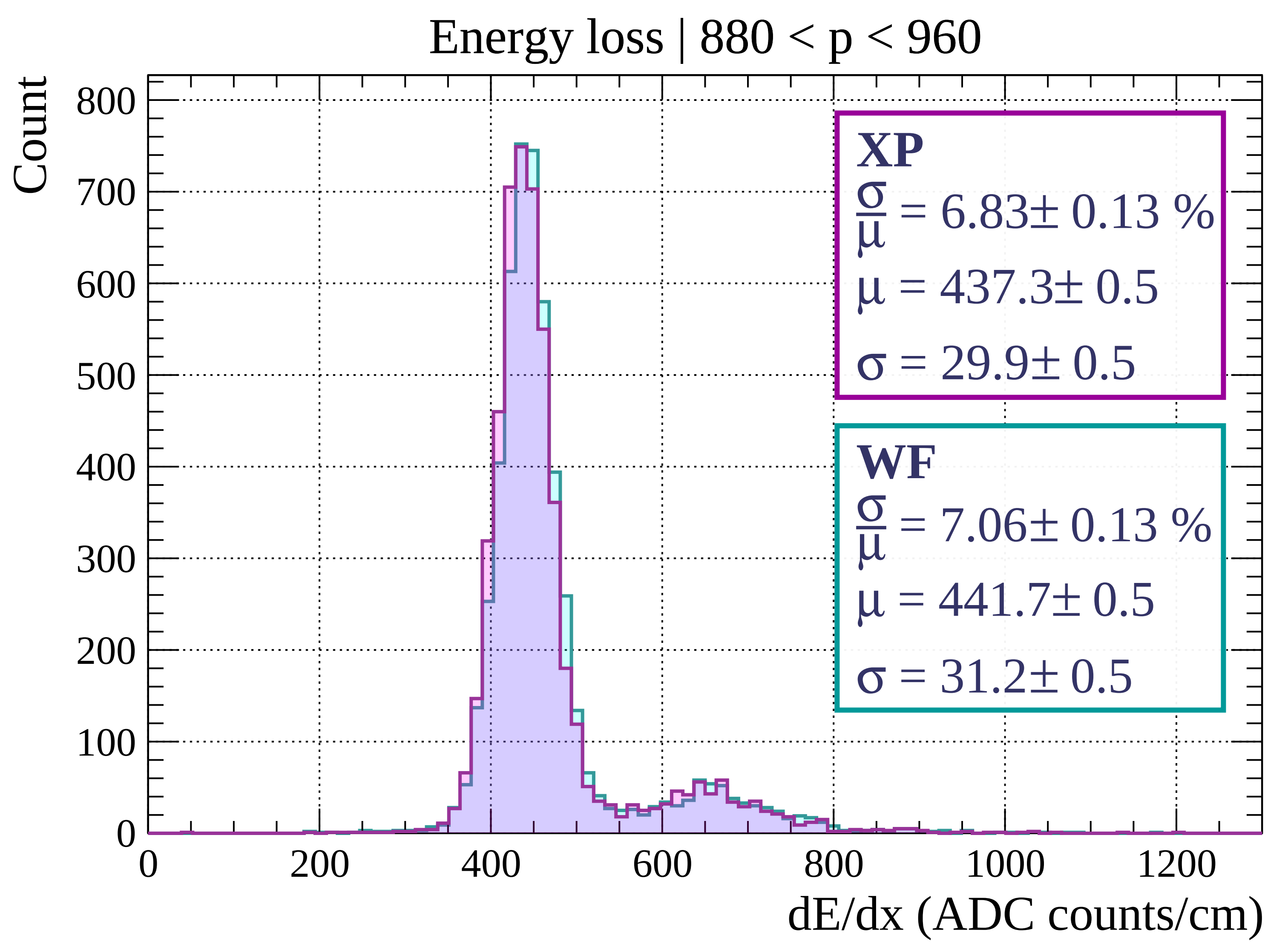
Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.92± 0.11 %
μ = 422.4± 0.4
σ = 29.2± 0.4
WF
$\frac{\sigma}{\mu}$ = 7.17± 0.12 %
μ = 425.5± 0.5
σ = 30.5± 0.5

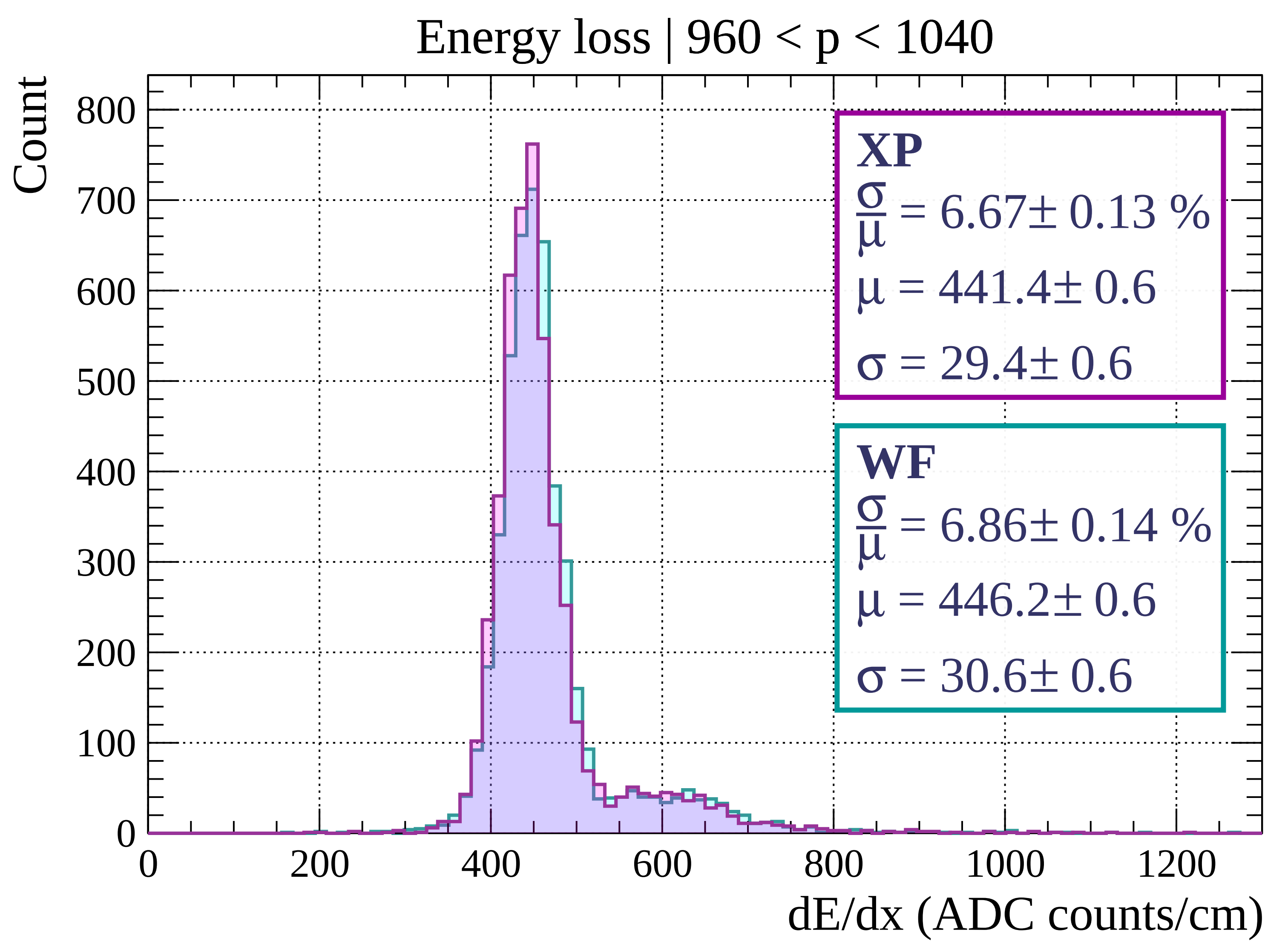


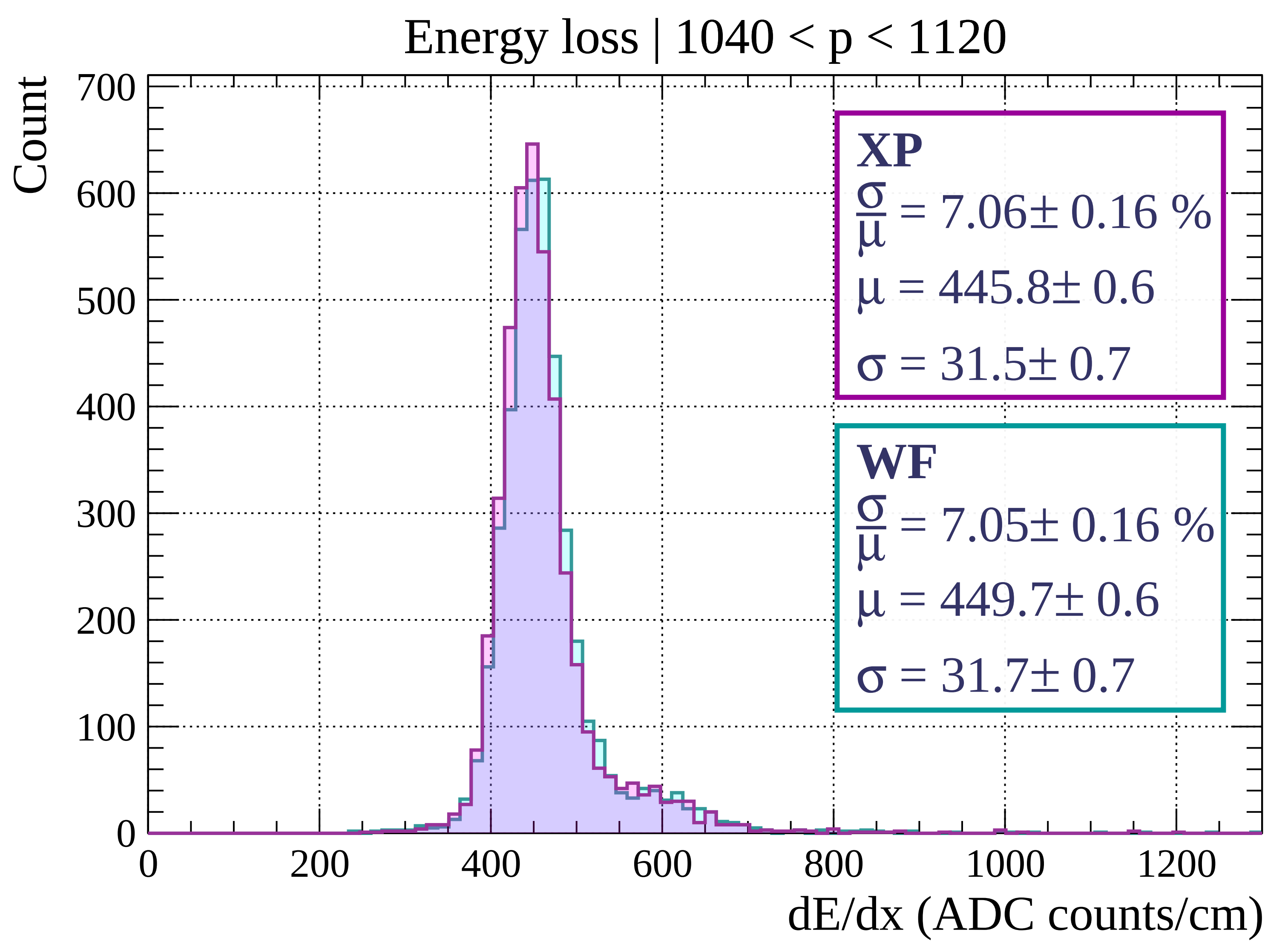
Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.12 \pm 0.11$ %
$\mu = 428.9 \pm 0.5$
$\sigma = 30.5 \pm 0.4$
WF
$\frac{\sigma}{\mu} = 7.23 \pm 0.11$ %
$\mu = 432.4 \pm 0.5$
$\sigma = 31.3 \pm 0.5$



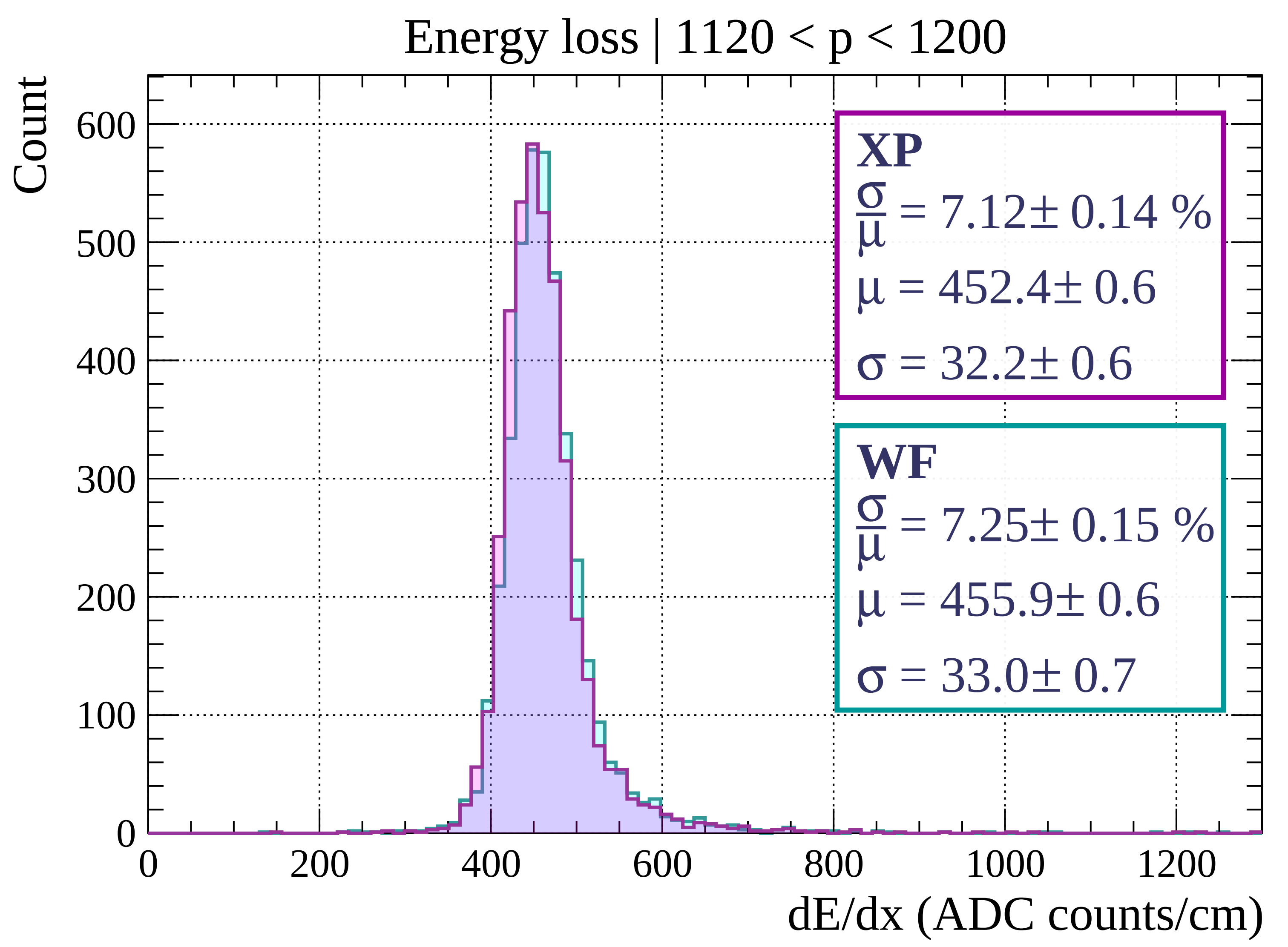
Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.95± 0.11 %
μ = 431.5± 0.5
σ = 30.0± 0.4
WF
$\frac{\sigma}{\mu}$ = 7.16± 0.14 %
μ = 435.1± 0.5
σ = 31.2± 0.6


Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.83± 0.13 %
μ = 437.3± 0.5
σ = 29.9± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.06± 0.13 %
μ = 441.7± 0.5
σ = 31.2± 0.5

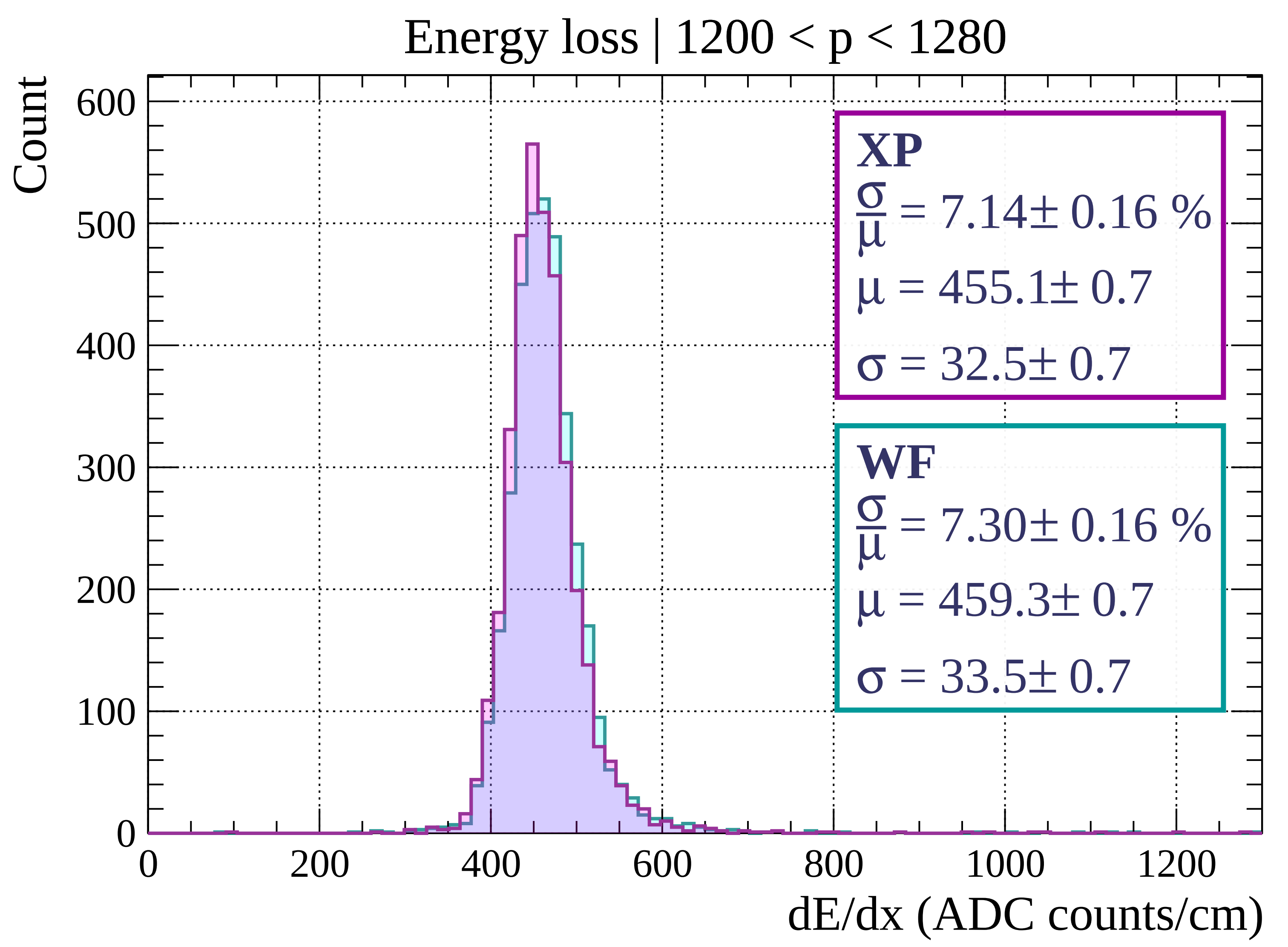

Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.67± 0.13 %
μ = 441.4± 0.6
σ = 29.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 6.86± 0.14 %
μ = 446.2± 0.6
σ = 30.6± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



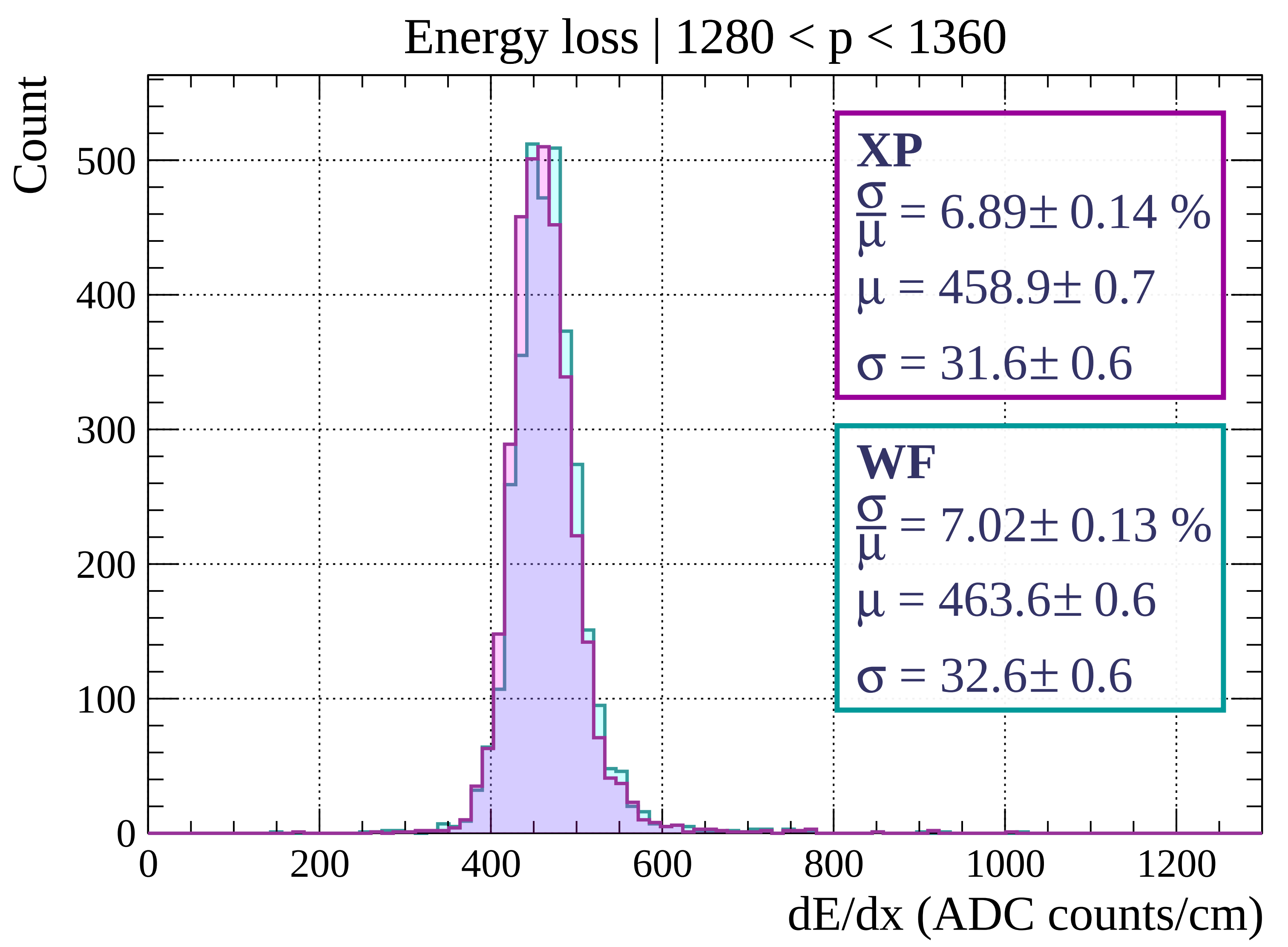
Energy loss | 1040 < p < 1120
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.06± 0.16 %
μ = 445.8± 0.6
σ = 31.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.05± 0.16 %
μ = 449.7± 0.6
σ = 31.7± 0.7



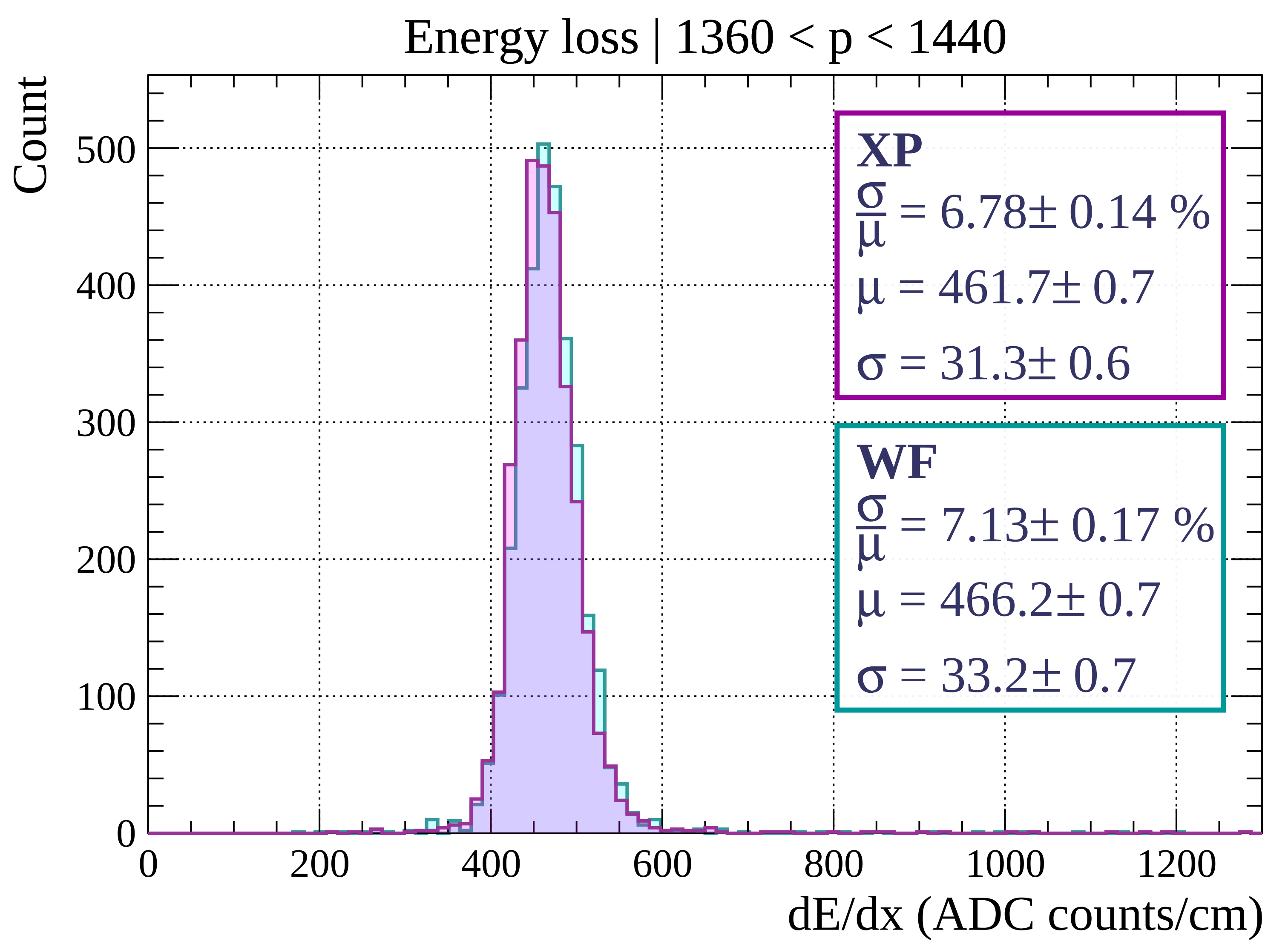
Energy loss | 1120 < p < 1200
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.12± 0.14 %
μ = 452.4± 0.6
σ = 32.2± 0.6
WF
σ/μ = 7.25± 0.15 %
μ = 455.9± 0.6
σ = 33.0± 0.7



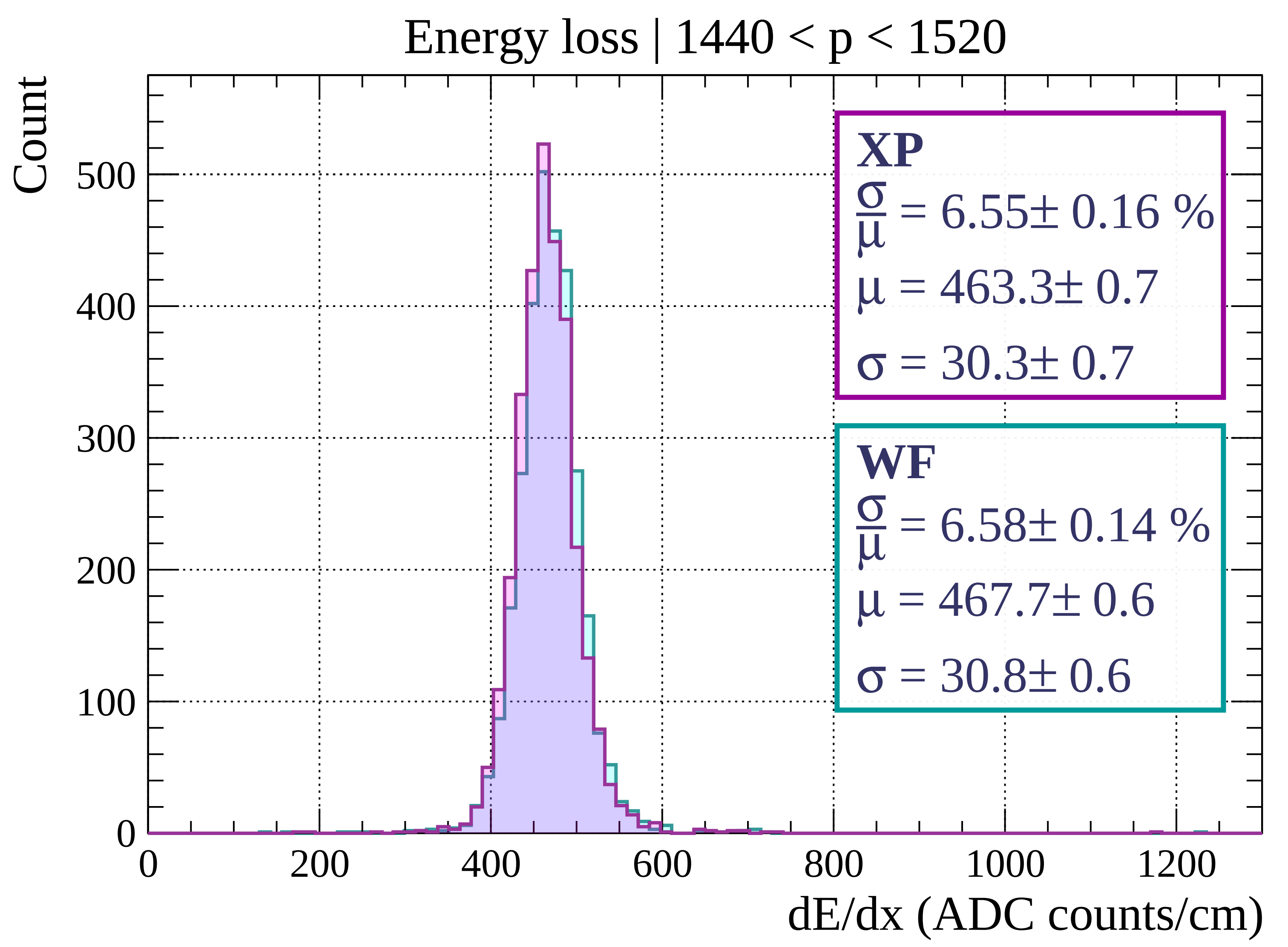
Energy loss | 1200 < p < 1280
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.14 \pm 0.16$ %
$\mu = 455.1 \pm 0.7$
$\sigma = 32.5 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 7.30 \pm 0.16$ %
$\mu = 459.3 \pm 0.7$
$\sigma = 33.5 \pm 0.7$



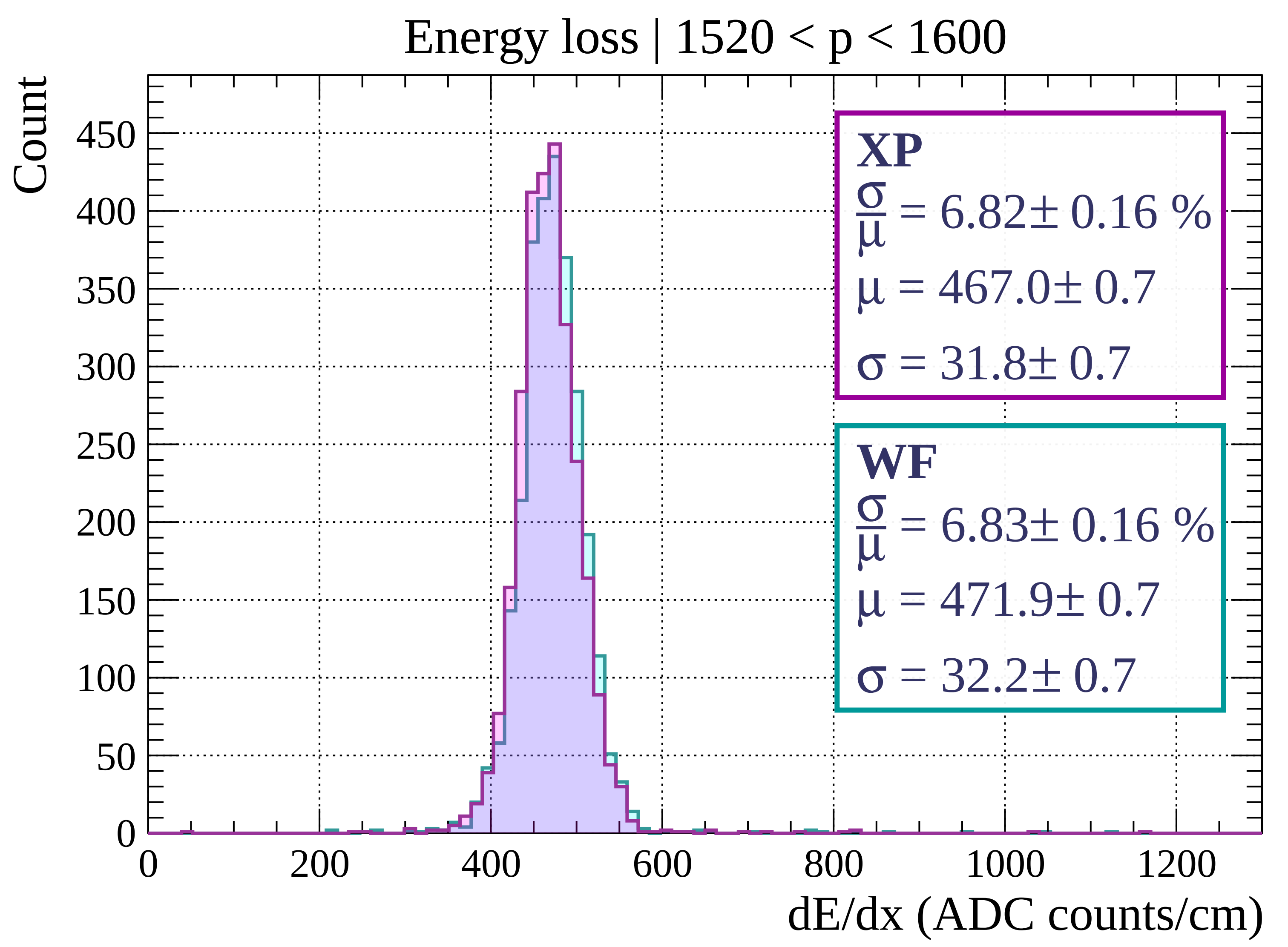
Energy loss | 1280 < p < 1360
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.89± 0.14 %
μ = 458.9± 0.7
σ = 31.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 7.02± 0.13 %
μ = 463.6± 0.6
σ = 32.6± 0.6

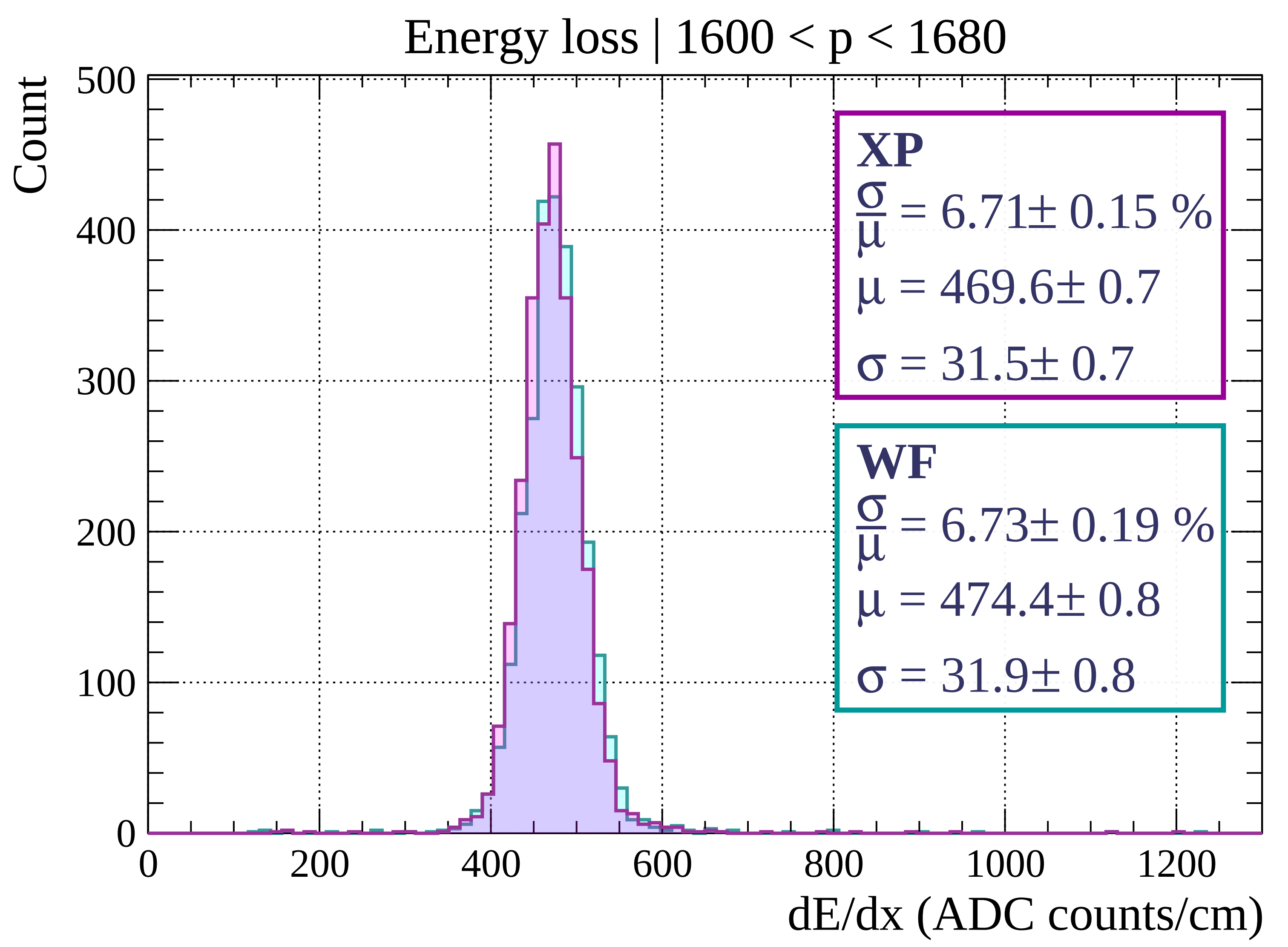


Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.78± 0.14 %
μ = 461.7± 0.7
σ = 31.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 7.13± 0.17 %
μ = 466.2± 0.7
σ = 33.2± 0.7

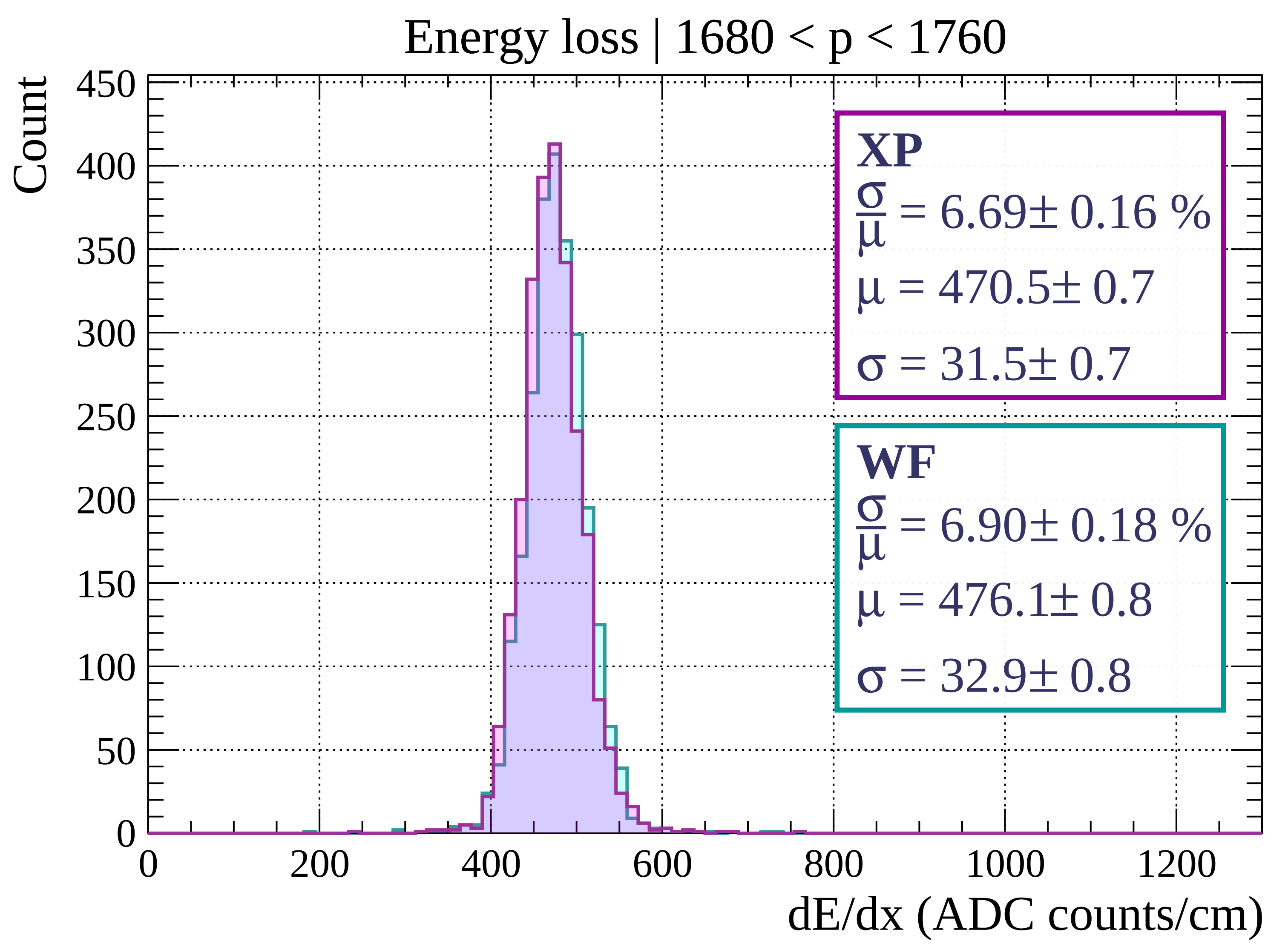


Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.55± 0.16 %
μ = 463.3± 0.7
σ = 30.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.58± 0.14 %
μ = 467.7± 0.6
σ = 30.8± 0.6

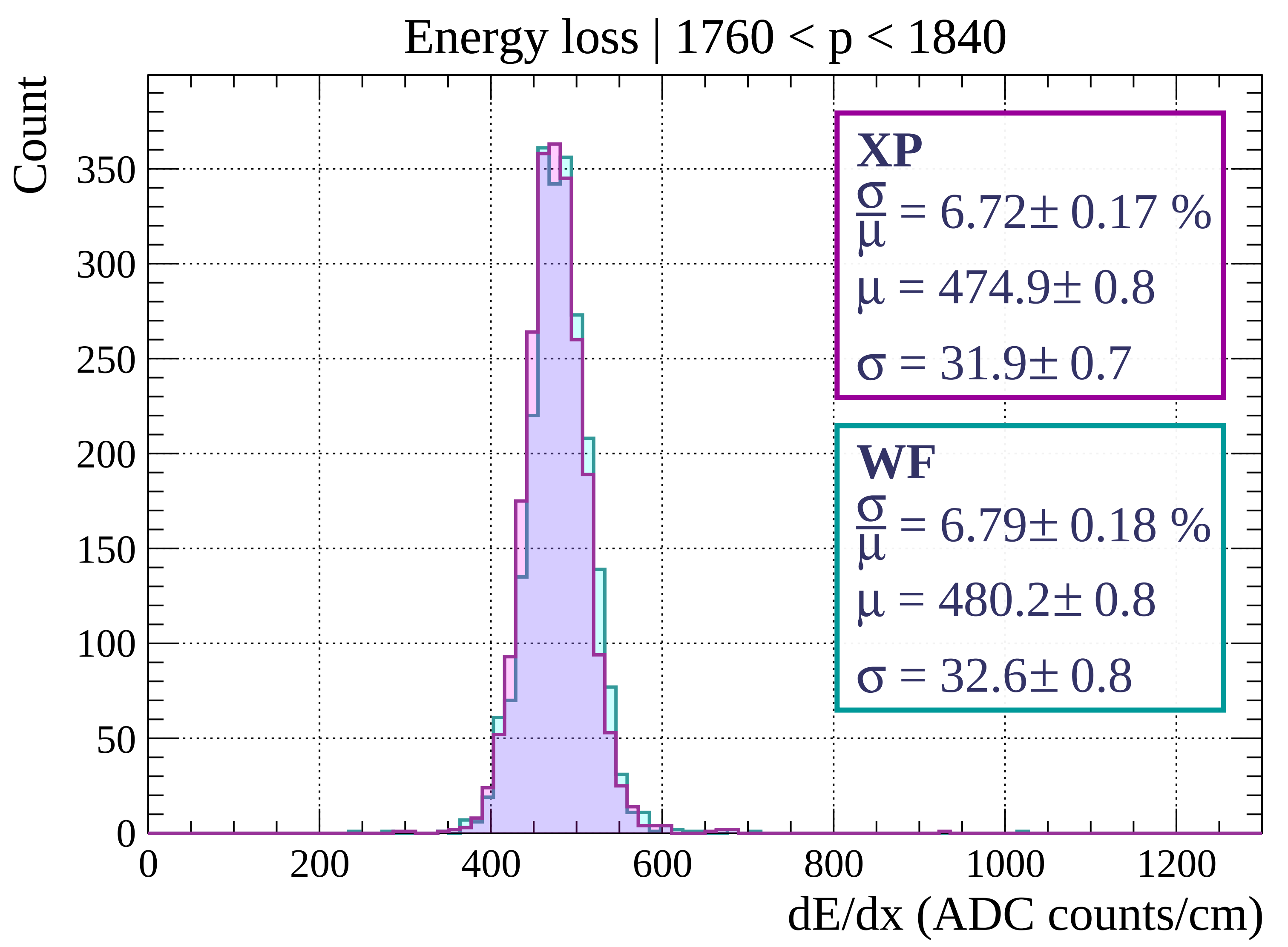

Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP
σ/μ = 6.82± 0.16 %
μ = 467.0± 0.7
σ = 31.8± 0.7
WF
σ/μ = 6.83± 0.16 %
μ = 471.9± 0.7
σ = 32.2± 0.7



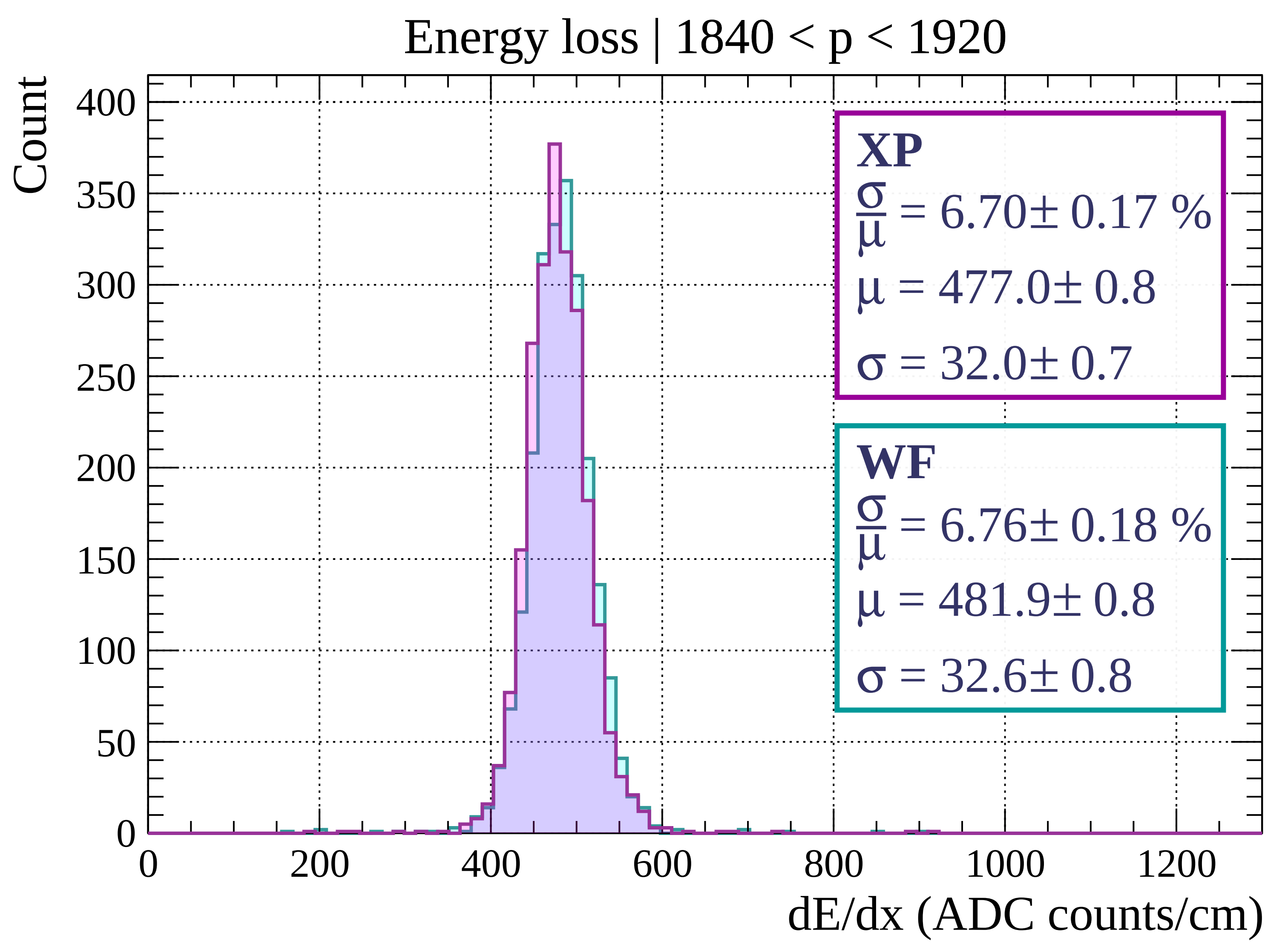
Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.71± 0.15 %
μ = 469.6± 0.7
σ = 31.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.73± 0.19 %
μ = 474.4± 0.8
σ = 31.9± 0.8

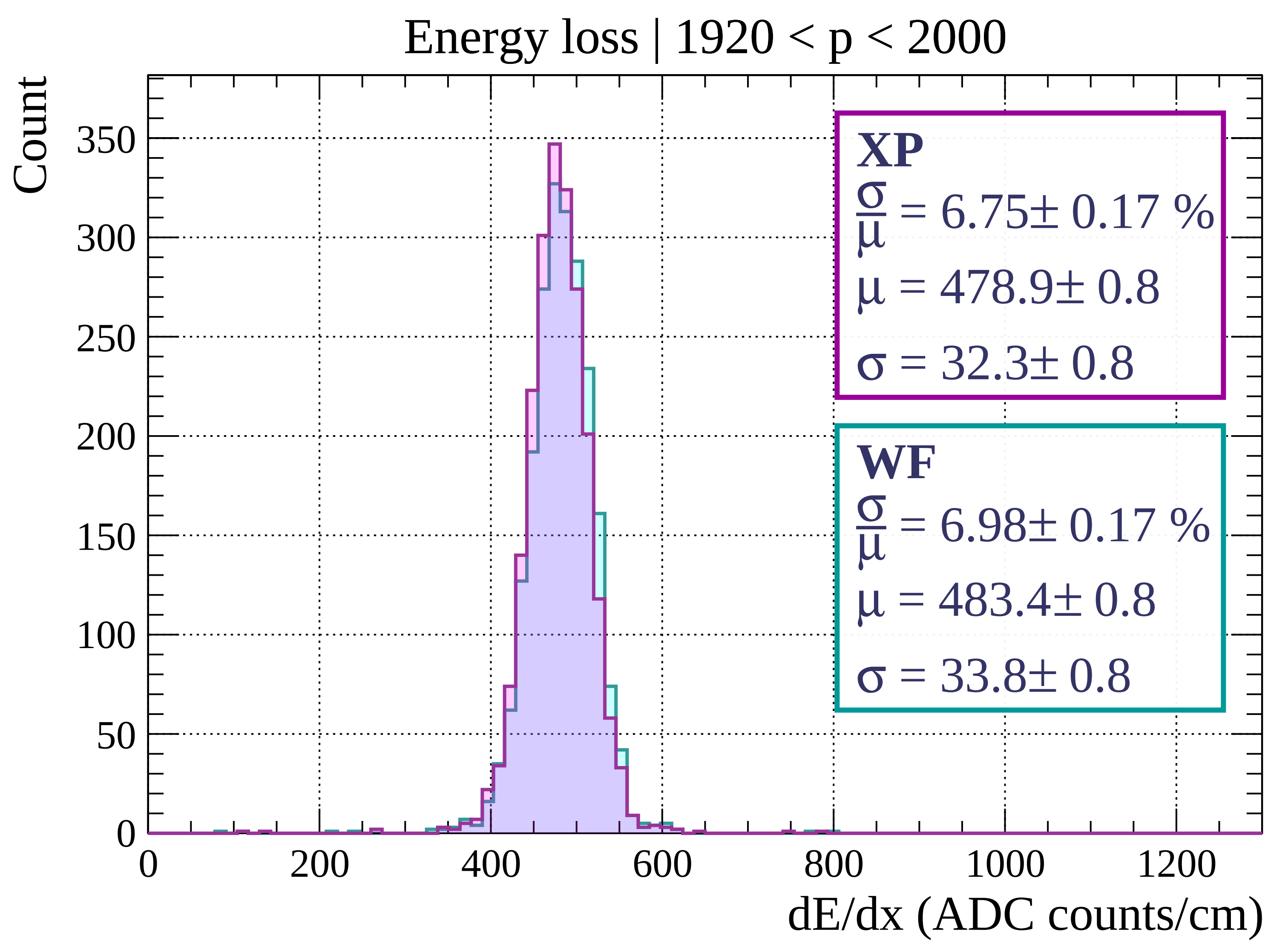


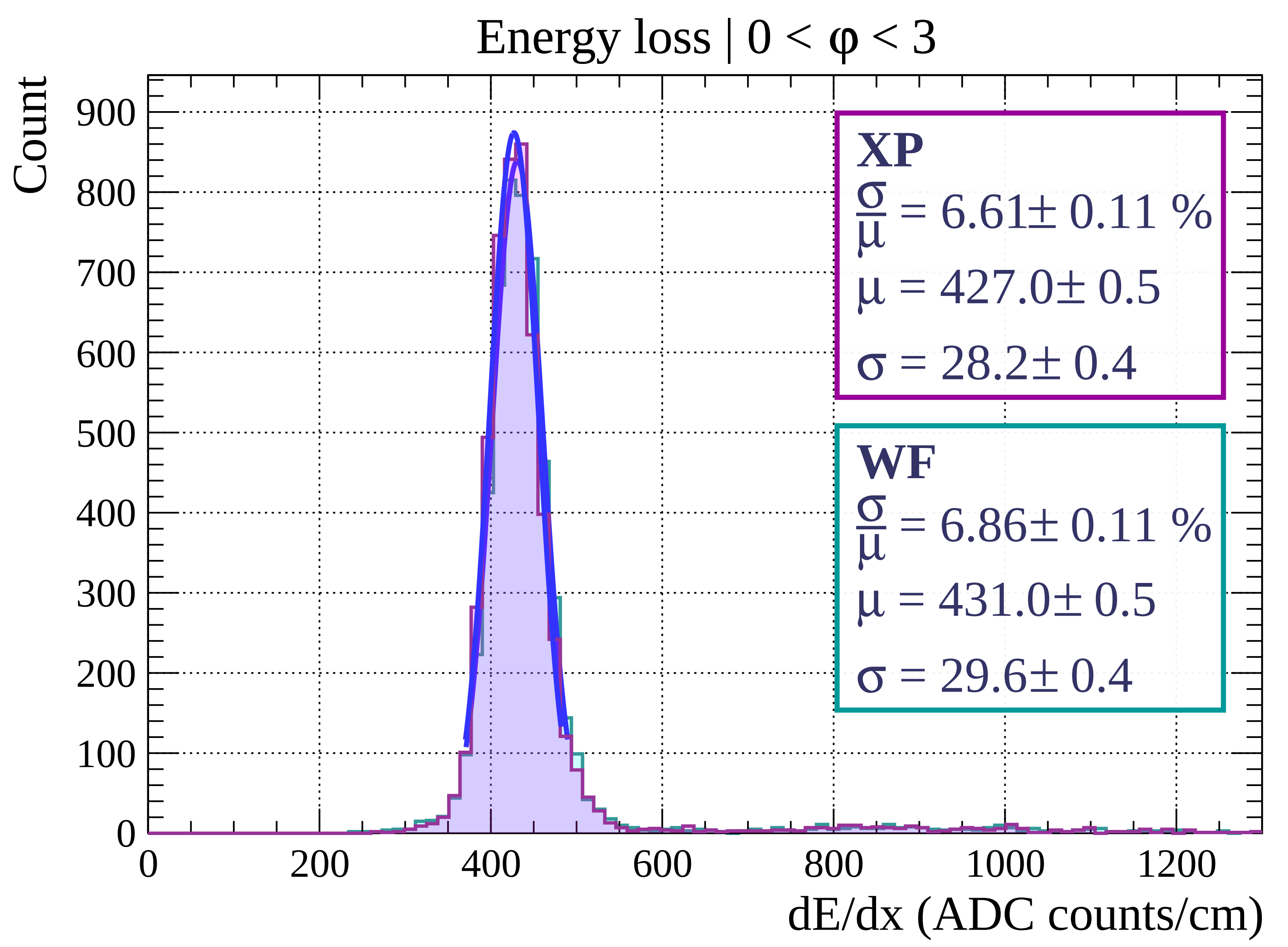
Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP
$\frac{\sigma}{\mu}$ = 6.69± 0.16 %
μ = 470.5± 0.7
σ = 31.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.90± 0.18 %
μ = 476.1± 0.8
σ = 32.9± 0.8

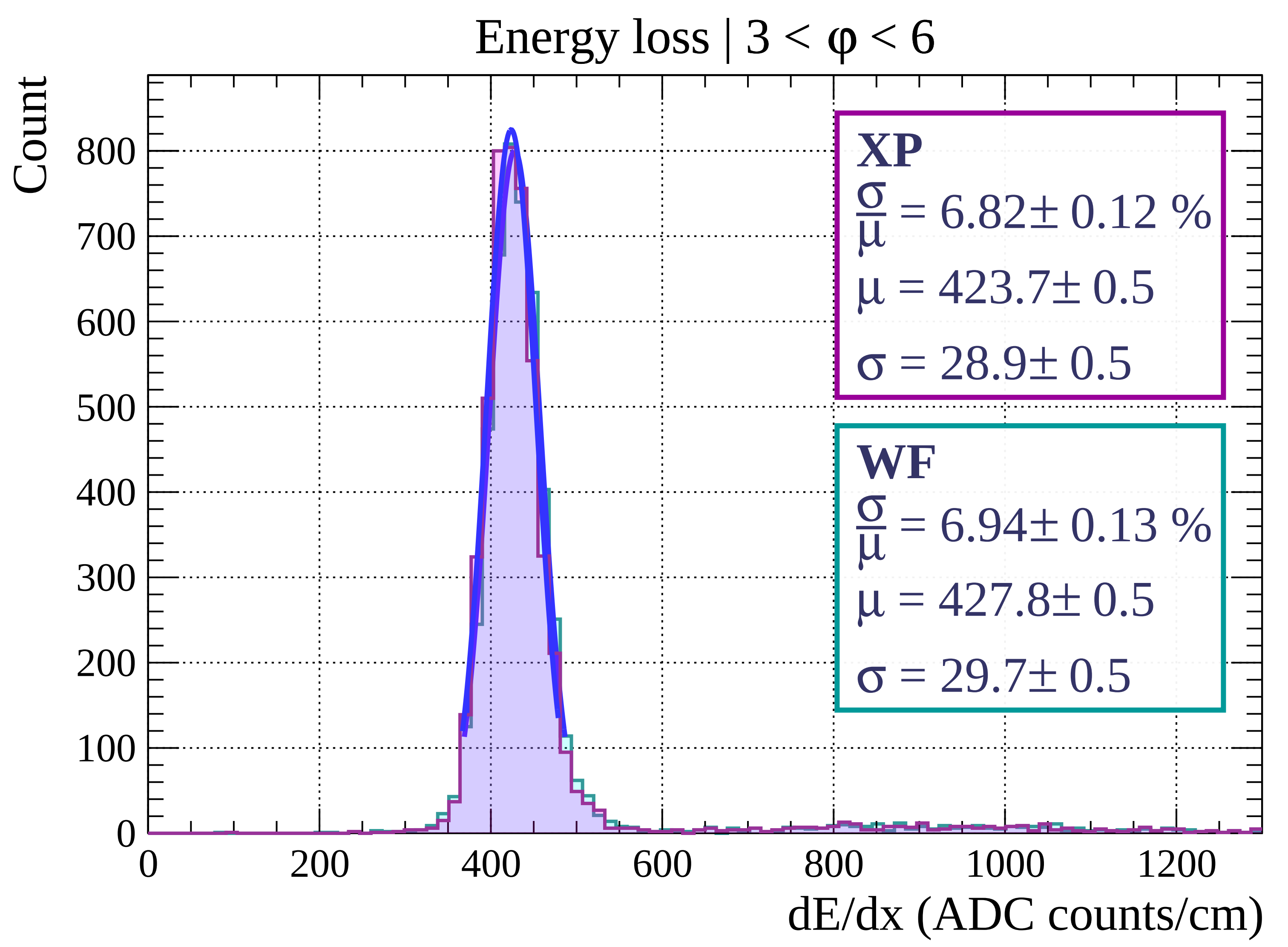


Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.72± 0.17 %
μ = 474.9± 0.8
σ = 31.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.79± 0.18 %
μ = 480.2± 0.8
σ = 32.6± 0.8

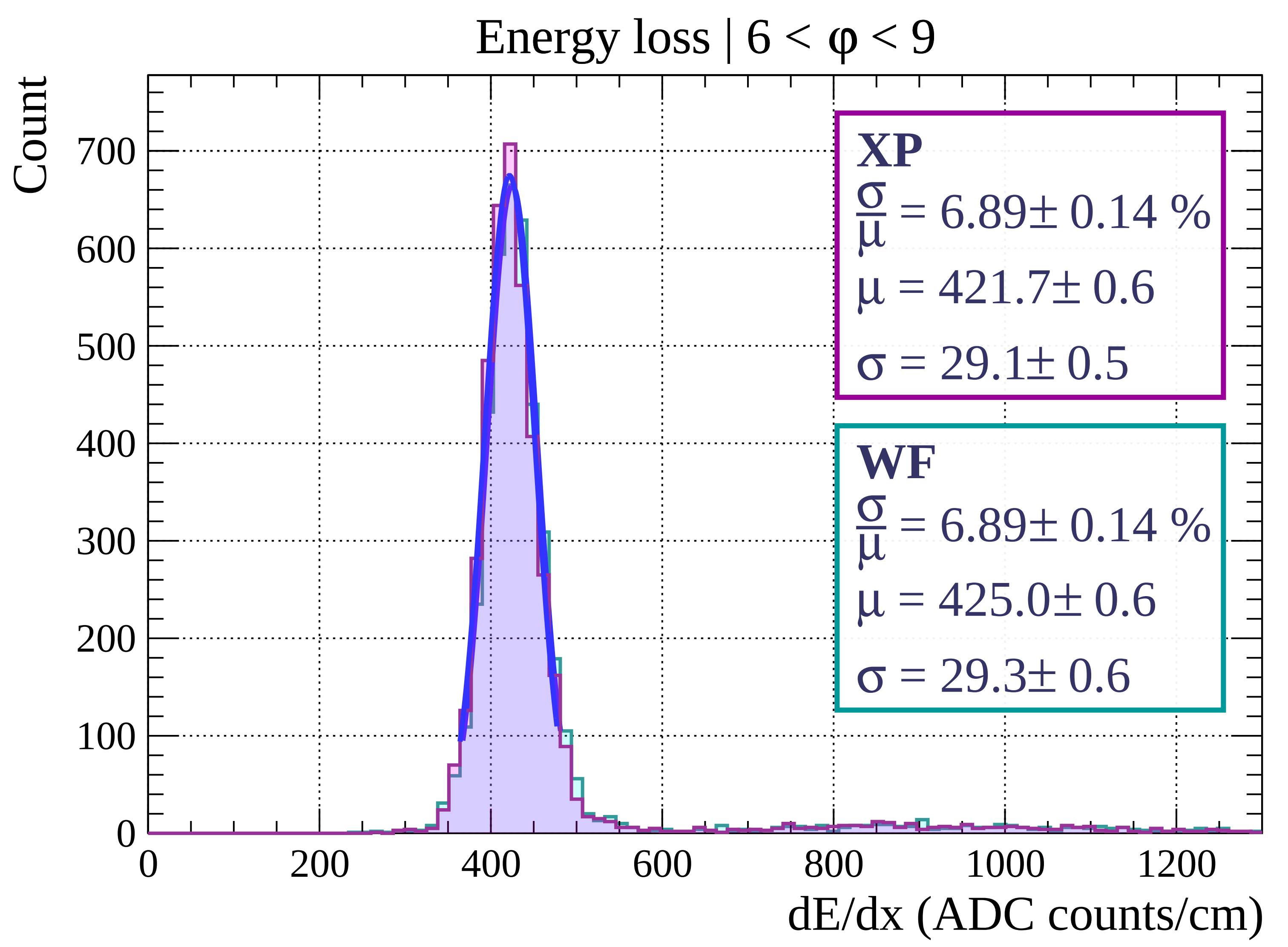

Energy loss | 1840 < p < 1920
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.70± 0.17 %
μ = 477.0± 0.8
σ = 32.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.76± 0.18 %
μ = 481.9± 0.8
σ = 32.6± 0.8

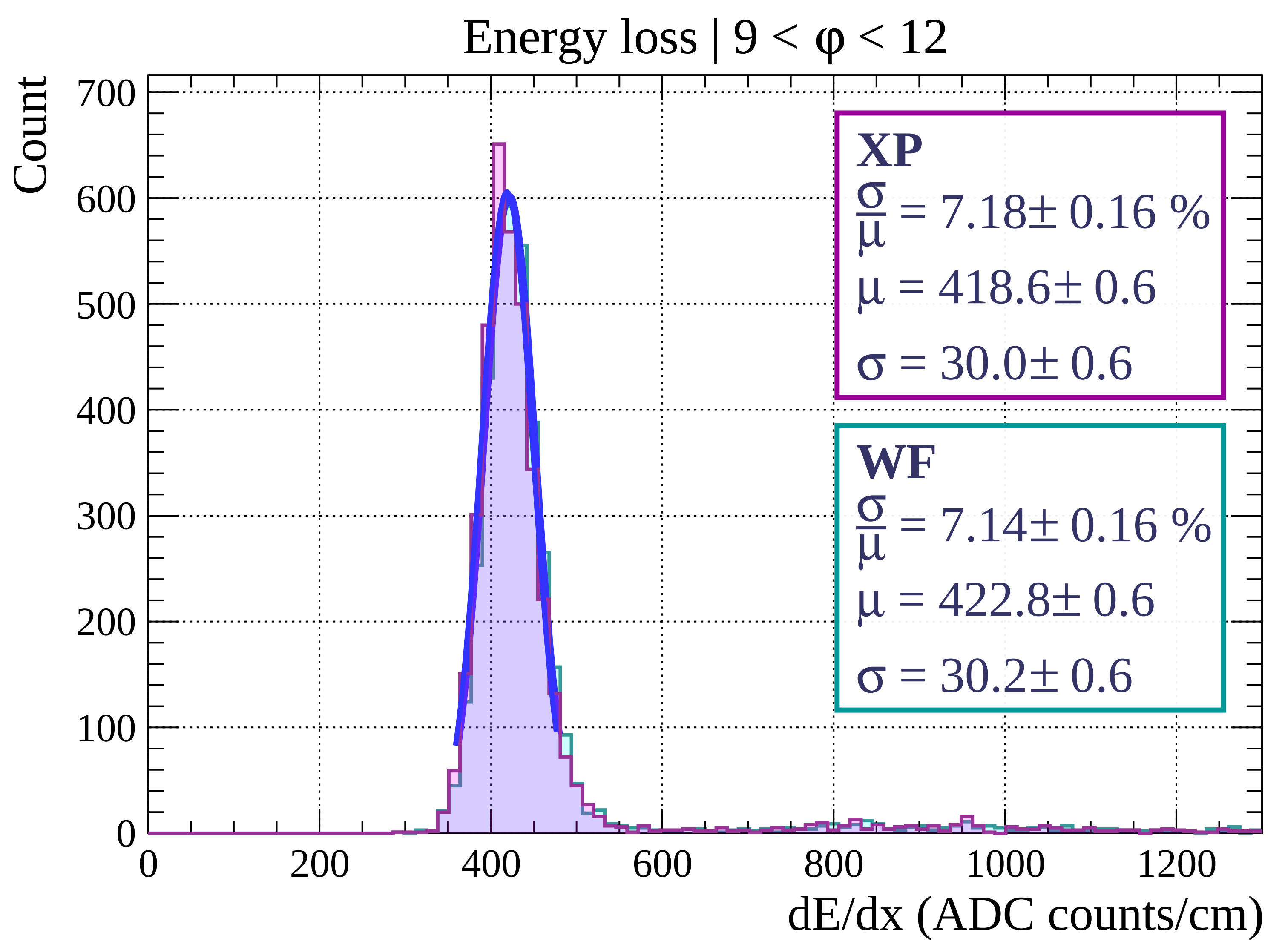

Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 6.75 \pm 0.17$ %
$\mu = 478.9 \pm 0.8$
$\sigma = 32.3 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 6.98 \pm 0.17$ %
$\mu = 483.4 \pm 0.8$
$\sigma = 33.8 \pm 0.8$


Energy loss | 0 < φ < 3
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.61± 0.11 %
μ = 427.0± 0.5
σ = 28.2± 0.4
WF
σ/μ = 6.86± 0.11 %
μ = 431.0± 0.5
σ = 29.6± 0.4

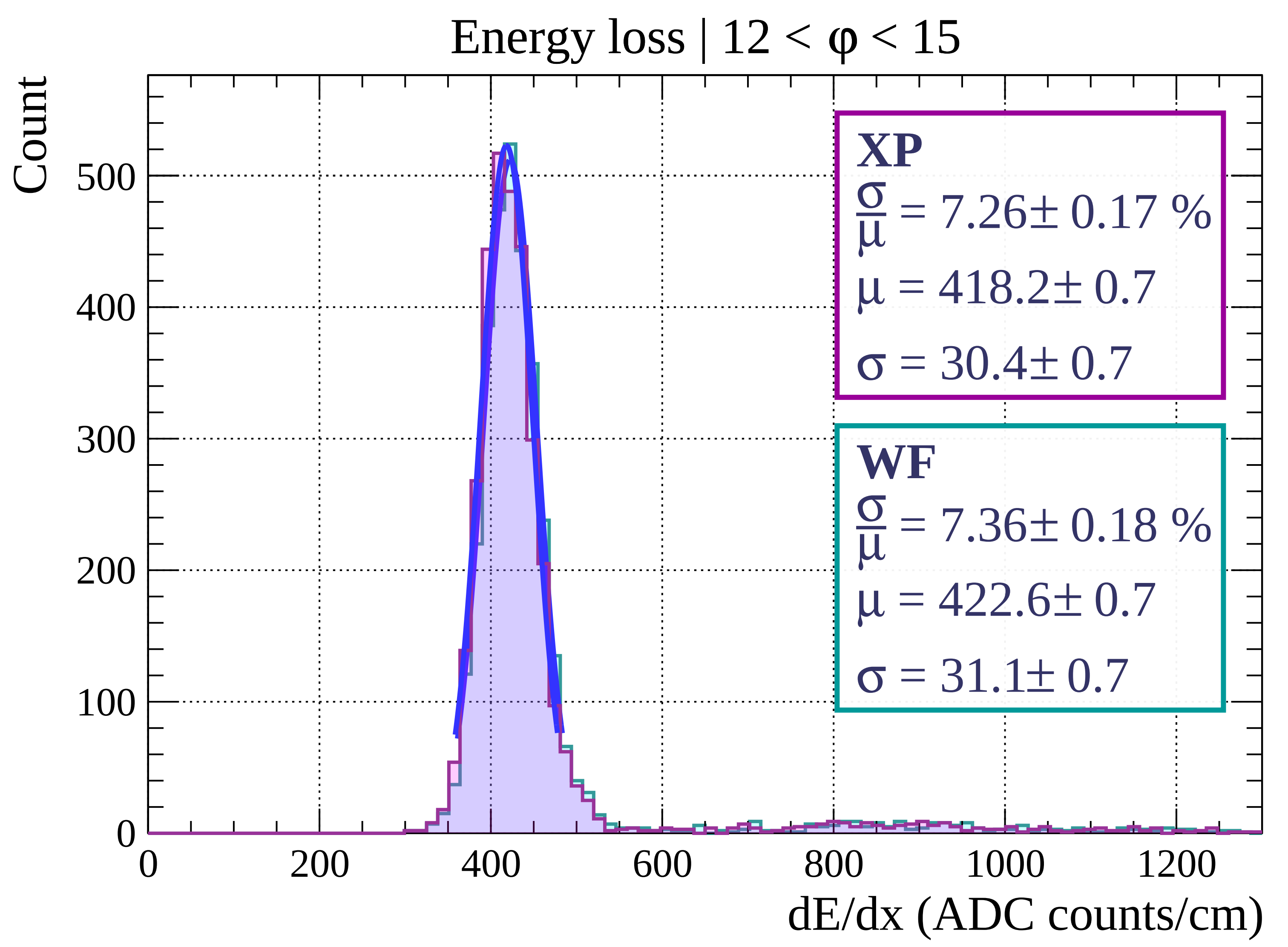


Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 6.82± 0.12 %
μ = 423.7± 0.5
σ = 28.9± 0.5
WF
σ/μ = 6.94± 0.13 %
μ = 427.8± 0.5
σ = 29.7± 0.5

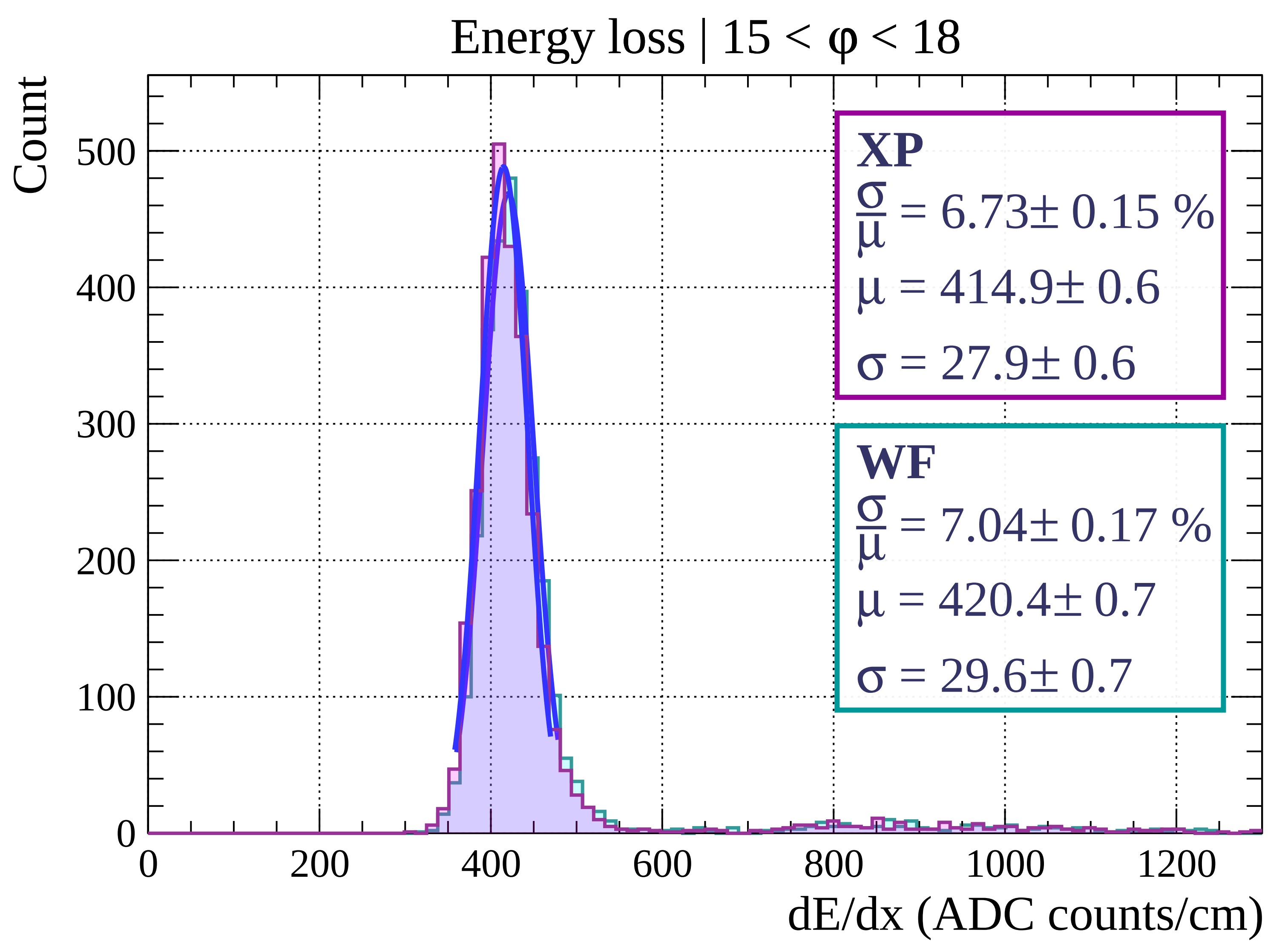

Energy loss | 6 < φ < 9
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.89± 0.14 %
μ = 421.7± 0.6
σ = 29.1± 0.5
WF
σ/μ = 6.89± 0.14 %
μ = 425.0± 0.6
σ = 29.3± 0.6



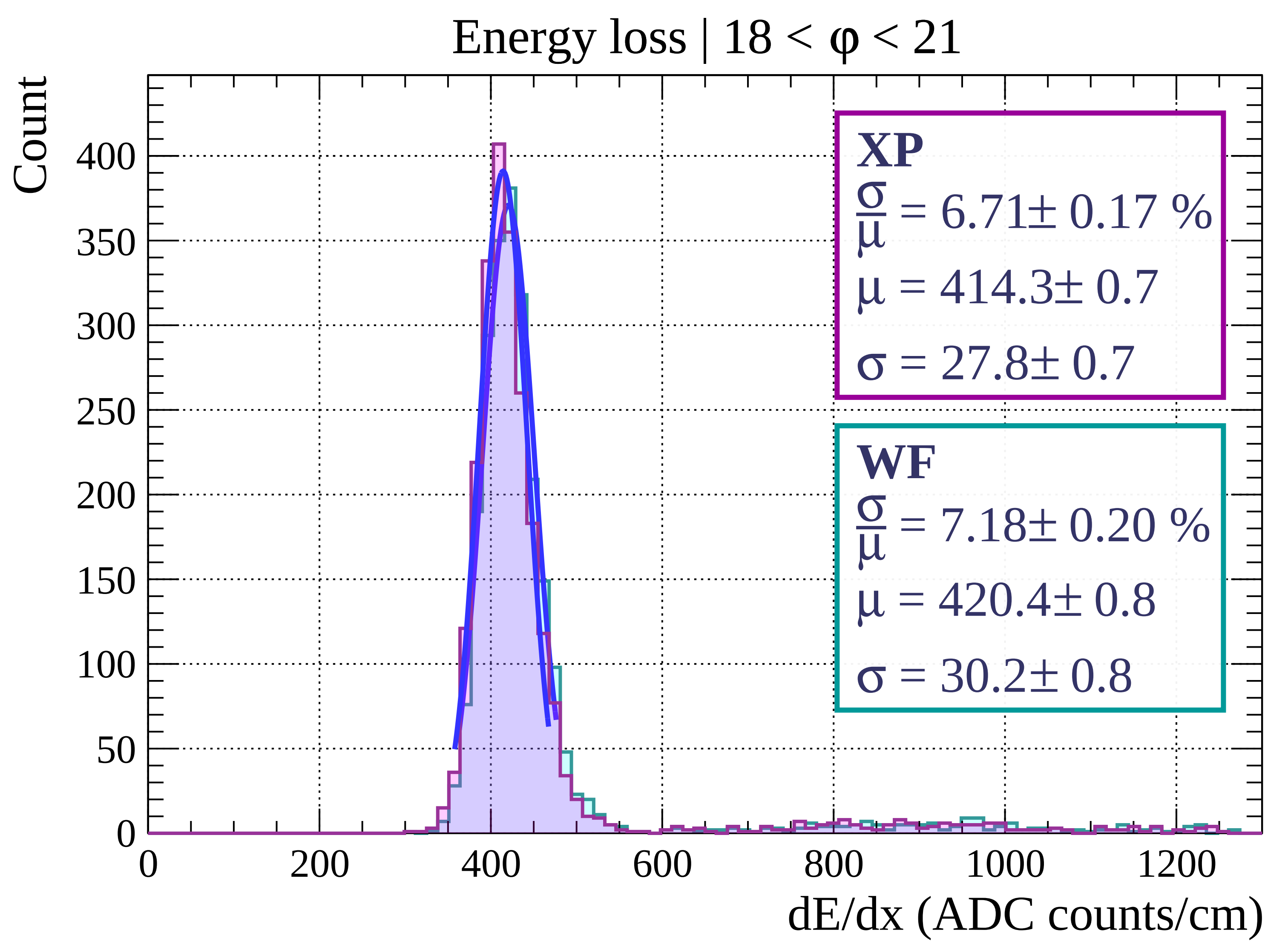
Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 7.18± 0.16 %
μ = 418.6± 0.6
σ = 30.0± 0.6
WF
σ/μ = 7.14± 0.16 %
μ = 422.8± 0.6
σ = 30.2± 0.6

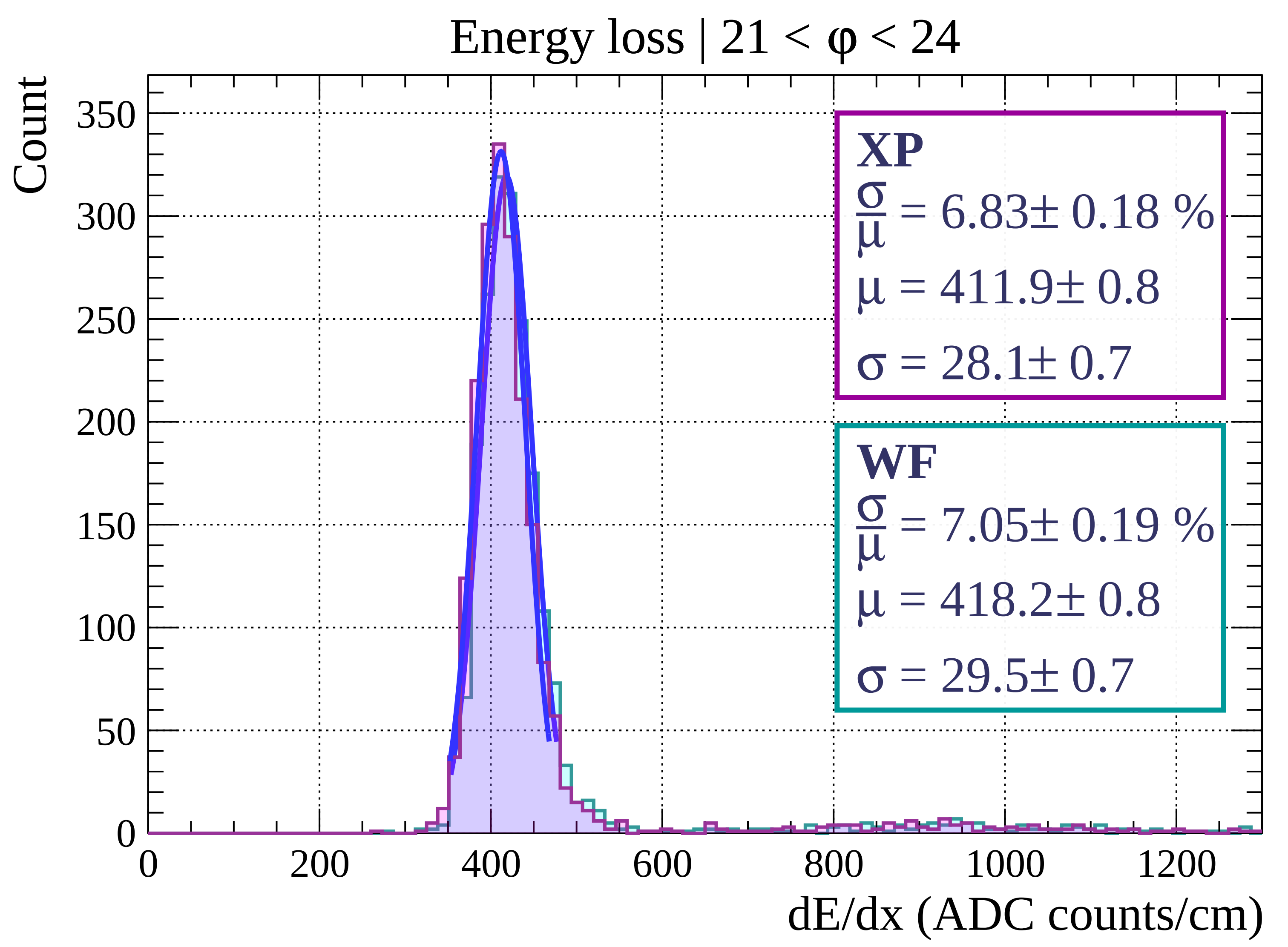


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 7.26± 0.17 %
μ = 418.2± 0.7
σ = 30.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.36± 0.18 %
μ = 422.6± 0.7
σ = 31.1± 0.7

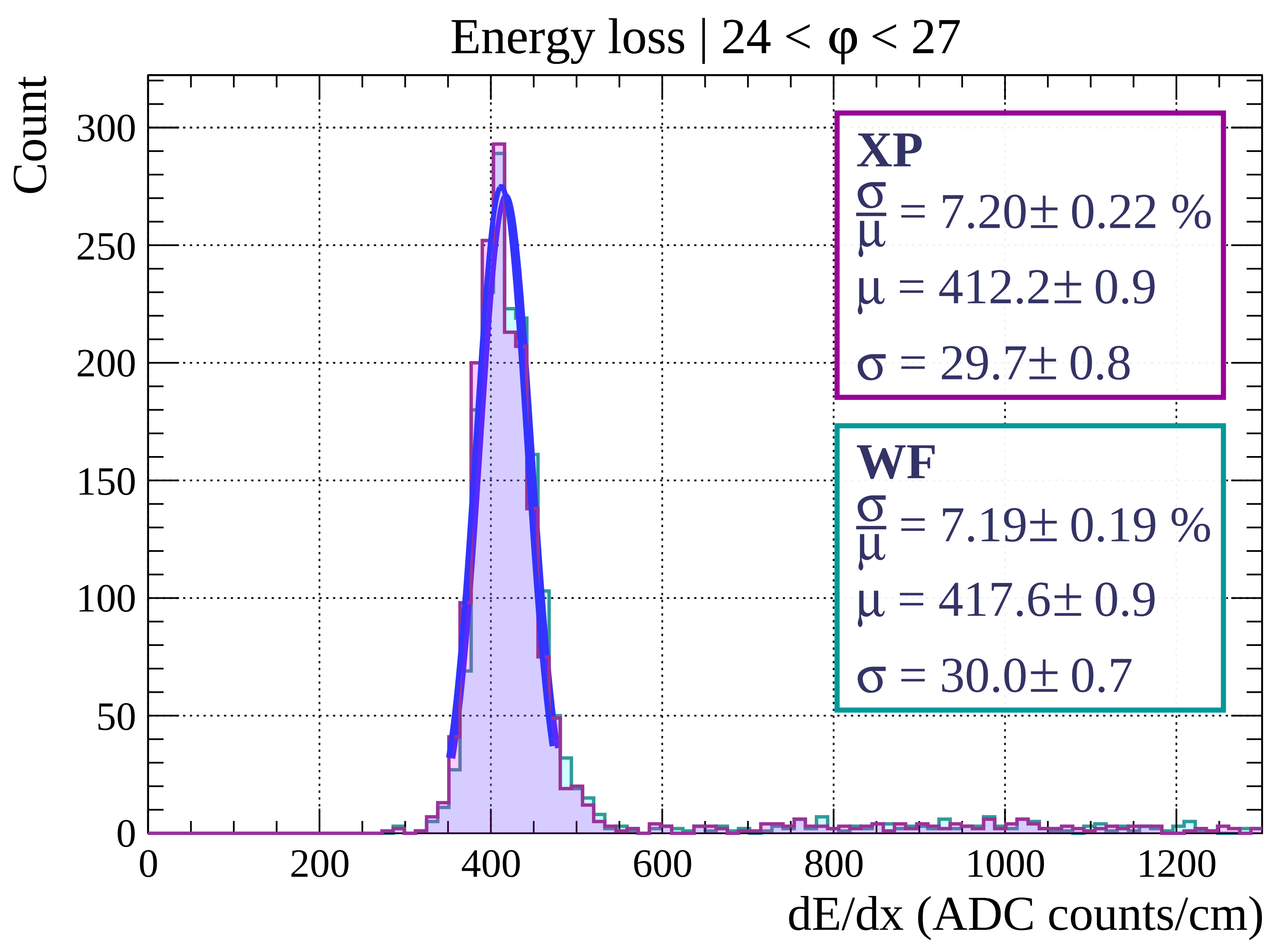


Energy loss | 15 < φ < 18
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.73± 0.15 %
μ = 414.9± 0.6
σ = 27.9± 0.6
WF
σ/μ = 7.04± 0.17 %
μ = 420.4± 0.7
σ = 29.6± 0.7

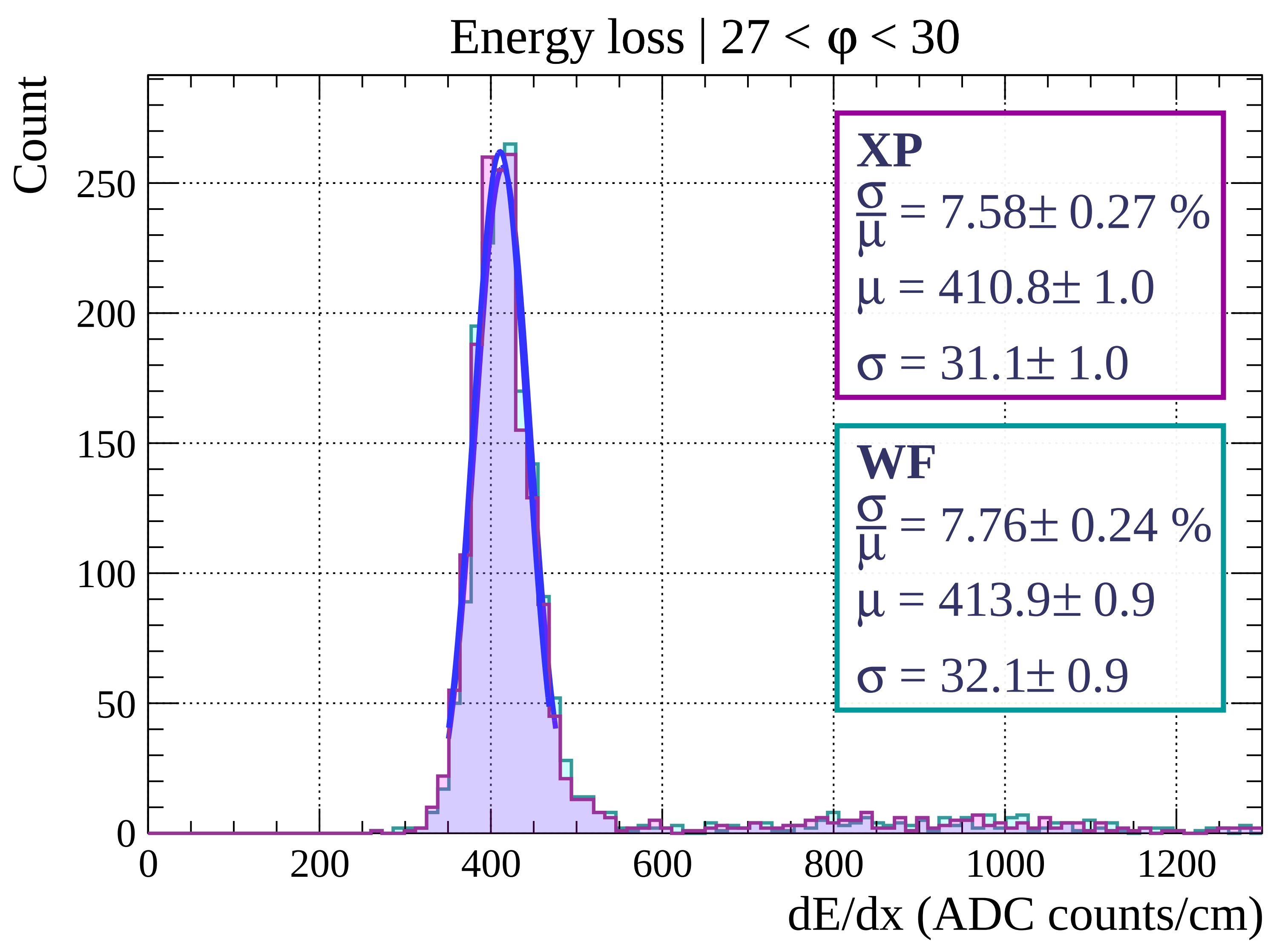

Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 6.71± 0.17 %
μ = 414.3± 0.7
σ = 27.8± 0.7
WF
σ/μ = 7.18± 0.20 %
μ = 420.4± 0.8
σ = 30.2± 0.8



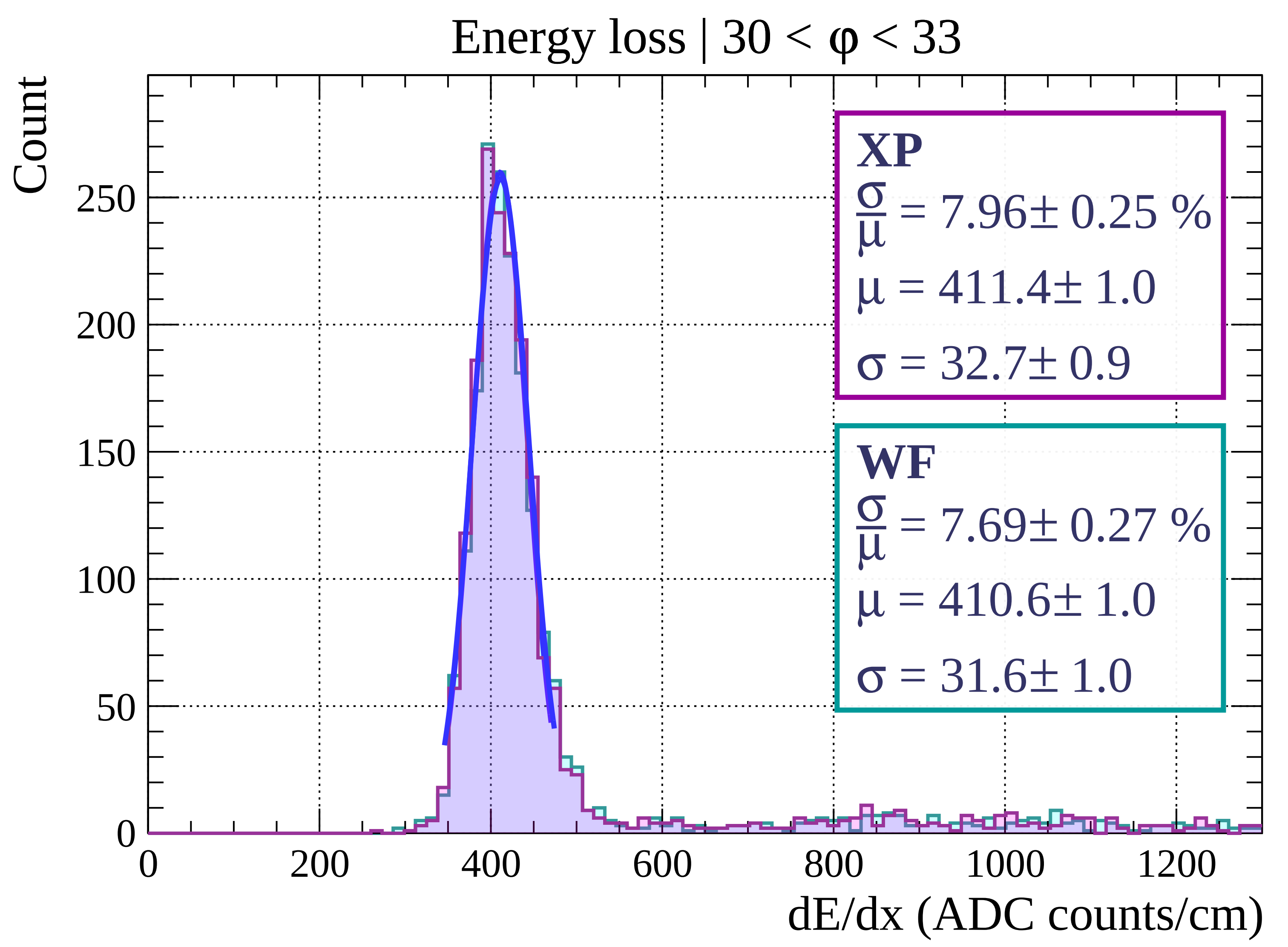
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 6.83± 0.18 %
μ = 411.9± 0.8
σ = 28.1± 0.7
WF
σ/μ = 7.05± 0.19 %
μ = 418.2± 0.8
σ = 29.5± 0.7



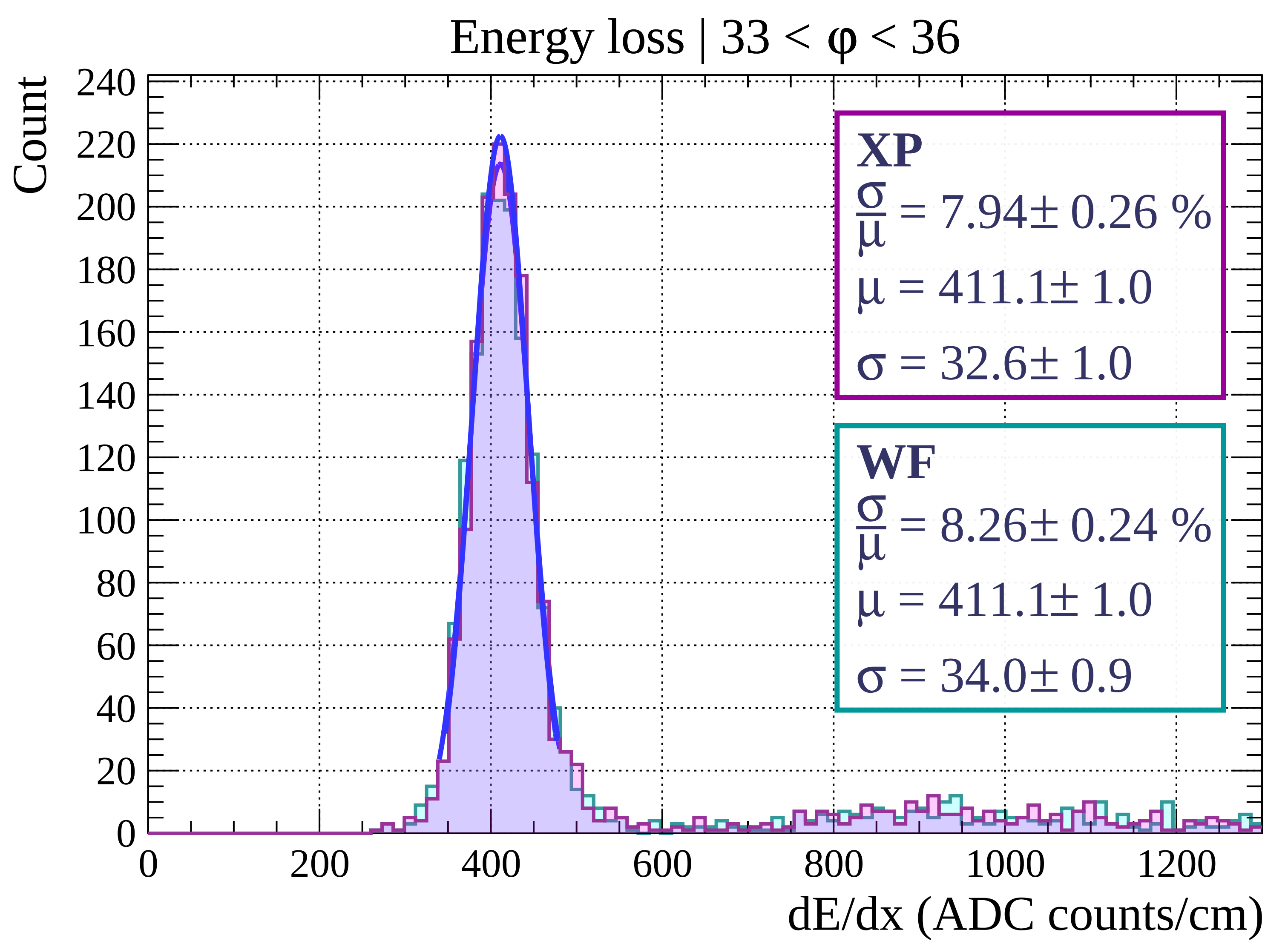
Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.20± 0.22 %
μ = 412.2± 0.9
σ = 29.7± 0.8
WF
σ/μ = 7.19± 0.19 %
μ = 417.6± 0.9
σ = 30.0± 0.7
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



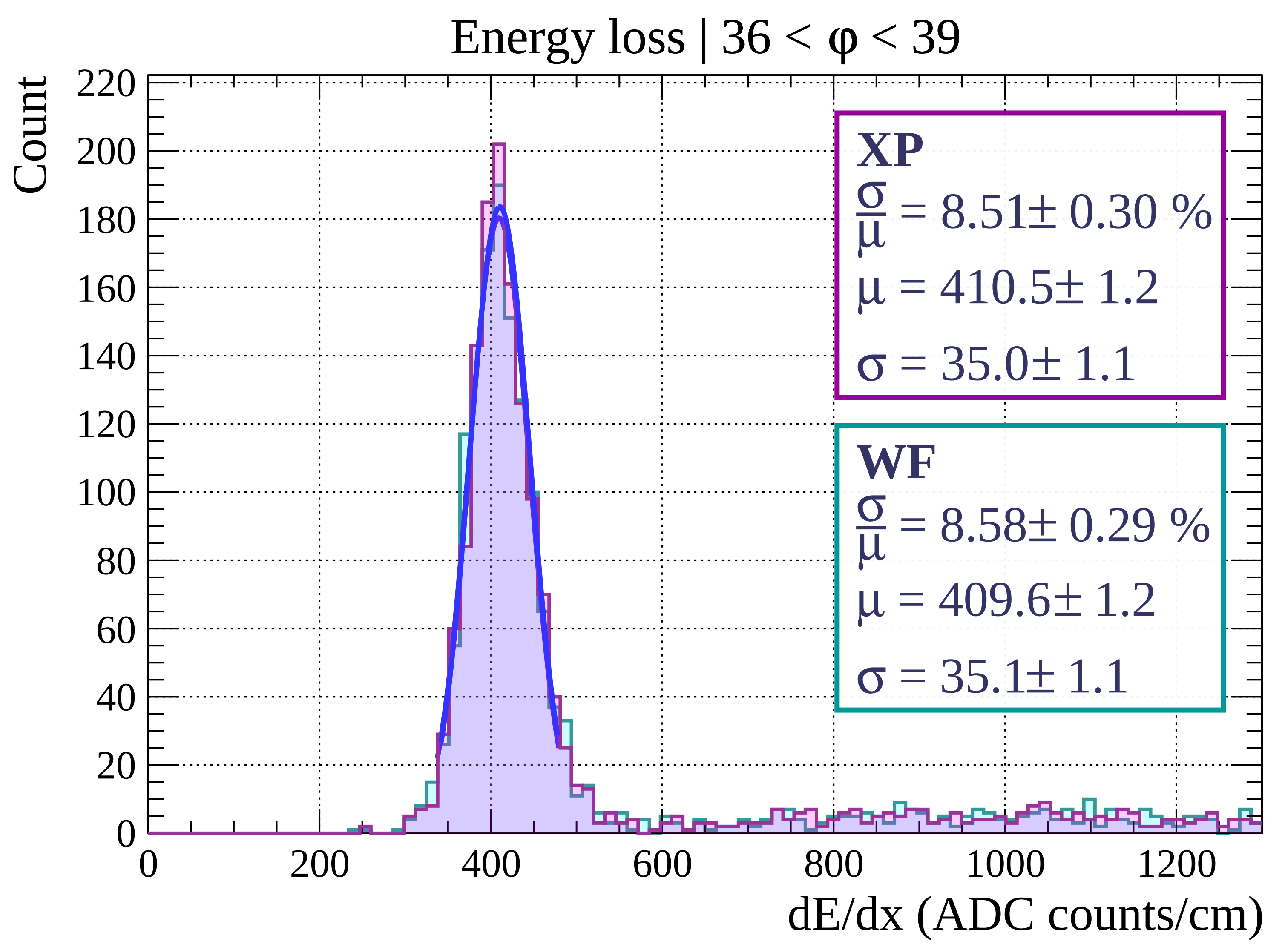
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 7.58± 0.27 %
μ = 410.8± 1.0
σ = 31.1± 1.0
WF
σ/μ = 7.76± 0.24 %
μ = 413.9± 0.9
σ = 32.1± 0.9

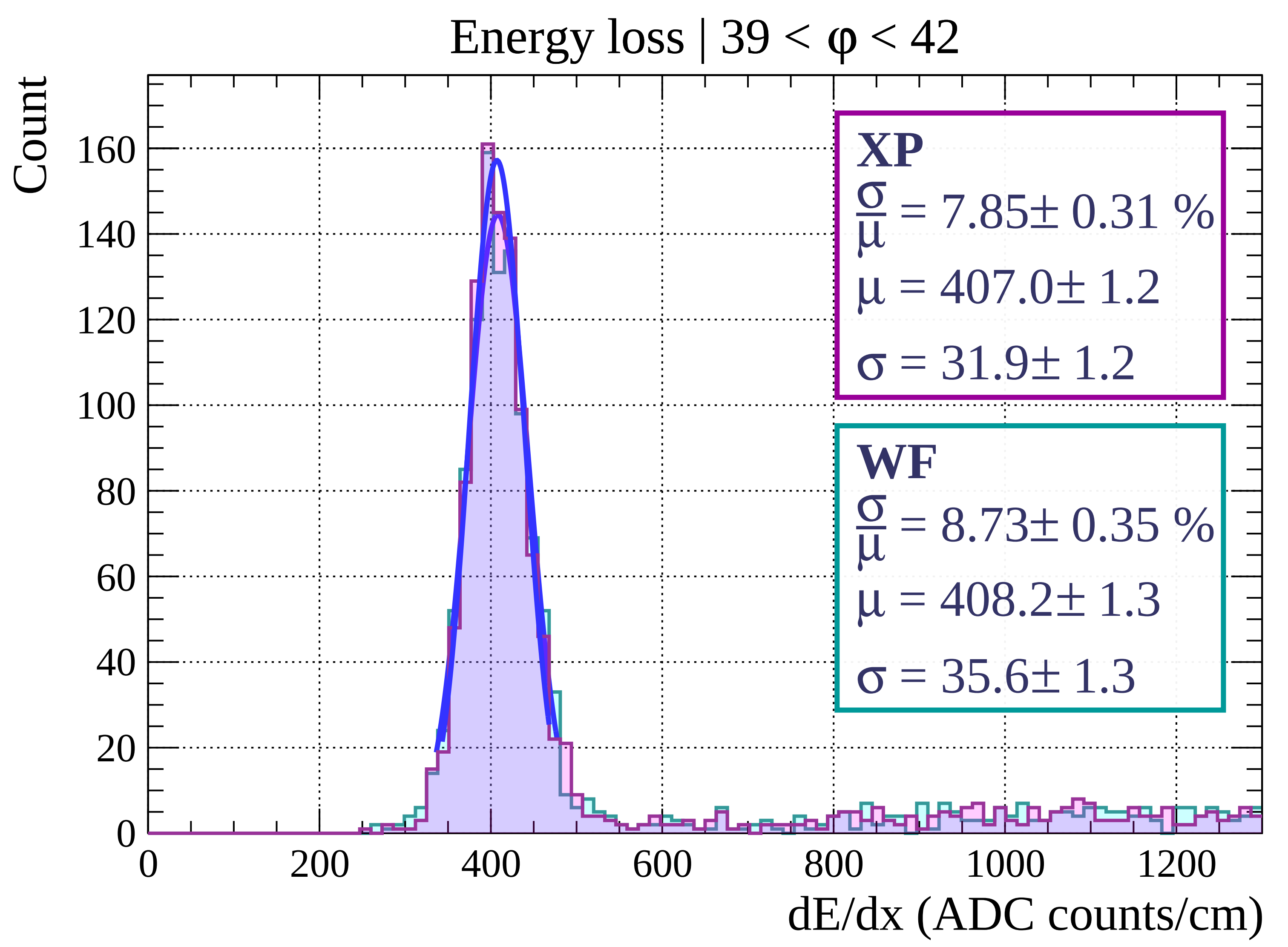


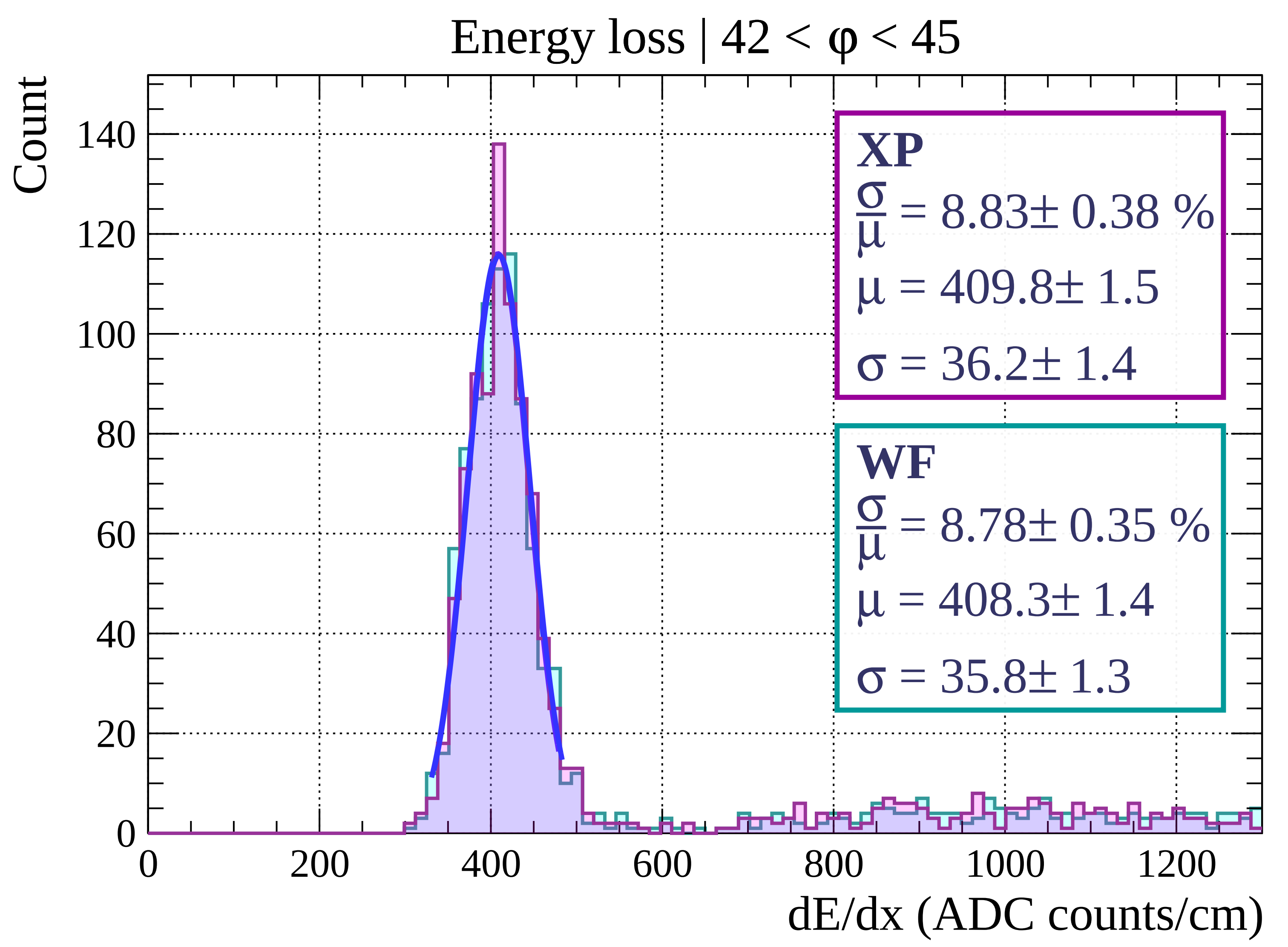
Energy loss | 30 < φ < 33
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.96 ± 0.25 %
μ = 411.4 ± 1.0
σ = 32.7 ± 0.9
WF
σ/μ = 7.69 ± 0.27 %
μ = 410.6 ± 1.0
σ = 31.6 ± 1.0



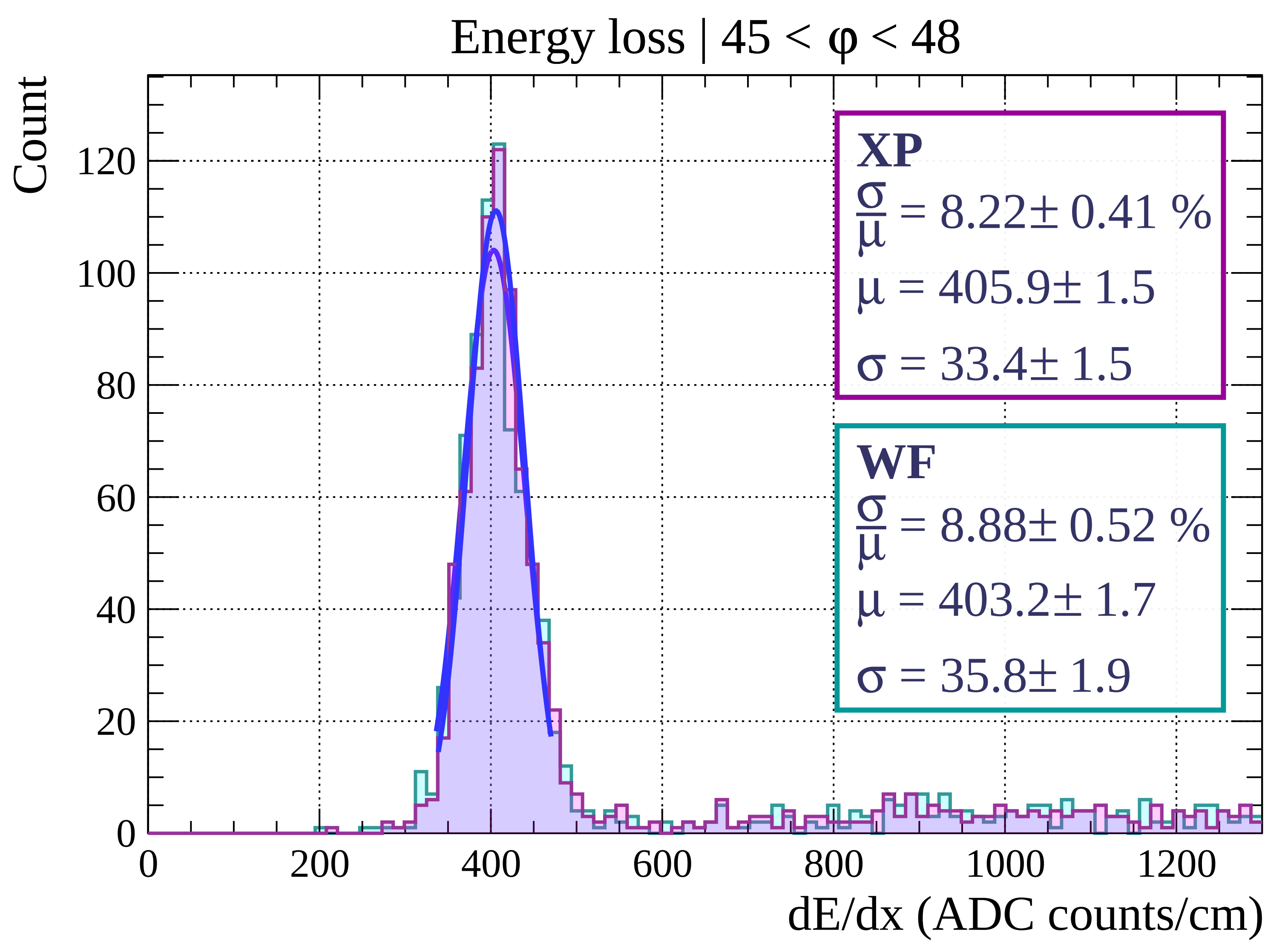
Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
σ/μ = 7.94± 0.26 %
μ = 411.1± 1.0
σ = 32.6± 1.0
WF
σ/μ = 8.26± 0.24 %
μ = 411.1± 1.0
σ = 34.0± 0.9


Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 8.51± 0.30 %
μ = 410.5± 1.2
σ = 35.0± 1.1
WF
σ/μ = 8.58± 0.29 %
μ = 409.6± 1.2
σ = 35.1± 1.1

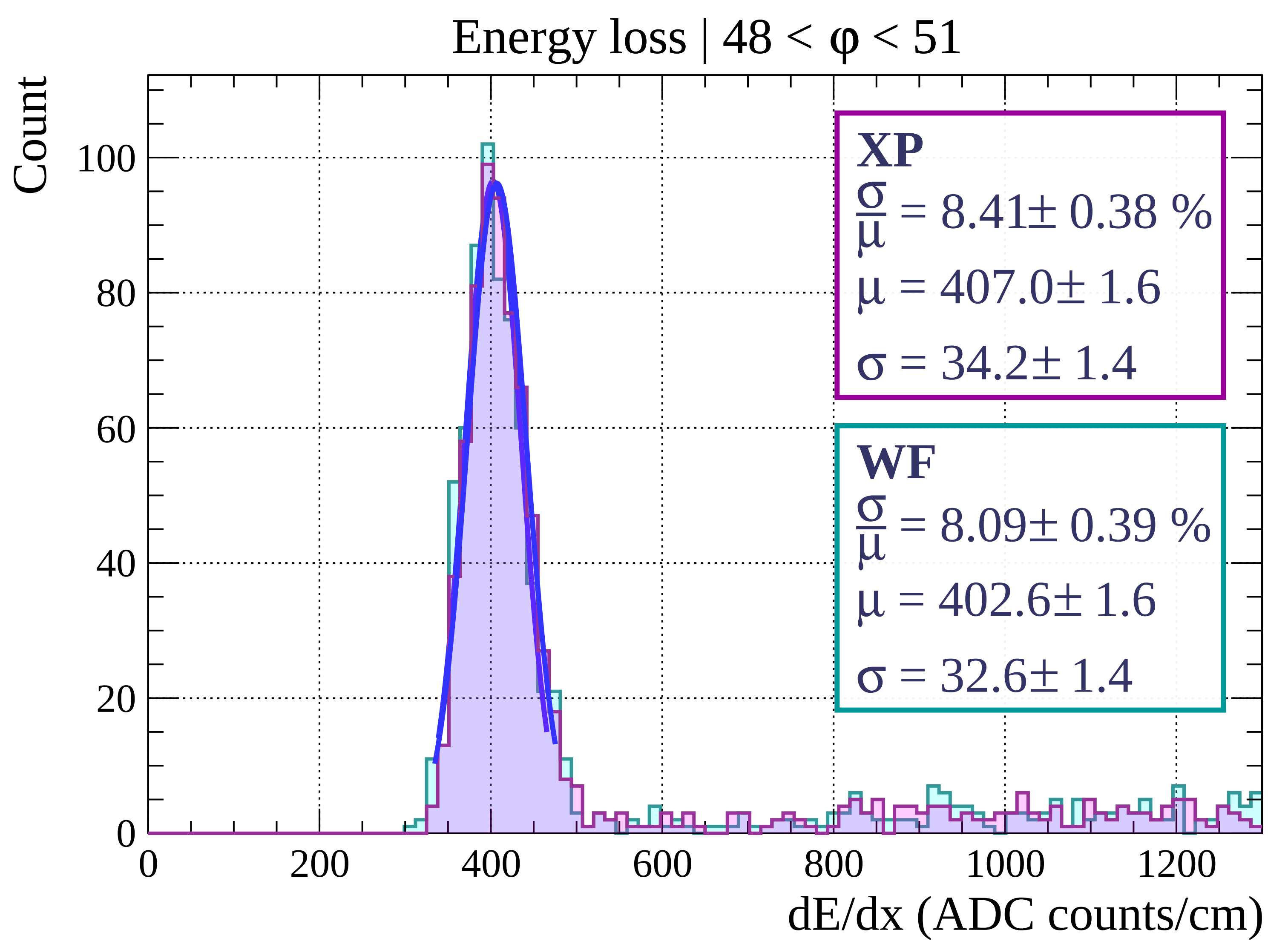

Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 7.85± 0.31 %
μ = 407.0± 1.2
σ = 31.9± 1.2
WF
σ/μ = 8.73± 0.35 %
μ = 408.2± 1.3
σ = 35.6± 1.3



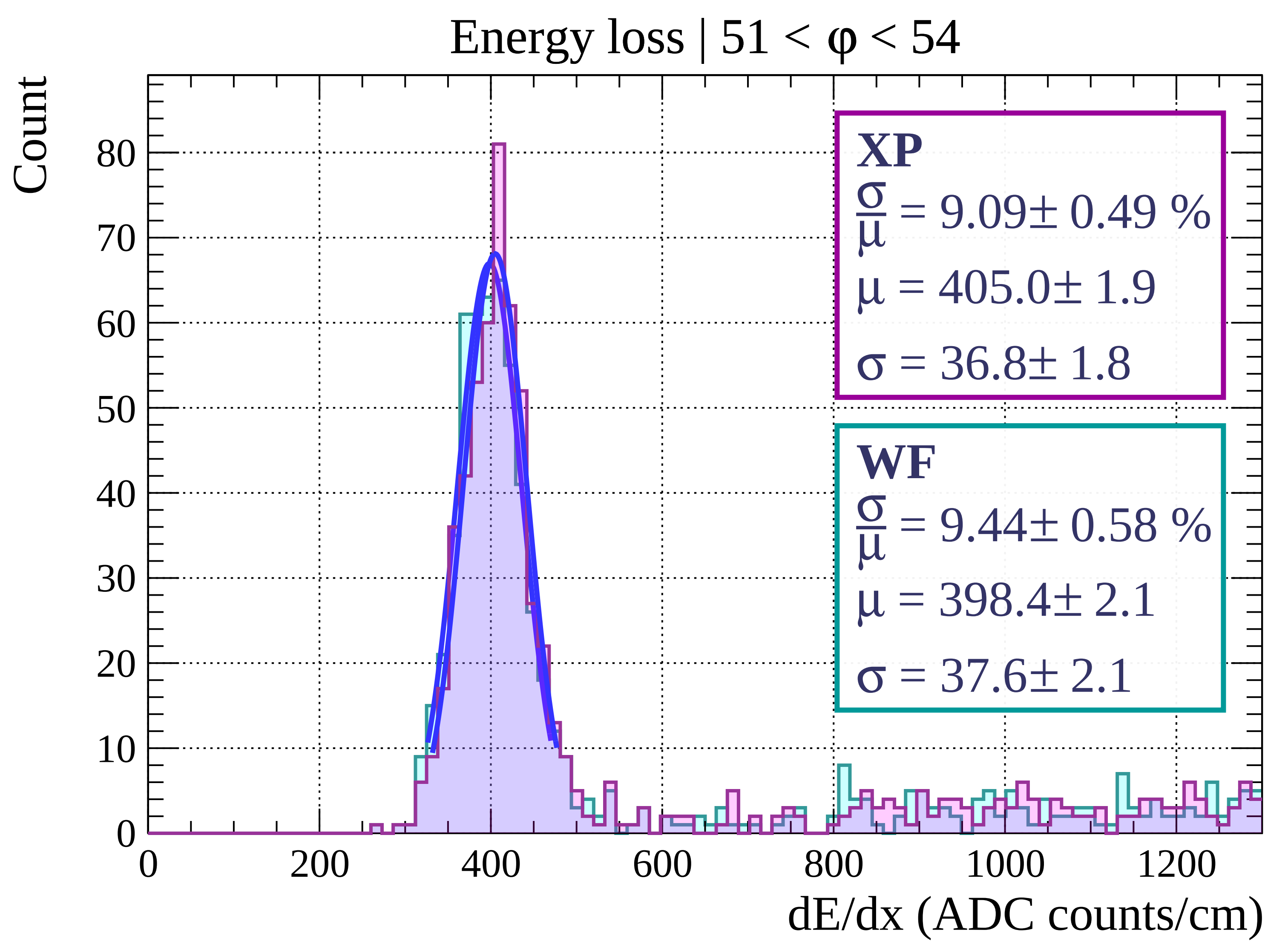
Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.83± 0.38 %
μ = 409.8± 1.5
σ = 36.2± 1.4
WF
σ/μ = 8.78± 0.35 %
μ = 408.3± 1.4
σ = 35.8± 1.3

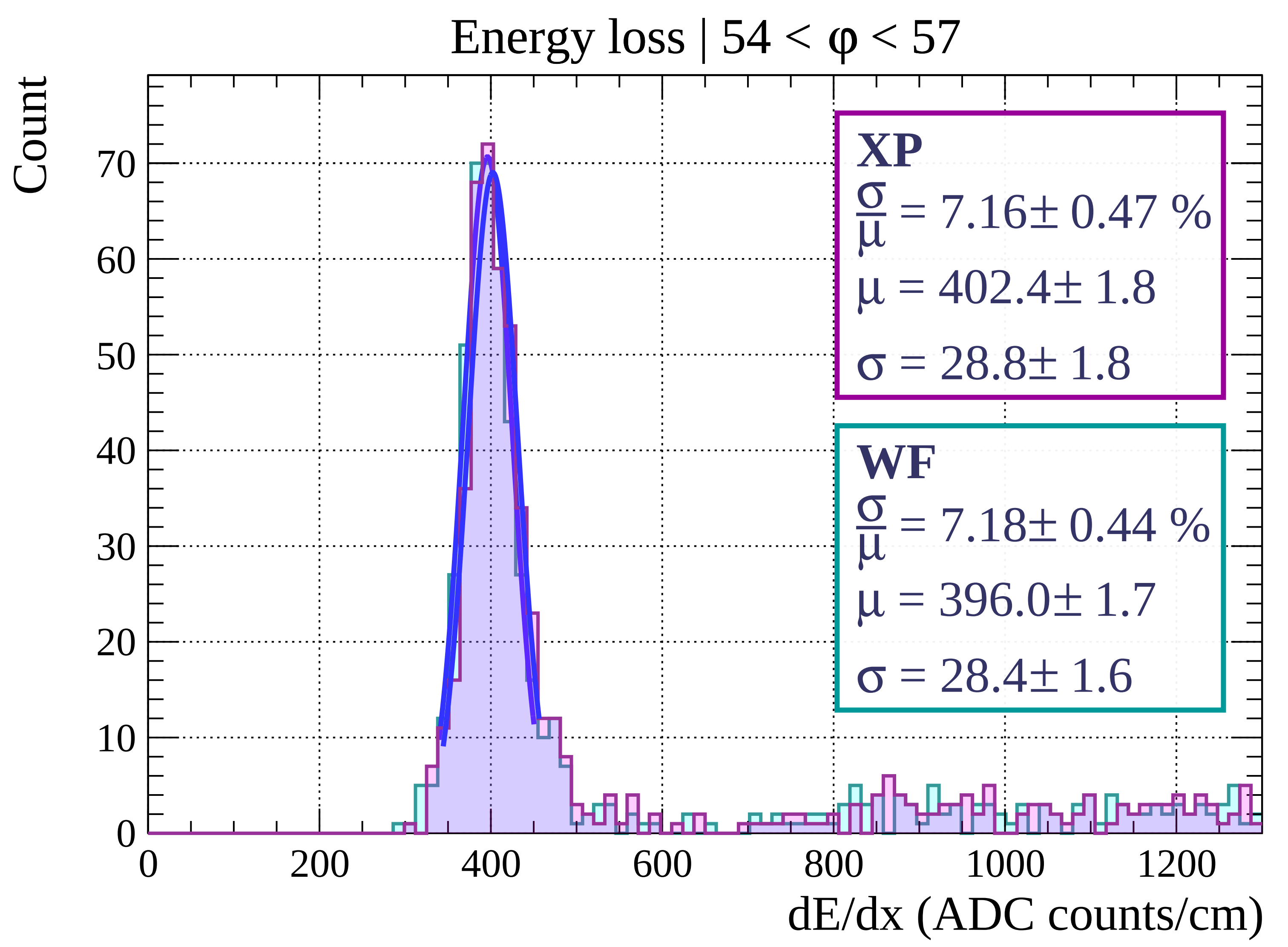


Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 8.22± 0.41 %
μ = 405.9± 1.5
σ = 33.4± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.88± 0.52 %
μ = 403.2± 1.7
σ = 35.8± 1.9

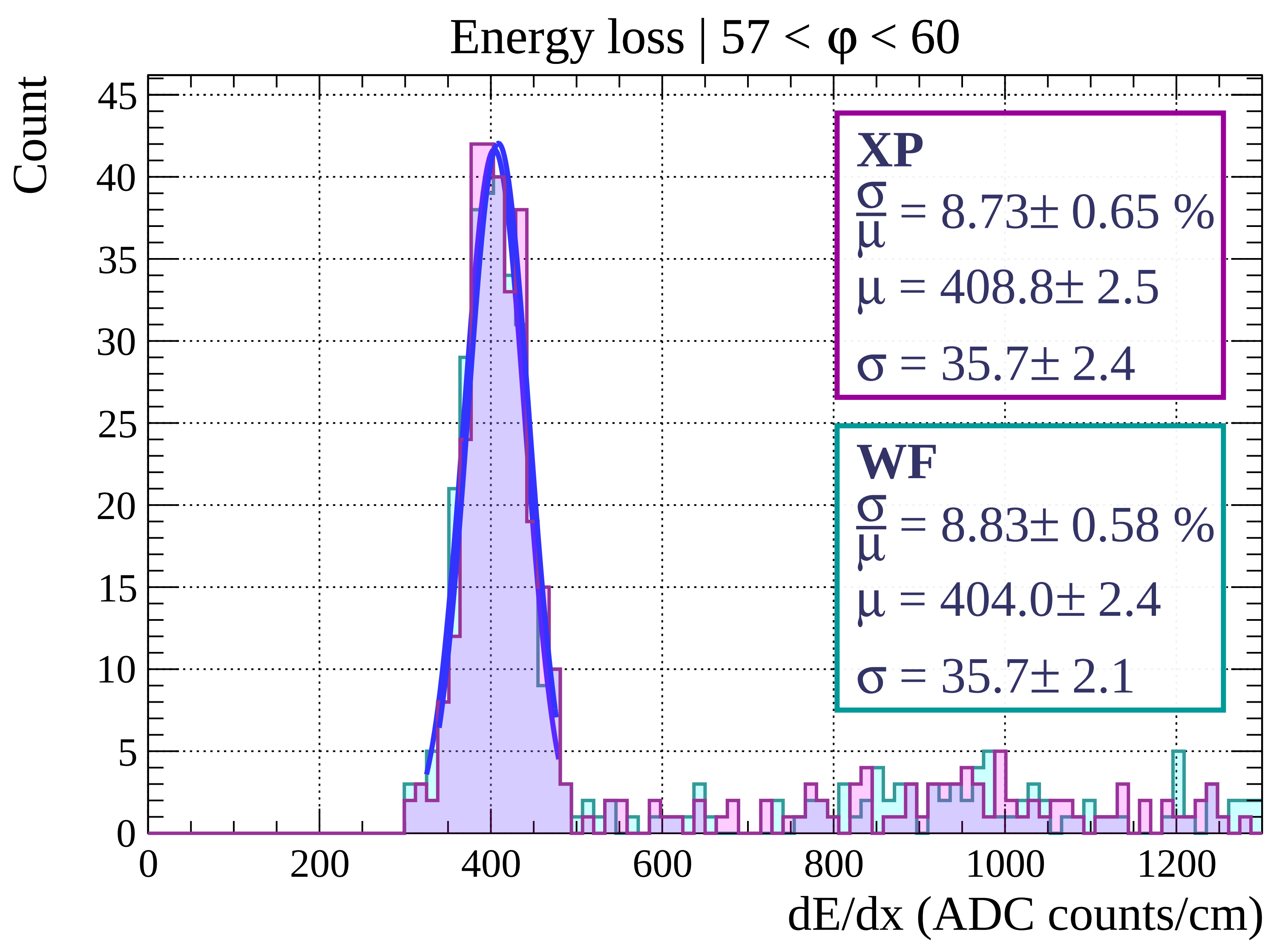


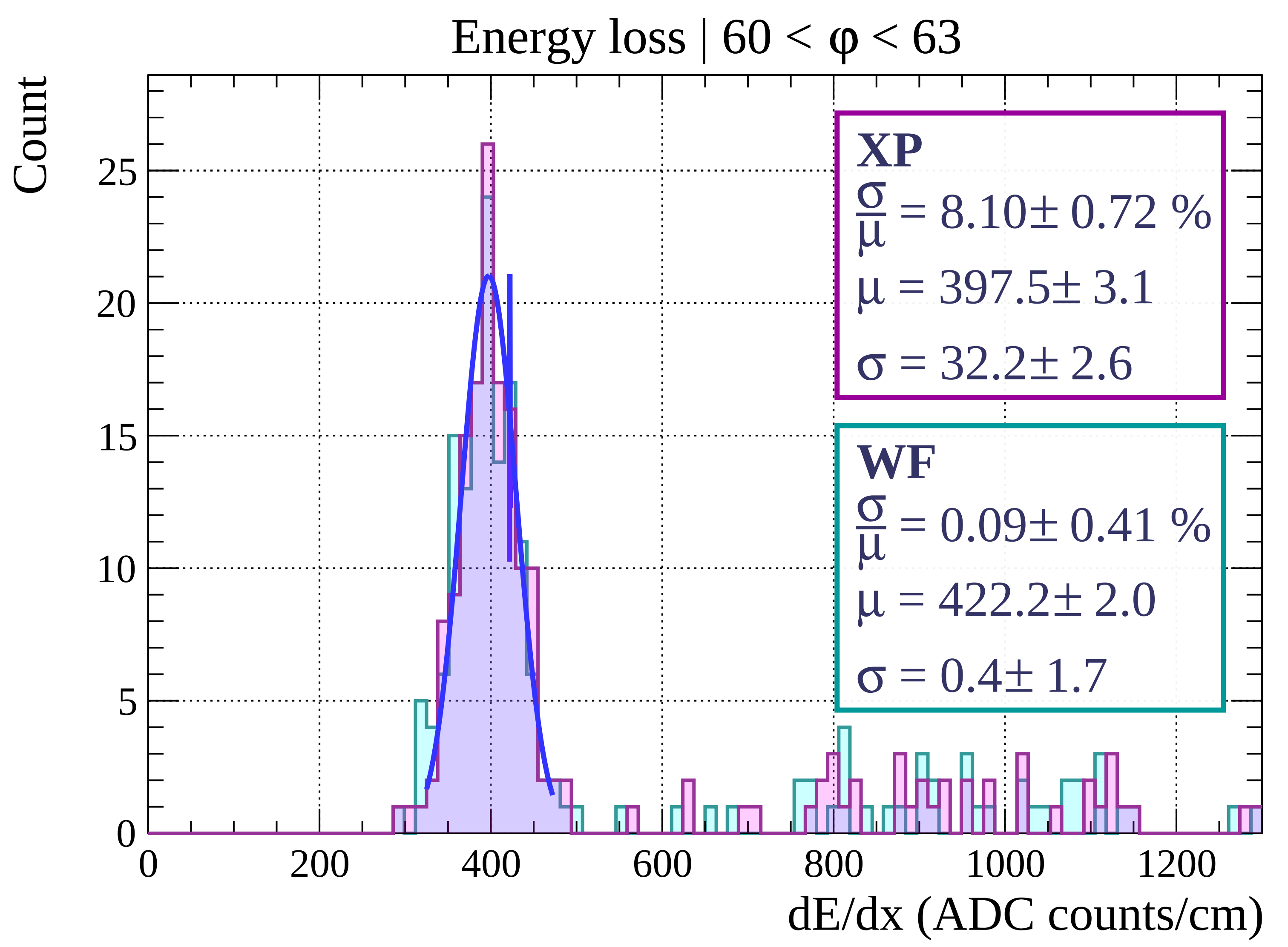
Energy loss | 48 < φ < 51
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.41± 0.38 %
μ = 407.0± 1.6
σ = 34.2± 1.4
WF
σ/μ = 8.09± 0.39 %
μ = 402.6± 1.6
σ = 32.6± 1.4


Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 9.09± 0.49 %
μ = 405.0± 1.9
σ = 36.8± 1.8
WF
σ/μ = 9.44± 0.58 %
μ = 398.4± 2.1
σ = 37.6± 2.1

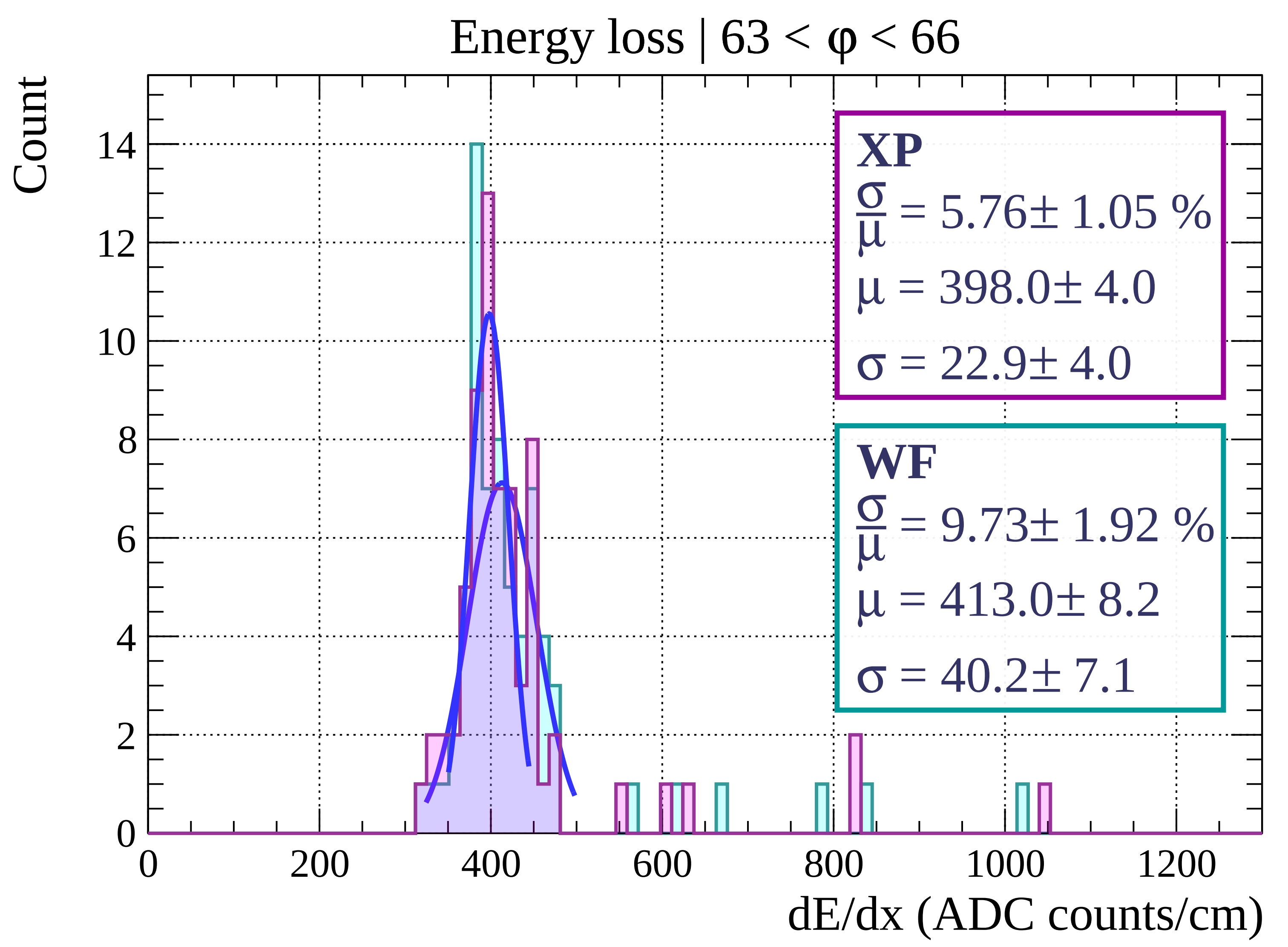


Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 7.16± 0.47 %
μ = 402.4± 1.8
σ = 28.8± 1.8
WF
σ/μ = 7.18± 0.44 %
μ = 396.0± 1.7
σ = 28.4± 1.6

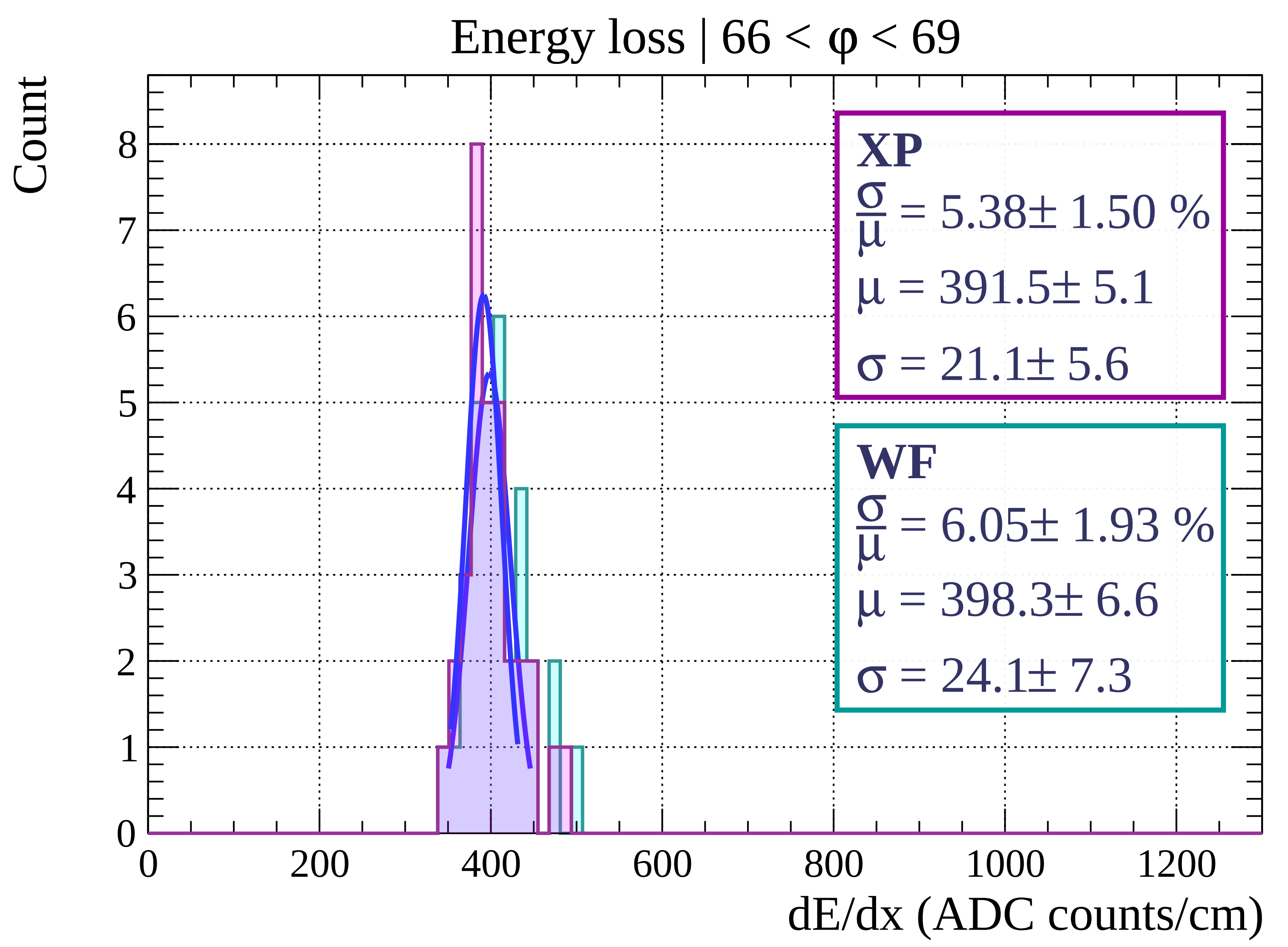

Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.73± 0.65 %
μ = 408.8± 2.5
σ = 35.7± 2.4
WF
σ/μ = 8.83± 0.58 %
μ = 404.0± 2.4
σ = 35.7± 2.1
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200



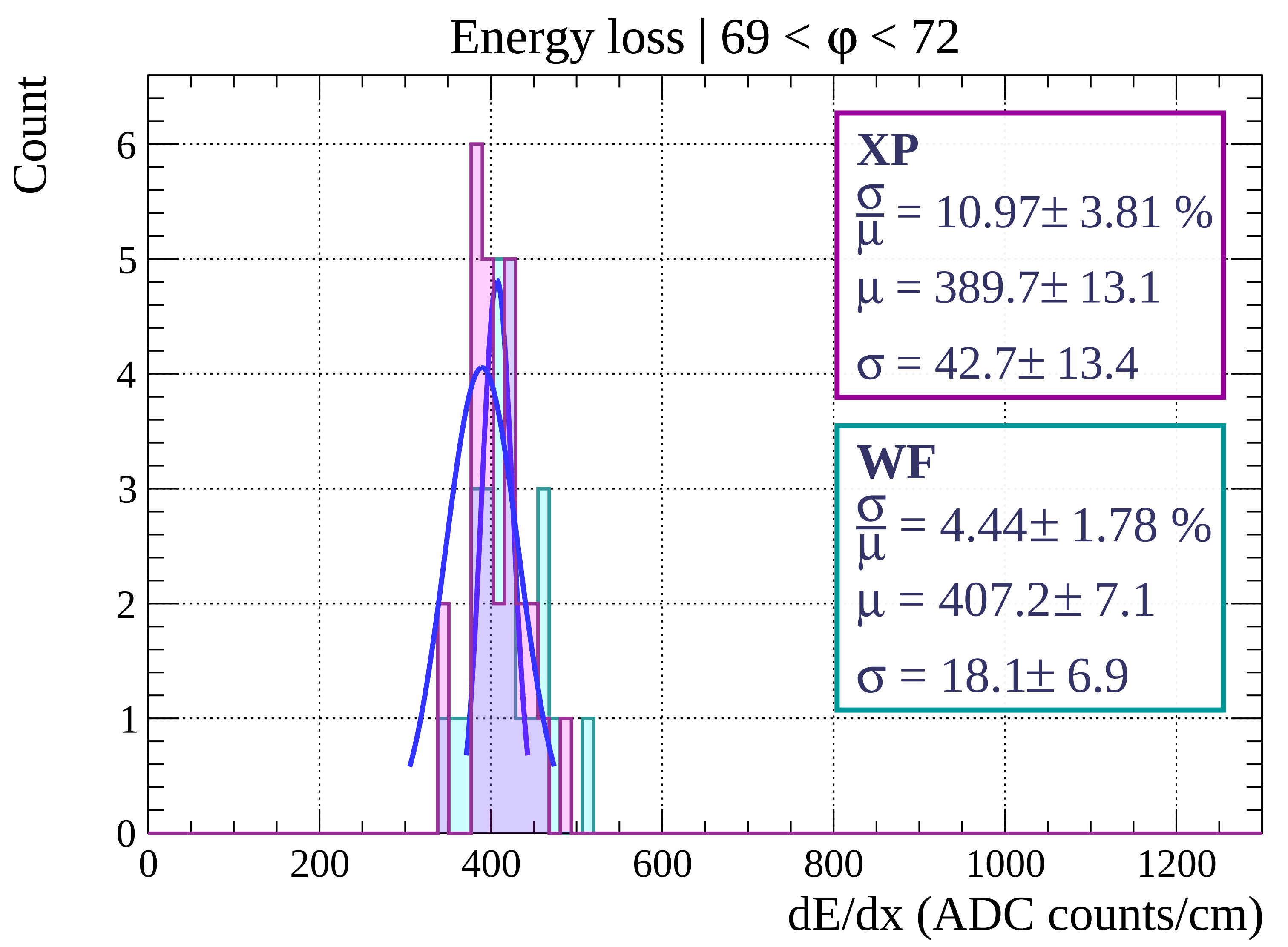
Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 8.10± 0.72 %
μ = 397.5± 3.1
σ = 32.2± 2.6
WF
σ/μ = 0.09± 0.41 %
μ = 422.2± 2.0
σ = 0.4± 1.7

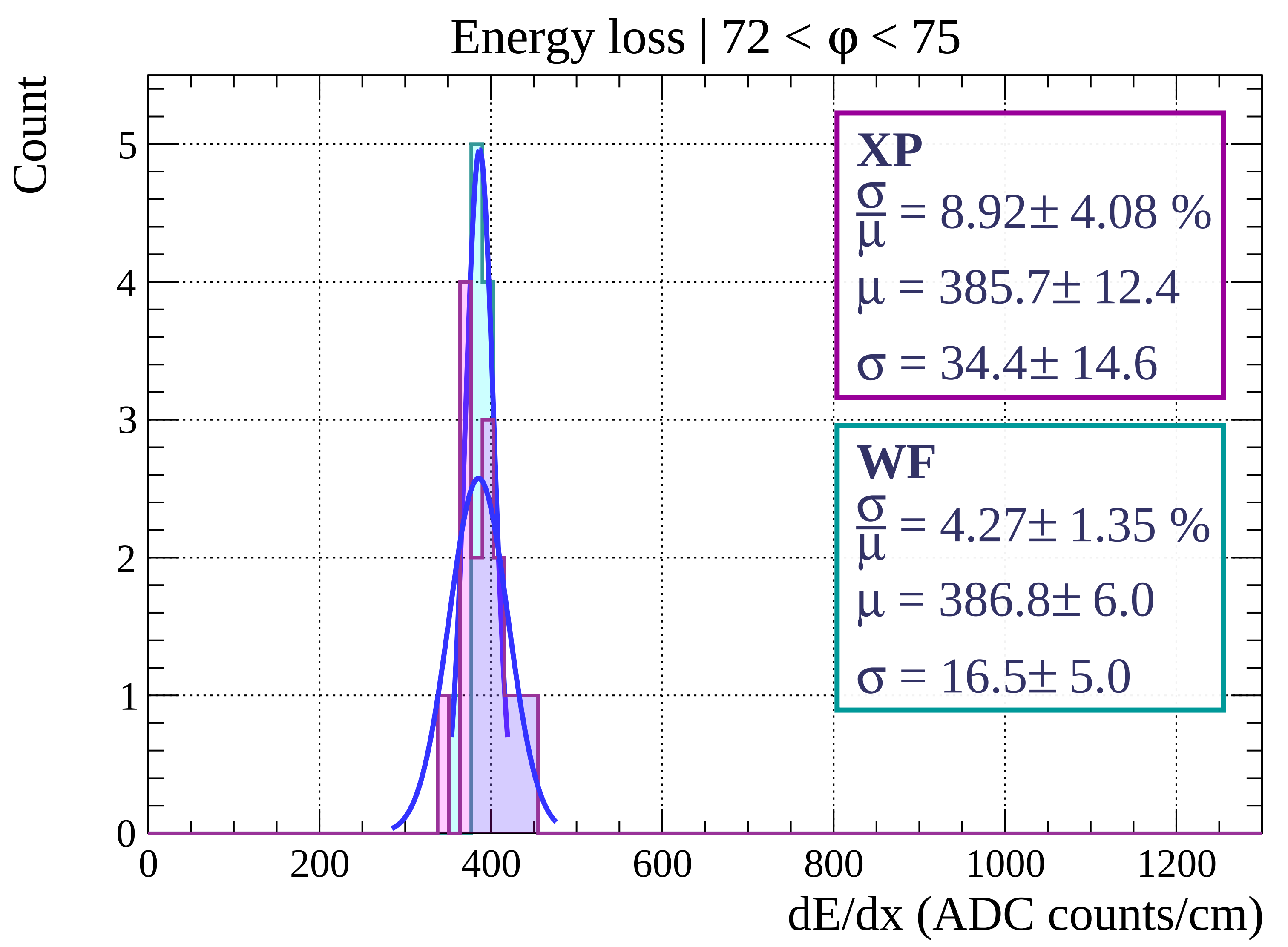


Energy loss | 63 < φ < 66
Count
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.76± 1.05 %
μ = 398.0± 4.0
σ = 22.9± 4.0
WF
σ/μ = 9.73± 1.92 %
μ = 413.0± 8.2
σ = 40.2± 7.1

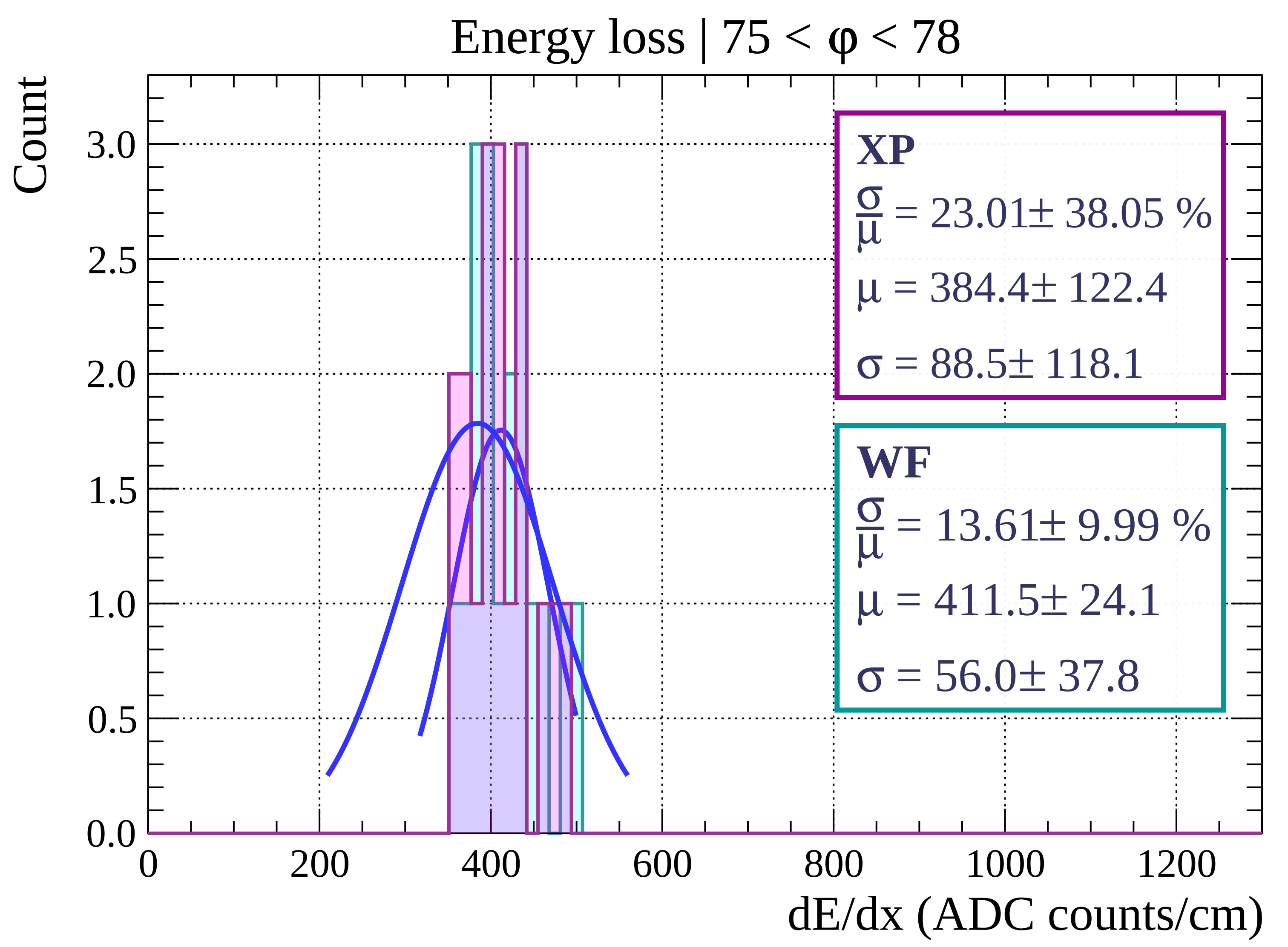


Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
$\frac{\sigma}{\mu}$ = 5.38± 1.50 %
μ = 391.5± 5.1
σ = 21.1± 5.6
WF
$\frac{\sigma}{\mu}$ = 6.05± 1.93 %
μ = 398.3± 6.6
σ = 24.1± 7.3

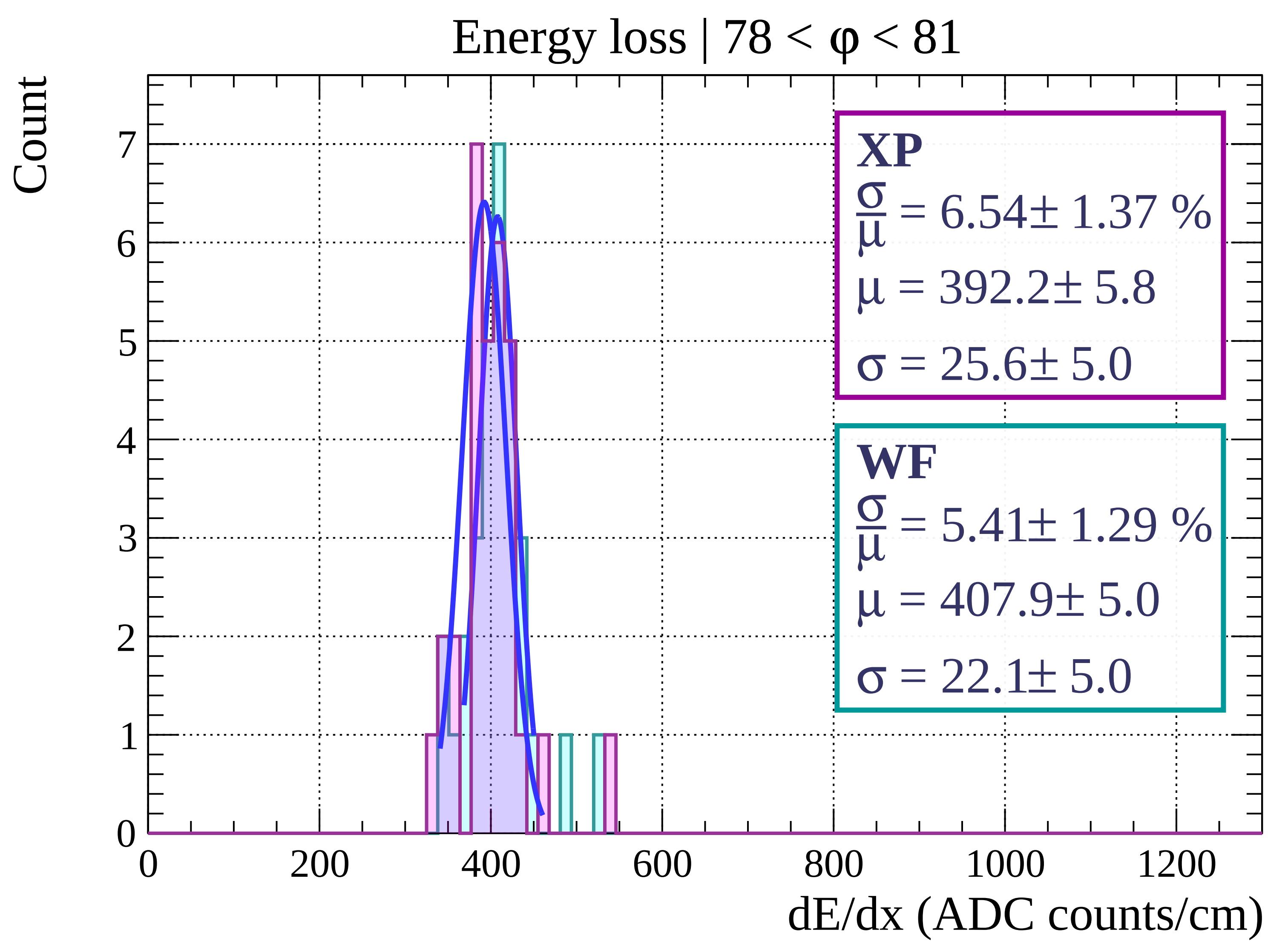

Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 10.97± 3.81 %
μ = 389.7± 13.1
σ = 42.7± 13.4
WF
σ/μ = 4.44± 1.78 %
μ = 407.2± 7.1
σ = 18.1± 6.9



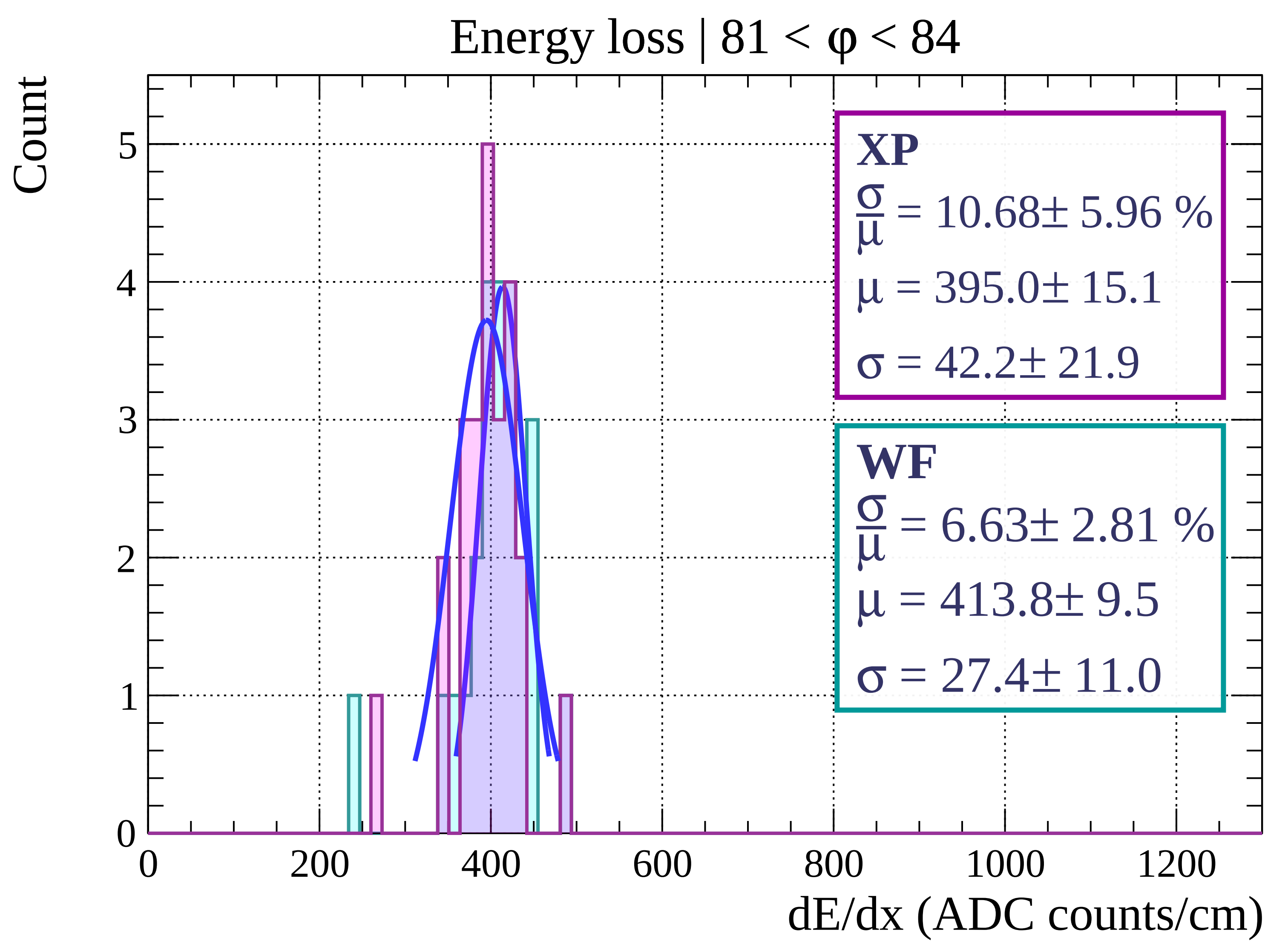
Energy loss | 72 < φ < 75
Count
0
1
2
3
4
5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.92± 4.08 %
μ = 385.7± 12.4
σ = 34.4± 14.6
WF
$\frac{\sigma}{\mu}$ = 4.27± 1.35 %
μ = 386.8± 6.0
σ = 16.5± 5.0

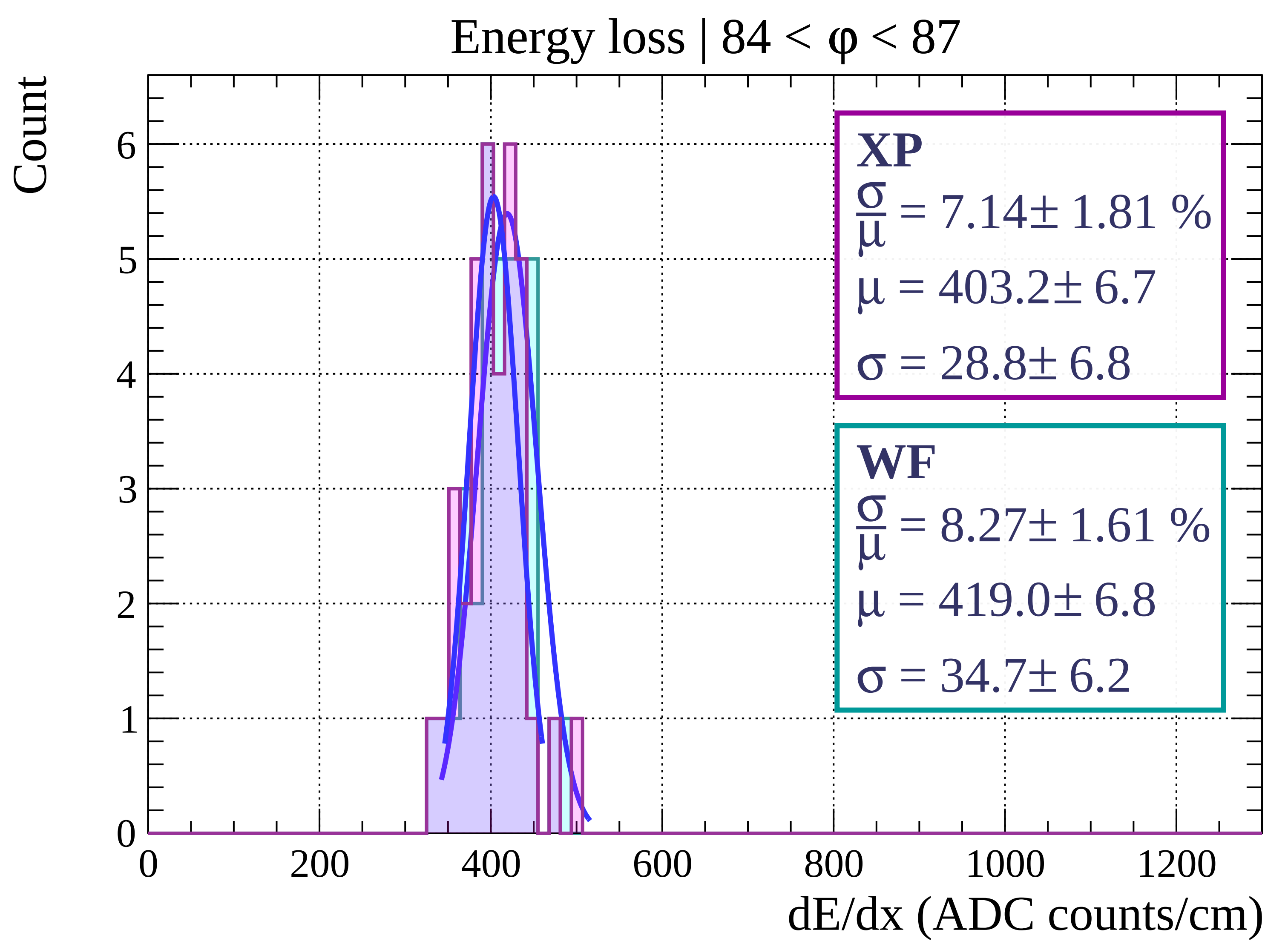


Energy loss | 75 < φ < 78
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 23.01± 38.05 %
μ = 384.4± 122.4
σ = 88.5± 118.1
WF
σ/μ = 13.61± 9.99 %
μ = 411.5± 24.1
σ = 56.0± 37.8

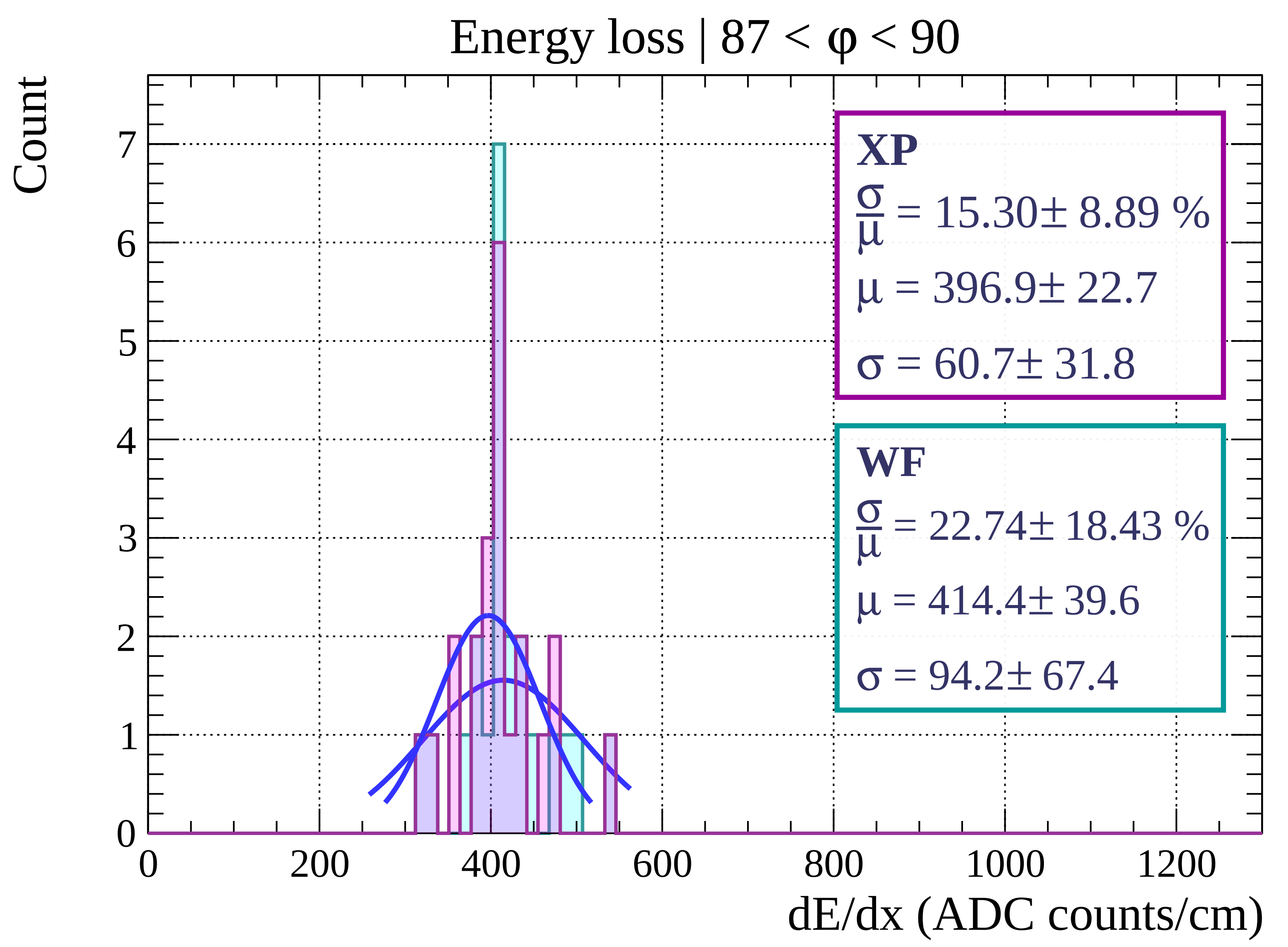


Energy loss | 78 < φ < 81
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 6.54± 1.37 %
μ = 392.2± 5.8
σ = 25.6± 5.0
WF
σ/μ = 5.41± 1.29 %
μ = 407.9± 5.0
σ = 22.1± 5.0

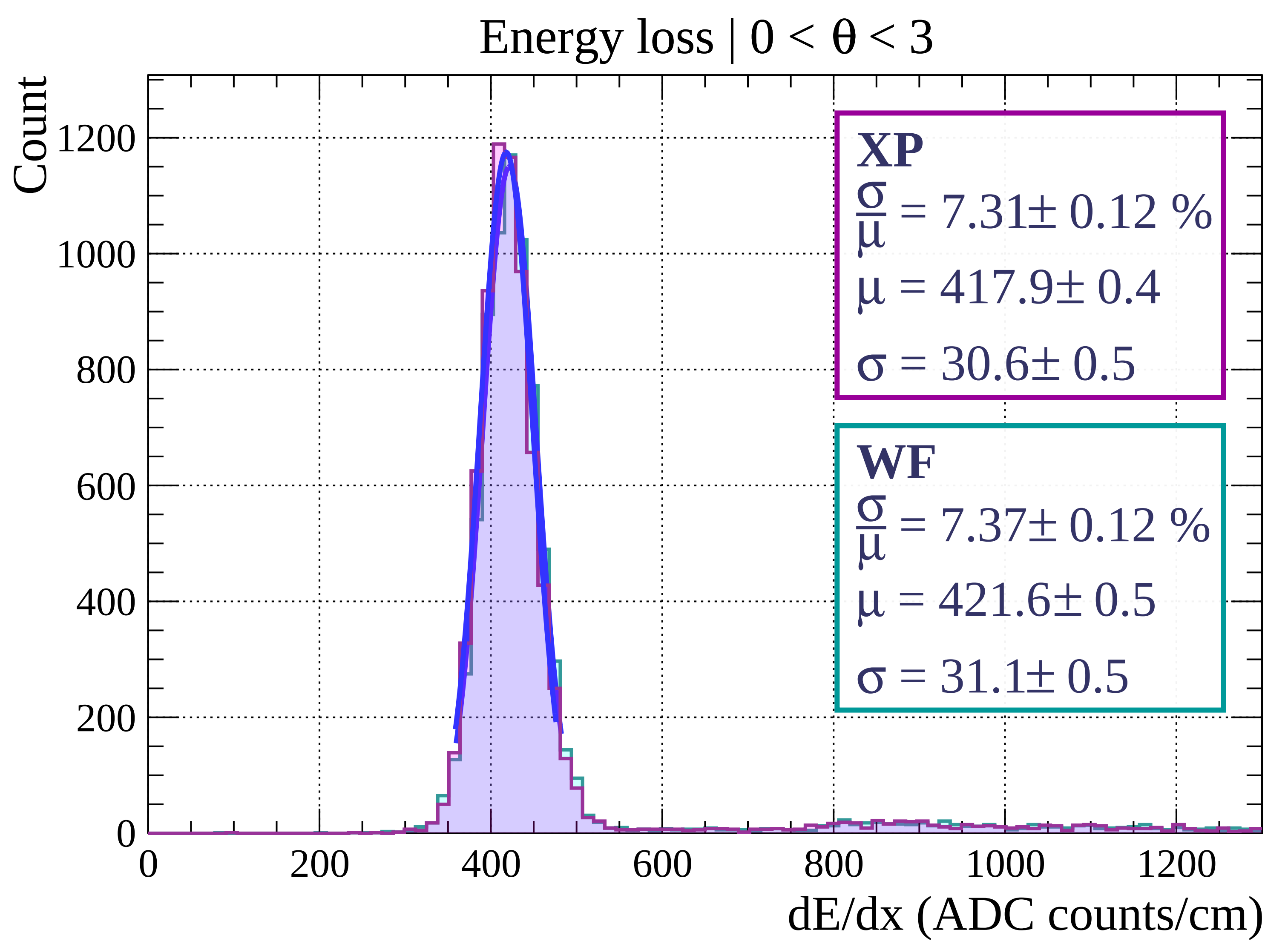

Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 10.68± 5.96 %
μ = 395.0± 15.1
σ = 42.2± 21.9
WF
σ/μ = 6.63± 2.81 %
μ = 413.8± 9.5
σ = 27.4± 11.0



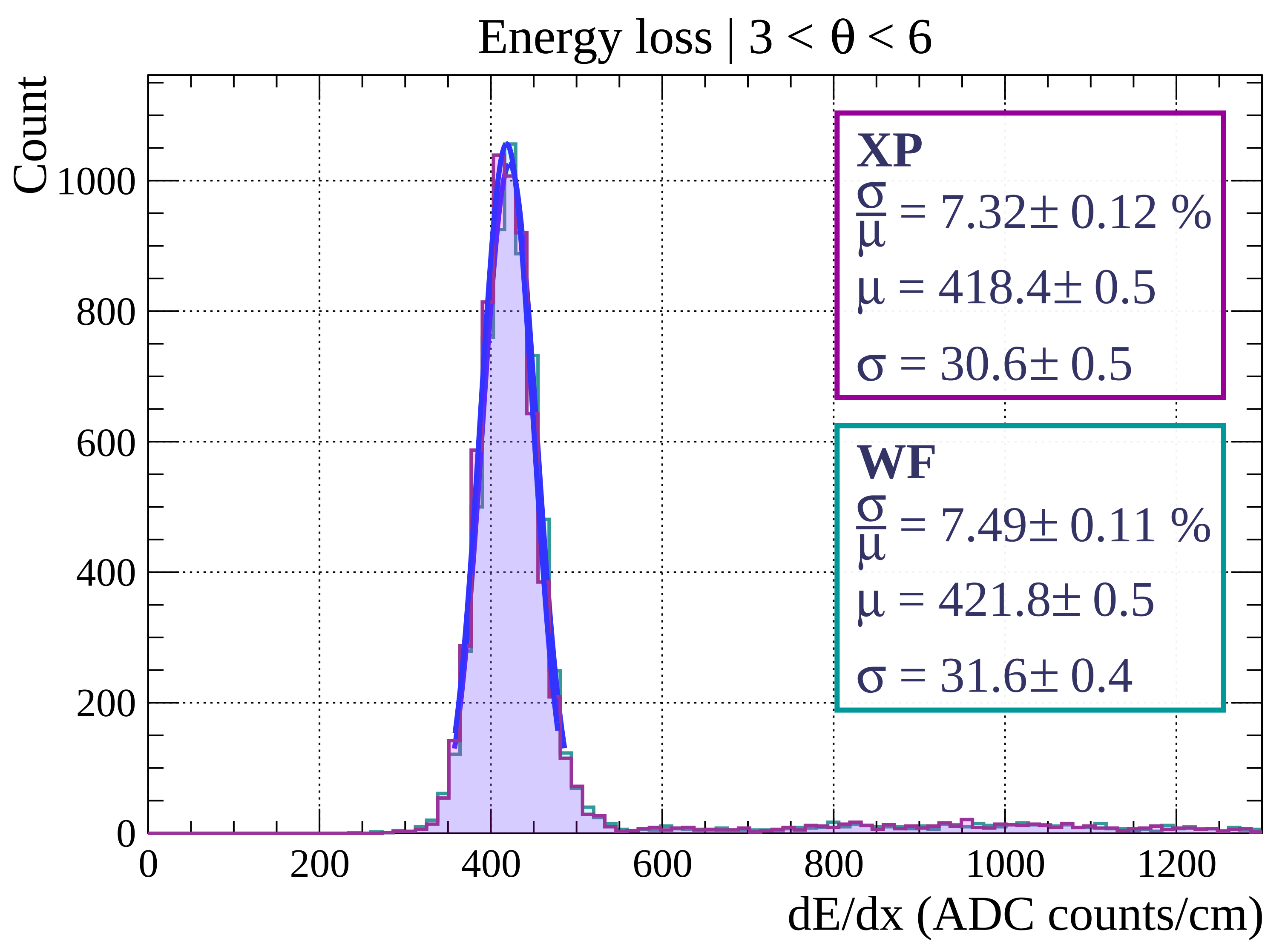
Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 7.14± 1.81 %
μ = 403.2± 6.7
σ = 28.8± 6.8
WF
σ/μ = 8.27± 1.61 %
μ = 419.0± 6.8
σ = 34.7± 6.2

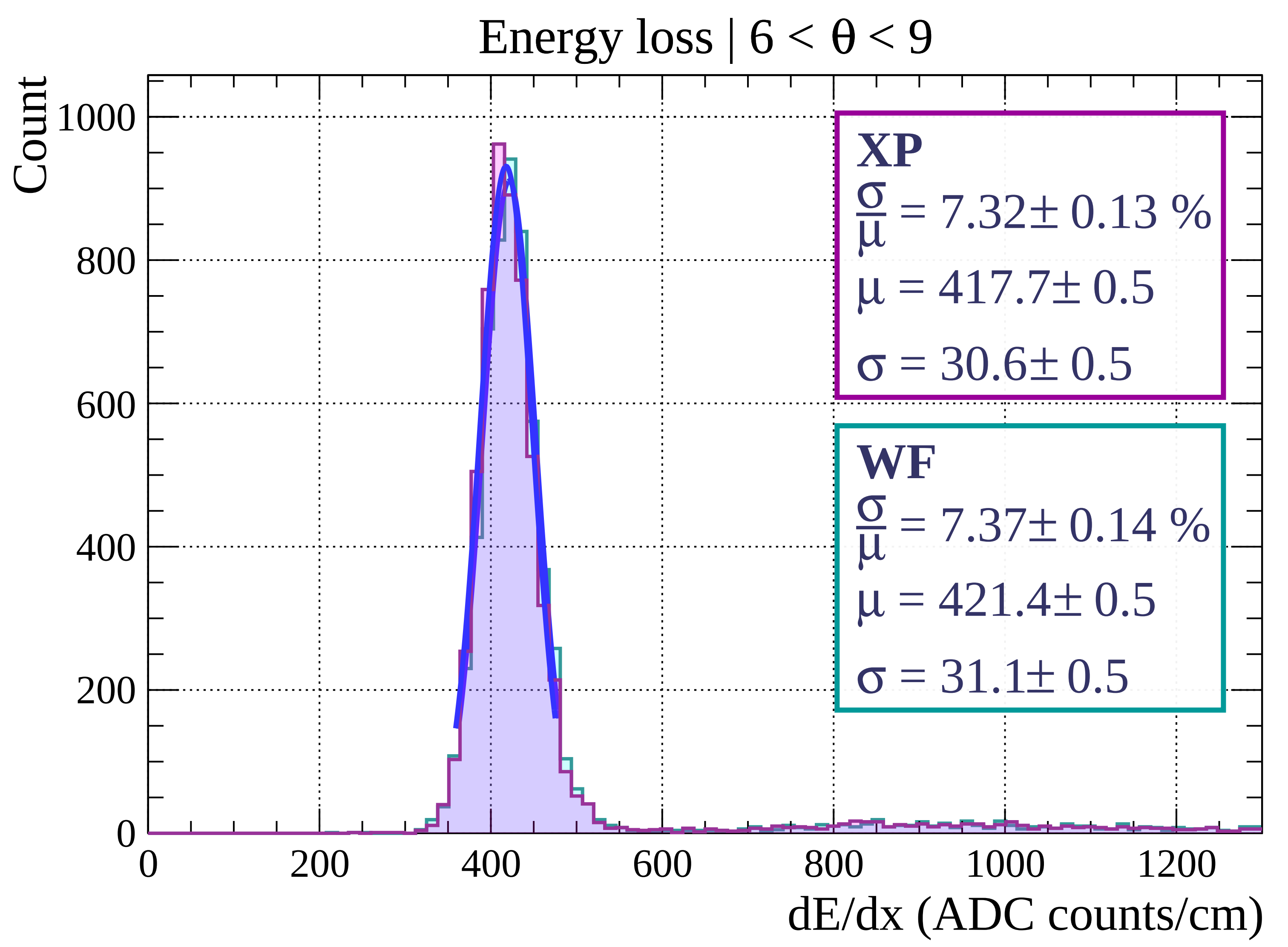


Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
σ/μ = 15.30± 8.89 %
μ = 396.9± 22.7
σ = 60.7± 31.8
WF
σ/μ = 22.74± 18.43 %
μ = 414.4± 39.6
σ = 94.2± 67.4

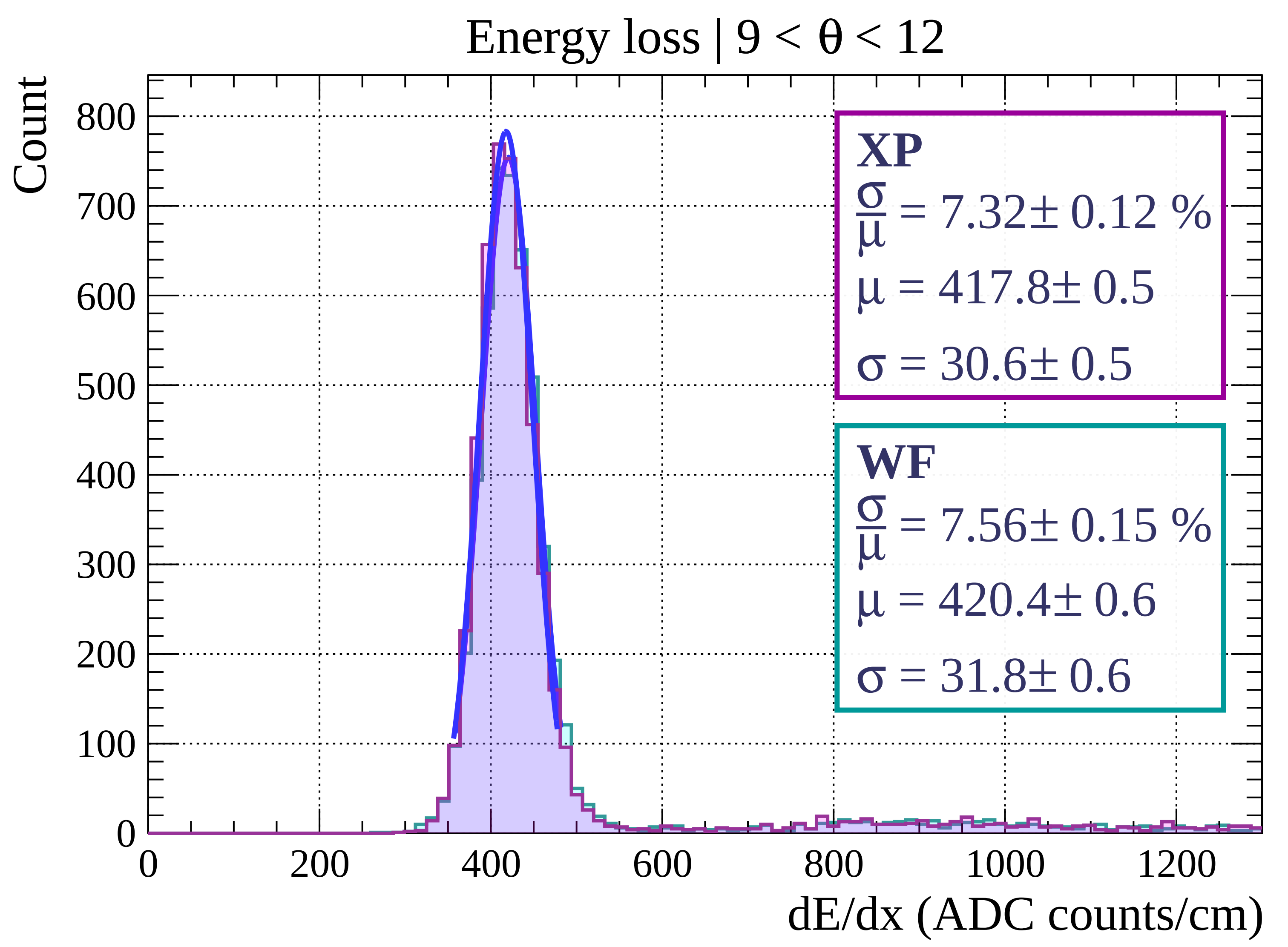


Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.31± 0.12 %
μ = 417.9± 0.4
σ = 30.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.37± 0.12 %
μ = 421.6± 0.5
σ = 31.1± 0.5

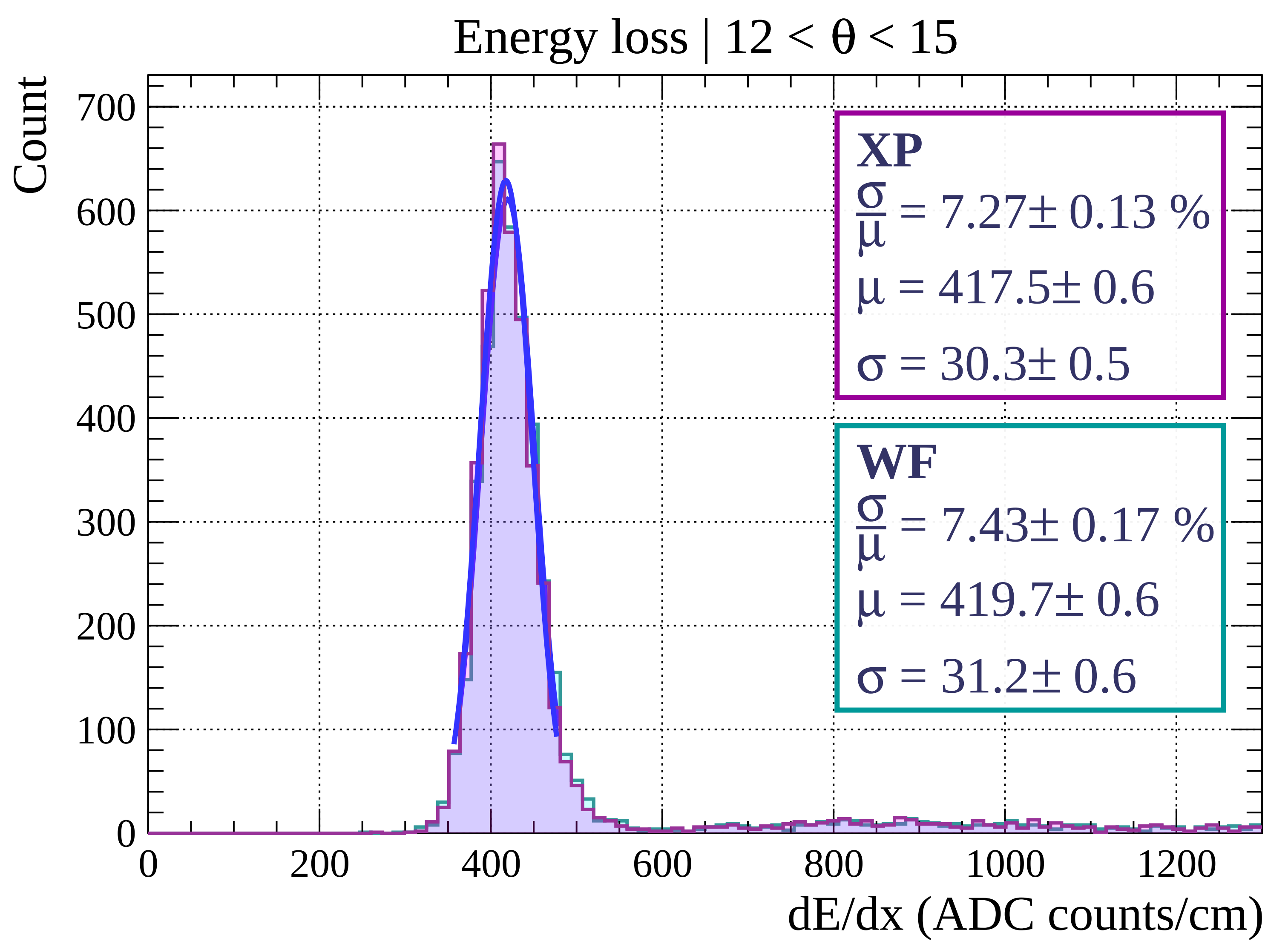

Energy loss | 3 < θ < 6
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.32± 0.12 %
μ = 418.4± 0.5
σ = 30.6± 0.5
WF
σ/μ = 7.49± 0.11 %
μ = 421.8± 0.5
σ = 31.6± 0.4

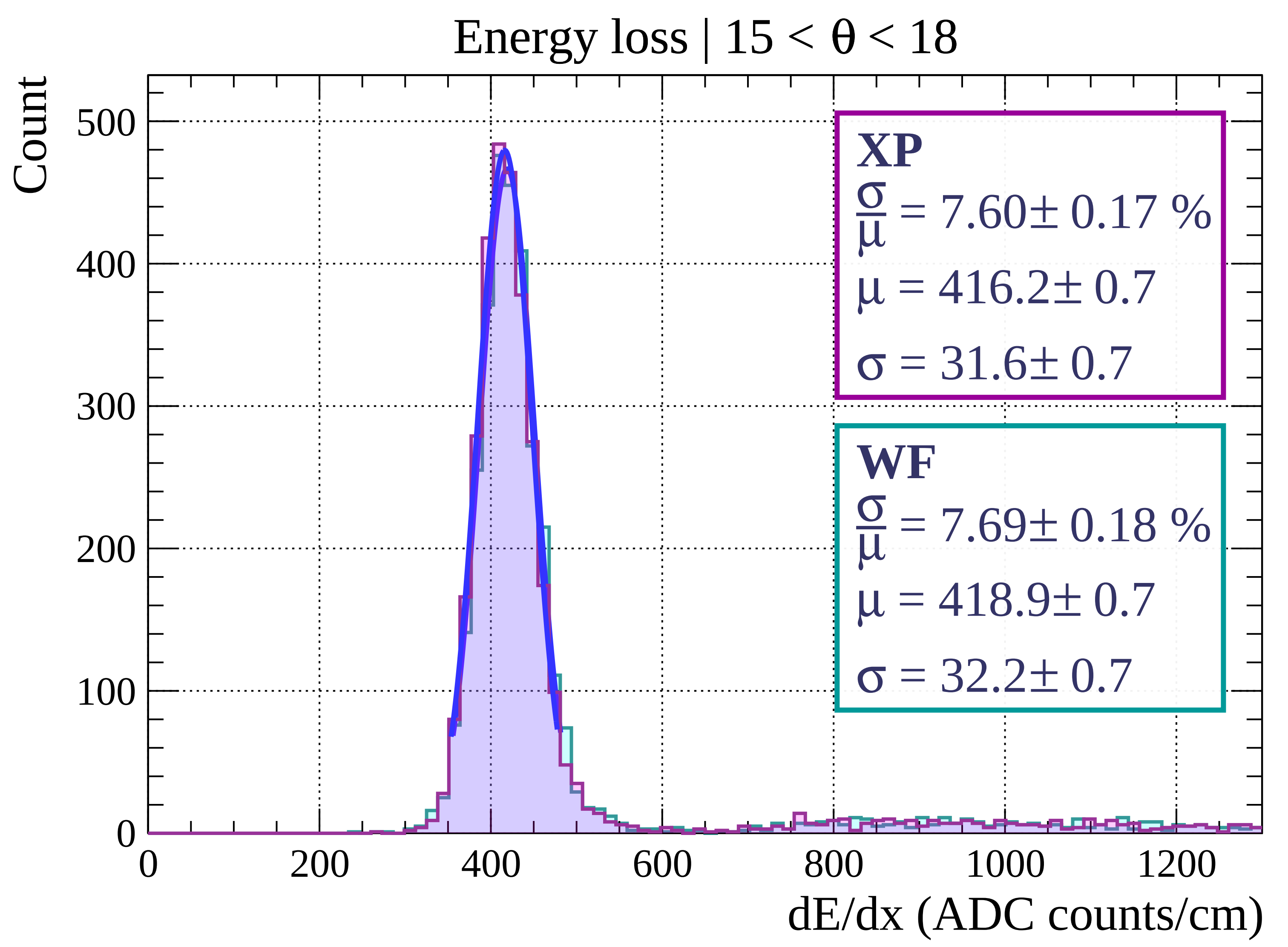


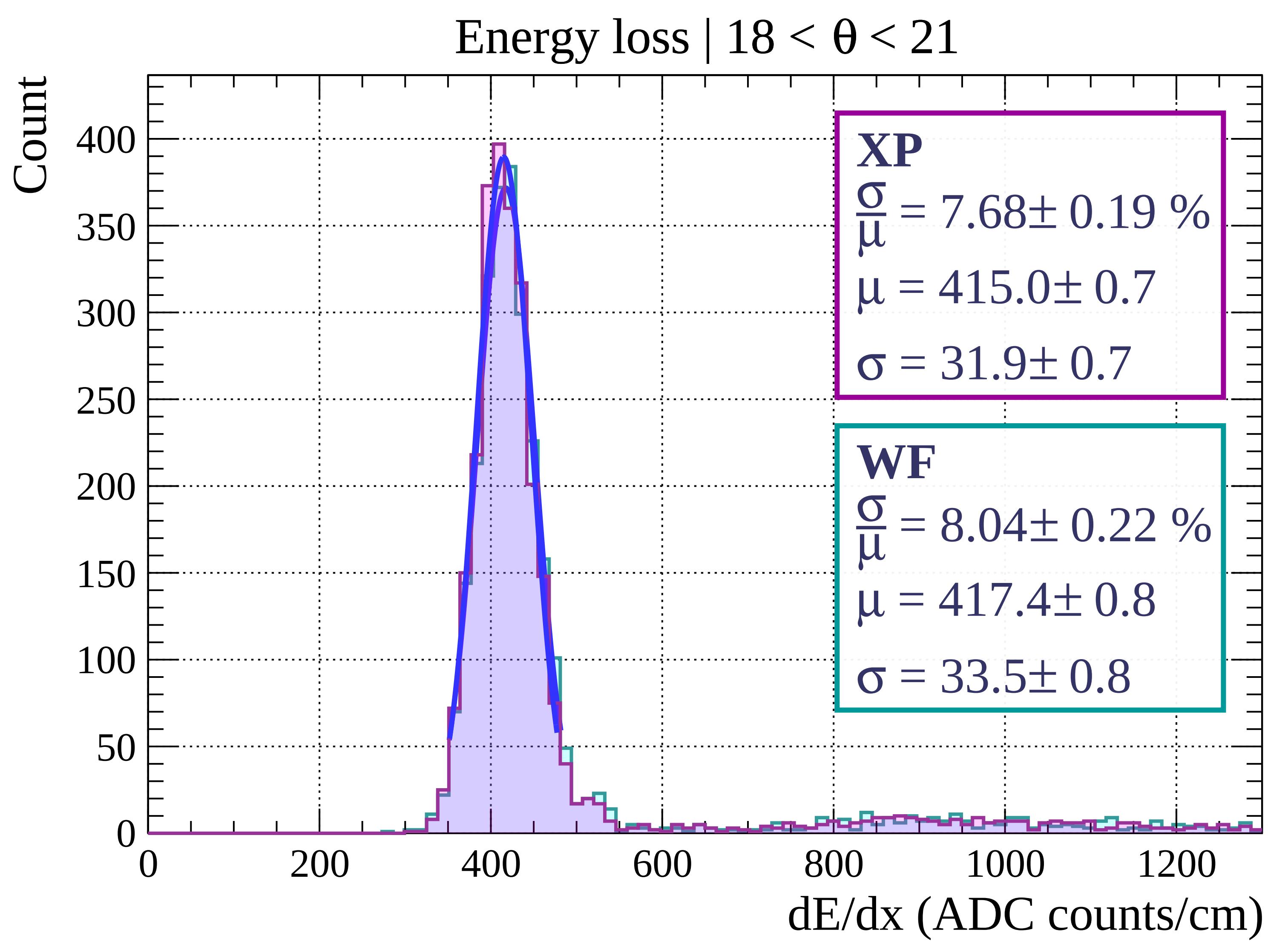
Energy loss | 6 < θ < 9
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.32± 0.13 %
μ = 417.7± 0.5
σ = 30.6± 0.5
WF
σ/μ = 7.37± 0.14 %
μ = 421.4± 0.5
σ = 31.1± 0.5



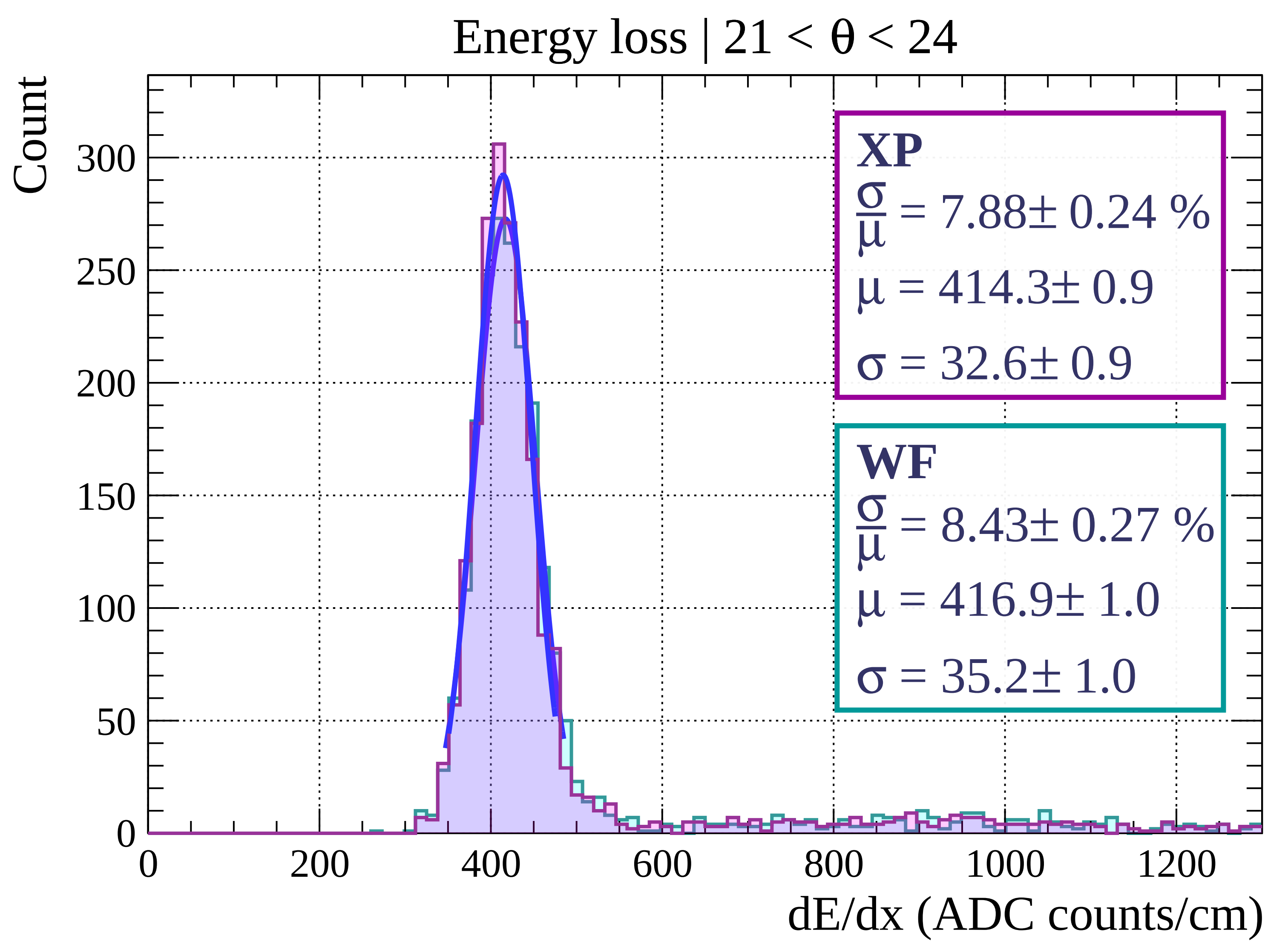
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.32± 0.12 %
μ = 417.8± 0.5
σ = 30.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.56± 0.15 %
μ = 420.4± 0.6
σ = 31.8± 0.6


Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.27± 0.13 %
μ = 417.5± 0.6
σ = 30.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.43± 0.17 %
μ = 419.7± 0.6
σ = 31.2± 0.6

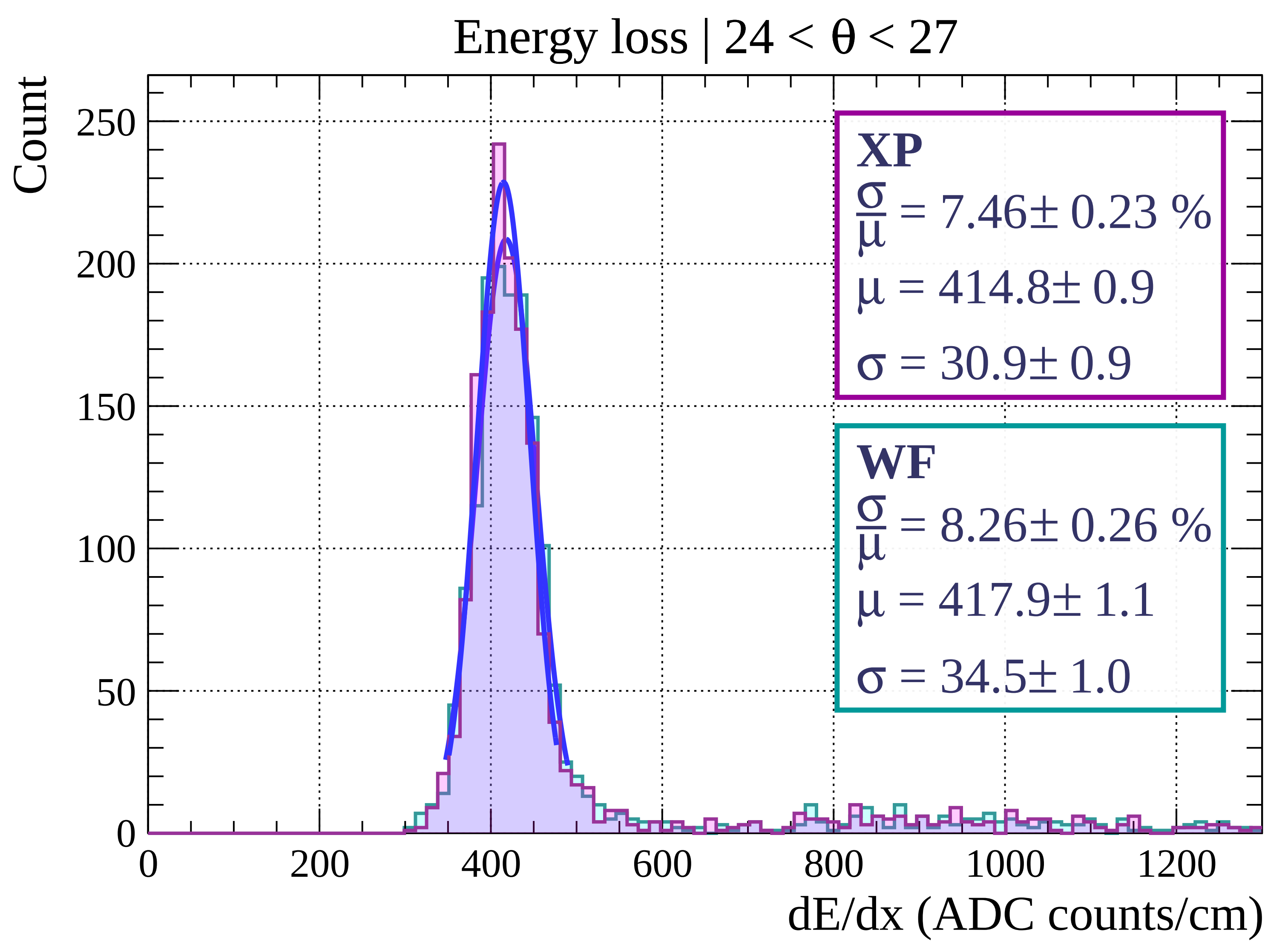

Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.60± 0.17 %
μ = 416.2± 0.7
σ = 31.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.69± 0.18 %
μ = 418.9± 0.7
σ = 32.2± 0.7

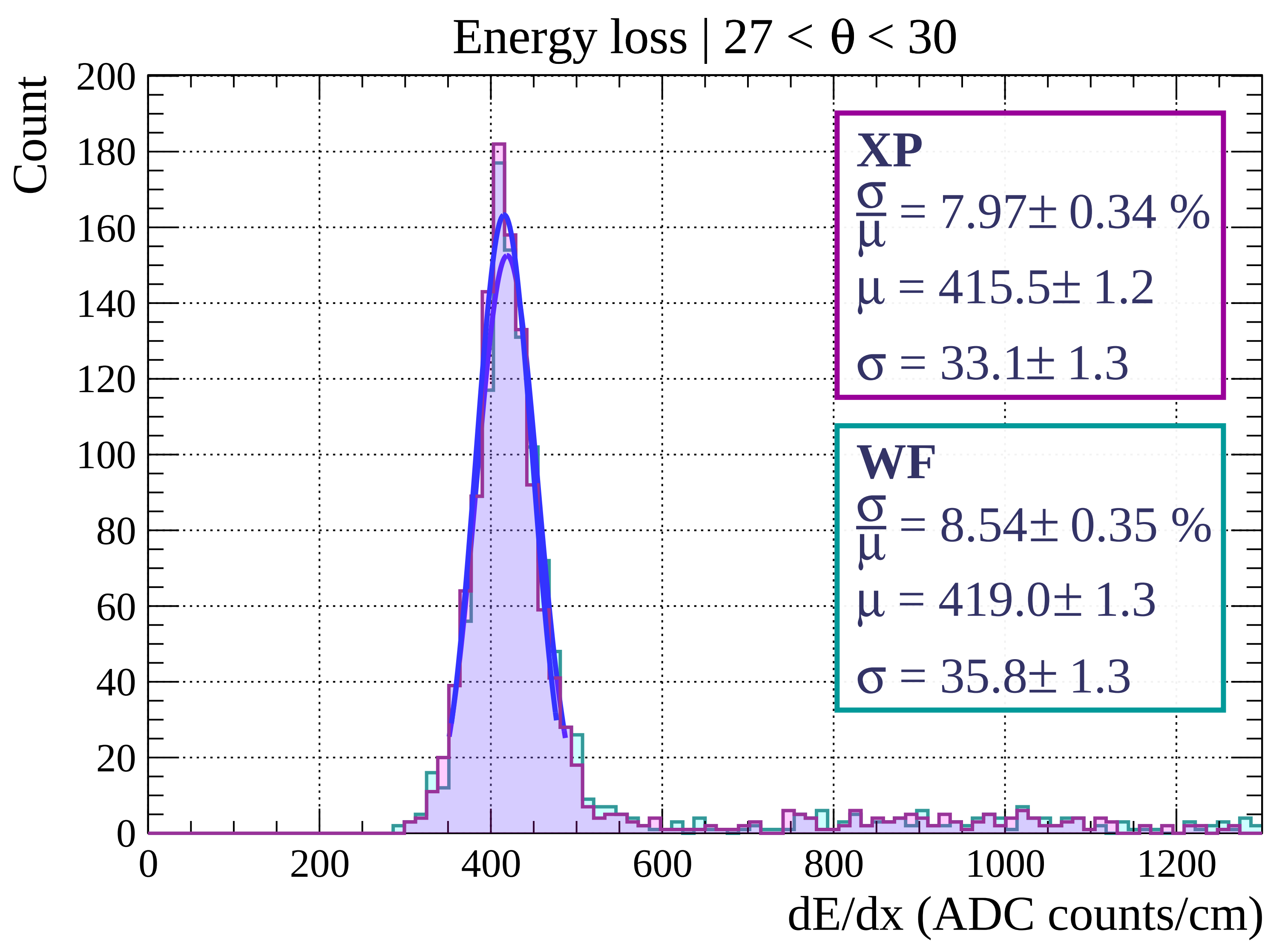


Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.68± 0.19 %
μ = 415.0± 0.7
σ = 31.9± 0.7
WF
σ/μ = 8.04± 0.22 %
μ = 417.4± 0.8
σ = 33.5± 0.8

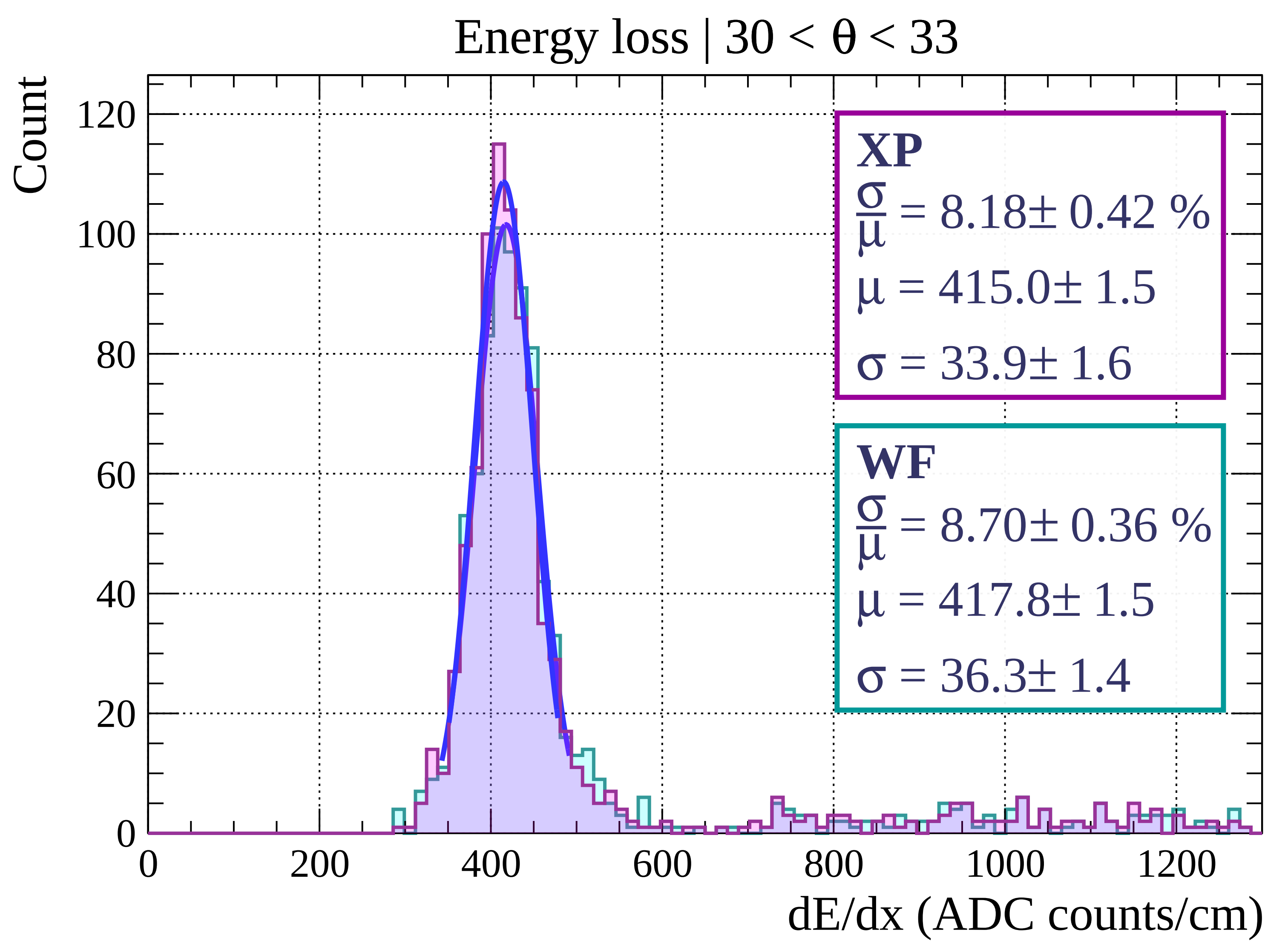


Energy loss | 21 < θ < 24
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.88± 0.24 %
μ = 414.3± 0.9
σ = 32.6± 0.9
WF
σ/μ = 8.43± 0.27 %
μ = 416.9± 1.0
σ = 35.2± 1.0

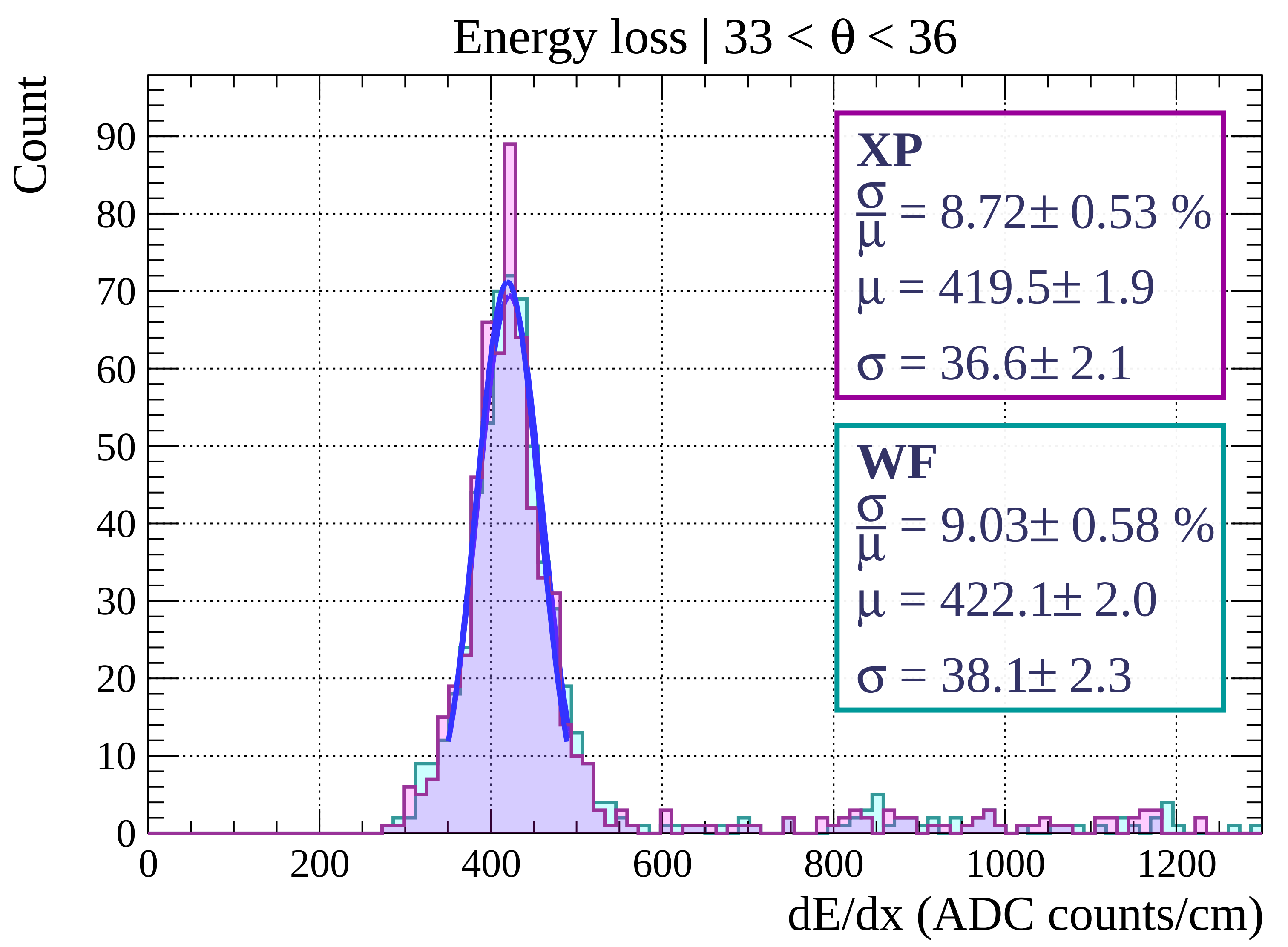

Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
σ/μ = 7.46± 0.23 %
μ = 414.8± 0.9
σ = 30.9± 0.9
WF
σ/μ = 8.26± 0.26 %
μ = 417.9± 1.1
σ = 34.5± 1.0

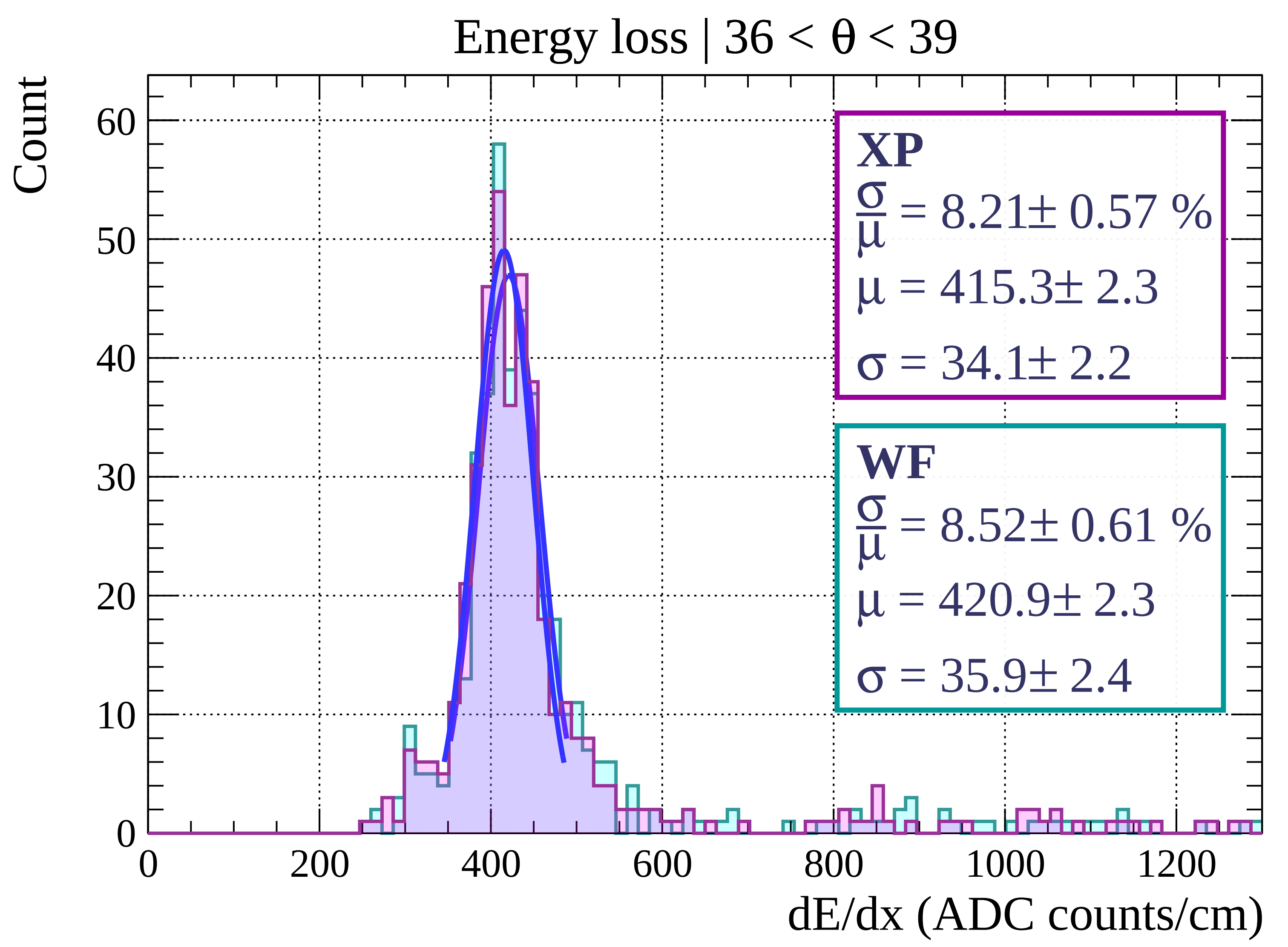


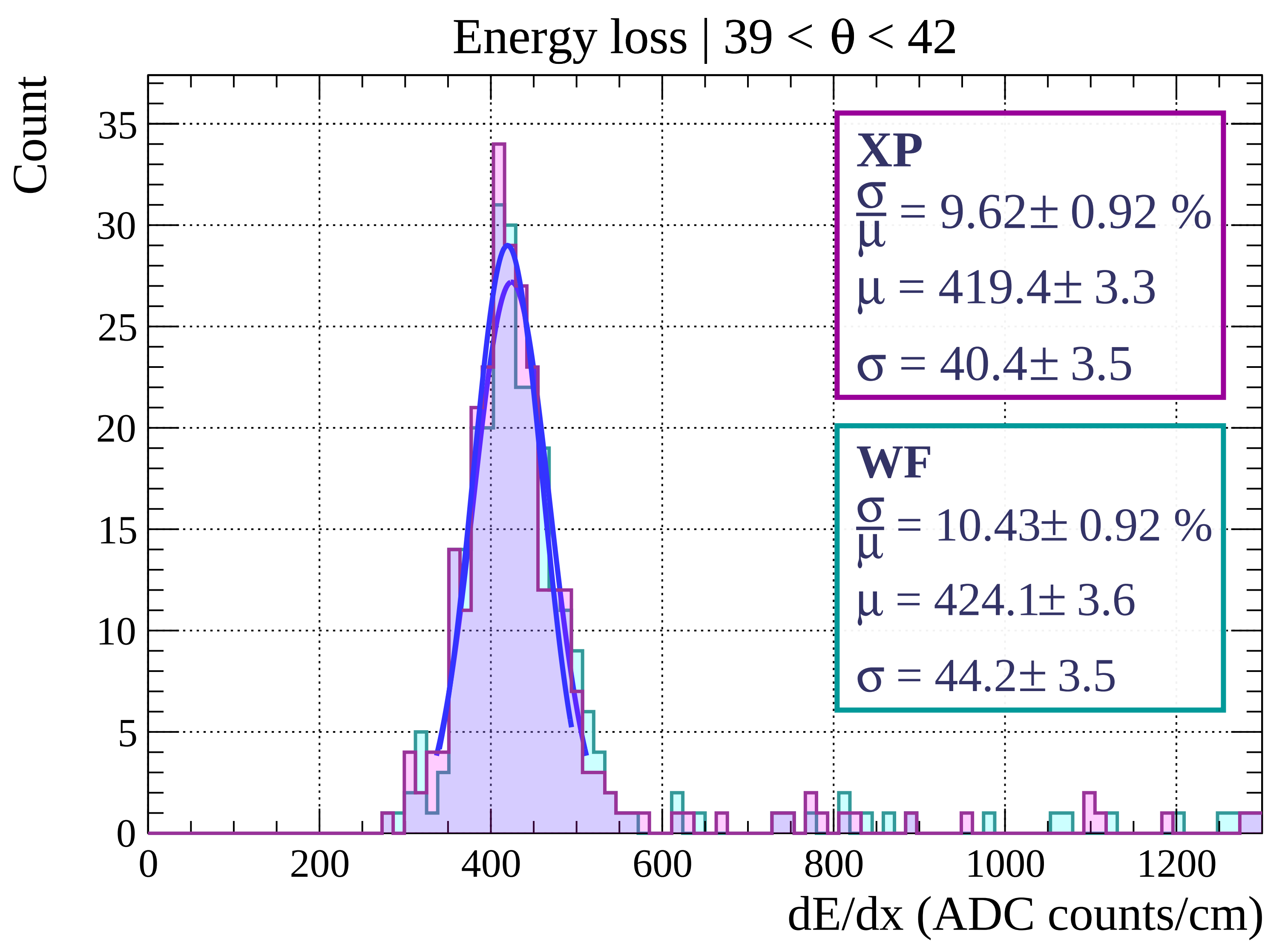
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.97± 0.34 %
μ = 415.5± 1.2
σ = 33.1± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.54± 0.35 %
μ = 419.0± 1.3
σ = 35.8± 1.3

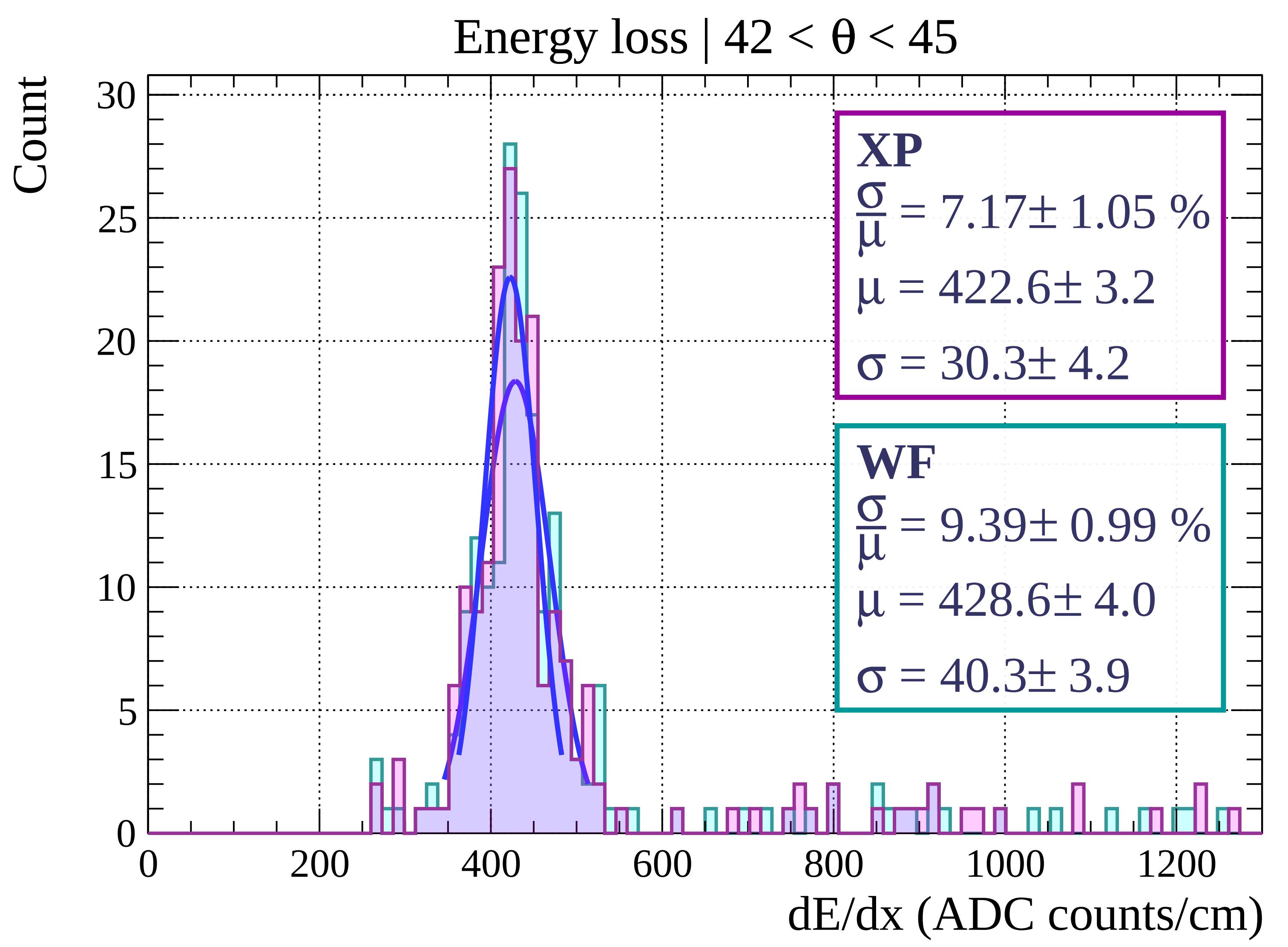


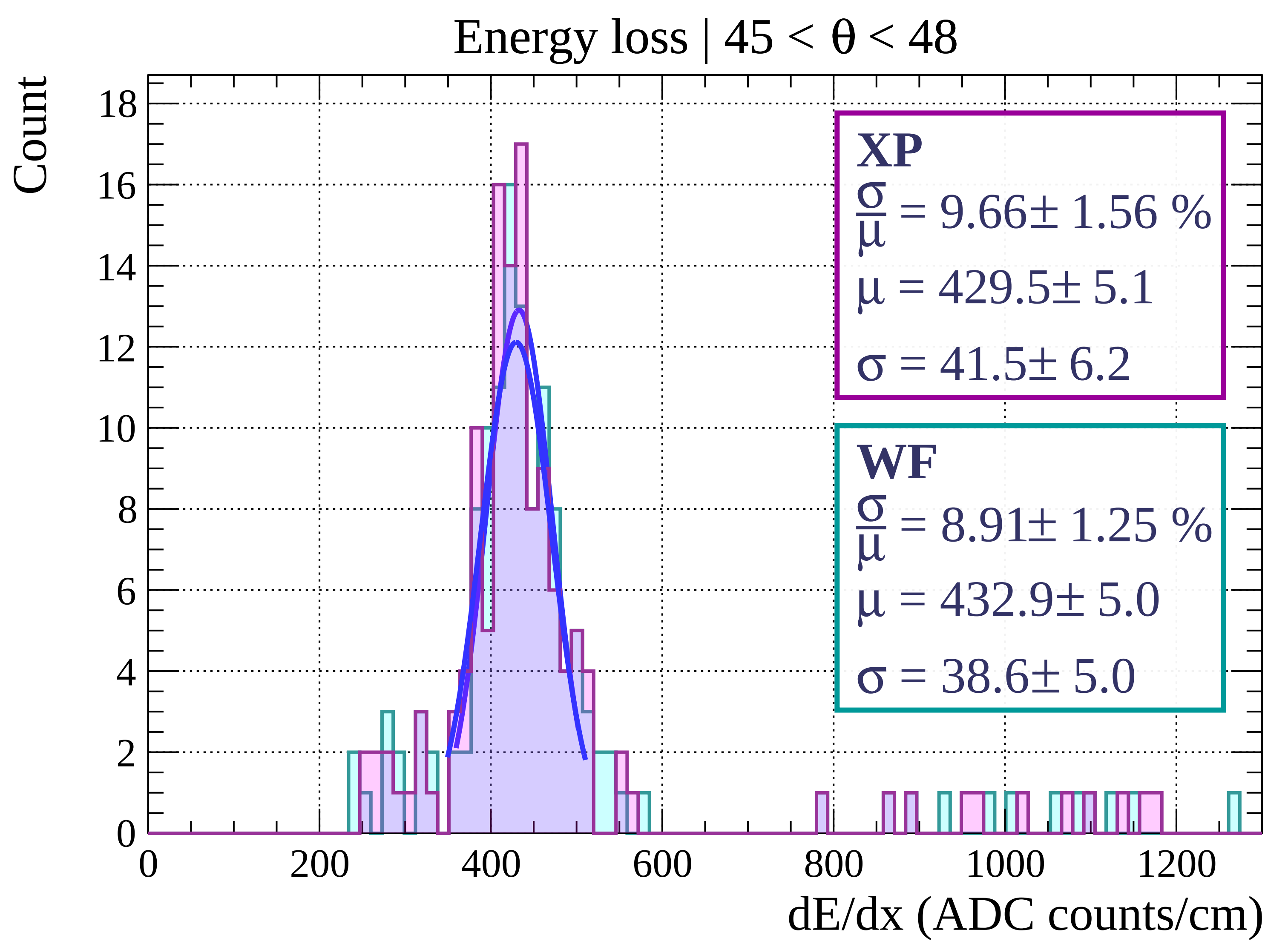
Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.18± 0.42 %
μ = 415.0± 1.5
σ = 33.9± 1.6
WF
σ/μ = 8.70± 0.36 %
μ = 417.8± 1.5
σ = 36.3± 1.4


Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 8.72± 0.53 %
μ = 419.5± 1.9
σ = 36.6± 2.1
WF
σ/μ = 9.03± 0.58 %
μ = 422.1± 2.0
σ = 38.1± 2.3

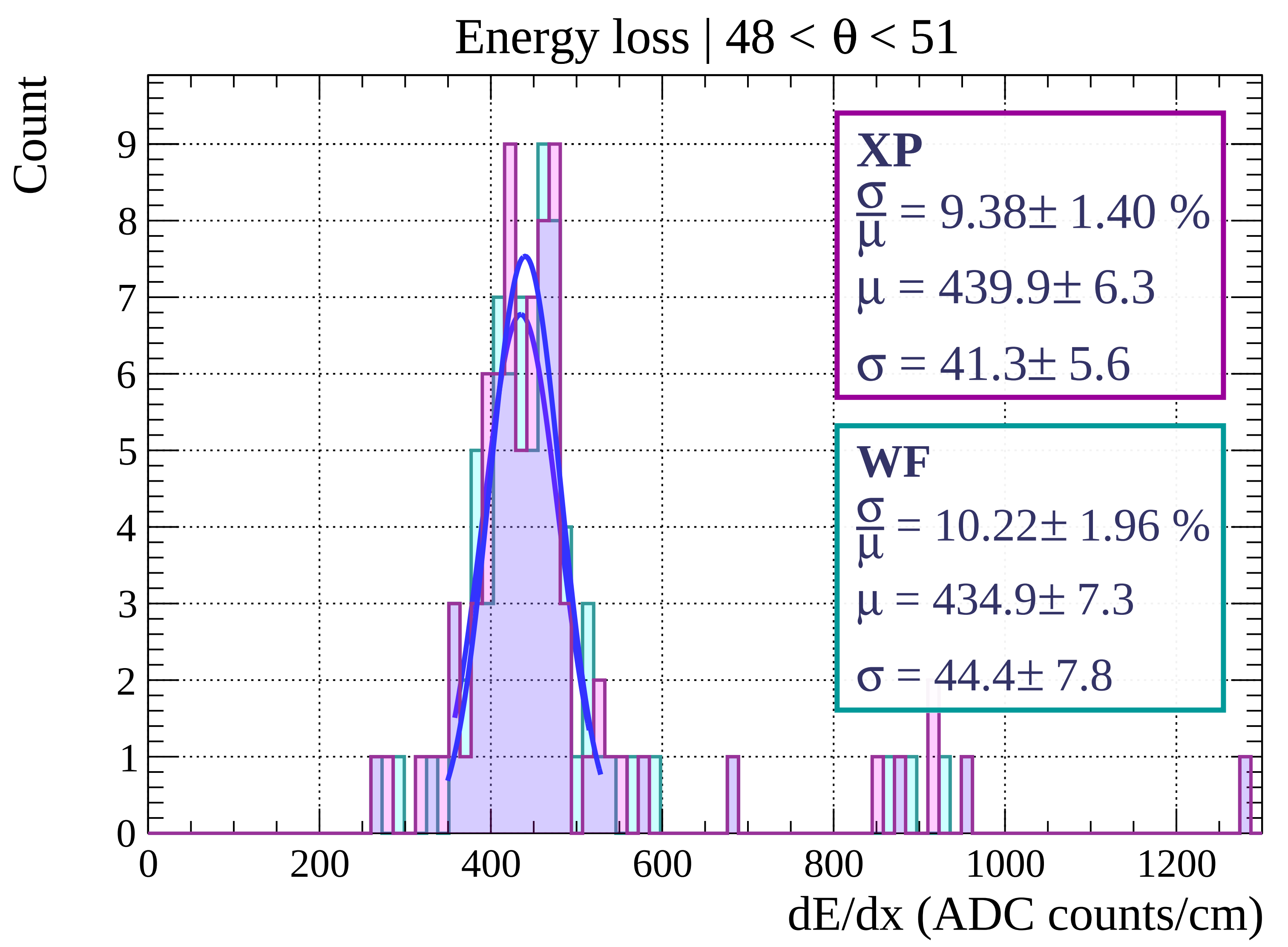

Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 8.21± 0.57 %
μ = 415.3± 2.3
σ = 34.1± 2.2
WF
σ/μ = 8.52± 0.61 %
μ = 420.9± 2.3
σ = 35.9± 2.4


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 9.62± 0.92 %
μ = 419.4± 3.3
σ = 40.4± 3.5
WF
σ/μ = 10.43± 0.92 %
μ = 424.1± 3.6
σ = 44.2± 3.5

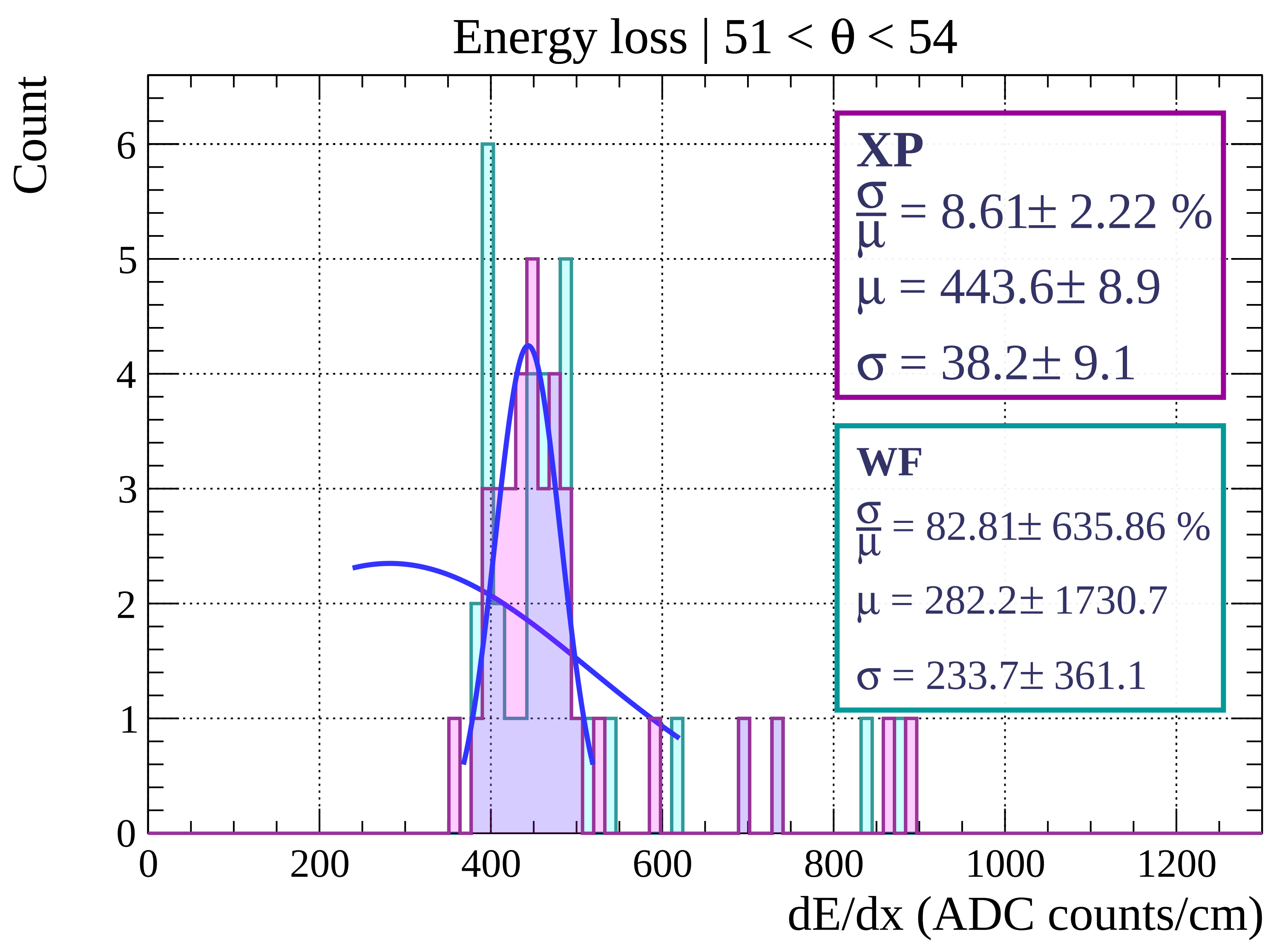

Energy loss | 42 < θ < 45
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.17± 1.05 %
μ = 422.6± 3.2
σ = 30.3± 4.2
WF
σ/μ = 9.39± 0.99 %
μ = 428.6± 4.0
σ = 40.3± 3.9
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200



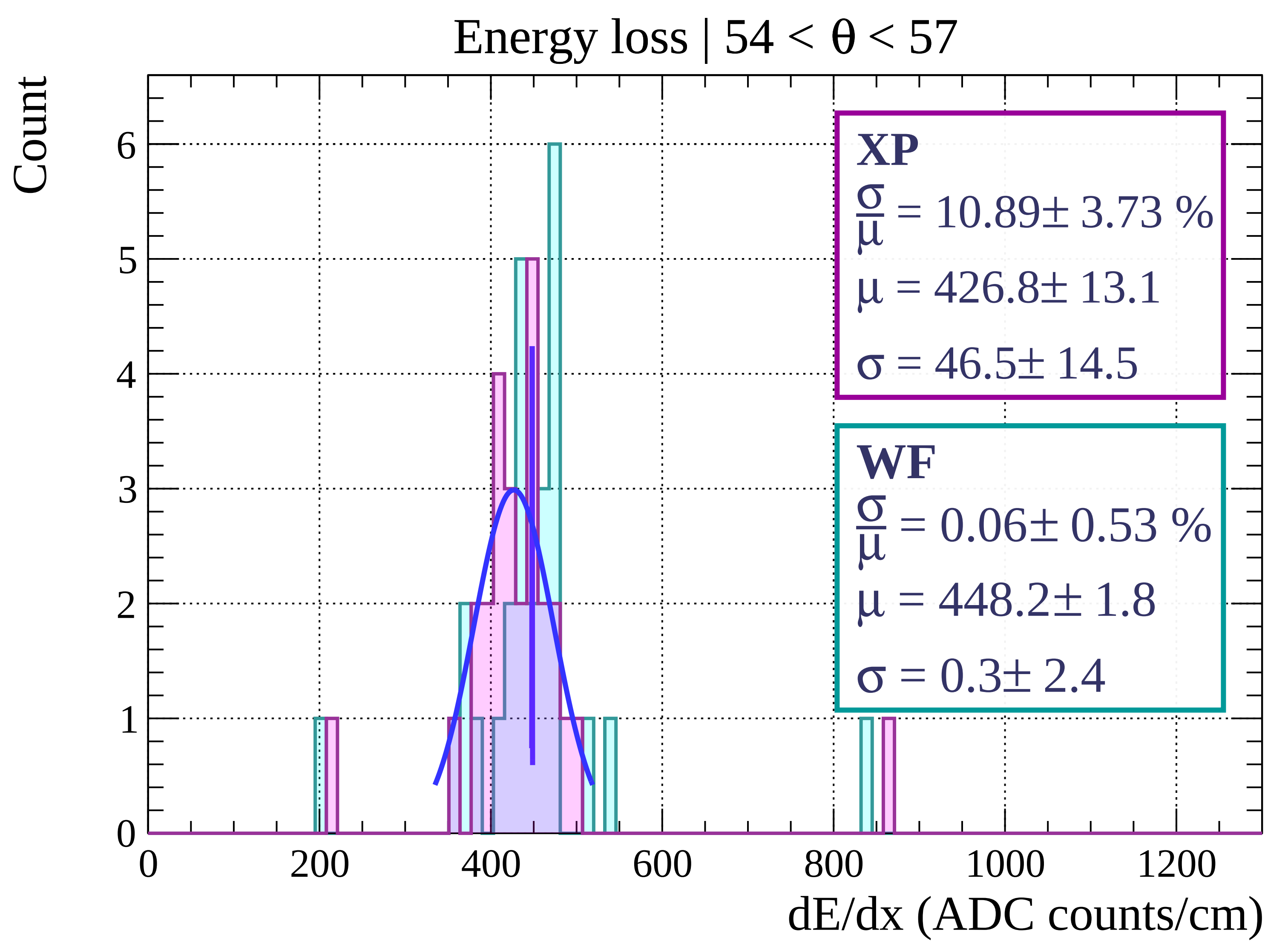
Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
$\frac{\sigma}{\mu}$ = 9.66± 1.56 %
μ = 429.5± 5.1
σ = 41.5± 6.2
WF
$\frac{\sigma}{\mu}$ = 8.91± 1.25 %
μ = 432.9± 5.0
σ = 38.6± 5.0



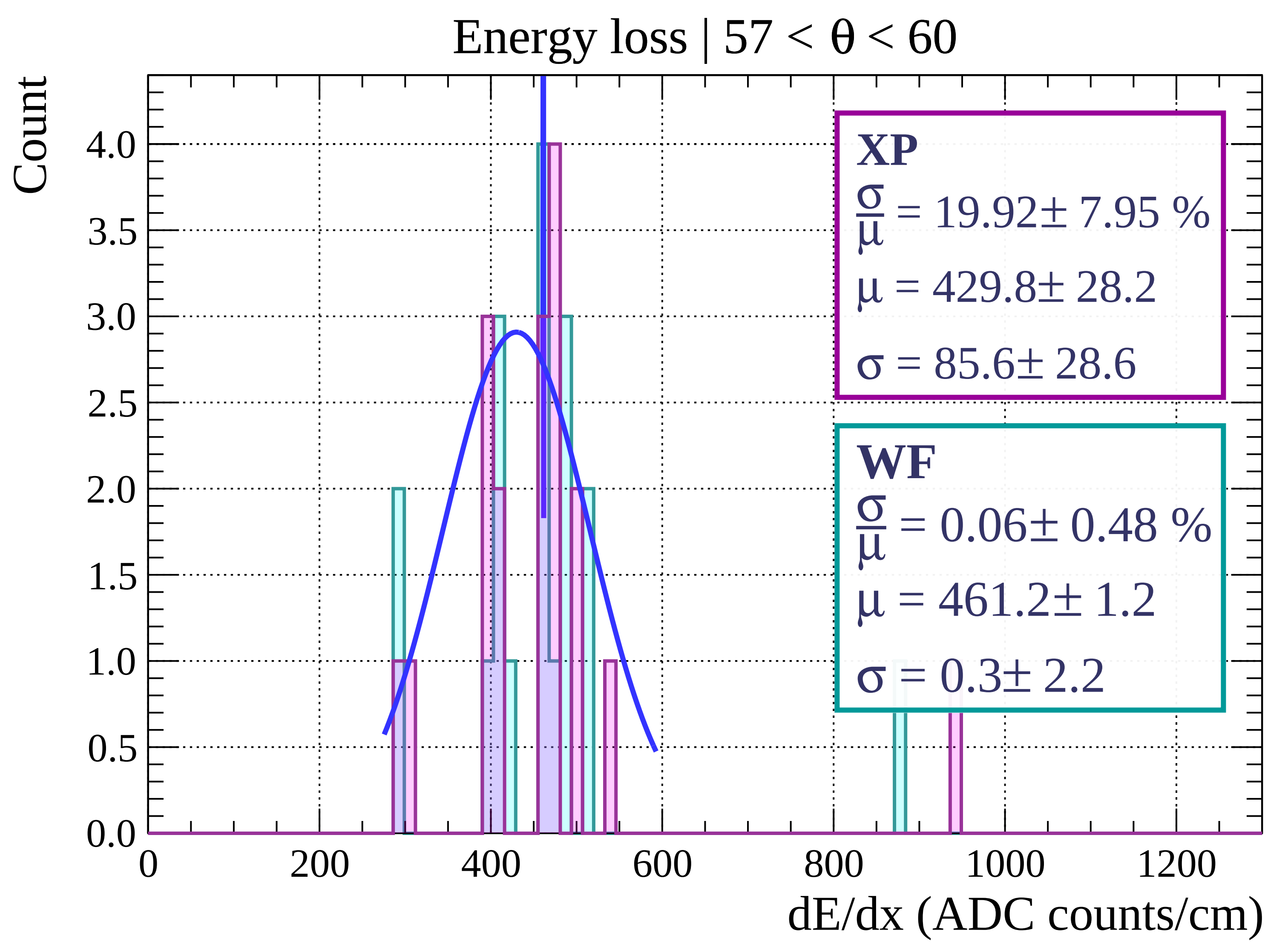
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 9.38± 1.40 %
μ = 439.9± 6.3
σ = 41.3± 5.6
WF
σ/μ = 10.22± 1.96 %
μ = 434.9± 7.3
σ = 44.4± 7.8



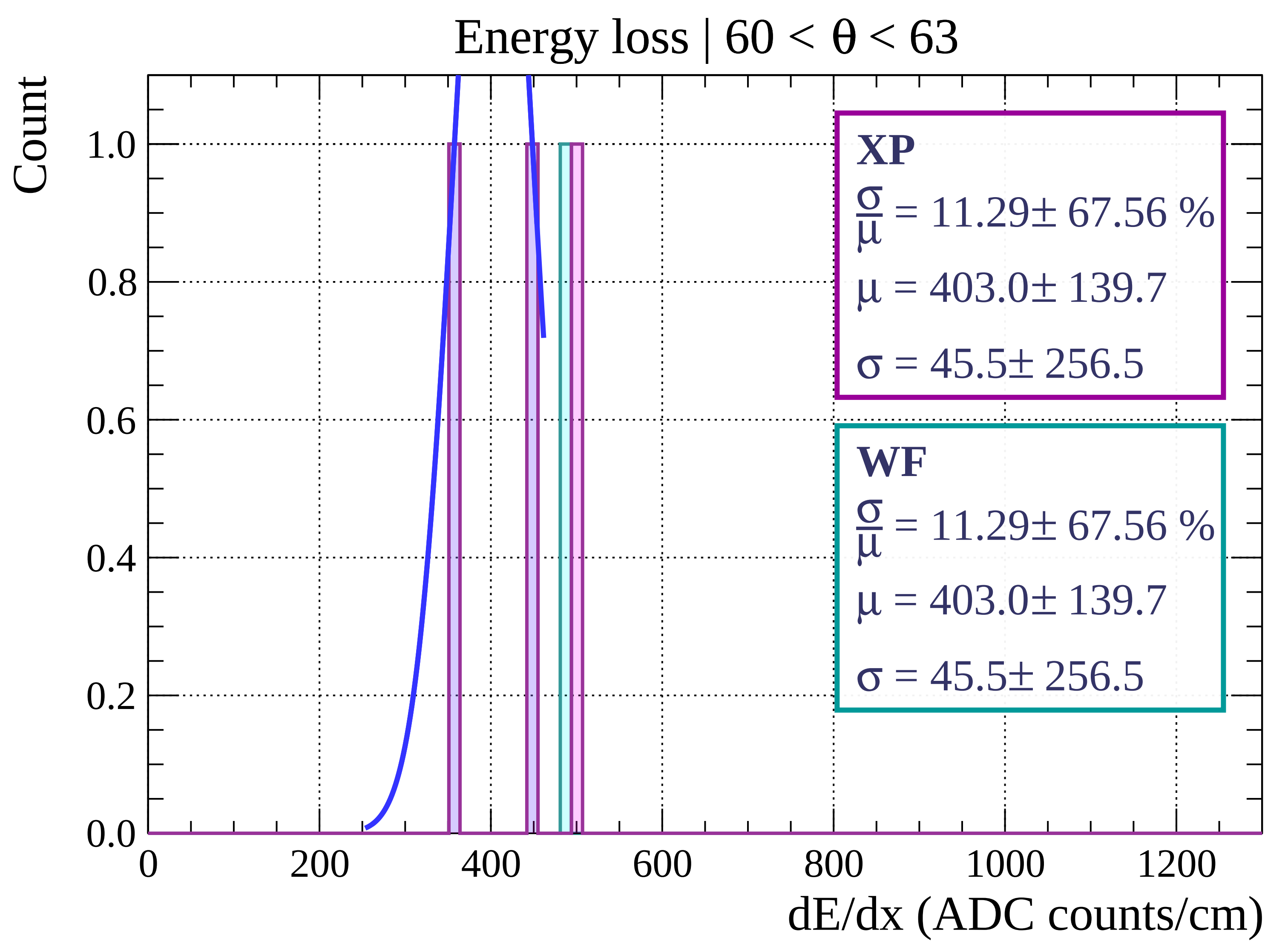
Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 8.61± 2.22 %
μ = 443.6± 8.9
σ = 38.2± 9.1
WF
σ/μ = 82.81± 635.86 %
μ = 282.2± 1730.7
σ = 233.7± 361.1



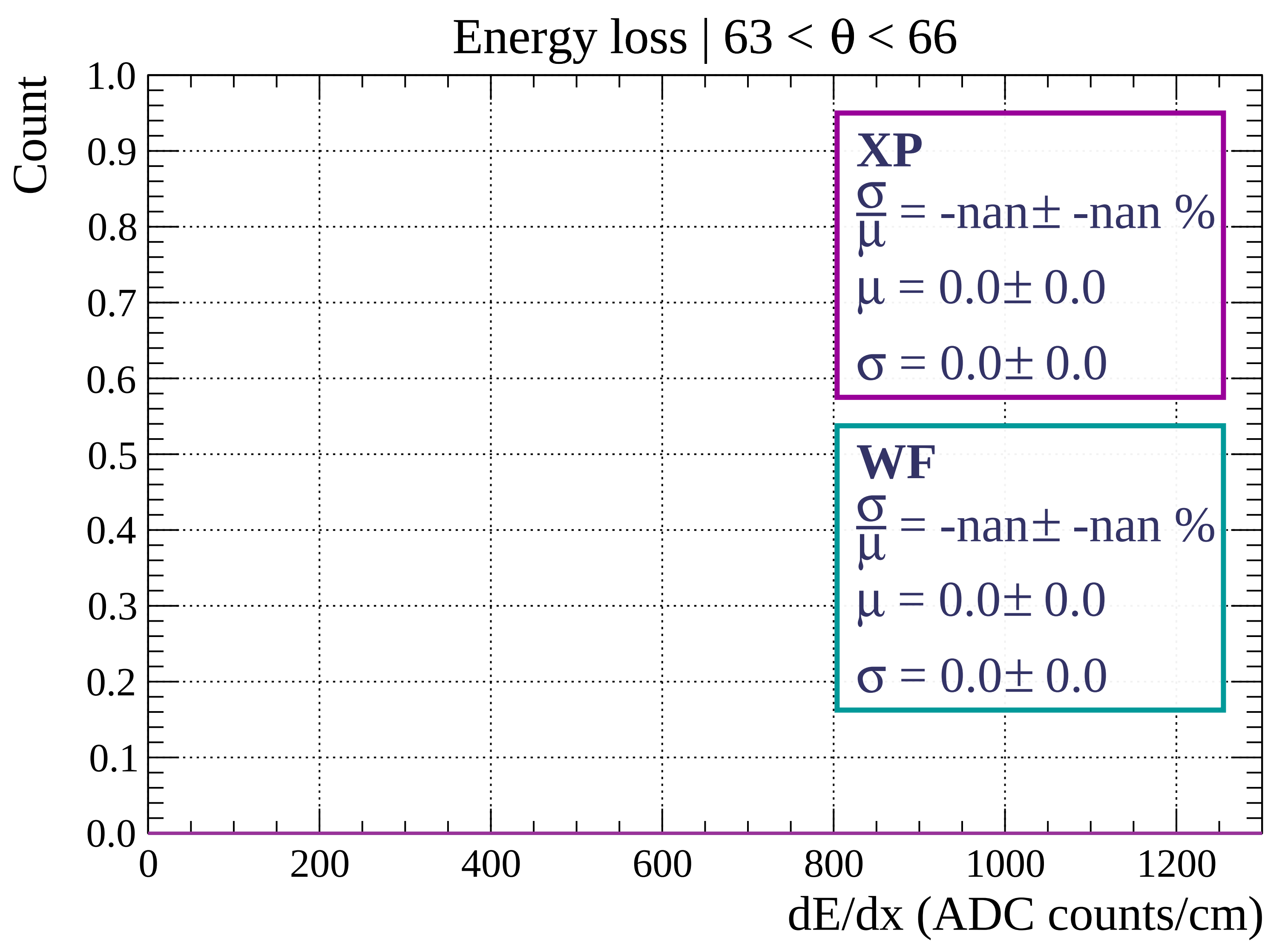
Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 10.89± 3.73 %
μ = 426.8± 13.1
σ = 46.5± 14.5
WF
σ/μ = 0.06± 0.53 %
μ = 448.2± 1.8
σ = 0.3± 2.4

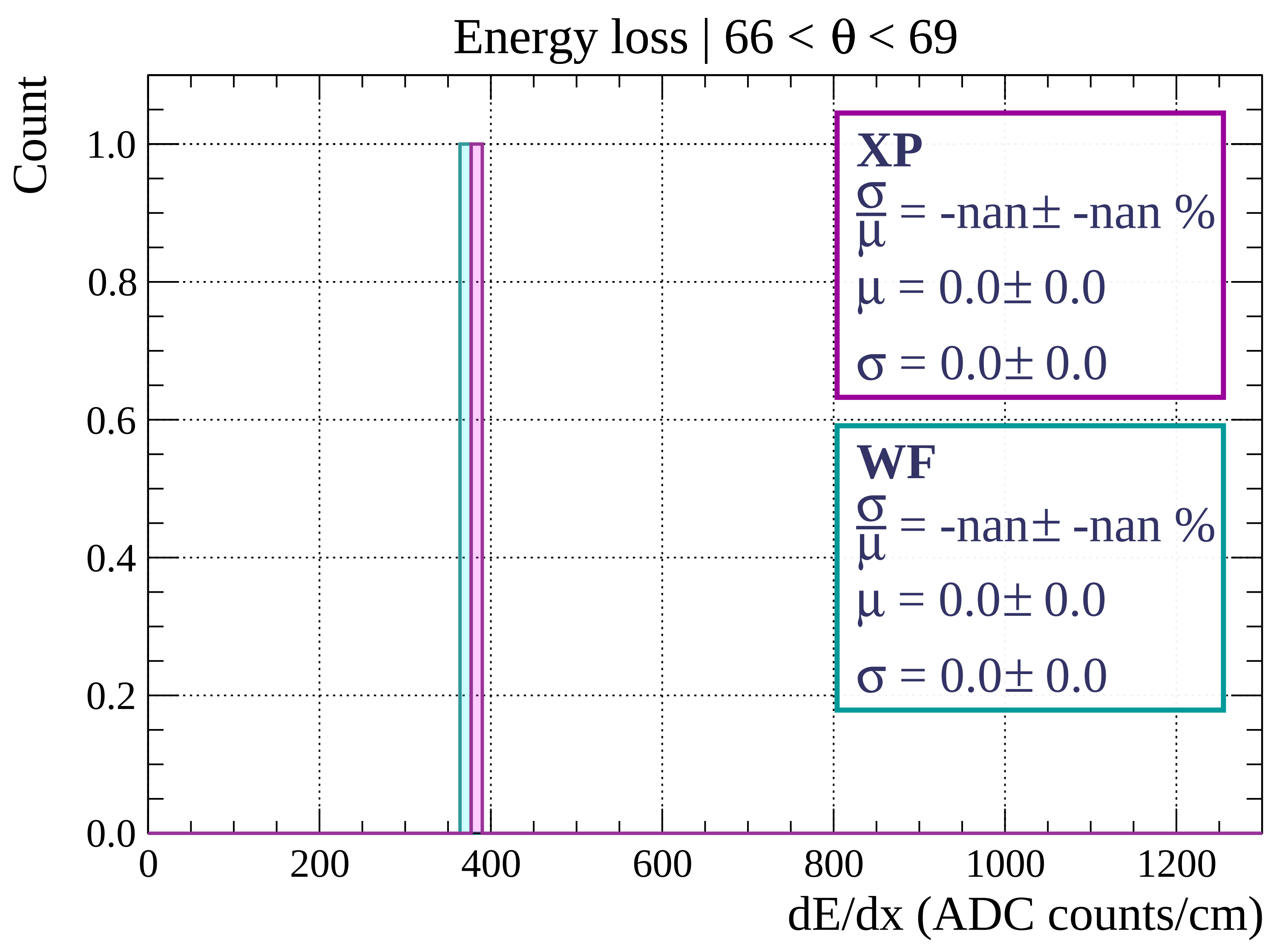


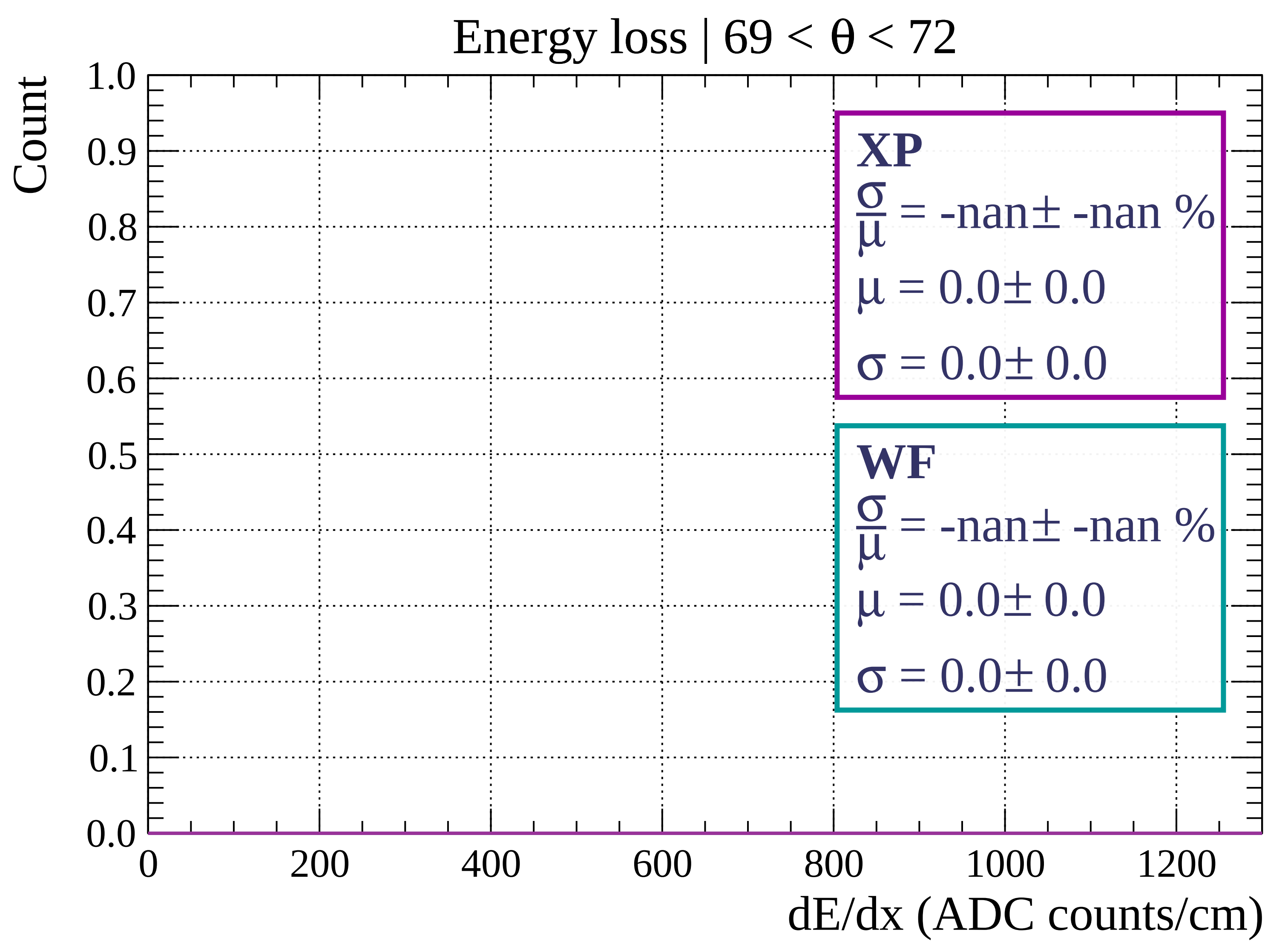
Energy loss | 57 < θ < 60
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 19.92± 7.95 %
μ = 429.8± 28.2
σ = 85.6± 28.6
WF
σ/μ = 0.06± 0.48 %
μ = 461.2± 1.2
σ = 0.3± 2.2



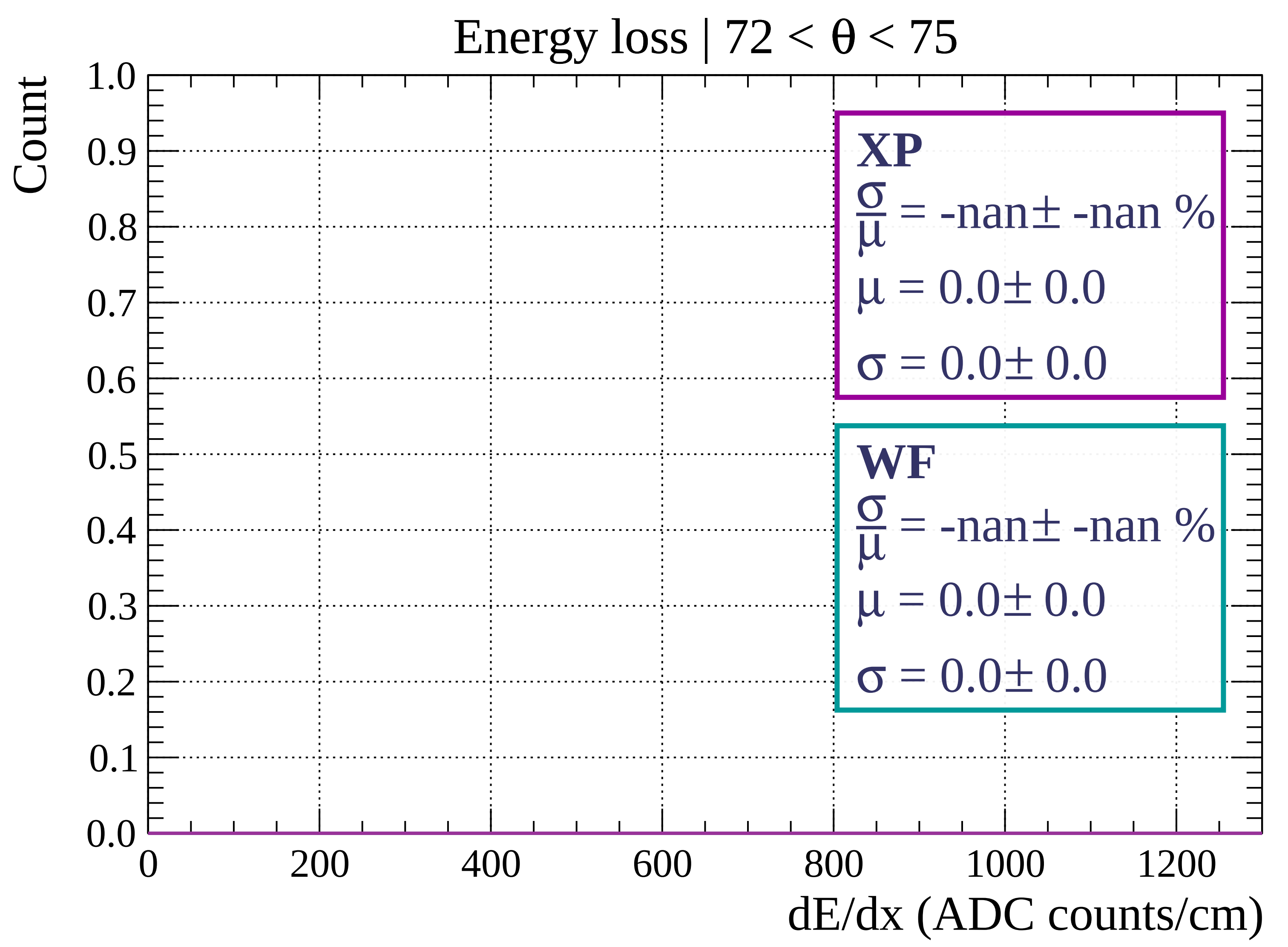
Energy loss | 60 < θ < 63
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.29± 67.56 %
μ = 403.0± 139.7
σ = 45.5± 256.5
WF
σ/μ = 11.29± 67.56 %
μ = 403.0± 139.7
σ = 45.5± 256.5


Energy loss | 63 < θ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

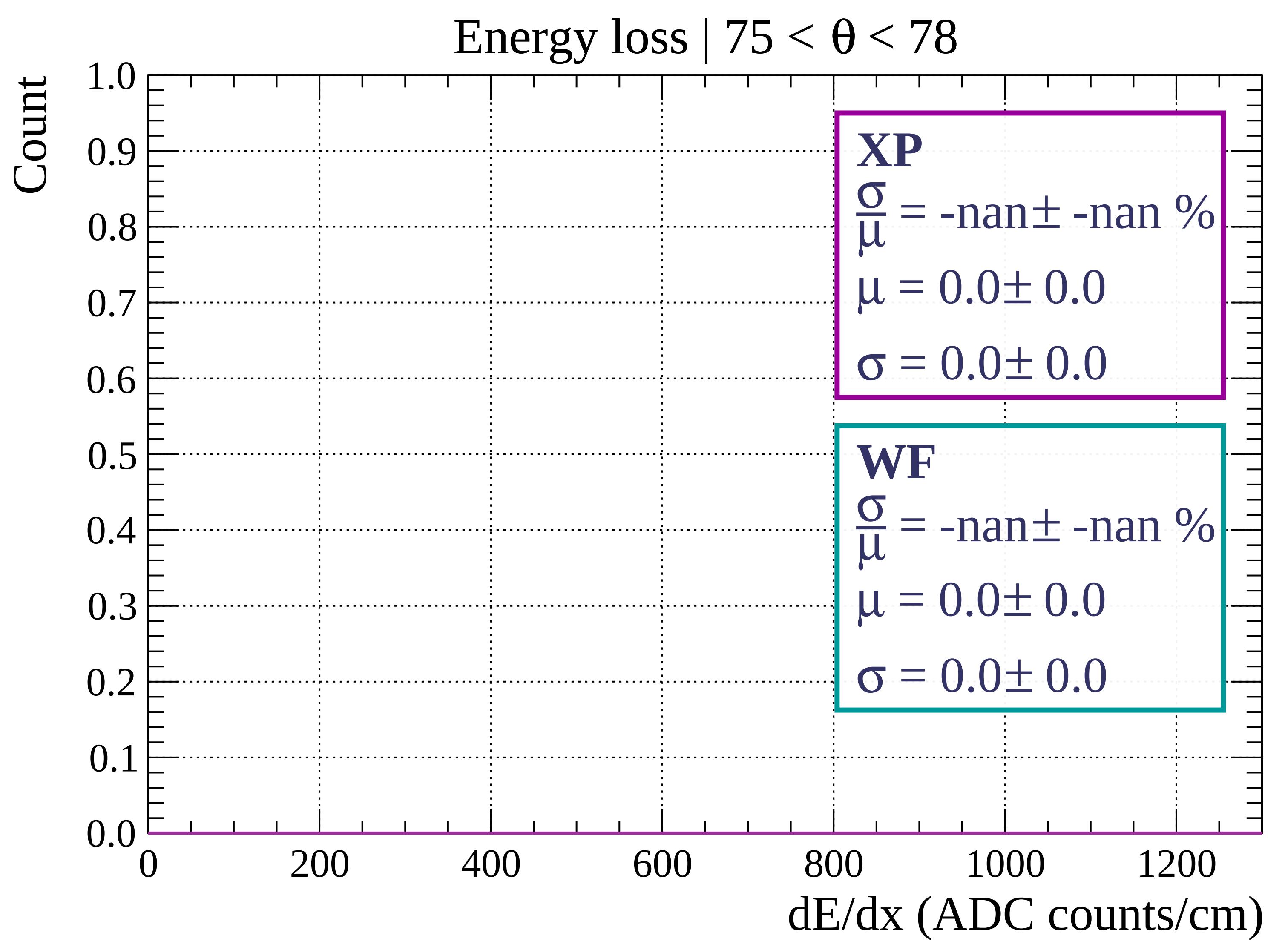

Energy loss | 66 < θ < 69
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

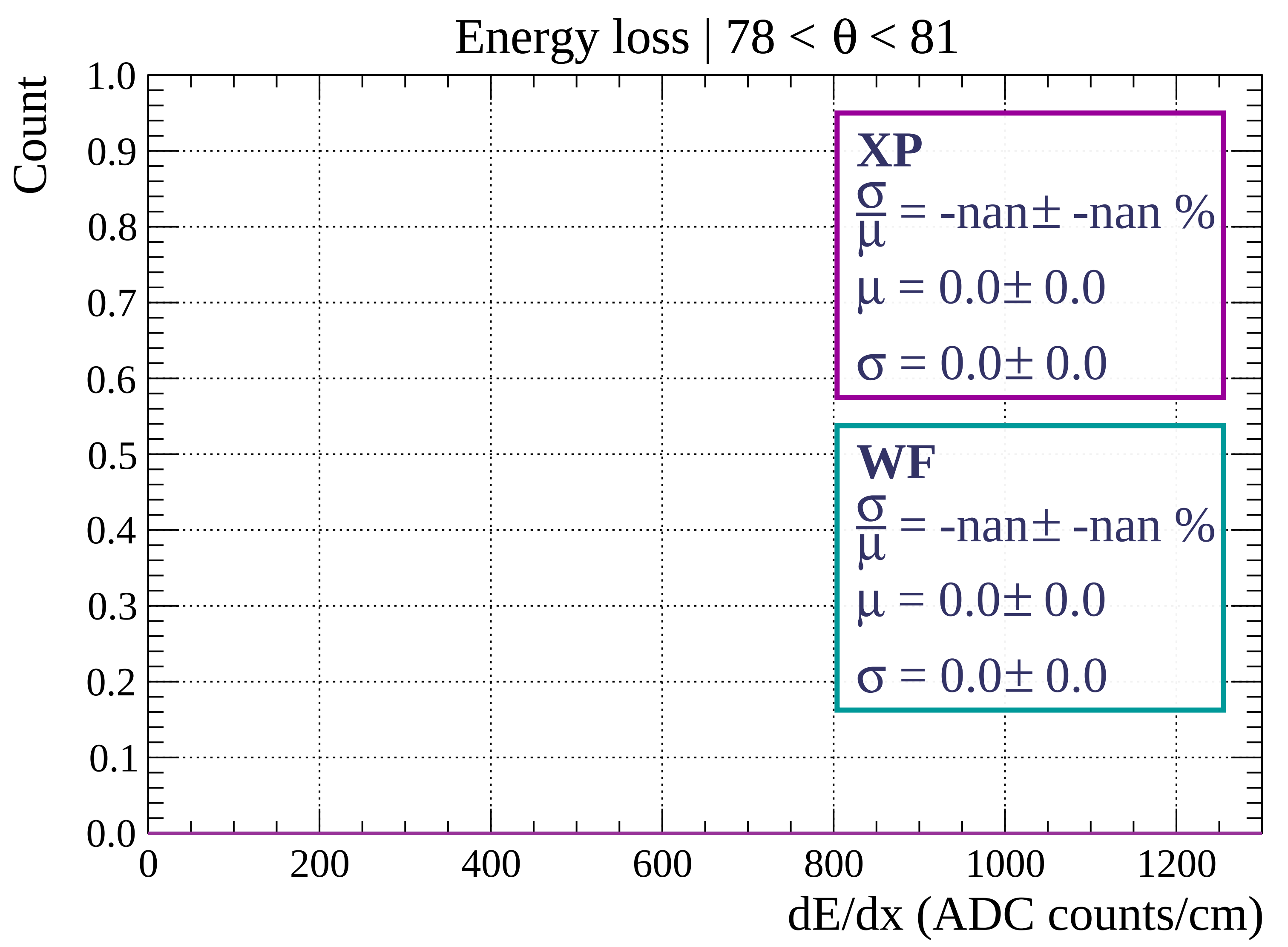


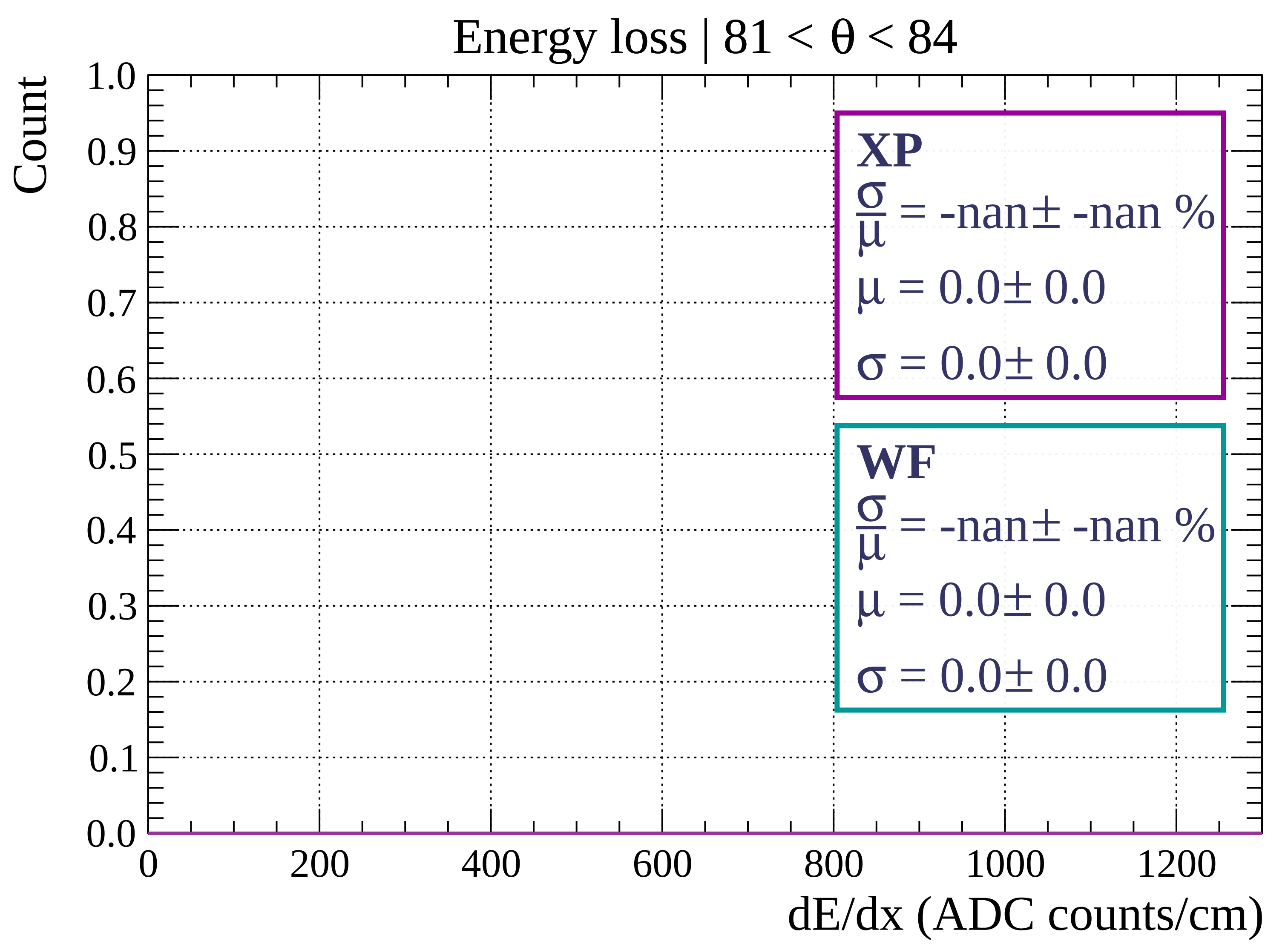
Energy loss | 69 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

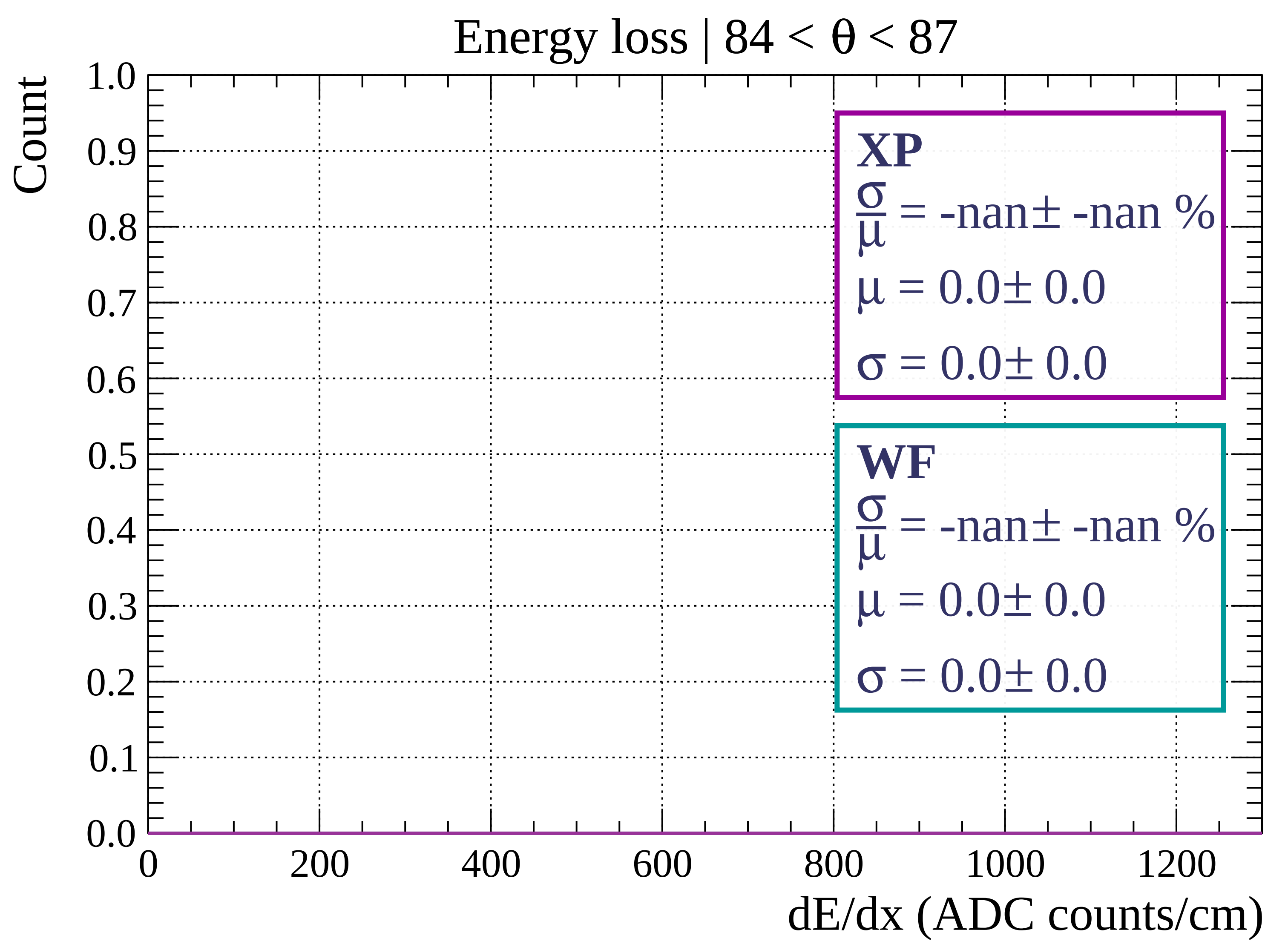


Energy loss | 72 < θ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

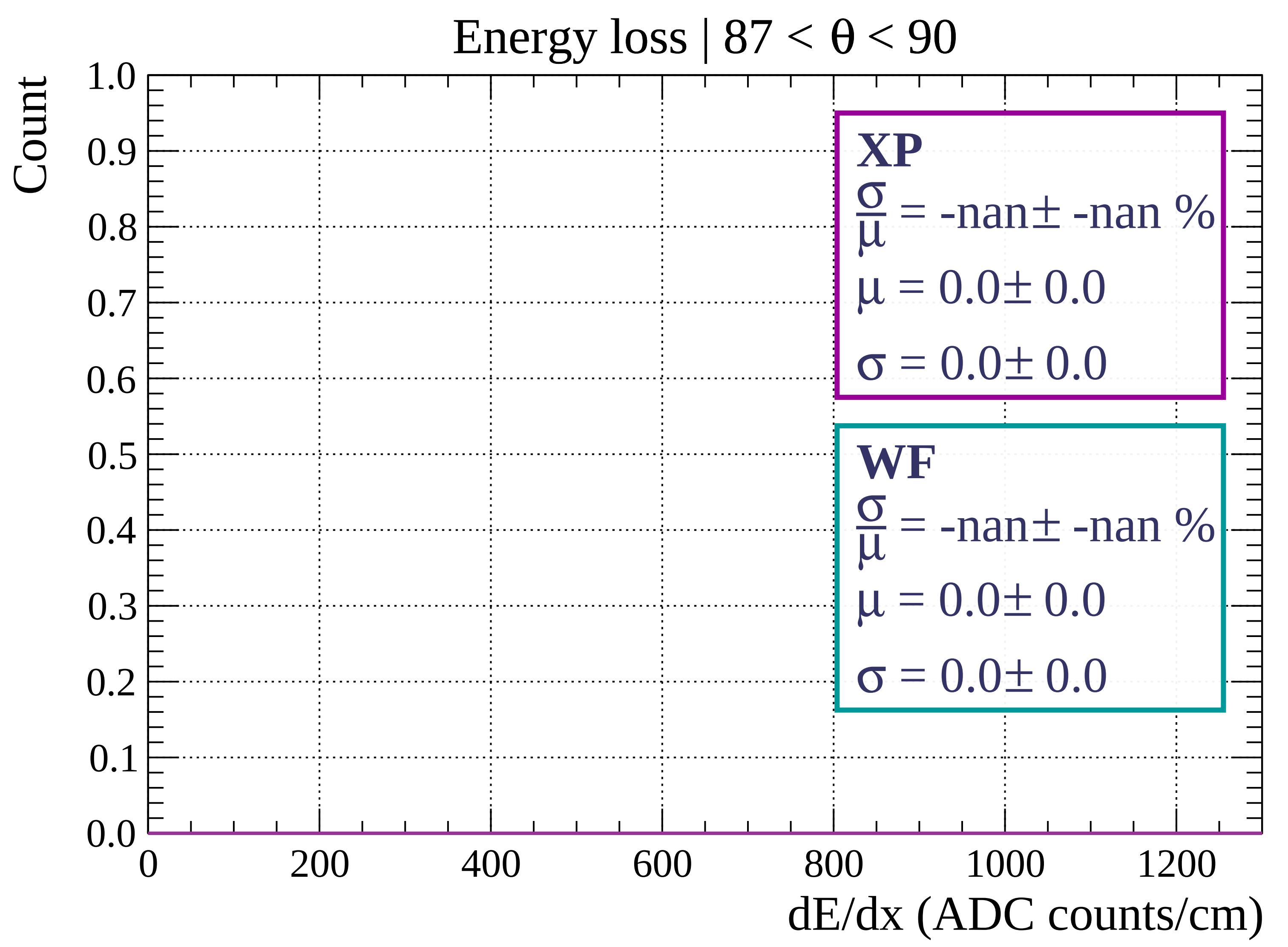

Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 78 < θ < 81
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

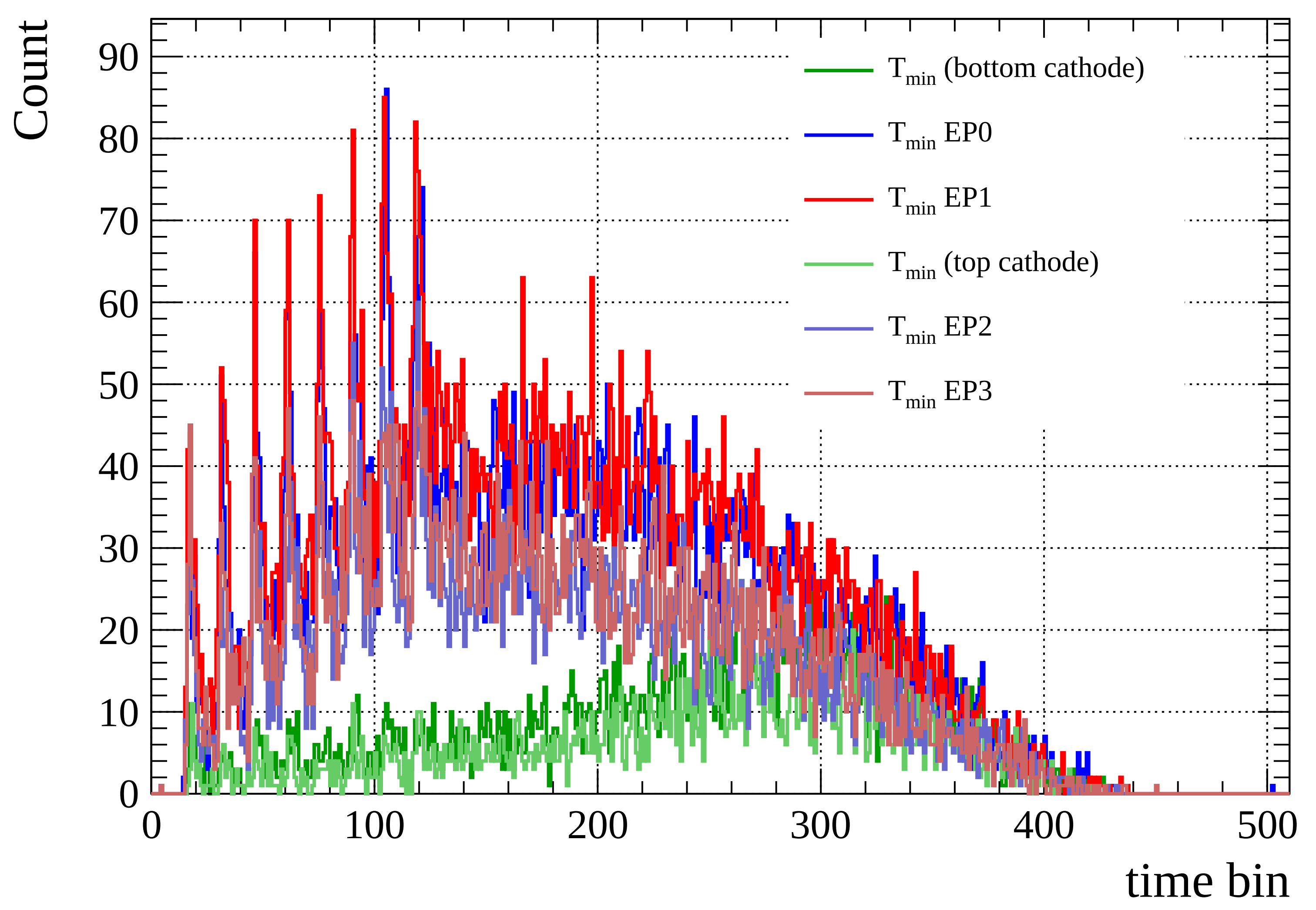

Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



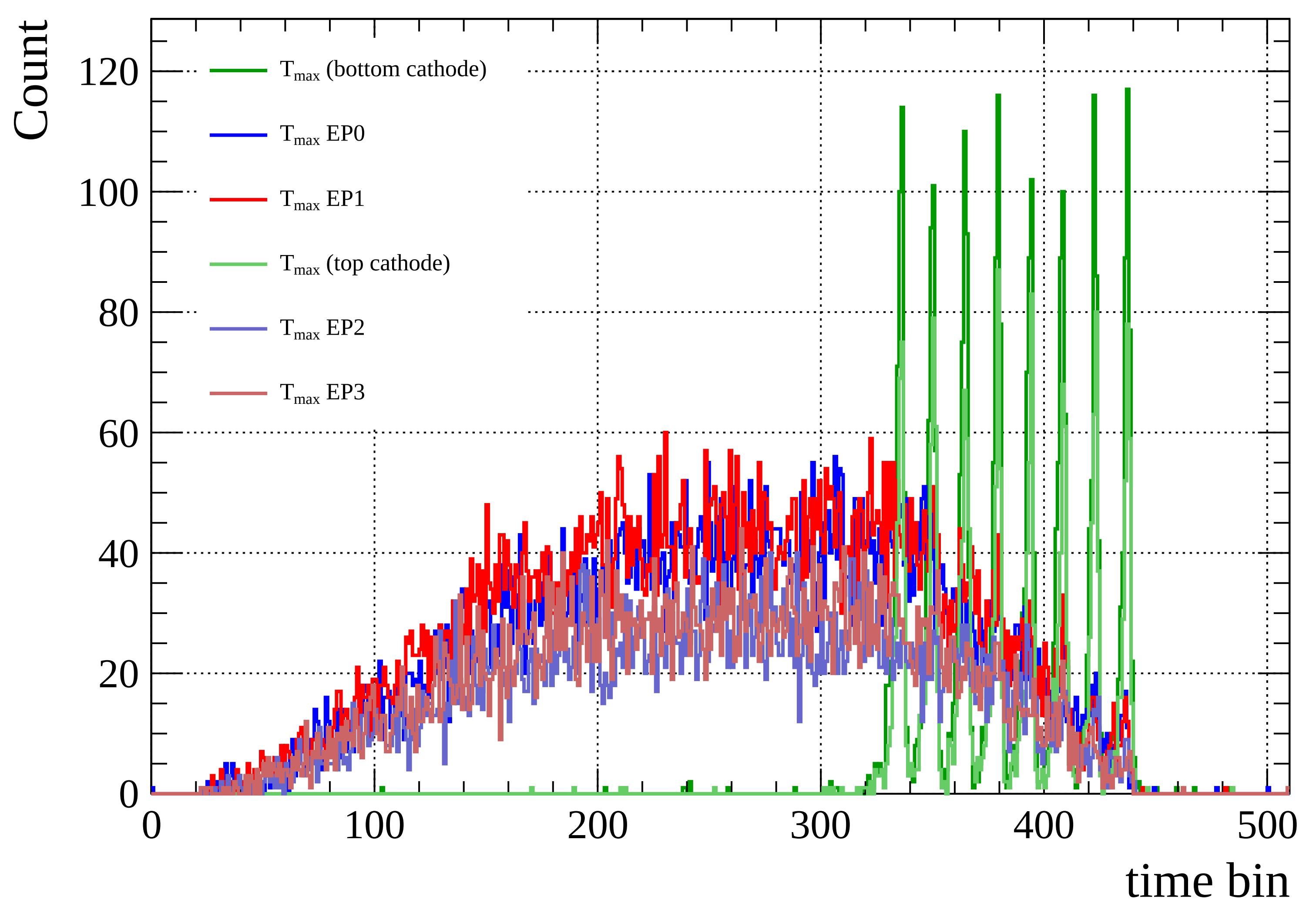
Energy loss | 84 < θ < 87
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 87 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3
0
10
20
30
40
50
60
70
80
90
0
100
200
300
400
500



Count
time bin
0
20
40
60
80
100
120
0
100
200
300
400
500
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3